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Latin *-tās* and Related Forms

A dissertation submitted in partial satisfaction of the

requirements for the degree Doctor of Philosophy

in Indo-European Studies

by

Moss Pike

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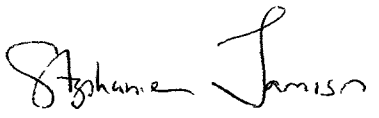
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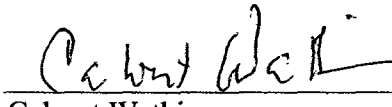
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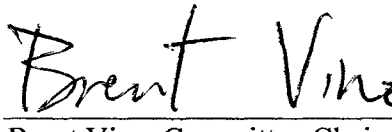
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LIST OF ABBREVIATIONS

Languages

Av.	Avestan	OE	Old English
Boeot.	Boeotian Greek	OFr.	Old Frisian
CLat.	Classical Latin	OHG	Old High German
Dor.	Doric Greek	OIr.	Old Irish
Etr.	Etruscan	OIc.	Old Icelandic
Fal.	Faliscan	OKh.	Old Khotanese
Fr.	French	OLat.	Old Latin
Gaul.	Gaulish	ON	Old Norse
Gk.	Greek	Osc.	Oscan
Go.	Gothic	OPer.	Old Persian
Gmc.	Germanic	OPr.	Old Prussian
It.	Italian	PDE	Present Day English
Hitt.	Hittite	Pg.	Paelignian
Hom.	Homeric Greek	PGk.	Proto-Greek
Khot.	Khotanese	PIE	Proto-Indo-European
Lat.	Latin	Praen.	Praenestine Latin
Latv.	Latvian	PT	Proto-Tocharian
Lith.	Lithuanian	Skt.	Sanskrit (post-Vedic)
LKhj.	Late Khotanese	Sp.	Spanish
Luv.	Luvian	SP	South Picene
Mars.	Marsian	TA	Tocharian A
Mc.	Marucinian	TB	Tocharian B
MP	Middle Persian	Umb.	Umbrian
Myc.	Mycenaean	Ved.	Vedic Sanskrit
MW	Middle Welsh	VOL	Very Old Latin
NP	New Persian	W.	Welsh
OAv.	Old Avestan	YAv.	Young Avestan
OCS	Old Church Slavic		

Linguistic Terms

AS	acrostatic
AK	amphikinetic
PK	proterokinetic

Authors and works are cited after *OLD* (for Latin), *LSJ* (for Greek), Monier-Williams for Sanskrit, and Bartholomae for Avestan.

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Pike, Moss. 2009a. "Homeric ἀνδροτῆτα 'manliness' and PIE '*-*tāt*-'", Paper presented at the 140th Annual Meeting of the American Philological Association (Philadelphia, PA).

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———— 2008a. "Lat. *voluptās* 'pleasure'/*volup* 'agreeably', Gk. ἐλπίς 'hope', and PIE *$\underset{\sim}{\mu}elh_1$- 'choose'", Paper presented at the 27th East Coast Indo-European Conference (Athens, CA).

———— 2008b. Review of James Clackson and Geoffrey Horrocks, *The Blackwell History of the Latin Language* (Blackwell, 2007). *UCLA Bulletin of Indo-European Studies* 13:53–55.

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——— 2006. "Mycenaean *qe-te-jo* and the Greek verbal adjective in -τέος", Paper presented at the 25th East Coast Indo-European Conference (Columbus, OH).

——— 2005. "Ovid and His Oral Poetry: Intertextual Relations between Ovid and Oscan *Defixiones*", Paper presented at the 136th Annual Meeting of the American Philological Association (Boston, MA).

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ABSTRACT OF THE DISSERTATION

Latin *-tās* and Related Forms

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Latin builds qualitative abstracts in *-tās*, and similar forms are found in Gk. *-τηϛ*, Ved. *-tāt-/-tāti-*, and Av. *-tāt-*, suggesting that **-tāt-* belongs to PIE. Descriptively, **-tāt-* built abstracts primarily to thematic adjectives from an early date, with subsequent reanalysis of the thematic vowel as a part of the suffix, as reflected in abstracts built to consonant stems, e.g. Lat. *auctōritās*. Moreover, some abstracts reflect collective semantics when built to a nominal base. While some synchronic studies of the *tāt*-suffix have been made, e.g. Paucker (1977) for Latin, Renou and Benveniste (1960) for Vedic, and Petersen (1922) for Greek, no serious work on the history of **-tāt-* has been done.

The study of PIE **-tāt-* goes as far back as Aufrecht (1852), who believed that the suffix **-tāti-* (cf. Ved. *-tāti-*) was older than "syncopated" **-tāt-*. He speculated that the

suffix *-*tāti*- reflects a combination of an underlying abstract in *-*tā*- with a *ti*-stem, since several *tāt*-abstracts correspond to abstracts in *-*tā*-, e.g. Skt. *gurutā*- 'weight' beside Gk. *βαρύτης* 'id.' and Lat. *gravitās* 'id.', etc. Though his survey was brief, subsequent work has largely accepted his view on the origin of *-*tāt*- and *-*tāti*- and their relationship.

Recently, however, Nussbaum (1996, 2004b) has uncovered a number of derivational patterns involving secondary *t*-stems, including formations built directly to case forms, as in *k^w*sp-én-t*- 'nighttime' (Hitt. *išpant*- 'night'), continuing a *t*-stem addition to the loc. sg. of the root noun *k^w*sep*- 'the dark' (Ved. *kṣáp*- and YAv. *xšap*- 'night'). Lat. *salūs* 'health' < *$s\underset{\circ}{l}h_2$-*ú*-h_1-*t*- likely reflects a similar formation, but with a secondary *t*-suffix added to an instr. sg. Such a derivational pattern may also underlie PIE "*-*tāt*-", based on its individualizing function, which could continue (as *teh_2-h_1-*t*-) the instr. sg. of an abstract in *-*teh$_2$*- subsequently "individualized" with a *t*-stem suffix.

If so, the history PIE *-*tāt*- becomes clear: it would indeed reflect a secondary addition to an original *teh$_2$*-abstract, but in a very different way than Aufrecht thought. A secondary *t*-stem origin of this sort underlying *-*tāt*- might shed light on other complex abstract suffixes found in Latin (and, in part, elsewhere), including -*tūs* (with comparanda in Celtic and Germanic) and -*tūdō*.

I. Introduction

I.A. PIE *-*tāt*-

Though the number of abstract-forming suffixes in Latin is quite large, the treatment of abstract formation in Latin from a historical perspective has been relatively neglected. Latin possesses all of the "simple" abstract formations found in PIE or with a clear basis in PIE, such as the feminine *-ia*, neuter *-um*, *s*-stems, etc.; but it also has some rather bizarre-looking complex suffixes like *-tās/-tātis*, *-tūs/-tūtis*, *-tiō/-tiōnis*, and *-tūdō/-tūdinis*, etc. that gain a relatively high degree of productivity in the literary language (cf. Quellet 1969 for statistics on abstract types in Latin literature). In fact, several of the suffixes continue to be quite productive in modern Romance languages, including *-tās*, which survives as Fr. *-té*, It. *-tà*, Sp. *-dad/-tad*, etc. Additionally, as has long been recognized, e.g. by Brugmann (1906:451), some of the abstract suffixes find potential comparanda in other Indo-European languages, suggesting that at least some of the complex suffix formations may belong to PIE.

One of the best known of the complex abstract suffixes in Latin is *-tās/-tātis*, which is universally taken to reflect an inherited PIE *-*tāt*-, following Brugmann (1906:451), though again, no one has had much to say about its makeup.[1] Of all the abstract suffixes in Latin, *-tās* is the third most numerically common with respect to the

[1] Though using "*-*tāt*-" for PIE reconstructions is technically incorrect, no judgments or opinions about its formation will be offered until **V.D.** Before the mechanisms behind the creation of *-*tāt*- can be explored in greater detail, the shorthand *-*tāt*- will be used throughout the following chapters with no bias toward its underlying structure.

number of attested abstracts (behind *-tiō* and *-tus*; cf. Quellet 1969:72). Similar-looking abstracts are also found in Greek in the suffix *-τηϛ*, *-τητοϛ* (Doric *-τᾱϛ*, *-τᾱτοϛ*), and in Indo-Iranian, with Ved. *-tāt-* (beside an *i*-stem variant *-tāti-*) and Av. *-tāt-*.[2] The suffix is found in some of the oldest material in each of the languages in which it is attested, and additionally, abstract word-equations are found in several languages that may be indicative of a common origin for the suffix (cf. Fig. 1), suggesting that **-tāt-* and perhaps even **-tāti-* belong to PIE. But the word-equations are not very convincing, in that the abstracts are relatively late and likely reflect inner-language creations (e.g., Gk. ὁλότηϛ 'wholeness' is not found until Aristotle), rather than formations of PIE date. Moreover, two of the word-equations are confined to Indo-Iranian. Despite the fact that **-tāt-* has always been assumed to be inherited from PIE, the relationship of the suffixes between languages has never been seriously questioned. The word-equations given in Fig. 1 will be discussed in further detail later in the appropriate subsections.

Fig. 1. Cross-language archaic **-tāt-* word-equations

Gk. ὁλότηϛ 'wholeness' :: Ved. *sarvátāt-* 'wholeness, health' :: OAv. *hauruuatāt-* 'health'
Hom. νεότηϛ 'youth' :: Lat. *novitās* 'novelty'
Hom. ἀνδροτήϛ 'manliness' :: OAv. *hu-narətāt-* 'talent'
Ved. *uparátāt-* 'superiority' :: YAv. *uparatāt-* 'id.'
Ved. *devátāt-/devátāti-* 'collection of the gods' :: YAv. *daēuuō.tāt-* '*Daeva*-hood'

[2] The *i*-stem variant *-tāti-* found in Vedic has been argued to reflect an inherited PIE suffix **-tāti-* beside plain **-tāt-*. The claim will be investigated in detail later, in **III.C.**; but before it can be determined whether **-tāti-* indeed belonged to PIE, only "**-tāt-*" will be used generically in the remainder of the introduction for both **-tāt-* and **-tāti-* for the sake of simplicity.

Descriptively, *-*tāt*- built qualitative abstracts to thematic adjectives, in addition to a small number of nouns. It became very productive in Latin and Greek, marginally productive in Avestan, and only slightly productive in Vedic. At a very early date, *-*tāt*- was specialized to thematic adjectives, though some of the older material shows that it could be used with athematic stems, including even some nouns, with which it was used in a collective sense. *-*tāt*- thus has an interesting history worth study, but when put beside other systematic treatments of nominal morphology and suffix formations, including Zucchelli (1960) on *-lo-*, Perrot (1961) on *-men* and *-mentum*, Serbat (1976) on instrumental suffixes, including Lat. *-*clo*-, Gaide (1988) on masc. *iō*-stems, and Meissner (2006) on *s*-stems, *-*tāt*- has been relatively neglected.

I.B. Previous Scholarship

Study of PIE *-*tāt*- goes back to Aufrecht (1852), who compared the suffixes in Lat. *-tās*, Ved. *-tāt-/-tāti-*, Av. *-tāt-*, and Gk. *-τηs*, *-τητοs*. He concluded that *-*tāt*- and *-*tāt*- reflect a combination of *-*tā*- plus *-*ti*-, with *-*tāt*- reflecting a syncopated version of *-*tāt*-. Aufrecht's idea is improbable, since it is hard to justify syncope of final *-*i*- in Latin, Avestan, and Greek, but not in Vedic (cf. **II.A.** for further discussion).

Bopp (1856:1131-40) also saw Ved. *-tāti-* as a composite suffix continuing *-*tā*- plus *-*ti*- but rejected Aufrecht's analysis and instead suggested that *-tāt-* might be older than *-tāti-*, continuing a *t*-stem added to *-tā-* (cf. Hom. *ἀγνώ-τ-* 'obscure'). According to

his line of thinking, the *i*-stem variant *-tāti-* found in Vedic must be secondary. He remarked that the base **-tā-* somehow might have been derived from the root *tan-* 'stretch', plus the abstract suffix **-t-*. While the root *tan-* can phonologically give a suffix **-tāt-* or **-tāti-* in Indic in the way Bopp conceived (cf. Ved. *uttānā-* 'stretched'), it cannot work for Latin and Greek.

Angermann (1870:122-6) did not see the use of a verbal root *tan-* as a suffix as likely and so rejected Bopp's theory. He, like Bopp, concluded that **-tāt-* itself must consist of **-tā-* plus **-ti-*, based on the fact that several abstracts in **-tā-* have comparanda in **-tāt-* (cf. Fig. 2 in **I.C.** below), e.g. Gk. *πλατύτης* 'width, breadth' (Xen.+) beside Skt. *pr̥thutā-* 'id.' (Kālidāsa), etc. Both Bopp and Angermann did not expand on their ideas that **-tāt-* somehow reflects a *t*-stem extension to an underlying *tā*-abstract. Brugman (1906:451) does not offer much more, adding that **-tāt-* and Ved. *-tā-* seem to reflect a combination or contamination of similar-functioning abstract suffixes.

Hence, the opinion that PIE **-tāt-* and **-tāti-* somehow continue **-tā-* plus **-t-* and **-ti-* has dominated the older thinking about the formation of the suffixes. But no serious exploration of the mechanics behind the composite formations was undertaken in any of the earlier treatments. Moreover, since the beginning of the study of **-tāt-* and **-tāti-*, it has always been taken at face value that Ved. *-tāti-* belongs in PIE along with **-tāt-*,

which by this way of thinking must oddly have had two very similar-looking suffixes with identical meanings.

More recent literature on *-*tāt*- has had just as little to say about the historical origin of *-*tāt*-, beyond repeating the long-held and unsupported idea that the suffix continues *-*teh*$_2$-*t*-, reflecting a secondary *t*-stem built to a *teh*$_2$-abstract, e.g. Mignot (1972:13), Rieken (1999:93-4), Ringe (2006:61-2), and Fortson (2010:125). Additionally, most modern treatments of *-*tāt*- fail to mention the *i*-stem material in Ved. -*tāti*- and the possibility that it reflects PIE *-*tāti*-. One exception is Neri (2003:58fn136), who suggests that *-*tāt*- and *-*tāti*- may reflect *-*teh*$_2$- plus a "hypercharacterizing" *-*ét*- and *-*ti*- that result in a suffix with collective and abstract semantics; but it is difficult to see how two different suffixes *-*ét*- and *-*ti*- could be added onto an existing abstract in *-*teh*$_2$- in such a way. Differently, Weiss (2009:304) conceives of *-*tāt*- in a way similar to "*-*tā*- + *-*t*-" though reflecting a "de-instrumental" formation *-*teh*$_2$-*h*$_1$-*t*-, with *-*t*- added to the instr. sg. of a *teh*$_2$-abstract, given the "presuffixal" lengthening in similar formations built with suffixes directly to case forms, as also implied by Rau (2009b:122), who offers *-*teh*$_2$(*h*$_1$)*t*- for *-*tāt*-. But neither Weiss's handbook treatment nor Rau's brief discussion provided a detailed analysis or argumentation for the de-instrumental hypothesis. No other significant treatments of PIE *-*tāt*- have been undertaken, and *-*tāti*- has largely been ignored.

In sum, the current thinking on *-*tāt*- is that it must continue *-*tā*- plus *-*t*-, with *-*tā*- reflecting either *-*teh*$_2$- or *-*teh*$_2$-*h*$_1$-, and likewise that *-*tāti*- (when not ignored) must continue *-*tā*- plus *-*ti*-. Thus, in the past 150 years since it was recognized that PIE likely possessed a suffix *-*tāt*-/*-*tāti*-, no serious progress has been made in understanding how *-*tāt*- and *-*tāti*- were formed, and so its implications for PIE morphology remain poorly understood.

I.C. Typology of Abstracts and Collective *-*tāt*-

Abstracts in *-*tāt*- generally express simple qualitative abstracts that reflect the general descriptive quality of an underlying base, much like the English suffix *-ness*, e.g. PDE *greatness* to the adj. *great* and *quickness* to the adj. *quick* . On occasion, however, the *tāt*-suffix found in both Latin, e.g. *cīvitās* 'state; city' (cf. **II.A.**) and Greek, e.g. *νεότης* 'youth' (cf. **IV.A.3.**), sometimes has collective semantics, in addition to forming qualitative abstracts, and similar uses are found in Indo-Iranian. Ved. *devátāt-/devátāti-* 'collection of the gods' and *vr̥kátāti-* 'collection of wolves', and *vasútāti-* 'collection of the Vasus' reflect collective abstracts in *-tāt-/tāti-* (cf. **III.A.3.**). Moreover, collectives in *-tāt-* are found in Iranian in OAv. *karapō.tāt-* 'class, group of *karapans*' and OAv. *kəuuītāt-* 'collection, group of *kauuis*' (cf. **III.A.4.**).

In addition to building qualitative abstracts, the *tāt*-suffix may have had a collectivizing function already in PIE. Typologically, collective and abstract nouns are very closely related, and semantic movement from one class to the other is very common, e.g. PDE *youth* and Hom. *νεότης* 'id.' (cf. Fortson 2010:132). Abstracts built to "count" nouns consisting of discreet countable units are one such class of abstract nouns that can acquire collective semantics rather easily, e.g. *cīvitās* 'state; city' built to the noun *cīvis* 'citizen' and Ved. *devátāt-/devátāti-* 'collection of the gods' built to the noun *devá-* 'god'. Hence, collectives tend to be formed to nouns rather than adjectives, but even abstracts built to adjectives can have collective semantics, as observed in instr. sg. *sarvátātā* 'en nombre total, au grand complet' (Renou 1960:11-13; cf. **III.C.1.a.**).

Collective semantics can easily develop out of an individualized abstract "the one who is *x*", when the word assumes the sense of "the collection or group of ones who are *x*". PDE *youth* 'condition of being young' reflects a similar development from qualitative abstract to individualized noun 'young person; a *youth*' to collective 'collective body of *youths*; the *youth*'. Such collective semantics sometimes shown by PIE **-tāt-* is indicative of the fact that **-tāt-* had a sort of individualizing function that is also perhaps reflected in its function in naming divine figures in Latin and Avestan. For instance, individualizations of underlying bases can be seen in Lat. *Voluptās*, beside Pg. *herentas*/Osc. **herentateís** '*Herentas*' (cf. **II.F.1.c.i.**), and YAv. *arštāt-*, the name of a

goddess "Uprightness" (cf. **III.E.**), where the *tāt*-suffix seems to be used to name the deity in the sense "the one who is or personifies *x*", where *x* is the base. But the examples do not necessarily imply that forming collectives is one of the features of **-tāt-* at the PIE level, given the semantic overlap between abstract and collective nouns.

I.D. *tā-* and *tāt*-abstracts

As suspected already from the beginning of the study of the suffix **-tāt-* (cf. **I.B.** above), it seems likely that PIE **-teh$_2$-* played a part in its development. *tāt*-abstracts share a close semantic and morphological relationship with abstracts in **-tā-* (Brugmann 1906:416, 418), and several *tāt*-abstracts are found next to *tā*-abstracts with similar meanings. Fig. 2 contains the word-equations alleged for abstracts in **-tā-* and **-tāt-*.

Fig. 2. Abstracts in **-tā-* beside **-tāt-*

-tā	*-tāt*
OPer. *aršta-* 'straightforwardness'	YAv. *aršt*āt- name of goddess "Uprightness"
Hom. *βιοτή*/Lith. *gyvatà*/OCS *života* 'life'	Gk. *βιότης* 'id.' (2nd century AD)
Ved. *devátā-* 'among the gods'	Ved. *devátāt-/devátāti-* 'collection of the gods', YAv. *daēuuō.tāt-* '*Daeva*-hood'
Skt. *gurutā-* 'weight' (Suśruta)	Gk. *βαρύτης* (Th.+)/Lat. *gravitās* 'id.' (Lucil.+)
Lat. *iuventa* 'youth' (Lab.+)	Lat. *iuventās* 'id.' (Lucr.+)
Skt. *pr̥thutā-* width, breadth' (Kālidāsa)	Gk. *πλατύτης* 'id.' (Xen.+)
Ved. *sūnr̥tā-* 'vigor'	OAv. *hunarətāt-* 'talent'/ Hom. *ἀνδροτής* 'manliness'
Ved. *vasútā-* 'wealth'	Ved. *vasútāti-* 'collection of Vasus'

It certainly seems to be the case that abstracts in *-*teh*$_2$- are somehow related to those built with *-*tāt*-, but many of the word-equations in Fig. 2 above are suspect, in that one of the members is late, e.g. Skt. *gurutā-* 'weight' (Suśruta) beside Gk. *βαρύτης* 'id.' (Th.+) and Lat. *gravitās* 'id.' (Lucil.+); or one of the members is very different in meaning, e.g. Ved. *vasútā-* 'wealth' beside Ved. *vasútāti-* 'collection of Vasus'. Hence, many of the assumed old word-equations are very poor, and so, as clear as the relationship appears to be, it cannot be absolutely assumed that *-*teh*$_2$- underlies *-*tāt*- in every case. Moreover, why a synonymous abstract in *-*tāt*- would be built beside an already existing abstract in *-*teh*$_2$- calls for explanation. As will be done with the *-*tāt*- : *-*tāt*- word-equations, *-*teh*$_2$- : *-*tāt*- word-equations will be handled individually later in relevant sections.

Several types of *teh*$_2$-formations belonged to PIE, including both deverbal and denominal abstracts, beside adjectives in *-*to*- (Brugmann 1906:414-20, Debrunner 1954:616-20).[3] If *-*tāt*- is indeed ultimately based on *-*teh*$_2$-, it must have been built to the denominal *teh*$_2$-abstracts, since none of the old *tāt*-abstracts are built to verbal roots. Latin only possesses a small number of denominal abstracts in *-ta* (Leumann 1977:335-

[3] Alongside formations in plain *-*to*-/*-*teh*$_2$- should be mentioned those in *-*eto*- and especially *-*eteh*$_2$-, which have received no attention in the discussion of *-*tāt*-/*-*tāti*-. Vine (1998a:10-11) notes that the various Greek formations in *-*eto*- have yet to be systematically classified in the standard handbooks, which can also be said, to a degree, of PIE *-*eto*-.

6), e.g. *iuventa* 'youth' (Lab.+) and *senecta* 'period of old age' (Pl.+), as does Indo-Iranian (Debrunner 1954:616-20), e.g. Ved. *agótā-* 'without cows', *bandhútā-* 'relationship', and YAv. *pauruuatā-* 'mountain'; but with the lone exception of *βιοτή* 'life', Greek entirely replaced denominal *teh*$_2$-abstracts with **-tāt-* already in Homer (cf. Schwyzer 1939:528, Buck and Petersen 1945:541).[4]

I.E. Problems with **-tāt-/*-tāti-*

Despite the fact that many *tāt-* and *tāti*-abstracts are found in the oldest layers of the languages in which the suffix is found, there are few convincing *tāt*-word-equations, if any, that can be securely dated to PIE. Still, rather than reflecting independent creations in the three IE branches in which **-tāt-* is found, the suffix most likely belongs to PIE, since independent innovations of such a suffix, all with similar functions, in Latin, Sanskrit, Avestan, and Greek is extraordinarily unlikely. Consequently, there are questions surrounding the *tāt*-suffix that may well have some potentially interesting implications the the understanding of PIE suffix-formation.

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[4] *βιοτή*, beside Lith. *gyvatà* 'life' and OCS *života* 'id.', point to a PIE *teh*$_2$-stem abstract **g^{w}ih$_3$-(u̯o)-téh$_2$-*. Also potentially belonging here may be Lat. *vīta* and Osc. **biítam**, which are ambiguous in that they could point back to **g^{w}íh$_3$-to-* or a *to*-adjective built to the *u̯o*-stem **g^{w}ih$_3$-u̯o-tó-* (cf. Untermann 2000:146-7). Under this view, however, **g^{w}ih$_3$-u̯o-tó-* cannot give Osc. **biítam** by vowel contraction. *βιοτή* and the related forms in Greek are rife with phonological problems, given that **g^{w}ih$_3$-u̯o-* should give *^{x}βῑϝο-* or *^{x}ζῶϝο-* in Greek, depending on the different alleged outcomes of **-ih$_3$-*. The problems with **g^{w}ih$_3$-u̯o-*, however, cannot be explored here, but cf. Klein (1988) and Vine (1998a:70fn168 with ref.) for more discussion. Cf. **IV.C.3.b.** on the *τηϛ*-abstract *βιότηϛ* 'life'.

The first issue is whether or not PIE had both plain *-*tāt*- and an *i*-stem variant in *-*tāti*- arguably found in Latin and Vedic and if not, how to explain the material reflecting *-*tāti*- secondarily. Scholars have dismissed the *i*-stem evidence in Latin as a secondary feature (cf. **II.C.**), but the Vedic *i*-stem data have usually been thought to continue a PIE formation (cf. **III.C.**). Consequently, it remains to be seen whether or not PIE could have built two suffixes *-*tāt*- and *-*tāti*- with ostensibly identical functions.

Additionally, *-*tāt*- from a very early date became specialized as an abstract suffix for thematic adjectives, especially in Latin and Greek, where the suffix became extraordinarily productive in later literature. Latin reflects a very high number of abstracts built to thematic adjectives, but some of the older material suggests that *-tās* could have been used with consonant-stem bases, including even consonant-stem nouns like *s*-stems (cf. **II.A.**). Most Vedic abstracts, aside from *devátāt-/devátāti-* built to the thematic noun *devá-*, are built to thematic adjectives (cf. **III.A.1.**), but the oldest Iranian abstracts reflect formations to a rather wide variety of base types, including both thematic and athematic stems (cf. **III.A.1.**), suggesting that the Indo-Iranian situation may have also been more variable than is reflected in later texts. And Greek abstracts in *-τητ-* are closely associated with thematic adjectives in Homer, but other base types are found too, including *u*-stems (cf. **IV.A.**). Thus, when and why *-*tāt*- became associated with thematic adjectives from such an early date deserves attention, in that the specialization

may or may not have occurred within PIE. Furthermore, why the suffix *-*tāt*- became associated with thematic stems is unknown as well.

Finally, from the perspective of PIE word formation, the suffix shape *-*tāt*- is bizarre: other suffixes of the shape -*CV̄C*- are virtually nonexistent. PIE possessed many -*CV̆C*- suffixes, including suffixes in *-*men*-, *-*tor*-, *-*u̯ent*-, etc. (cf. Fortson 2010:123-6), but no other suffix similar to invariable *-*tāt*- can be identified.[5] Hence, it may make sense to think of *-*tāt*- as a complex suffix built from other suffix material in a similar way to PIE *-*mer*-/*-*men*-, *-*ser*-/*-*sen*-, and *-*ter*-/*-*ten*-, etc., as continued in Hittite, which are built from simple suffix units. If *-*tāt*- indeed reflects a complex suffix, the underlying suffix constituents remain to be identified. The likelihood that abstract nouns in *-*teh*$_2$- may be one constituent must also be evaluated. In addition, the justification for building a PIE suffix of the shape *-*tāt*- from smaller units must be uncovered.

I.F. Goals and Objectives

What will follow in the study will be a language-by-language treatment of the material reflecting *-*tāt*- and *-*tāti*-. Italic (cf. **II.**) will be examined first, given that it

[5] It has been assumed, e.g. by Brugmann (1906:453), that Latin abstracts in -*tūs*/-*tūtis*, e.g. *virtūs* 'manliness, courage' (Andr.+), *senectūs* 'old age' (Pl.+), etc., share an origin with Celtic masculine abstracts in *-*tūt*-, as in OIr. -*tūt*-, e.g. *oítiu* 'youth', *sentu* 'age', and MW -*dit*/*tit*, e.g. *gwendid* 'weakness', Gothic -*dūþs* < *-*tūti*-, e.g. *ajukdūþs* 'eternity', *mikildūþs* 'greatness', etc., and perhaps even YAv. *gaδōtūt*- 'band of robbers' (but Hoffmann 1967:188 rejects the connection, thinking -*ū*- is secondary beside an original -*ī*-). Just as *-*tāt*- is too unique to be independently innovated in multiple languages, the suffix *-*tūt*- likely also belongs to PIE. Given the parallel structure between *-*tāt*- and *-*tūt*- it is likely that they share a common derivational pattern, but a comprehensive study of *-*tūt*- is well beyond the scope of the present work (cf. **VI.C.** for further discussion).

may present some of the biggest challenges. Latin *-itās*, with a "linking vowel" generalized from *-tās*, becomes one of the more productive abstracts suffixes in the language relatively quickly, but there are no old texts in Latin or anywhere else in Italic comparably old as the Rig Veda or Homer that have usable material for the study of the *tāt*-suffix. Old-looking abstracts will thus have to be identified and examined to see if they may offer evidence for inherited features of **-tāt-*. The *i*-stem evidence found in gen. pl. *cīvitātium*, etc., will also be discussed to see whether it gives support for an *i*-stem variant in PIE **-tāti-* (cf. **II.C.**). Latin also builds a fair number of abstracts to consonant-stem bases with plain *-tās*, including *s*-stems, e.g. *tempestās* 'weather; storm', which may be relatively old (cf. **II.F.**). Lastly, Italic seems to have used *tās*-abstracts as theonyms for personified divine figures, e.g. *Voluptās* and Pg. *herentas* (cf. **II.F.11.a.**), which may reveal something about the semantic features of the suffix both in Italic and possibly even in PIE.

Indo-Iranian **-tāt-* and **-tāti-* will be treated after the Italic material (cf. **III.**), focusing primarily on the data for the Vedic *i*-stem **-tāti-* found in abstracts like *devátāti-* 'collection of the gods', and whether or not the *i*-stem evidence must be projected back to PIE (cf. **III.C.**). Already in the Rig Veda, **-tāt-*/**-tāti-* has been generalized to thematic bases, while the Avestan material, particularly the Young Avestan abstracts, reflects a wider variety of base types than Vedic. But the oldest abstracts in Indo-Iranian do not

differ significantly from the standard de-adjectival qualitative abstract use of *-*tāt*-/*-*tāti*- found elsewhere.

Archaic Greek texts contain abstracts in -*τηϛ* (cf. **IV.**), including some potentially very old-looking forms like *ἀνδροτῆτα* 'manliness'. Just as in Vedic, Hom. -*τηϛ* is largely suffixed to thematic stems, and the abstracts do not offer much in the way of unusual semantics worthy of comment. But unlike any examples found in Vedic (cf. **III.A.5.**), a few Homeric forms show deviant accentuation, with accent on the suffix rather than the base or the thematic vowel, e.g. *δηϊοτήϛ* 'battle strife' (cf. **IV.C.3.c.**). Furthermore a number of bases of derivation are without clear etymologies. Hence, some of the Greek examples will be treated in detail, focusing on the role -*τηϛ* may have had within the Homeric poetic tradition.

PIE *-*tāt*- will be dissected in the next to last chapter of the study (cf. **V.**), based on the data obtained in the previous chapters. The goal will be to uncover the underlying components of such a complex suffix to better understand whether it indeed belongs to PIE, and if so, how it was formed. The question of why it then became productive and why it chiefly became associated with thematic bases will also be addressed, including the question of whether or not the thematic association was a feature of the suffix already in PIE.

Finally, the study will be closed in the final chapter with a summary of findings, as well as providing some directions for further study (cf. **VI.**). By coming to an understanding about the formation and use of *-*tāt*-, it may also be possible to arrive at a

better understanding of PIE derivational morphology and derivation that may provide some insights into the analysis of other similarly complex morphemes found in Latin and elsewhere and also identify some possible directions for further research.

II. Latin *-tās* and Other Potentially-Related Italic Forms

II.A. Descriptive Facts

Generally, the Latin suffix *-tās* formed feminine qualitative from adjectives, e.g. *lībertās* 'freedom' (Naev.+; beside Fal. **loifirtato**, on which cf. **II.D.1.** below) to *līber* 'free', *vēritās* 'truth' to *vērus* 'true, real', etc.; a few official terms from nouns, e.g. *aedīlitās* 'aedileship' to *aedīlis* 'aedile (Roman magistrate)', *cīvitās* 'state; city' to *cīvis* 'citizen', *hērēditās* 'inheritance' to *hērēs* 'heir', etc.; and a small group of technical terms having to do with harvesting or seasons, e.g. *autumnitās* 'autumn' to *autumnus* 'autumn' (Leumann 1977:373-5). The *tās*-suffix constituted one of the more productive means of abstract formation within Latin, with a high level of productivity attested throughout the history of the language, even in the earliest literature. Additionally, unlike Indo-Iranian and Greek, Latin *tās*-abstracts never acquired a specialized "technical" sense (cf. Ved. loc. sg. *devátātā* 'among the collection of the gods' discussed in **III.C.1.b.**). Rather, a similar technical semantic function was assumed by the more technically nuanced suffixes in *-tudō* and *-itia* that rivaled *-tās* later in Latin (Brachet 2002:79-80).[6]

Abstracts in *-tās* were sometimes found alongside abstracts built with the other suffixes *-tūdō*, *-ēdō*, *-tia*, or *-tiēs*, with several abstracts built to the same base found in

[6] Cf. Aul. Gel. (17.2.19-20), who reports that abstracts in *-tūdō*, e.g. *sanctitūdō* 'sanctity' beside *sanctitās* 'id.', were thought to be more "dignified" and archaic than corresponding in abstracts in *-tās*.

later Latin, e.g. *dūritās* : *dūritia* : *dūritiēs* 'harshness' (cf. Paucker 1877:159-67, Mikkola 1964:124-30 for lists of such alternations). Alternations in suffix choice are a result of "experimentation" on the part of Roman authors wishing to play with the variety of abstract suffixes in Latin and reflect certain stylistic subtleties. The abstract suffix *-tūdō*, for example, was felt to be more archaic and thus carried archaic nuances when used; but it was almost completely replaced by *-tās* already in Classical Latin. Lucretius, for example, began to replace older abstracts with forms in *-tās*, e.g. *differitās* 'difference' for *differentia*, *maximitās* 'greatness' for *magnitūdō* and *pestilitās* 'plague' for *pestilentia*, given the metrical advantage *-itās* has within the hexameter (Hofmann and Szantyr 1965:743-4).[7]

Synchronically, the suffix *-itās*, with generalized "*-i-*", made qualitative and relatively unmarked abstracts from adjectives and became productive very early in Latin;[8] but as can be seen already in the oldest data, some forms in plain *-tās* did not make simple qualitative abstracts: *aevitās/aetās* 'lifetime, age', along with Osc. **aítateís**/Pg. *aetatu*, *egestās* 'destitution', *tempestās* 'period of time, season; storm', and *voluntās* 'will' may well belong to an older stratum of *tās*-abstracts, given that they do not behave as simple qualitative abstracts, and that they, moreover, fail to show the

[7] Even in prose the metrical shape of *-itās* was used to advantage in clausulae, as in Cicero's *patrimonium dignitatis* (*De Orat.* 3.6), with its "cretic plus double trochee" clausula –⏑––⏑–⏓ (Mankin 2011:44).

[8] Judging from *auctōritās* found in the *XII Tables*, the generalization of *-itās* occurred very early date, already by the middle of the 5th century. But evidence from the *XII Tables* is always suspect, since it was subject to "modernizations" of sorts when recopied.

generalized suffix form *-itās* found in the majority of *tās*-abstracts. From an early point, any original abstract meaning has been concretized, e.g. *tempestās* 'period of time, season; storm' from *tempus* 'time' and *hērēditās* 'inheritance' from *hērēs* 'heir', and especially in the technical/harvest terms *autumnitās* 'autumn' from *autumnus* 'autumn' and *aestās* 'summer' from *aestus* 'glow'. Concrete substantivizations are also found in divine names, if the semantics of *tempestātēs* 'the weather gods' (cf. **II.F.1.b.**) and Pg. *herentas* 'Herentas (= Voluptas)' reflect anything old (cf. **II.F.11.a.**).[9] Some collectivizing uses are found too, as in *cīvitās* 'state; city' from *cīvis* 'citizen', which find echoes both in Indo-Iranian, in Ved. *devátāt-* 'collection of the gods', *vr̥kátāt-* 'pack of wolves', and OAv. *kauuītāt-* 'class, group of *kavis*' (cf. **III.A.3.** on Vedic abstracts and **III.A.4.** on Avestan abstracts); and to a lesser extent in Greek, in *νεότης* 'youth; collection of youths' (cf. **IV.A.3.**).[10]

II.B. Previous Scholarship

As mentioned earlier in **I.B.**, no extensive studies of PIE **-tāt-* have been undertaken, and a historical analyais of the Latin *tās*-suffix likewise remains yet to be

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[9] Cf. the more-widespread use of **-ta* (< PIE **-teh$_2$*) in naming divinities, e.g. *Maiesta*, *Mātūta*, *Morta*, *Segesta*, etc. (Stolz 1895:532). *-tās* will be seen to have a similar individualizing function discussed below in **II.F.11.a**.

[10] Vine (2006c) has recently argued that *cīvis* may continue an old *i*-stem abstract, beside the *o*-stem adjective in Ved. *śéva-* 'auspicious, dear, friendly'. *cīvitās*, then, reflects a "renewal" with *-tās* built to the abstract noun (Vine 2006c:152). It will be shown below that *-tās* can indeed be usd to build abstract renewals of this sort to underlying abstract nouns (cf. on *honestās* below in **II.D.3.**).

done. Before beginning the discussion of *tās*-abstracts in Italic, it will be helpful to expand on the theories introduced in **I.B.** regarding its PIE heritage. Previous scholarship on the Latin *tās*-suffix goes back to Aufrecht (1852), who first made the comparison among Lat. *-tās*, Gk. *-τηϛ*, Ved. *-tāt-/-tāti-*, and Av. *-tāt-*, connecting the abstracts with the suffix *-tā-* (cf. **III.A.1.** and **III.A.2.** for Aufrecht's treatment of Indo-Iranian *-tāt-* and *-tāti-* and **IV.A.** for Gk. *-τηϛ*). He made nearly all of the observations mentioned above that would characterize all subsequent studies of the suffix, namely, that a "linking" vowel, continuing either **-o-* or **-i-*, preceded the *-tāt-* suffix, which was affixed to adjectives in a majority of the abstracts, but that *-tās* is also found with a number of consonant-stems, chiefly *s-* and *n*-stems (1852:160-1). Additionally, Aufrecht noticed (1852:161) that a small number of *tās*-abstracts were built to nouns rather than adjectives, e.g. *aedilitās* 'aedileship', *famulitās* 'slavery', *plēbitās* 'rank of a *plebs*', etc., and that some of the abstracts are collective nouns, e.g. *cīvitās* 'state', *rūsticitās* 'country occupations', etc., or concrete nouns, e.g. *hērēditās* 'inheritance', rather than true qualitative abstracts. But though he recognized the differences in abstract base types, he made no attempt to ascertain their age relative to each other and thus open the question of the potential significance older *tās*-abstracts may have for the history of the suffix. He did suspect that the *i*-stem alternant **-tāti-* found in Vedic (cf. **III.C.**) was older than plain **-tāt-* and represented a combination of the abstract suffixes **-tā-* and **-ti-*; **-tāt-*,

consequently, somehow continued a prehistoric loss of final *-*i*- (Aufrecht 1852:163). He also suspected some sort of link between -*tās* and the similar-looking abstract suffix -*tūs* (1852:161), which he saw as a replacement for original -*tās* in a number of forms, e.g. *iuventūs* 'youth', *senectūs* 'old age', *servitūs* 'slavery', etc. (cf. **VI.C.** for further directions with -*tūs*).

Walter (1861) cited the prominence of the abstract suffixes *-*tā*- (as in Lat. -*ta*, Ved. -*tā*-, and Gk. -$\tau\eta$) and *-*tŭ*- (as in Ved. -*tū*-, Gk. -$\tau\upsilon\varsigma$ and -$\sigma\upsilon$ as in -$\sigma\upsilon\nu\eta$), alongside parallels like *iuventās* beside *iuventa*, elaborating on the supposed connection between the two suffixes *-*tāt*- and *-*tūt*-. He took them both back to feminine abstract suffixes *-*tā*- and *-*tū*-, respectively, each enlarged with *-*ti*-. Assuming that Ved. *-*tāti*- continues the oldest form of the suffix and moreover that its accentuation pattern is old, whereby the preceding vowel always carries the accent, Lat. -*tās* also ultimately reflects *-*tāti*-, he thought. Latin -*tās*, by this idea, has undergone a sort of syncope of final -*i*- due to the effect of the accent, as is also the case for Gk. -$\tau\eta\tau$- and Av. -*tāt*- (Walter 1861:159). Although syncope is found in most *ti*-stems within Latin, e.g. nom. sg. *mentis* (Enn.) > *mēns*, etc. (cf. Weiss 2009:243n12), the evidence for an *i*-stem *-*tāti*- in Vedic is misleading (cf. **III.C.**), and potential support within Latin for *-*tāti*- in some

gen. pl. forms in *-tātium* is hardly compelling (cf. **II.C.** below for discussion of **-tāti-* in Latin).

Vine (2006b) has recently demonstrated that the PIE accent may have persisted in Italic longer than previously thought, surviving in some form into Latin. That PIE accent was still present in early Italic is suggested by "Thurneysen-Havet's Law", whereby PIE **-o̯u̯-* > *-a̯u̯-* in syllables preceding the PIE accent, e.g. Lat. *caveō* 'beware of' < **ko̯u̯h$_1$-éi̯o/e-*. Likewise, it may be questioned if a similar accent-induced vowel loss may have led to syncope of the sort suggested by Aufrecht and Walter, and furthermore, whether the same sort of syncope occurred in a similar context with the primary verbal endings in **-mi*, **-si*, **-ti*, etc. The sort of accent conditioning governed by Thurneysen-Havet's Law happened very early in Italic, before the stress accent developed and cannot have anything to do with loss of final **-i-* in **-tāti-*. Moreover, syncope of **-i-* in the primary verb endings **-mi*, **-si*, **-ti* is a different phenomenon from similar-looking syncope in nominal forms, given that they tended to be sentence-final, with a weaker prosodic status than nouns (Weiss 2009:146-7). VOL *tremonti* '*tremunt*' (*Car. Sal.*), if a real form, shows that the loss of **-i* happened within Latin, so syncope of **-i-* in **-tāti-* in the way conceived by Aufrecht and Walter must have occurred independently in Latin, Avestan, and Greek, which seems highly unlikely, especially since Avestan and Greek do not regularly syncopate **-i-* in final syllables.

Bopp (1856:1136-7) followed Aufrecht (1852:161) in thinking that * *tāt-*/**-tāti-* was founded upon original **-tā-*, and offered several possibilities regarding their relationship: **-tāti-* may continue **-tā-* plus **-ti-*, as Aufrecht thought; or on the other hand, **-tāti-* may continue a sort of reduplicated combination of **-tā-tā-* > **-tāti-*, in which "the *tāti* is a simple phonetic extension of *tā*; so that *ti* is properly only the repetition of *tā*, with the weakening of the *ā* to *i*".[11] Bopp (1856:1136-7), however, rightly decided against Aufrecht's ideas, instead favoring the hypothesis that **-tāt-* reflects **-tā-* plus a simple *t*-suffix. The *i*-stem variant **-tāti-*, found only in Vedic to his mind, was "an off-shoot or later growth", and **-tāt-*, which is found in Vedic, albeit late, is indeed the oldest version of the suffix, given the fact that at least in Greek and Avestan, "final *-i* has never allowed itself to be displaced in the classes of words in *i*, which are common also to the Sanscrit". As will be shown later in **II.C.**, Bopp's idea that **-tāt-*, not **-tāti-*, was inherited from PIE was indeed correct.

Müller (1876:366) also followed Aufrecht's (1852:161-2) view that **-tāti-* reflects **-tā-* plus **-ti-*, but turned attention to the collective semantics of the suffix, which led him to agree with an idea stemming from the Indian grammatical tradition in thinking that **-tāti-* ultimately derived from the verbal root *tan-* 'stretch' (i.e. PIE **tenh*$_2$-)

[11] Bopp (1856:1132) also compared **-tāt-* to the similar-looking **-tūt-*, but thought it to be a "weakened" form of **-tāt-*. There is no basis for phonological "weakening" of this sort, and so such an idea can be discarded.

extended with the abstract suffix *-*ti*-. Pāṇini (*Aṣṭ*. 4.4.143) said of -*tāti*- that it sometimes carries the significance of "making, maker", as is the case with masc. dual *śáṃtātī* '(two) happiness makers, augmenters of happiness' (RV 1.112.20a), and based on Pāṇini's statement, Bopp (1856:1139-40), in his parenthetical "Remark", also discussed the possibility of such a derivation of -*tāti*- from the root *tan*- in positive terms (although he opted to see the *ti*-extension in this case as an agent marker), admitting that the -*tāti*- found in Vedic could be different in origin from *-*tāt*- in Latin, Greek, and Avestan.

Clearly, though, the *seṭ* character of the root *tan*- (< PIE *$tenh_2$-) precludes it from underlying the suffix *-*tāt*-/*-*tāti*-.[12] Furthermore, as it will be argued later (cf. **III.C.**), the *i*-stem evidence for Vedic -*tāti*- is at best marginal, and evidence points to a single PIE suffix *-*tāt*- that cannot have arisen from "*tan*- plus -*ti*-" in any way to give the familiar forms in Latin, Indo-Iranian, and Greek, without overcoming severe phonological obstacles, not to mention semantic ones.

"Fads" of this sort, by which suffixes are derived from roots themselves, are beginning to find favor again in modern scholarship. Olsen (2009:192), for example, has recently argued that the suffix *-*tūt*- is based on a root noun *tuh_2- form the root *$teu̯h_2$- of Ved. *tavīti*- 'is strong'. Olsen (2009:192n8) also adds that she considers *-*tāt*- to be a

[12] The laryngeal in *tan*- is clear in Gk. *ταναός* 'long', but sometimes *tan*- appears *aniṭ*, e.g. Ved. *tanóti* 'extend'. The PIE root *$tenh_2$- 'stretch' and its possible analysis with an h_2-extension will be revisited below as a parallel to *$u̯elh_2$-.

contamination of *-*tah*$_2$- and *-*tuh*$_2$-*t*-; but contamination is impossible, on the grounds that the suffix *-*tūt*- has comparatively little influence vis-à-vis *-*tāt*-, given the relative productivity of each suffix. Though found in Latin, e.g. *iuventūs* 'youth', *senectūs* 'old age', etc., *-*tūt*- is not found in Greek or Indo-Iranian. Rather, it is found in Celtic and Germanic, neither of which has *-*tāt*-, raising the question why more examples of *-*tāt*- beside *-*tūt*- are lacking.

Paucker's (1877) synchronic study of Lat. -*tās* remains the most thorough of its kind, and though he had nothing to say about its origin, he provided an exhaustive collection of all the *tās*-abstracts attested in Latin, including dates of attestation and counts of abstracts built with -*tās* that alternate with others of the same base built with the suffixes -*tūdō*, -*ēdō*, -*tia/-tium*, -*tiēs*, -*mōnia/-mōnium*, and -*ia/-ium*. Paucker's detailed collection of *tās*-abstracts revealed that 93.5% of the forms in Latin are built to thematic adjectives.

Meyer-Lübke (1893:321-43) likewise surveyed several abstract suffixes in Latin from a synchronic perspective, including -*tās*, with particular focus on the differing adjective stems underlying the abstracts, e.g. *k*-stem adjectives making abstracts in -*citās*, *r*-stem adjectives making abstracts -*ritās*, adjectives in -*stus* making abstracts in -*stitās*, etc. Outside of his survey of the relationship between the abstract nouns and their

adjectival bases, Meyer-Lübke has nothing to offer on the historical formation of the *tās*-suffix itself.

Brugmann (1906:452-3) first considered that Italic possesses an *i*-stem variant from **-tāti-*, based on forms like gen. pl. *cīvitātium*. He compared the alternation between *-tāt-* and *-tāti-* to the same sort seen in **noct-* and **nocti-*, in which an old consonant-stem paradigm has merged with an *i*-stem paradigm. Though Brugmann posited a PIE **-tāti-*, Lat. gen. pl. in *-tātium*, however, cannot be used as evidence in support of it, given the large number of secondary *i*-stem forms created within Latin (cf. **II.C.** below on Lat. *-tātium*). Brugmann (*loc. cit.*), moreover, viewed a number of abstracts in *-itās* as having been remodeled from forms in plain *-tās*, e.g. *asperitās* ← **aspertās*, *posteritās* ← **postertās*, etc. He categorized the *tās*-abstracts according to the stem types of the underlying adjectives, with *o*-stems, *i*-stems, nasal and liquid stems, adjective stems ending in a heavy syllable, e.g. *capāx* → *capācitās*, and *s*-stems, though his categorization of the abstracts is purely synchronic.

Brugmann took note of some of the problematic abstracts in Latin but did not elaborate on why the forms are problematic historically, nor did he attempt to develop a relative chronology of forms. He offered that *voluntās* could continue **velonti-tās*, or it could be built to a verb **voluntāre*, much like his idea that *satiās* 'abundance' (Pl.+) is based on the verb *satiāre* 'satisfy' (Lucr.+), though such a deverbal derivation is correctly

rejected by Leumann (1977:374). He also thought *aestās* must continue **aidhs-tāt-*, based on the comparative evidence in Greek and Indo-Iranian (1906:446; cf. **II.F.2.** below for more on *aestās*). Brugmann also saw a problem in *lībertās*, which does not follow the customary patterning with *-itās* he claimed for other adjectives in *-er*, like *ācer* beside *ācritās* for older **ācertās*, in his mind. Following Sommer (1900:227), *lībertās* is based on the *to*-adjective *lībertus*, which must continue **loufero-to-* or **loufere-to-* (Brugmann 1906:404). While his treatment of the abstracts is brief, it nonetheless influenced those that followed. Many of the problems Brugman identified are still unsolved and will be investigated later in greater detail.

Mikkola's (1964) survey of abstract nouns in Latin focused on the semantic relationship various abstract suffixes share with each other and how they were used by Laitn authors. While Mikkola provided useful and very detailed lists and frequency charts by author for the abstract suffixes in Latin, including case usage for each suffix, he did not have anything to say about the historical problems with a number of *tās*-abstracts, nor was he concerned with the origin of the forms.

Leumann (1977:373-5) divided the *tās*-abstracts into two groups: *-itās/-ietās* and postconsonental *-tās*. The former class continues abstracts built to *o*- and *i*-stem bases in Latin, as well as consonant stem bases built with the generalized suffix *-itās*, which became the productive form of the suffix early on. He also considered *autumnitās* 'autumn', *aestās* 'summer', *oleitās/olīvitās* 'olive harvest', and *fīcitās* 'fig harvest' as

members of a small class of exceptions that reflect specialized technical semantics for harvesting. The class is particular only to Latin and does not have parallels in other languages.

The consonant-stem class constitutes the more interesting set of forms, and as such Leumann (1977:373-5) commented on several of the more problematic forms in it. Within the group, many different base types are found, too, including thematic adjectives. *lībertās* is built to the thematic adjective *līber* 'free', and Leumann (1977:98, also following Sommer 1900:227) assumed a complicated development **leudherotās* > **loufrotās* > **loifr̥tās* > *lībertās* (discussed in greater detail below in **II.D.1.**), unless *lībertās* is simply formed on the model of *paupertās* (for which Leumann provides no justification). A number of *tās*-abstracts continue haplologized forms, e.g. *honestās* < **honestitās*, **honestitātem*, etc. and *voluntās* < **velonti-tās*, **velontitātem*, etc. (unless from **velont-tās*, with inner-Latin simplification of a late **-tt-* > *-t-*; cf. 1977:198).[13] Moreover, several abstracts in *-estās* and *-ustās* beside adjectives in *-estus* and *-ustus* are found in Latin, and so Leumann thought *tās*-abstracts were built to the *tus*-adjectives, which gave the attested forms in *-stās*, *-stātis*, etc. by haplology from nom. sg. **-sto-tāt-s*, gen. sg. **-sto-tāt-es*, etc. Because of the close association between abstracts in *-tās* and adjectives in *-tus*, some of the older abstracts gave rise to new back-formed adjectives in

[13] Abstracts built from *nt*-participles are also found in Avestan, Leumann (1977:374) noted, e.g. *uxšiiąstāt-* 'waxing'; for more on the abstracts built to present participles, cf. Benveniste (1960) and **III.A.2.**

-tus, like *lībertus*, *ūbertus*, and *maiestus*, and so the *-tās* : *-tus* relationship will be investigated below (cf. **II.D.1.a.** and **II.F.1.**).

Most recently, Brachet (2002) has examined the relationship between certain *tās*-abstracts and their corresponding adjectives in *-tus*, with special attention paid to the relationship between *lībertās* 'freedom' and *lībertus* 'freedman'. He is not as concerned with the historical formation of **-tāt-* as much as he is with the inner-Latin semantic development of the association some abstracts in *-tās* share with adjectives in *-tus* (cf. **II.D.2.** below).

Given the number of potentially interesting *tās*-abstracts in Latin and the questions surrounding their formations, it will prove worthwile to study them in great detail, including related Sabellian material. What is to follow will treat the supposed *i*-stem inflection of the suffix **-tāt-* within Italic, as such a **-tāti-* may have a parallel in Indic. Next, the various base types underlying the *tās*-abstracts will be evaluated, divided into thematic and non-thematic forms. A thorough examination of the various base types, with special attention paid to the diachronic details and the relative ages of the abstracts, could well reveal how the *tās*-suffix worked in early Italic and help reveal how it was formed in PIE.

II.C. *i*-Stem Variants

Along with Indic, Latin shows some *i*-stem variants reflecting **-tāti-* beside plain **-tāt-*, e.g. gen. pl. *aetātium* (Liv.+), *cīvitātium* (Cic.+), *hērēditātium* (Cic.+), HEREDITATIVM (CIL 13.1810; Lugdunum, late 2nd c. CE), TEMPESTATIVM (CIL 8.2609, 2610; early 2nd c. CE), etc. (cf. Neue and Wagener 1902:408-10). The sort of variation with **-tāt-* beside an *i*-stem **-tāti-* is also found in other stem types already in PIE, e.g. Gk. *νύξ*, *νυκτός* 'night' beside Ved. *náktis* 'id.', **dek̑m̥t-* 'decade' beside **dek̑m̥ti-* 'id.', etc. (cf. Brugmann 1906:451), and thus could be suggestive that the alternation between **-tāt-* and **-tāti-* also belonged in PIE, as thought Müller (1876:366), Stolz (1895:553), Brugmann (1906:451), and Sommer and Pfister (1977:390). In fact, even though *i*-stem inflection with**-tāti-* in Indic is far more common than **-tāt-* (cf. Fig. 5 in **III.A.**), relatively few of the Latin abstracts show *i*-stem inflection, and the forms are all late; no gen. pl. in *-ium* seems to exist before Cicero, nor does any other *i*-stem evidence, for that matter: no abl. sg. *-ī* is ever found, and moreover, though plurals to *tās*-abstracts are rare, no acc. pl. *-īs* is attested. In fact, Varro (*Lat. Lang.* 8.66) makes mention of the *i*-stem variant *cīvitātium*, but does not say anything about a *ˣcīvitātī* or *ˣcīvitātīs*:

> *nam sine reprehensione vulgo alii dicunt in singulari hac ovi et avi, alii hac ove et ave; in multitudinis hae puppis restis et hae puppes restes; item quod in patrico casu hoc genus dispariliter dicuntur*

> *civitatum parentum et civitatium parentium, in accusandi hos montes fontes et hos montis fontis.*
>
> Without blame, some people generally say in the (abl.) singular "*ovi*" and "*avi*"; others say "*ove*" and "*ave*". In the (nom.) plural (some say) "*puppis*" and "*restis*" beside "*puppes*" and "*restes*". But it is likewise the fact that in the gen. pl., "*civitatum*" and "*parentum*" beside "*civitatium*" and "*parentium*" are said. In the accusative, "*montes*" and "*fontes*" (are said) beside "*montis*" and "*fontis*".

The only real support for *i*-stem inflection is found in the gen.pl., e.g. *cīvitātium* (Neue and Wagener 1902:408), and so the evidence for any original *i*-stem inflection is cast into doubt. Priscian's (*IG* 7.16.77) comments on the *i*-stem inflection of the *tās*-abstracts in the abl. sg. and gen. pl. also do little to bolster the credibility of an old *i*-stem variant *-*tāti*- in Latin:

> *in "as" quoque Latina mutant e ablativi in i et assumunt "um" : "a civitate civitatium", "a probitate probitatium". in his tamen frequenter solet fieri concisio i in genetivo plurali: "a civitate civitatium" et "civitatum", "a probitate probitatium" et "probitatum".*
>
> Also in Latin forms ending with *-ās*, they make a change from the ablative in *-i* and take up [the gen. pl.] in *-um*, e.g. from *civitas* (is) *civitatium* and from *probitate* (is) *probitatium*. In these forms, however, omission of *-i-* in the genitive plural frequently occurs, e.g. from *civitas* (is) *civitatium* beside *civitatum* and from *probitate* (is) *probitatium* beside *probitatum*.

Priscian also says that the *i*-stem in the gen. pl. *-tātium* found beside plain *-tātum* is never found in other case forms.

Consequently, the lack of reliable *i*-stem evidence in the inflection of *-tās* makes the assumption of an original suffix **-tāti-* in Latin both unlikely and unnecessary. To Leumann (1977:375), the variant suffix with *i*-stem inflection in Latin is a "Neuerung" and does not reflect anything old (cf. a similar conclusion for Indic **-tāti-* in **III.C.**). The *i*-stem forms in *-tātium* owe their existence to the prehistoric spread of *i*-stem inflection into consonant-stems. Just as gen. pl. *noctium* was remade with a clearly secondary *i*-stem *-ium* at a very early date (cf. *NIL* s.v. 1. **neg^{w}-* with n. 7), so must have the abstracts in *-tātium* been analogically formed. The fact that they seldom inflected in the plural, moreover, could easily have aided in the creation of *-tātium*, on the model of the large number of other historical *ti*-stems that made a gen. pl. in *-tium*, e.g. *mentium* 'of minds'/MENTIVMQUE (*HD* 016279; Cissa, early 1st c. CE), *montium* 'of mountains', *partium* 'of parts'/PARTIVM (*CIL* 6.10230; Rome, 1st c. BCE), etc.

II.D. Thematic Bases

Synchronically within Latin, a *-tās* abstract could be built to any adjective, either second declension or third declension. With the exception of *lībertās* and the abstracts thought to be built to *to*-adjecives, *o*-stem adjectives of the second declension make their

abstracts with *-itās* from **-o-tās*, with weakening of the thematic vowel, e.g. *novitās* 'newness' < **neu̯-ó-tāt-*; cf. formally identical Gk. *νεότης* 'youth' (but cf. **I.A.**). Likewise, abstracts built to *i̯o*-stems surface with *-ietās*, from **-i(i̯)o-tās* after **-i̯o-* was weakened to **-i̯i-* and then dissimilated to **-i̯e-*, e.g. *pietās* 'piety' (cf. Leumann 1977:90 for the dissimilatory change of **-i̯o-* > **-i̯e-* in Latin). As mentioned above, the abstracts made to thematic stems make up 93.5% of the *tās*-abstracts found in Latin (Paucker 1877:157).

The majority of the abstracts that reflect *-itās* are uninteresting historically, in that they simply show the productive formation in **-o-tās*; but *lībertās* does not show *-itās* poses some interesting questions. All other *er*-adjectives in Latin that make *tās*-abstracts are built with *-itās*, as is *ācerītās* 'force' (Acc.+), *integritās* 'purity' (Cic.+), *pulchritās* 'beauty' (Caecil.+), etc. As such, one may question why no **lībritās* or *līberitās* exists within Latin (cf. however later LIBERITAS in CIL 10.2151; Puteoli, late 2nd/early 3rd c. CE). Likewise, some of the abstracts thought to be built to *to*-adjectives and shortened by haplology, including *honestās*, *tempestās*, and *egestās* mentioned by Leumann (1977:374), seem to continue older material, given that later abstracts in *-titās* are found that never underwent hapology, e.g. *castitās* 'purity' (Cic.+). *lībertās* and *honestās*, etc.

are then of great interest for the history of *-tās* in Latin, in that they may reflect early formations in plain *-tās* before the generalization of *-itās*.

II.D.1. *lībertās* 'freedom'

lībertās 'freedom' (Pl.+) is built to *līber*, an adjective in **-ero-* built to the verbal root **h₁leud^h-* 'rise, grow', with comparanda in Ved. *róhati* 'ascend', Hom. ἤλυθον 'came', Go. *liudan* 'grow, increase', etc. (*LIV* s.v., *NIL* s.v.). The *ero*-adjective, moreover, is found in Osc. *lúvfreís*, Pg. *loufir* (but cf. Stuart-Smith 2004:57 for a different reading), and Gk. ἐλεύθερος 'free' (cf. Untermann 2000:437-8).[14] After Leumann (1977:98), the consensus on *lībertās* assumes it to be the product of syncope of the vowel before *-r-*, ultimately going back to **leu̯d^herotāt-*, which developed to **lou̯frotāt-* (in his notation), after the second syllable was lost by syncope (cf. Leumann 1977:97 for examples of syncope before *-r-*), and then, with subsequent loss of the new second syllable in **lou̯frotāt-*, **loi̯frtāt-*, with **-ou̯-* > **-oi̯-*, whence *lībertās*.[15]

On the other hand, Nishimura (2008:154n57) questions Leumann's reluctance to consider syncope after *-r-* in the third syllable of **leu̯d^herotās*. Instead, he suggests that

[14] Ven. *libertos* 'freedman' and *louderobos* 'to/for the children' are usually placed with the other forms built to the root **h₁leud^h-*. *libertos* is most likely a borrowing from Latin, but *louderobos* is very likely inherited.

[15] Cf. Weiss 2009:191-2 for a relative chronology of sound changes. First, **-d^h-* > **-ð-*, and then **-eu̯-* > **-ou̯-*. **-ou̯-* must have been unrounded to **-oi̯-* (cf. Weiss 2009-141-2) after the *RUbL* Rule changed **-ð-* to *-b-* (on which cf. Weiss: 2009:75-6), based on OLat. *loebertatem* (P. *ex Fest.* 108 L). But as Weiss (2009:142n35) notes, the change **louferos* > **loiferos* was prehistoric, whereas *lubet* > *libet* happened much later, suggesting that the two changes are typologically similar. **-ou̯-* > *-oi̯-* may have been a diffused change.

remote syncope after *-r-* is indeed possible in **leu̯dʰerotās* (cf. 2008:143-6 for discission of syncope of this type), and that the resultant **leu̯dʰertās* would give *lībertās* (and Fal. **loifirtato** with raising of *-er-* to *-ir-*; cf. n. 21) without problem. If so, the development **leu̯dʰerotās* > **leu̯ðerotās* > **lou̯ðerotās* (cf. Ven. *louderobos*) > **lou̯ðertās* with syncope > **lou̯bertās* > **loi̯bertās* must be assumed. Such a development, though, must depend on *-u̯-* (rather than vocalic *-u-*) triggering the *RUbL* change of **-ð-* to *-b-*. *ūber* 'rich' (Pl.+) might support such a development **-u̯ð-* > **-u̯b-*, if it reflects *o*-grade **h₁ou̯H-dʰr-* with loss of **H* by the "Saussure Effect" (*EDLIL* s.v.), as seems to be likely, based on Gk. *οὐθαρ* 'udder'. Even if **lou̯bertās* ultimately gave **līfertās*, however, it is thinkable that it could have been analogically remodeled to *lībertās*, beside the adj. *liber* and the noun *lībertus*.[16]

Nishimura (2008:154n57) also offers the possibility that *lībertās* continues **līber-tās* (as opposed to the **leu̯dʰerotās* suggested above), with a direct suffixation of **-tās* to the adjective *līber*, after final syllable syncope had taken place in the adjective **līberos* > **lībers* > **līb̥rs*.[17] Fal. **loifirtato** could also reflect the same independent development, if it is not simply a borrowing from Latin (cf. **II.D.1.b.** below). It is certainly plausible that

[16] In fact, it is certainly the case that *lībertās* developed in parallel to *līber* and even *lībertus*, so analogical pressure would have been exerted by each of the forms to maintain a consistent root shape.

[17] Cf. below on the unlikely possibility that *lībertās* can continue a syncopated **liber-itās*, built with *-itās*, rather than plain *-tās*.

lībertās continues a relatively later formation **līber-tās*, but such a formation would also assume that plain *-tās* could still build abstracts by the time **līb̥rs* > *līber*. Both Indic and Greek had generalized **-otās* already in the earliest attestations of the abstracts, but it may be the case that Lat. *-tās* was not generalized to *-itās* in Latin until relatively late. Otherwise, **līberotās* > *ˣlīberitās* might actually be expected, since Latin hase several other abstracts in *-eritās*, e.g. *āceritās* and even the late analogical LIBERITAS. It is consequently very difficult to motivate a late formation in plain *-tās* after the generalization of *-itās* without an analogical model, but no apparent model exists, and so **līber-tās* is thus preferable to **līber-otās*.

Hence, it is very difficult to determine whether *lībertās* and **loifirtato** continue an early **leu̯dʰerotās* (with various syncopations) or a somewhat more recent **līber-tās*. It seems most likely that *lībertās* continues **leu̯dʰerotās*, with syncope before *-r-*; however, other evidence that suggests plain **-tās* could be appended to adjectival stems directly will be explored below (cf. the possibility that *voluptās* continues **velh₁p-tās*), leaving open the possibility that *lībertās* reflects *līber-* plus plain *-tās*. If **līber-tās* is the source of *lībertās*, it would be significant that abstract nouns could have been built with plain *-tās* to adjective stems without the need for any sort of linking vowel at a relatively late date in Latin.

II.D.1.a. *lībertus* 'freedman'

The abstract *lībertās* is likely one of the older *tās*-abstracts in Latin, continuing either **leu̯dʰero-tās* or less likely **līber-tās*. Beside *lībertās* and *līber* is *lībertus* 'freedman' (Pl.+), thought to be related derivationally to *lībertās*, given the association between *tās*-abstracts and *tus*-adjectives. Leumann (1977:335-6, 374) thought *lībertus* had to be a back-formation to *lībertās* on the model of the pair *honestās* ~ *honestus* (as were also *ūbertus* 'abundant', *maiestus* 'dignified' built from their respective *tās*-abstracts), and his opinion remains as the *communis opinio*, repeated by de Vaan (*EDLIL* s.v. *līber*). Brachet (2002:87-8) also arrived at the same conclusion, based on his semantic argument that *lībertus* must reflect use of the *-tus* suffix "pour désigner une qualité obtenue au terme d'un procès, par opposition à la même qualité présentée comme d'emblée inhérente à l'objet", comparable to the formation of *vetustus* beside *vetus* and *vetustās* (discussed later in **II.F.1.e.**).

Rix (1994:90-1) expanded on Leumann's idea, assuming a Proto-Italic *tās*-abstract **loiβerotāt-* on which a possessive adjective in **-to-* was built.[18] The adjective

[18] Interestingly, that *tās*-abstracts seem to be attracted to *to*-adjectives is also true in Indo-Iranian and Greek; but it is the complex morpheme **-to-tāt-*, not Rix's **-tāt-to-*, that is found elsewhere (cf. especially OAv./YAv. *amərətatāt-/amərətāt-* < **n̥-mr̥-tó-tāt-* and Gk. *ἀνδροτῆτα*, possibly continuing **h₂nr̥-tó-tāt-*, which will be treated in greater detail in **IV.C.3.a.**). It may well be coincidental that some of the oldest-looking material continues a putative **-to-tāt-*, but the pattern may also be inherited and of great importance in the development and generalization of **-otāt-*.

**loiβerotāt-to-* was subsequently reduced by haplology to **loiβeroto-* and thus Lat. *lībertus*. Rix added that the adjective **loiβerotāt-to-* belonged to Proto-Italic, since an older form should have given Lat. **leiberotāsso-* > *ˣlībertāsus* (*vel sim.*), with the expected Italic outcome of an original PIE dental cluster *-*TT*- > *-ss*-. But the existence of Rix's **loiβerotāt-to-* is difficult to accept for multiple reasons.

Brachet (2002:87-8), to begin with, rejected **loiβerotāt-to-* because *-*tāt-to*- lacks any trace in Latin.[19] Even more problematic for Rix's idea, however, is the fact that there is no support for denominal *to*-adjectives in Latin built to *t*-stems. The best examples are formations to thematic stems and some *i*- and *u*-stems (typically showing presuffixal lengthening), e.g. *barbātus* 'bearded' (Pl.+) to *barba* 'beard' and *aurītus* 'listening' (Pl.+) to *auris* 'ear', or *s*-stems, e.g. *honestus* 'honored' (Pl.+) to *honor* 'honor' (cf. Weiss 2009:292-3), and the suffix *-mentum* is also thought to continue a possessive derivative in *-*to*- to a *men*-stem (cf. Weiss 2009:313-4). But *-*tāt-to*- assumed for **loiβerotāt-to-*

[19] Generally, derivatives directly to *tās*-abstracts were rare, though a few are allegedly found. According to Leumann (1977:299), there existed a synchronic pattern within Latin whereby the adjectival suffix in *-ārius* could be appended to *-tās* abstracts, forming an adj. in *-tārius* by haplology from the resulting *-tāt-ārius*: *hērēditās* 'inheritance' (Cic.+) → **hērēditāt-ārius* > *hērēditārius* 'of inheritance' (Cic.+), *voluntās* 'will' (Pl.+) → **voluntāt-ārius* > *voluntārius* 'of one's accord' (Cic.+), *voluptās* 'pleasure' (Pl.+) → **voluptāt-ārius* > *voluptārius* 'concerning pleasure' (Pl.+). Given the general lateness of adjectives in *-tārius*, though, it could be the case that they are simply mechanical replacements of *-tās* with *-tārius*. A few abstracts with the compound suffix *-tātiō* are found later in Latin, e.g. post-Classical *dēhonestātiō* 'disgrace' (Tert.), *salūtātiō* 'greeting' (Cic.+), but such forms are just *tiō*-abstracts built to fourth principal parts of verbs, rather than derivatives to *tāt*-abstracts in *-*tāt-iō*-.

would require an entirely isolated formation in *-*to*- to -*tās*, none of which seem to exist in Latin.[20]

Additionally, the pattern of deriving a *to*-adjective from an underlying adjective is not old, unlike the pattern of deriving *to*-adjectives to *s*-stem nouns (on which cf. **II.F.1.**), so *lībertus* cannot continue and old *$*h_1leu̯d^hero$-*to*-. *lībertus* thus has all the appearances of being relatively young and secondary, likely formed by analogy to some of the older *tus*-adjectives with corresponding *tās*-abstracts like *honestās* and *honestus*, which did make a relatively old possessive adjective in *-*to*- (cf. **II.D.3.**), just as Leumann (1977:335-6, 374) suspected, e.g. *honestās* : *honestus* :: *lībertās* : *x*, where *x* = *lībertus*. Semantically, the adjective *lībertus* was substantivized as *lībertus* 'freedman' and *līberta* 'freedwoman'.

So, while *lībertās* seems to be one of the older abstracts in Latin reflecting plain -*tās*, *lībertus* is relatively young, formed off the semi-productive pattern whereby abstracts in -*tās* correspond with *to*-adjectives in -*tus*.

II.D.1.b. Fal. loifirtato 'Libertatis'

Faliscan gen. sg. **loifirtato** 'Libertatis' (Ve 253 = G 25), alongside **[l]o̯ifirta̯** 'freedwoman' (Ve 276a = G 25), **loferta** 'id.' (Ve 322a = G 121), and Ven. *libertos* and

[20] Cf. another derivative to a *tās*-abstract in the Mc. adj. *herentatia*, built to Pg. *herentas*/Osc. **herentateís** (Untermann 2000:320), which shows the normal pattern of building an adjective to a consonant-stem with *-*i̯o*-.

likely *]firtati* '*Libertati*' (cf. Vine 1993:170), correspond exactly with Lat. *lībertus* and *līberta* (for the most recent discussion of the Faliscan inscriptions, cf. Bakkum 2009:189, 424-5).[21] Rix (1994:95-6) suggested that it is certainly within the realm of possibility that *līberta* < (*Transponat*) *$h_1leu̯d^hero$-(tā)tā > PItal. *$lou̯\delta ero(tā)tā$ > Proto-Latino-Faliscan *$lou̯\beta er(o)(tā)tā$ > *$loi\beta er(o)tā$ was inherited into both languages, but adds that, while it may be the simplest interpretation, it is not necessary: the Faliscan material may simply be a loanword or a calque, with **f** spelling for Lat. *-b-*.[22] Traditionally, Faliscan is not thought to show syncope (cf. Nishimura 2008:140-2 with refs.), so it would be extremely surprising for **loifirtato**, etc. to show the same phonological developments as Lat. *lībertās*, if indeed dating back to a common Proto-Italic *$lou̯\delta ero(tā)tā$. With Lat. *lībertus* likely reflecting a recent, analogical creation and, moreover, a technical term of importants in Roman social law that could have easily spread into non-Roman areas, the latter of Rix's options, namely that **loifirtato**, etc. is simply a loanword, is preferable.

II.D.2. Abstracts in *-tās* and Adjectives in *-tus*

Latin somewhat productively built *to*-adjectives to *s*-stem nouns (Brugmann 1906:403), and 7 of the adjectives in *-stus* correspond to abstracts in *-stās* (cf. Fig. 3).

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[21] The dialectal change *-*erC*- > -*irC*- in **loifirtato** is also found in Praen. **merc-/merq-** beside Lat. *Mercurius*, likely reflecting a non-urban Latinism (cf. Vine 1993:161-73 and Adams 2008:89-91).

[22] Cf. also Fal. **efiles** '*aediles*' (Ve 264) with **f** spelling Latin *b* (Rix 1994:96n36).

Fig. 3. Latin abstracts in *-tās* beside adjectives in *-tus*

angustās 'narrowness' (Acc.)	*angustus* 'narrow' (Pl.+)
honestās 'honor, integrity' (Sis.+; later *honestitās*)	*honestus* 'honorable' (Pl.+)/ Pg. *hanustu* 'Honesta'
lībertās 'freedom' (Pl.+)	*lībertus* 'freedman' (Pl.+)
maiestās 'dignity' (Andr.+)	*maiestus* 'dignified' (hapax, Petr.)
tempestās 'period of time; storm' (*XII Tab.*+)	*intempestus* 'at the wrong time' (Enn.+)
venustās 'charm, grace, beauty' (Pl.+)	*venustus* 'charming' (Pl.+)
vetustās 'age, antiquity' (Pl.+)	*vetustus* 'ancient' (Pl.+).

There is accordingly a close derivational connection between *-stās* and *-stus* that developed fairly early in Latin, as has long been recognized (cf. Meyer-Lübke 1893:333-4, Stolz 1895:553, Brugmann, 1906:418, and Leumann 1977:335-6). All of the forms show the same vocalism in both root and suffix, and thus likely share derivational patterns too.

Brachet (2002:83) says of the relationship between adjectives in *-tus* and abstracts in *-tās*:

> En synchronie, la relation entre *honestus/uenustus* et *honestas/uenustas* peut se lire dans les deux sens: *honestas* et *uenustas* désignent les qualités respectives dénotées par *honestus* et *uenustus*, mais *honestus* et *uenustus* peuvent aussi s'interpréter respectivement comme signifiant "pourvu d'*honestas* et "pourvu de *uenustas*". Aussi peu nombreuses soient-elles, ces formations ont fait émerger un modèle morphologique qui associe par

permutation adjectif en *-tus* et substantif en *-tās*, et permet de pallier l'absence de dérivé en *-tus* sur abstrait en *-tās*.

Based on the pattern, as Brachet notes, the standard view repeated in all the handbooks is that new *to*-adjectives could be built as *Rückbildungen* to *tās*-abstracts or vice-versa, with a new *-tās* abstract built to older adjectives in *-tus* from relatively early on, e.g. as argued above for the noun *lībertus* 'freedman' back-formed analogically from *lībertās*. The derivational pattern has consequently become the standard explanation for the more obscure members of the *tās*-abstract : *tus*-adjective pair, but the question of which of the forms are older remains unanswered.

Rix (1994:90-1) has explored the relationship in the greatest detail, projecting it back to Proto-Italic. To him, some of the *to*-adjectives were formed directly to *tās*-abstracts, e.g. **loiβerotāt-to-* > *lībertus* (the lone substantive in *-tus* in the group), while some of the *tās*-abstracts were built to their *to*-adjectives, e.g. **honesto-tās* > *honestās*.[23] But as seen above, it is highly unlikely that **loiβerotāt-to-* existed in Latin, and thus at least some adjectives in *-tus* should have an independent derivational history from abstracts in *-tās*. The possibility that *honestās*, along with the other abstracts in Fig. 3,

[23] Though Accius' *honestitūdō* may be used in support of a *ti*-stem **honesti-* underlying **honesti-tās*, *honestitūdō* is more likely to continue **honesto-tūdō* by analogy to *pulchritūdō*, according to *WH* (s.v. *honōs*), built to the adjective *honestus*, with customary vowel weakening.

reflects a haplologized **honesto-tās*, with *-tās* added to the *to*-adjective rather than the *s*-stem base of *honor* (which will be explored in **II.D.3.a.**), will be addressed below.

II.D.3. *honestās* 'honor'

honestās 'honor' (beside a personified military god "Honor"; on which cf. below) is based on the *s*-stem *honor* 'honor' (OLat. *honōs*), which *WH* (s.v. *honōs*, with refs.) connected with OIr. *gen-*/MW *gwen-* 'smile' and Gk. *γάνυμαι* 'brighten up; be pleased'. But OIr. *gen-*/MW *gwen-* point to PCelt. **g^{w}eno-* (*EDPC* s.v.) that can neither phonologically nor semanically underlie *honor*, and *γάνυμαι* < **gh$_2$-n-u-* is a nasal-infix present to PIE **geh$_2$u-* 'be bright, be glad', which precludes any connection to the Celtic material (*DELG* s.v.). Differently, Debrunner (1933:195) reconstructed **g^{wh}onōs/*g^{wh}enōs* with "Sabine" *h-* for *honor*, beside Ved. *āhaná-* 'swelling, distended' and Gk. *εὐθενής* 'flowering' (with which also belongs Lith. *ganà* 'enough' and OCS *goněti* 'be sufficient'; cf. *EWAia* s.v. *āhanás-*, Stüber 2002:189-91); but *WH* (s.v. *honōs*) rejects the connection, on the grounds that such a dialectal origin of the initial consonant in *honōs* would be awkward. As de Vaan (*EDLIL* s.v. *honōs*) concludes, *honor* can only reflect **g^{h}on-* or **ĝ^{h}on-*; but based on Pg. *hanustu*, which implies **g^{h}n̥H-os-* (cf. **II.D.3.c.** below), the root could be *seṭ* **g^{h}enH-/*ĝ^{h}enH-*. There are no other known connections to such a root, and so the root etymology remains unclear.

Alongside *maiestās* (on which cf. **II.F.6.** below), *honestās*, with *e*-grade of the *s*-stem suffix, beside masc. *honor* 'honor', with consistent *o*-grade, seems to be the only example of a derivative in *-tās* or *-tus* to an animate *s*-stem with an *e*-grade suffix, as Meissner (2006:152) noted, and may reflect something old. The noun *honor* is representative of a well-known class of deverbal *s*-stem abstracts in Latin (Weiss 2009:307-8), but the root is without connection to any verbal material in Latin; there is no verb *ˣhoneō (vel sim.).[24] As noted in both *DELL* (s.v. *honōs*) and *WH* (s.v. *honōs*), *honor* beside *honestus* should point back to a neuter noun **honos, -eris*, but no such form is found in Latin. In principle, neut. **honos, -eris* cannot be excluded, though, since neuter *s*-stems coexisting beside animate *s*-stems do have parallels in Latin, e.g. *decus* 'dignity' (Pl.+) and *decor* 'beauty' (Naev.+). More crucially, but not mentioned by the etymological dictionaries, is the fact that such a neuter noun should turn up as **henos*, with *e*-grade in the root.

Nussbaum (2003), in his study of the "*bonus*" rule in Latin, whereby **-e-* > *-o-* in the environment *#(C)u̯_NV*[+back], suggests that *honor* may indeed reflect an older **henōs* or **enōs*. Because *honor* is never spelled without initial *h-*, it is likely

[24] The fact that *honor* lacks a synchronic verb, though, is not a serious problem by itself, given other isolated animate *s*-stems in Latin, e.g. the thematized s-stem underlying *aurōra* 'dawn', beside Ved. *uṣás-* 'id.' and Gk. *ἠώς*, *sūdor* 'sweat', etc.

etymological, and so the simplest starting point is **henōs*. Discounting a large number of Greek loanwords, initial *he-* in Latin is comparatively rare, limited only to the sequences *heCe-/heCi-* and *heCC-*. The general lack of initial *he-* raises the question whether the sequence **heCV-*[+back], or at the very least **heNV-*[+back], was also subject to the "*bonus*" rule, even if lacking the labial element **-u̯-*. If so, *honor* can point back to a neuter *s*-stem **g^{h}enH-os/es-* or **g^{h}n̥H-os/es-*, with either full or zero grade in the root (cf. below on Pg. *hanustu*).

honestās, as mentioned above, represents an older stratum of *tās*-abstracts within Latin, and there are two approaches for analyzing its derivational history, namely, a formation in *-tās* to the *to*-adjective in *honestus* with the invocation of haplology or a direct suffixation to the noun *honor*. The first option that *honestās* continues Rix's haplologized **honesto-tās*, with the abstract derived to the *to*-adjective, places it in the thematic derivation category. *tās*-abstracts are found in other languages that seemingly are built to *to*-adjectives, e.g. OAv. *amərətatāt-* 'immortality' < **n̥-mr̥tó-tāt-*, beside haplologized *amərətāt-* (cf. n. 18 above). If an old **honesto-tās* in Latin can be established, it might suggest that the pattern whereby *-tās* could build abstracts to adjectives in **-to-* also belongs to Latin.

Secondly, *honestās* might continue a direct suffixation with *-tās* to **hones-*, the weak stem from the underlying neuter *s*-stem noun **gʰenH-os/*gʰenH-es-* (on which cf. **II.D.3.c.** below). Such a derivational pattern, with *-tās* added directly to the base without a thematic vowel, has already been alleged for *lībertās* < **līber-tās* (if not continuing **leu̯dʰero-tās*) and will find other parallels below. Most importantly, **hones-tās* would show that *-tās* could be added directly to nouns in Latin, in addition to adjectives, which is typologically very unusual for abstracts built with **-tāt-*, based on its use in other languages.

II.D.3.a. *honestās* < Haplologized **honesto-tās*

**honesto-tās* is rejected by Brachet (2002:81-2), on the grounds that the chronology does not permit it: other *tās*-abstracts derived from adjectival stems in *-tus* surface in *-titās* without problem, with no observed haplology. He cites some other *tās*-abstracts built to stems in *-tus* in support of *-titās* forms in Latin that have not undergone haplology, e.g. *angustitās* 'narrowness' (hapax; also *angustās* in Acc.[25]) to *angustus* 'narrow' (Pl.+) and *beātitās* 'happiness' (Cic.+) to *beātus* 'happy' (Pl.+). Additionally, Brachet continues, haplology need not be assumed to affect **-to-tāt-es*, **-to-tāt-ei̯*, **-to-tāt-m̥*, etc. when other similar sequences survive unaffected in Latin, as do abstracts in -

[25] But the reading of *angustate* in Accius' *Alphesiboea* fragment is uncertain. The MSS variously transmit *angustatem* and even *angustitate* and *angustitatem* (cf. Ribbeck 1897:169).

titūdō, *-titūdinis*, etc. (cf. *honestitūdō* already in Acc.) that are never haplologized. The examples provided by Brachet, however, are all relatively late and reflect the early productive use of *-itās* for plain *-tās* and only suggest that *honestās* must have been formed before the generalization of the suffix as *-itās*, thus leaving open the possibility that *honestās* can indeed continue **honesto-tās*.

Despite Bracchet's reservations, it is certainly plausible that a paradigm like **honesto-tāt-s*, etc., could be subject to haplology, but there is no trace of **-titāt-* in old-looking material in Latin (or in Faliscan and Sabellian, for that matter).[26] If haplology did indeed occur, it must have occurred at a very early date, well before the historical period. But without any clear evidence of haplology, such a development cannot be assumed and other solutions should be explored in detail. If fact, it will be demonstrated that the older-looking Latin abstracts in *-tās* reflect abstract formations in which *-tās* is added directly to the stem, before the generalization of *-itās*.

II.D.3.b. *honestās < *hones- + -tās*

honestās could simply reflect **hones-tās*, with *-tās* added directly to the base **hones-* before the generalization of *-itās*, since an old formation **honesitās* would give *ˣhoneritās*/*ˣhonertās*. If so, such a formation would contradict the expected de-adjectival

[26] Haplology is in fact well-supported in Latin, e.g. OLat. *dīxtī* 'you (sg.) have spoken' beside CLat. *dīxistī*, *sēmodius* 'a half-*modius*' (a grain measurement) < **sēmimodius*, etc. (Leumann 1977:234-5, Mercado 2001).

derivation, with the *tās*-suffix appended directly to a noun. As will be demonstrated, several parallels for the use of plain *-tās* exist within Latin, where it could apparently be added directly to adjectival or nominal bases early on, without a connecting vowel, e.g. *lībertās* discussed above, and *tempestās*, *voluptās*, etc. discussed below.

Moreover, the *e*-grade of the suffix in *honestās* is expected from a PK derivational base, rather than AK **honos-* with *o*-grade. The *o*-grade in *honor* and other similar formations, e.g. *venustās*, is suggestive of later derivatives to the AK paradigm, after its vocalism was leveled in the nouns earlier in Latin, whereas other old-looking plain *tās*-abstracts built to neuter nouns reflecting PK ablaut, e.g. *tempestās* to *tempus*, *-oris* 'time', for instance, also reflect older formations from the oblique stem with *e*-grade of the suffix. Hence, **hones-tās* must reflect an archaism, with plain *-tās* added to an athematic stem, and so belongs with the other athematic *tās*-abstracts below, rather than the thematic abstracts after the assumption that it reflects **honesto-tās*.

If *-tās* was used to derive an abstract from a nominal base **hones-*, the question arises of why Latin would derive an abstract noun from another abstract noun. Such a denominal derivational pattern is found in a few old abstracts in other languages but only with collective semantics, e.g. Ved. *devátāt-/devátāti-* 'collection of the gods' (cf. **III.C.1.**). *honor* 'honor accorded by rank, special privilege' acquired a somewhat

technical sense as an official designation (*OLD* s.v.), but *honestās* 'honorableness' seems to only have been used as a pure abstract (cf. also concrete Greek *s*-stem abstracts beside qualitative abstracts in *-τηs*, e.g. ἦδοs 'vinegar; pleasure delight' beside ἡδύτηs 'sweetness'; Meissner 2006:111). Perhaps, much like *cīvitās* can be thought of as a renewal of the old *i*-stem abstract **ḱeiu̯i-* (cf. n. 10 above), *honestās* represents more of a true qualitative abstract as a renewal of sorts of the older *s*-stem abstract *honor*, with its more nuanced and technical semantics (cf. below for parallel developments in *venustās* beside *Venus*, etc.).

**hones-tās* is accordingly preferable to **honesto-tās*, but perhaps it does not reflect an old derivation, given that a non-collective denominal use does not seem to be a feature of PIE **-tāt-*; it may be a relatively recent formation created by analogy, e.g. *-tās* :*-tus* :: *x* : *honestus*, where *x* = *honestās*. If so, some analogical model must have already been well established at the time of the formation of *honestās*. *libertās* : *lībertus* could have served as the necessary model, depending on the relative chronology of formation, though *lībertus* is likely relatively recent and so it is difficult to know whether the *-tās* : *-tūs* pattern was already in place at the time of formation of *honestās*. The analogical argument, however, can only go so far before it becomes circular, so before it can be established that analogy drove the formation of *honestās* and even some of the other

forms in Fig. 3, clearly older abstracts built to *s*-stem nouns with corresponding adjectives in *-tus* must be identified.

II.D.3.c. Pg. *hanustu* 'Honesta'

Pg. *hanustu* (Pg 9), a fem. nom. sg. teh_2-stem noun, is unclear in both meaning and etymology, though has been suspected to reflect a deity "Honesta" and thus belongs with Lat. *honor* (Untermann 2000:317), but it must continue a different *s*-stem base than *honestās* and *honestus*. *hanustu* contains *o*-grade of the *s*-stem suffix, after generalization of the weak stem of the AK *s*-stem noun **honos-*, as is found in Latin in some other *to*-derivatives to *s*-stems, e.g. *onustus*, *venustus*, etc.[27] Of greater interest is the fact that *hanustu* also must show zero grade of the root, with *-a-* by "Untermann's law", by which preconsonantal **-n̥-* in initial syllables > *-a-*, elsewhere *-en-*, e.g. Osc. **anter**, Umb. *ander* < **n̥-ter* (cf. Meiser 1986:69-70).

In Latin, the clear traces of isolated AK *s*-stems also reflect zero grade in the root, as in *cruor* 'bloodshed' (Acc.+) < **kruh₂-ōs*, beside OIr. *crú* 'id.', Ved. *kraviṣ-* 'raw meat', Gk. κρέας 'id.', Lith. *kraũjas* 'blood', OE *hrēaw* 'raw', etc. (cf. *EDLIL* s.v. *cruor*).

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[27] The suffix vocalism in *onustus* and *venustus* is likely just copied from the nominative. The distribution of the suffix vocalism of *to*-adjectives built to *s*-stem nouns that lack corresponding *tās*-abstracts reveals both *e*-grade, e.g. *fūnestus* 'deadly, fatal' (Acc.+), *modestus* 'modest, calm' (Pl.+), and *scelestus* 'wicked' (Pl.+), and *o*-grade, e.g. *arbustus* 'planted with trees' (Cic.+) beside *arbustum* 'grove' (Naev.+), *augustus* 'solemn, venerable' (Enn.+), and *rōbustus* 'oaken; strong' (Pl.+). Neither suffix type was preferred in any way in Latin and simply reflects the relative age of formation, with *e*-grade representing an older layer of formations.

And so, *hanustu* must continue the weak stem with zero grade of the root from an AK noun *g^{h}énH-os-/*g^{h}ṇH-s-́ 'having honor' of the same neuter PK *s*-stem noun presumedfor Lat. **hones*-, which shows the typical possessive characteristics associated with AK derivatives from PK inflection (Weiss 2009:262). AK *g^{h}énH-os-/*g^{h}ṇH-s-́, must also be the source of Lat. *honor*, which continues the leveled full grade *g^{h}enH-os-, with *g^{h}enH-os- > *g^{h}onH-os- by the modified "*bonus*" rule above, beside Pg. *hanustu*, which must reflect zero-grade *g^{h}ṇH-os-.

II.D.4. Remaining Exceptions to *o*-stems Making Abstracts in *-itās*

The remaining thematic adjectives that make abstracts in plan *-tās* rather than generalized *-itās* can be explained without difficulty, in that the abstracts all have analogical sources (cf. Leumann 1977:374). Seeing that no such adjective *viduerus/*viduer* or *vidueris is found, *viduertās* 'bereavement', found in Cato's Mars Prayer (*Ag.*141) in the sequence *viduertatem vastitudinemque calamitates intemperiasque* must be a secondary analogical formation, based on the model of its antonymic counterpart *ūbertās* 'richness, fullness' (Cic.+; Leumann 1977:374, after Wackernagel 1897:36), even though *ūbertās* is attested later. *pūbertās* 'puberty, virility' does not appear until Cicero and may likely also be a secondary creation based on its phonetic affinity with *ūbertās*. *paupertās* 'moderate circumstances, poverty' is found

already in Plautus, and belongs in the same class of secondary formations as *viduertās*; in all likelihood it is also analogical, perhaps even modeled on *ūbertās* (discussed below in **II.F.10.** with the other athematic base types).

II.F. Athematic Bases

Just as *tās*-abstracts built to second-declension adjectives reflect *-itās* from **-o-tās*, abstracts built to third-declension adjectives (and some abstracts to *i*-stem nouns like *cīvis*) reflect a suffix in *-itās* from **-i-tās*, e.g. *brevitās* 'shortness', *facilitās* 'ease', *fidēlitās* 'loyalty', *gravitās* 'heaviness', etc.[28] **-i-tās* from *i*-stem adjective bases fell together with **-o-tās* from thematic adjective bases after the effects of vowel weakening, giving *-itās*, from which the suffix shape *-itās* was extracted as a generalized abstract suffix that was then used with consonant stems. Already in the earliest attested literature, the reanalyzed *-itās* replaced plain *-tās* as the productive suffix used to build some obviously secondary creations as, for instance, *auctōritās* 'authority' (already *XII Tab.*), Accius' *ponderitātem* 'weight' (Blänsdorf 1995:89), etc. The remaining abstracts showing plain *-tās* built to consonant-stem bases are few in number and fall into specific

[28] Very early in Latin, PIE consonant-stem adjectives merged with *i*-stem and *u*-stem adjectives into the third declension, inflecting as *i*-stems synchronically. In principle, a very old abstract built to a *u*-stem adjective before its conversion to an *i*-stem would appear as **-itās* ← **-u-tās*, falling together with the other abstracts from thematic and *i*-stem adjectives; but it seems clear enough that no Latin abstracts continue any direct formations to PIE *u*-stems, unlike some examples in Indo-Iranian, e.g. Ved. *vasútāti-* 'collection of Vasus' to *vásu-* 'good' (cf. **III.D.1.**), YAv. *pourutāt-* 'multiplicity' to *pouru-* 'much, many'; and Greek, e.g. Hom. βραδυτής 'slowness' to βραδύς 'slow' (cf. **IV.A.**).

categories.[29] The abstracts, in fact, contain some of the older material found in Latin, with forms continuing old *s*-stems, e.g. *egestās* 'destitution' and *maiestās* 'majesty'; an *n*-stem in *iuventās* 'youth'; and even some *nt*-participles, e.g. *voluntās* 'will' and Osc. **herentas** 'Herentas'.

II.F.1. *s*-Stem Nouns and Adjectives in *-stus*

Latin attests a large number of adjectives in *-stus* that continue possessive *to*-adjectives built to *s*-stem nouns. The pattern of adjective formation in *-stus* was moderately productive not only in Latin but even seemed to exist already in PIE (Stüber 2002:31), e.g. **$h_2em\hat{g}^h$-os-to-* > Lat. *angustus* 'narrow' and OFr. *angost* 'fear' beside **$h_2em\hat{g}^h$-os-* > Lat. *angor, -ōris* 'anxiety' (Cic.+), with further evidence for an adjectival suffix **-os-to-* found in Slavic (cf. Vaillant 1974:376-7) and Anatolian (cf. Melchert 1999a:366). Thus, it would seem to be the case that the Latin adjectives in *-stus* could be quite old.

Latin possessed 10 *tās*-abstracts built to *s*-stem bases: *aestās* 'summer' (Pl.+), *angustās* 'narrowness' (hapax; Acc.), *egestās* 'destitution' (Pl.+), *honestās* 'reputation, esteem' (Cic.+), *maiestās* 'dignity' (Andr.+), *pesestās* '*significare videtur pestilentiam*, plague' (P. *ex Fest.*, p. 231 L), *potestās* 'power' (*XII Tab.*+), *tempestās* 'period of time, season; storm' (*XII Tab.*+), *venustās* 'charm' (Pl.+), and *vetustās* 'antiquity' (Pl.+).

[29] A few even show both suffix shapes: *facilis* → *facultās* next to *facilitās* (Leumann 1977:374).

Despite Meyer-Lübke's (1893:328-9) assurance that the formations continue *-itās* built directly to the *s*-stem, with syncopation of *-i-* in **-s-itās* > *-stās*, this clearly cannot be the case. As argued above, a form like **honesitās* would be subject to rhotacism, giving in all likelihood **honeritās*, and then **honertās* by syncope, perhaps. One could, of course, push Meyer-Lübke's argument further, claiming that **hones-itās* was syncopated to *honestās* before rhotacism, which began sometime in the middle of the 4th century, given that evidence points to the generalization of *-itās* already by the middle of the 5th century, based on the clearly secondary *auctōritās* already in the *XII Tab.* Additionally, it seems to be the case that, once the suffix *-itās* was generalized, it became morphologically "protected" against deletion, in that the linking vowel *-i-* was maintained in all derivatives, even where analogy would have facilitated its loss.[30] As mentioned above, for example, obviously late-looking abstracts to adjectives in *-er* always surface in *-eritās* (cf. even LIBERITAS). If *honestās* were indeed built with *-itās* after the suffix shape became grounded in Latin abstract morphology, **honeritās* could well be the expected outcome.

[30] For a more thorough discussion of analogical factors preventing syncope, cf. Nishimura 2008:135 with n. 29, where he notes that the idea that *-itās* was analogically preserved is already proposed by Ciardi-Dupré (1901:199). If the connecting vowel before *-itās* was indeed syncopated, syncope must have happened very early, well before the generalization of *-itās*.

Hence, given the pattern of adding plain *-tās* directly to stems that was already well established by the time of the formation of *honestās*, it is preferable to reject Meyer-Lübke's **honesitās* (and likewise **līberitās* underlying *lībertās*) in favor of the more archaic formation **hones-tās*, with plain *-tās* added to the *s*-stem before generalization of *-itās*. Consequently, the abstracts in *-stās* without analogical origin that were built to *s*-stems must reflect some of the older *-tās* material in Latin.

Brachet (2002:82-3) suggests that *egestās*, along with *tempestās* (to *tempus*, *-oris* 'period of time'), *honestās* (to *honor*, *-ōris* 'respect, esteem'), *aestās* 'summer heat', *venustās* 'charm, grace, beauty' (to *venus*, *-eris* 'grace, charm'), and perhaps even *vetustās* 'age, antiquity' (to the adjective *vetus*, *-eris* 'old'), may be ultimately based on nouns rather than adjectives.[31] All of the putative nominal bases, moreover, are *s*-stem nouns in Latin, with the exception of *aestās*, which lacks a synchronic base. A denominal pattern of *tās*-abstract formation has important implications for the history of the *tās*-suffix in Latin, since the other branches that have denominal *tās*-abstracts only possess a small number of examples, and the examples, moreover, reflects collective semantics (cf. **III.A.3.** for Vedic, **III.A.4.** for Avestan, and **IV.A.2.** and **3.** for Greek collective abstracts).

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[31] Leumann (1977:374) does not believe the adjective *vetus* is old, but rather a back-formation to *vetustās*. Brachet (2002:83-6) rightly rejects the idea, based on the lack of any analogical model for the extraction of an *s*-stem adjective. The problem will be treated in greater detail below.

Brachet (2002:82-3) does not comment in detail on the fact that the ablaut grades of the suffix in the *s*-stem nouns from which the *tās*-abstracts are derived do not all coincide.[32] The disparity in ablaut underlying *tempestās* beside *tempus, -oris* (n) and *honestās* beside *honor* (m) is significant, in that **tempestāt-* and **honestāt-*, which reflect *e* grade of the *s*-stem suffix, cannot have been derived directly from the nouns as they appear in Latin (but cf. *temperī* 'at the right time' below). Some of the abstracts in *-stās* may indeed originate as backformations from the *to*-adjectives built to the same underlying nouns, e.g. *honestās* (cf. discussion above); but others may be quite old. If so, the variation in suffixal vocalism beween synchronic base and abstract may be one piece of evidence reflecting their antiquity. Thus, each of the abstracts in *-tās* in Fig. 3 above will be treated in detail below to discover if they have anything to add to the history of *-tās* in Latin.

II.F.1.a. *egestās* 'destitution'

egestās lacks a synchronic *to*-adjective **egestus*, and therefore cannot be assumed to continue a back-formation of any sort to a *to*-adjective. Rather, it is thought to continue a very old abstract built directly to the unattested *s*-stem **eg-os/*eg-es-*, indirectly visible in the adjective *egēnus* 'destitute of' (Pl.+) < **eges-no-* (*EDLIL* s.v.

[32] Brachet (2002:82n9) does provide a cryptic footnote regarding the discrepant vowel grades: "[i]l y a entre *hone*stus et *venu*stus| la même différence qu'entre, p. ex., *onu*|stus| et *scele*|stus|. On ne voit pas ce qui la motive."

egeō). The *s*-stem **eg-os/*eg-es-* points back to a root that has good support elsewhere: OIc. *ekla* 'lack, want', OHG *eko-rōdo* 'only', TA/TB *yäk-* 'neglect, be careless about', and probably also Osc. *egmo* 'case, question' (Untermann 2000:198). Adjectives in **-no-* are closely associated with *to*-adjectives, as is also seen with *venēnum* 'poison' (Pl.+) < **u̯enh$_1$-es-no-* (cf. *EDLIL* s.v.) beside *venustus*, so it is thinkable that **eg-es-tās* may have been built beside the adjective **eg-es-no-*, rather than a *to*-adjective. Moreover, the suffix **-no-* is a well-known member of the Caland system (cf. Nussbaum 1976), beside stative verbs, as in Lat. *egēre* 'need, want' (Pl.+), giving further support for an old PIE *s*-stem **eg-os/*eg-es-*. *egestās*, then, most likely points back to **eg-es-tās*, an abstract built with plain *-tās* and reflects one of the older abstracts in *-tās* in Latin. Brachet (2002:82) is at least partially correct in thinking that *egenus/egestas* and *honestus/honestas* are the oldest of the *-tās* : *-tus* pairs. But other *tās*-abstracts in Latin show similar morphology to *egestās* and *honestās* and may also be old, including *tempestās* 'period of time, season; storm'.

II.F.1.b. *tempestās* 'period of time, season; storm'

tempestās 'period of time, season; storm', first appearing in the *XII Tab.* in the sense 'time', is often found in archaic Latin prose.[33] Though *tās*-abstracts in Latin, including **-tāt-* in other languages, made no old-looking plurals (Leumann 1977:375), the dat. pl. TEMPESTATEBVS is found in one of the Scipio inscriptions (CIL I^2 9.6), dating to the mid-3rd BCE, in the phrase DEDET.TEMPESTATEBVS.AIDE.MERETO['he deservedly dedicated a temple to the Tempestates'. *tempestās* could be a back-formation to *tempestus*, but the word occurs only in Paulus *ex Festo* (p. 499 L). The adjective *intempestus* 'at the wrong time' (Enn.+) and *tempestīvus* 'seasonable' (Pl.+), though, may provide indirect evidence for an older *to*-adjective;[34] but given relatively early dates of attestation and adaptation to the name of a relatively important set of deities, *tempestās* might be better taken as archaic.

Underlying *tempestās* and the neut. noun *tempus, -oris* 'period of time' (Pl.+) is the *s*-stem noun **temp-os/*temp-es-* 'stretch, measure', in turn based on the verbal root **temp-* 'stretch, string', with which belongs Lith. *tem̃pti* 'pull, stretch', OIc. *þǫmb* 'bow', TA *cämpäṣ*/TB *campäṃ* 'is able' (*EDLIL* s.v. *templum, tempus*). **temp-os/*temp-es-* is

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[33] Beside *tempestās* is Varro's single instance of the *-tūs* abstract *tempestūs* at *Ling. Lat.* 7.51, which he glosses as '*libri augurum pro tempestate tempestutem dicunt supremum augurii tempus*'.

[34] Adjectives in *-tīvus* are regularly built to *to*-adjectives (cf. Nussbaum 1975a, Weiss 2009:299), suggesting an underlying **tempesto-*.

a standard neuter *s*-stem verbal abstract built to the root, with generalization of **-os-* in Latin. Possibly reflecting *e*-grade of the suffix, though, are the adverbs *temperī* 'at the right time' (Pl.+; beside *temporī* also in Pl.) and *temperius* 'earlier' (Cic.+), the adj. *tempestīvus* 'at the proper time, ripe' (Pl.+), and the verb *temperāre* 'to restrain oneself, modify, blend' (Pl.+), all of which could have been formed to **temp-es-* before **-os-* in the suffix was leveled.[35] *tempestās* must also reflect a relatively old formation in *-tās* to the stem *tempes-*, in which the meaning of *tempus* 'period of time' was specialized with the *tās*-suffix to mean something like 'specific (bad) period of time; storm'.

Tempestas, usually pluralized as gods of the weather, is an old figure closely associated with Mars in archaic Roman religion (Carandini 2002:207). In addition to their epigraphic testimony, the gods also figure in Augustan literature, e.g. Ovid *Fast.* 6.193-4, which refers to the very same temple of Tempestas in *CIL* I^2 9.6 dedicated by Scipio in 259 BCE in the epitaph above after surviving a storm and may even be a direct reference to the earlier dedication, with the phrase *Tempestas meritam delubra* used similarly to TEMPESTATEBVS.AIDE.MERETO:

> *te quoque, Tempestas, meritam delubra fatemur*
> *cum paene est Corsis obruta classis aquis*

[35] N.B.: The value of *temperius*, etc. above for *e*-grade is minimized by the fact that it is ambiguous. It could indeed reflect *e*-grade of the *s*-stem suffix, but it could also reflect *o*-grade, with **-o-* > *-e-*. In Latin, all short vowels ended up as *-e-* in a medial open syllable before **-r-* (Weiss 2009:117-8), hence both **temp-es-ei̯* and *temp-os-ei̯* would give *temperī*.

I confess you too, *Tempestas*, are deserving of sanctuaries,
when our ship was nearly destroyed in Corsican waters.

A similar passage is found in the *Aeneid* (5.772-3):

tris Eryci vitulos et Tempestatibus agnam
caedere deinde iubet solvique ex ordine funem

Then he ordered them to slaughter three calves for Eryx and
a lamb to the *Tempestates* and the cable to be duly loosened.

And another occurs in Horace (*Ep.* 10.23-4):

libidinosus immolabitur caper
et agna Tempestatibus

A capricious goat and a lamb
will be sacrificed to the *Tempestates*.

Mankin (1995:192), in his commentary on Horace's *Epodes*, notes that the sacrifice of a lamb in this passage is frequently associated with the wind cult in Greek, and because of the well-known connection, scholars have frequently translated *tempestātēs* as 'winds', comparing the use in Latin to Greek epic, e.g. ἀνέμοισι 'wind' at Hom. *Il.* 23.194 and ἀνέμους at Hes. *Theog.* 378. Lat. *tempestās*, however, never really means 'wind' alone in a sense comparable to Greek Boreas and Notus. The literary passages, of course, could well reflect Hellenisms, with the lines modeled on Greek practice; but the evidence suggests that Roman religion may have had its own deities known as the *Tempestates* independent of Greek influence. And even if *tempestās* reflects a Hellenism, it is interesting that *Tempestās* was named using a *tās*-derivative, in

that parallels exist for personifications of *tās*-abstracts for theonyms in both Latin and Indo-Iranian.[36]

II.F.1.c. *venustās* 'charm, grace, beauty'

venustās (Pl.+) is related to the feminine *s*-stem theonym *Venus, -eris* 'Venus' (Naev.+), also found in Osc. dat. sg. *ϝενζηι* '*Venerī*' (*ST* Lu 31).[37] Also underlying the *s*-stem is the well-attested root *$^*(H)\underset{\sim}{\mu}enh_1$- 'desire', e.g. Ved. *vánas-* 'love', *vánati* < *$^*\underset{\sim}{\mu}enH$- and PPP *-vāta-*, desid. *vívāsati* 'wishes, likes' < *$^*\underset{\sim}{\mu}\underset{\circ}{n}H$-, Go. *wunan* 'be delighted', etc. (cf. *EDLIL* s.v. *vēnor, venus*, *DELL* s.v.).[38] *Venus, -eris*, though feminine, must continue a neuter *s*-stem *$^*\underset{\sim}{\mu}en$-os-/*$\underset{\sim}{\mu}en$-es- 'charm', to which the abstract *venustās* reflects *o*-grade from the strong stem.

Unlike *tempus*, with generalized *o*-grade of the *s*-stem suffix, *Venus* kept the original suffix ablaut, as was not unusal for some neuter *s*-stems, e.g. *onus, -eris*, etc. The abstract *venustās*, consequently, suggests derivation to *o*-grade of the suffix, which can only have been found in the strong stem of the noun. For this reason, *venustās* is highly unlikely to be archaic; it seems to be based on the nom. sg., as are the other

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[36] Cf. Lat. *Honestās* and Pg. *herentas*/Osc. **herentateis** (discussed below) and Avestan names built with *-tās* in **III.E.**, e.g. YAv. *arštāt-*, the name of a goddess "Uprightness".

[37] Osc. *ϝενζηι* is thought to be a Latin borrowing (Untermann 2000:837-8 with refs., *EDLIL* s.v. *venus*). The spelling *νζ* continues prerhotacized *$^*\underset{\sim}{\mu}enesē\underset{\sim}{i}$ giving *$^*\underset{\sim}{\mu}ensē\underset{\sim}{i}$ by syncope.

[38] Kloekhorst (2008:999) adds the initial laryngeal *$^*(H)\underset{\sim}{\mu}enh_1$-, based on Hitt. *ú-* in the verb *$\underset{\sim}{\mu}en$-^{zi}/$u\underset{\sim}{\mu}an$-* 'copulate'. If the Hittite form is cognate, moreover, it would also identify the final laryngeal as *$^*h_1$ (*EDLIL* s.v. *vēnor*, cf. LIV s.v. *$^*\underset{\sim}{\mu}enH$-).

obviously secondary derivatives, e.g. *venustus* 'charming' (Pl.+) and *venustulus* 'charming' (Pl.), unlike the old-looking *i̯o*-adjective *venerius* 'of Venus; erotic (later)' (Pl.+), the denominal verb *venerāre/venerārī* 'worship, pay homage' (Naev.+), and the noun *venēnum* 'poison' (Pl.+) < **u̯en-es-no-*, which are all likely older than *venustās*, all reflecting the oblique stem with *e* grade of the suffix.[39] It would seem to be the case that, based on the semi-productive pattern by which *tās*-abstracts could be built to *s*-stems, both *venustās* and *venustus* were built directly to the nom. sg. of the noun **u̯en-os-/*u̯en-es-*, constituting a *tās*-abstract noun designating the quality "charm; desire" and a possessive adjective in *-stus* based on that quality.

II.F.1.c.i. Theonyms in *-tās*

Though *venustās* lacks the appearance of being as old as *honestās* or *tempestās*, the three abstracts all seem to show the same pattern, namely, deriving an abstract noun in *-tās* directly from *s*-stem abstract, which was not a prevalent pattern of word-formation in Indo-European. None of them reflect the semantics of a true qualitative abstract built to an adjective, but rather reflect specialized uses of the base *s*-stem noun. *honestās* 'honor' seems to have been a generalized version of the nuanced *honor* 'a specific honor, *tempestās* 'bad period of time; storm' is a specialization of *tempus* 'period of time', and

[39] Other than *venēnum*, the data are ambiguous in reflecting *e*- or *o*-grade in the *s*-stem suffix (cf. on *temperī* above in n. 35).

venustās 'charm' reflects the quality that the goddess Venus possessed. Additionally, *honestās* and *tempestās*, along with *Lībertās*, Pg. *herentas*/Osc. **herentateís**, and several other examples of divine figures in *-tās*, were personified as divinities, which perhaps reveal that the *tās*-suffix had a special association with divine names. No other complex abstract suffix, e.g. *-tiō*, *-tūdō*, *-tūs*, etc., had a similar function. While Fal. **loifirtato**, etc. and Osc. **aítateís**/Pg. *aetatu* (on which cf. below) can be interpreted as borrowings from Latin, Pg. *herentas*/Osc. **herentateís**, unless they are just calques on Lat. *voluntās*, must reflect Sabellian creations and so might provide evidence that the "individualizing" function of **-tās* in naming abstract divinities may belong to Proto-Italic and not just Latin (cf. **III.E.** for similar uses in Indo-Iranian). If so, the shared function may also imply that **-tāt-* belongs to Proto-Italic too, rather than just Latin (cf. **II.F.4.** below Osc. **aítateís**).

II.F.1.d. Pg. ***herentas*****/Osc. herentateís 'Herentas'**

Pg. *herentas*/Osc. **herentateís** reflect abstracts in **-tāt-* that signify a deity "Herentas", in a sense similar to Latin 'Voluntas, Venus' (Untermann 2000:319-20). An Oscan dedicatory inscription to "Herentas" (*ST* Cm 10) from Herculaneum attests the abstract Osc. **herentateís** twice:

> [a]**herentateís.súm**
> [b]**l(úvkis).slabiis.l(úvkieís).aukíl.meddíss.túvtíks.herentateí.herukinaí.prúffed**

A similar dedicatory inscription from Frentani (*ST* Fr 6) on the opposite coast from Herculaneum is also found: [a]**herettates.súm** [b]**agerllúd**. No Latin parallels for this sort of dedication with a deity "Voluntas" are known, making the likelihood that it is a borrowing into Sabellian very unlikely. Even line 7 of the famous Paelignian "Prismu Petieðu" inscription (*ST* Pg 9) *lexe.lifar∧dida∧uus∧deti∧hanustu∧herentas*, heavy with Latin influence (cf. below on *aetatu*, also appearing in line 5 of the inscription), seems to contain non-Latin thematic material.[40] Several attestations of 'Herentas' are known, suggesting the god was relatively important, e.g. Pg. nom. sg. *herentas* (*ST* Pg 9), Osc. gen. sg. **herentateís** (*ST* Cm 10)/ **herettates** (*ST* Fr 4), dat. sg. **herentateí** (*ST* Cm 10)/**heretatei** (*ST* Cm 11).[41]

The abstracts belong with the root *$\hat{g}^h$er- 'enjoy, be excited', which serves the same verbal role in the Sabellian languages as *volō* (< *$\underset{\sim}{u}elh_1$- 'wish, want'; cf. discussion of *voluntās* below) does in Latin. *$\hat{g}^h$er-, moreover, is found in Latin too in the verb *horior* 'encourage, urge' (Enn.+), and likely also in the goddess name *Herem* (Enn.) and an epithet of Juno *Heriem* (Gel.).[42] The root also has a small number of well-represented

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[40] Cf. Mercado (2007) for a discussion of the poetics and metrical qualities of the inscription, and Vine (1993:366-71) for issues connected with complex punctuation found here, including the interpretation of the verbal form *lexe*.

[41] Untermann (2000:319 with refs.) takes **herettates** as a misspelling for **here<n>tates** and adds that the spelling of **heretatei** without **-n-** is unproblematic, given that nasals were frequently not written before consonants (cf. Buck 1928:71).

[42] For the vocalism of *horior*, *LIV* (s.v. 1. *$\hat{g}^h$er-) posits a full-grade *$\hat{g}^h$ér-$\underset{\sim}{i}e/o$-, based on the Vedic and Umbrian comparanda following Nussbaum (1975b), that was either remodeled after *hortārī* or the phonological outcome of zero-grade *$\hat{g}^h\underset{\circ}{r}$-$\underset{\sim}{i}e/o$-. Untermann (2000:321-4) notes that *o*-grade in *horior* could

cognates outside of Italic in Ved. *háryati* 'be pleased', OAv. *zara-* 'aim, goal', Gk. χαίρω 'be pleased', OHG *ger* 'eager, zealous' and *gerno* 'eager, readily', and OIr. *gor* 'pious' (*EDLIL* s.v. *horior*).

*$\hat{g}^{h}$er- is well-attested in Sabellian, found in the verbs Osc. **heriiad**/Umb. **heri** 'wish, want', adverbs in Umb. **heris**...**heris** 'either...or', interrogative Umb. *pisher* 'whoever', and the abstract nouns Pg. *herentas*, Osc. **herentateís**, beside the denominal adjective in Mc. *herentatia* 'of the goddess Herentas' (cf. *EDLIL* s.v. *horior* and Untermann 2000:321-4 for a complete list of attested forms).[43] Nussbaum (1975b:282-3) showed that Osc. **heriiad**/Umb. **heri** must go back to a full-grade *i̯e/o*-present *$\hat{g}^{h}$er-*i̯e/o*-, while Lat. *horior* continues a zero-grade *$\hat{g}^{h}$r̥-*i̯é/ó*-, which can reflect a PIE athematic present *$\hat{g}^{h}$er-/*$\hat{g}^{h}$r̥- later renewed with **i̯e/o*-, giving both the full-grade and zero-grade *i̯e/o*-presents reflected above (Nussbaum 1975b:282).

Pg. *herentas*/Osc. **herentateís** must assume as its base *$\hat{g}^{h}$er-*n̥t*-, an *n̥t*-participle built to the root present, rather than *$\hat{g}^{h}$er-*i̯ent*- built to the *i̯e/o*-present. Corroborating

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be reflective of an abnormal causative *$\hat{g}^{h}$or-*i̯e/o*- (which perhaps is found in Ven. *horeionte*, cf. *LIV* s.v. 1. *$\hat{g}^{h}$er- n. 8), but *horior* is better interpreted as reflecting zero-grade *$\hat{g}^{h}$r̥-*i̯e/o*- (*EDLIL* s.v.).

[43] Osc. PN **heírens** (*ST* ZO 1)/*hηιρενσ* (*ST* Lu 5) and the gentilicium *herenneis* (*ST* Sa 35)/**heírennis** (*ST* Cm 6), may somehow belong with *$\hat{g}^{h}$er-. **heírens** (beside Lat. *Herennius*) reflects a suffix in **-en(n)o-* (Lejeune 1976:83), to which *herenneis* continues a *i̯o*-derivative (Lejeune 1976:103). **hereiis** (*ST* Cm 14) reflects **-i̯o-* directly to the root (Lejeune 1976:144), and may also go with Etr. *Erie*, which could be a misspelling for *Herie*, and Fal. *Fere*, beside Lat. *Herius* (Marchesini 2007:85). The source of the diphthong in the Oscan form vis-à-vis **hereiis** and Lat. *Herius* is mysterious, and it cannot have anything to do with *$\hat{g}^{h}$er-, leaving the connection with the root unlikely.

*$\hat{g}^h$er-n̥t- is Umb. **herinties**, an abl. pl. of a pres. pple. to Nussbaum's mind (1975b:286), in that it must reflect an Umbrian remodeling of *herent- (with 3 sg. pres. **heri** < *herīt) to *herint- on the analogy 3 sg. pres. -āt : pres. pple. -ant- :: -ĕt : -ent- :: -ĭt : *x*, where *x* = -int-.[44] Driving the remodeling is the fact that a present participle of the shape *herent- would not seem to belong to the verbal paradigm *$\hat{g}^h$er-i̯e/o-, after the renewal of present *$\hat{g}^h$er-/*$\hat{g}^h$r̥- to *$\hat{g}^h$er-i̯e/o- > *heri̯o- in Sabellian, as is likely reflected in the Hesychian gloss 'Εριέντης· 'Αφροδίτης ἐπώνυμον, a loanword into Greek from Oscan (Untermann 2000:319). Nussbaum (1975b:296-7n27) had the same idea, taking 'Εριέντης to reflect Osc. *herient-, which was also remodeled from an older anomalous *herent- on the analogy 3 pl. pres. -ant : pres. pple. -ant- :: 3 pl. pres. -ient (e.g., as in **fiiet** '*fiunt*') : *x*, where *x* = -ient.[45]

With *$\hat{g}^h$er-n̥t- > *herent- 'wishing, wanting' established in Proto-Sabellian, the origin of Pg. *herentas*/Osc. **herentateís** becomes clear: it continues a formation in -tās built directly to the old participle stem *herent- before it was remodeled.[46] *herent-tās

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[44] Meiser (1986:112) offered another approach for **herinties**, suggesting that the pres. pple. *her-i̯ent- was remodeled to *her-int- after internal syncope reshaped the present stem of the verb from *her-i̯e/o- to *heri-.

[45] Nussbaum (1975b:296-7n27), as typological parallel for the remodeling of Osc. *herient- from *herent-, offers the remodeling of the Doric present participle in -ουτες to -εντες after the 3rd pl. pres. in -εντι.

[46] Thus far, participial stems have not been identified as derivational bases for abstracts in -tās, but as will be seen later (cf. **III.A.1.** and **III.A.2.**), support for such deparicipial abstracts will also be found in Indo-Iranian, e.g. Ved. *gr̥bhītátāti-* 'captivity' is built to a *to*-participle *gr̥bhītá-* 'seized' and YAv. *nərəfsaṣtāt-* 'waning' to *narp-* 'wane' (cf. Renou and Benveniste 1960:19). There is no archaic evidence for *-tās used with *to*-participles in Latin (cf. on *honesto-tās*), since *-to-tās, if actually built early in Proto-Italic, was

could easily have meant something along the lines of "desiring (adj.)" → "that which is characterized by desire" → "Desire". Phonologically, the inner-Sabellian development of **herenttās* > Pg. *herentas*/Osc. **herentateís**, with simplification of **-tt-* to *-t-* is trivial.[47] Proto-Sabellian dental clusters **-tt-* seem to have been simplified to *-t-*, at least in Umbrian, based on the testimony of Umb. **titu** 'let one give' < **dedatōd* < **di-dh$_3$-tōd* and Umb. *prevendu* 'let one turn' < **praį-u̯end^h-e-tōd* (Untermann 2000:175-8 and 835-6, respectively, following Meiser 1986:180).[48] Whether or not Pg. *herentas*/Osc. **herentateís** represents a parallel but independent formation or a calque on Lat. *Voluptās* is open to question (on which, cf. **II.F.11.a.** below).

II.F.1.e. *vetustās* 'age, antiquity'

The final clear *s*-stem of the group in Fig. 3 above, the abstract *vetustās* (Pl.+), is built to the Latin adjective *vetus, veteris* 'old' (with old nom. sg. *veter* in Enn.), beside the *to*-adjective *vetustus* 'ancient' (Pl.+). Leumann (1977:269, 374) believed *vetus* to be a backformation to the abstract *vetustās*, though Nussbaum (1986:145n15) and more recently Brachet (2002:83) rightly dismissed the likelihood that such a backformation

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haplologized to plain *-tās*. No other early abstracts in *-tās* reflect *-titās, vel sim.*, that points directly back to **-to-tās*.

[47] Rix (1994:90) observes that a dental cluster of the sort underlying **-nt-tāt-* of PIE date would yield **-ss-* in Italic, and so **ĝ^{h}er-ent-tāt-* > **herenssāt-* > **herēsāt-* (likewise Lat. **u̯elh$_1$-ont-tāt-* > **u̯olōsāt-*). The abstracts, evidently, are no earlier than Proto-Italic.

[48] Umb. 3 pl. pres. subj. **tertu/teřtu**, 3 sg. pres. pass. **teřte**, and 3 sg. fut. imp. *dirstu*, spelled with **-ř**/rs-, are thought by Meiser (1986:180) to have been imported from other parts of the verbal paradigm where **-ř**/rs- is phonologically regular, e.g. subj. **teřa**, and fut. *dirsust* (cf. Meiser 1986:218-26 for a discussion of the phoneme **ř**).

could result in a morphologically isolated adjective in *-us, -eris* without gender-marking in the singular (other than nom./acc. animate pl. *veterēs* vs. neut. *vetera*). Now, *vetus* is generally thought to continue an *s*-stem **u̯et-os/*u̯et-es-*, based on Myc. *we-to* 'year', Hom. ϝέτος 'id.' (Stüber 2002:187), along with themaized **u̯et-s-ó-* in Ved. *vatsá-* 'calf'. Its Sabellian counterparts in Umb. **vesune** and Mars. *uesune* (and perhaps Osc. **vezkeí**) are believed to continue **u̯etsōnā-* 'lady of the year' (*EDLIL* s.v. *vetus*; but with more skepticism, cf. Untermann 2000:852-3).[49] A thematic derivative **u̯et-us-ó-* was also built from the *s*-stem, which is found in Lith. *vetušas* 'old' and OCS *vetŭxŭ* 'id.' (*EDLIL* s.v.). Just as *venustās* continues a relatively late *tās*-abstract built to the strong stem of the noun *Venus*, the abstract *vetustās* (and likewise the *to*-adjective *vetustus* 'ancient') could be viewed in the same way. Unlike *Venus*, though, *vetus* as an adjective would seem to fit the normal de-adjectival pattern of *tās*-abstract formation, and thus would seem to be a relatively recent creation. But *vetus* possesses a rather murky prehistory, and the real picture underlying this adjective may be much more complicated than previously thought.

Vine (2009:210-12) has recently demonstrated that the PIE stem **u̯et-es-* may ultimately go back to the gen. sg. of a root noun **u̯et-* 'year', rather than a true PIE *s*-stem, given that the evidence for an old stem **u̯et-os-* is not as strong as it is usually

[49] Untermann (2000:853-4, with refs.) makes no claims about Osc. **vezkeí**, other than citing the possible connection to Lat. *vetus*.

thought.[50] The gen. sg. of the root noun *$\underset{\frown}{u}$et-es (possibly also extended to *$\underset{\frown}{u}$et-es-i) was primarily used adverbially, e.g. Myc. *we-te-i-we-te-i* 'every year', which was later reanalyzed as a new *s*-stem noun. Lat. *vetus*, then, may actually be based on a *u*-stem adjective *$\underset{\frown}{u}$et-u- 'old', if it served as the base for an *s*-stem derivative *$\underset{\frown}{u}$et-u-s-, whence the thematic derivative *$\underset{\frown}{u}$etus-ó- found in the Balto-Slavic and Iranian forms (Nussbaum 1986:145).[51] Adding strength to such an *s*-stem hypothesis may be PN FETVSIA on the Bernardini tomb cup (Praeneste, mid-7th c. BCE), reflecting *$\underset{\frown}{u}$etusii̯ā- (cf. the Lat. PN *Veturius/Vetusius* possibly reflecting *$\underset{\frown}{u}$etus-i̯o-, much like *temperius* and *venerius*).[52] The new *s*-stem noun *$\underset{\frown}{u}$et-u-s- ultimately gave rise to the adjective *vetus*. If so, *vetus* and *vetustās* both reflect an etymological *-*u*-, instead of the assumed weakened strong stem *-*o*- of an *s*-stem.

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[50] Supporting the root noun *$\underset{\frown}{u}$et- 'year' are Hitt. *wett-/witt-* 'id.' (c.) and PIE *pér-ut(i)* > Gk. πέρυσι(ν) 'last year', Ved. *parút-* 'id.', etc. (cf. Vine 2009:210-11 with nn. 18, 19).

[51] Cf. a similar derivational pattern in Ved. *tápu-* 'hot' → *tápu-ṣ-* 'hot' → a neuter substantive *tápu-ṣ-* 'heat' as an adjectival abstract, but in the case of *$\underset{\frown}{u}$et-u-s- evidence for the underlying *u*-stem *$\underset{\frown}{u}$et-u- is absent.

[52] A fair amount of debate surrounds FETVSIA, centering around the question of whether it is a real Latin word or an Etruscan borrowing. Based on the lack of *e*-grade in the *s*-stem suffix that is found in other *i̯o*-adjectives, e.g. **tempesi̯os* and **venesi̯os*, Hartmann (2005:65) is of the opinion that FETVSIA is based on the Etruscan PN *Vetu* rather than an old **vetusi̯os* from the adj. *vetus* with no connection to Lat. *vetus*, which could well be the case. Weiss (*forthcoming*), however, notes that Lat. *Veturius* was connected with the rite of *Mamurius Veturius*, the "Old Mars", celebrating the incarnation of the old year, so a connection to *$\underset{\frown}{u}$etus- cannot be so easily dismissed, leaving open the possibility that FETVSIA indeed continues a *i̯o*-derivative to Lat. *$\underset{\frown}{u}$et-u-s- (cf. Vine 2008:28n3 for a similar interpretation of *vetus* and its possible connection to *Veturius*).

The abstract *vetustās* could be an old formation, built to the noun *vetus*; or it could be relatively recent, built to the adj. *vetus*, after it was reanalyzed from the noun *vetus*. It would seem to be the case, though, that the *to*-adjective *vetustus* must have been built to the old noun *vetus*, before its conversion to an adjective. The abstract *vetustās* could thus reflect a relatively recent formation based on *vetustus*, in much the same way as the pair *venustās/venustus* was built to the noun *Venus*.

Upon examining the *s*-stem abstracts in Fig. 3, it indeed seems that much of the material is quite old. Not all of the abstracts and adjectives reflect archaic formations, e.g. *venustās* and probably *vetustās*, and none of them can be securely dateable to PIE; but they still represent some of the older *tās*-abstracts in Latin, belonging to a period when abstracts were still formed with plain *-tās*, rather than *-itās*, which was generalized later in all subsequent abstract formations. *tās*-abstracts could evidently be directly backformed to adjectives in *-stus*, which likely contine PIE possessive adjectives in **-to-* built to *s*-stem nouns. And finally, *-tās* seems to have had a special function in naming divinities, with the abstract used as a theonym. The remaining athematic abstracts in plain *-tās* will now be examined, in the event that they too may reveal more detail about the history of *-tās* in Latin and Italic.

II.F.2. *aestās* 'summer'

aestās 'summer' (Pl.+) and its related noun *aestus*, *-ūs* 'heat, fervor, swell' (Naev.+) are based on the root *$*h_2ei̯d^h$-* 'heat', also found in Ved. *inddhé* 'ignites', Gk. αἴθω 'kindle, burn' (*LIV* s.v.), and also OIr. *áed* 'fire', OE *ād* 'pyre', and OHG *eit* 'id.' (*EDLIL* s.v. *aedēs*). It has been assumed that *aestās* shows the regular Latin reflex of PIE *$*-d^ht-$ > *-st-* via the intermediate stage *$*-zd^h-$, e.g. *īnfestus* 'hostile' < *$*n̥-g^{wh}ed^h-to-$ (cf. Leumann 1977:168, followed by Meiser 1998:124), rather than *-ss-*, the usual outcome of PIE dental clusters. Based on the assumption *$*-d^ht-$ > *-st-* (for which *aestās* is also the prime example), Leumann (*loc. cit.*) took the abstract *aestās* back to *$*h_2ei̯d^h-to-tāt-$, with *$*-tāt-$ built to a *to*-adjective *$*h_2ei̯d^h-to-$ (beside a *tu*-stem *$*h_2ei̯d^h-tu-$ in *aestus*) and simplification by the same syncopation he argued for in **honestotāt* (cf. **II.D.3.a.** above). But the solution is unsatisfactory, in that haplology is the usual "emergency" solution for such forms, when no better solution is apparent. Without any direct evidence for haplology, it should be rejected in favor of exploring other options.

On the other hand, Sihler (1995:203) noted that the development of *$*-d^ht-$ in Latin is actually uncertain. Leumann (1977:168) and Meiser (1998:124) both hang the assumption that *$*-d^ht-$ > *-st-* mostly on the hypothetical *$*h_2ei̯d^h-to-$, given the lack of other ironclad examples. Potential parallels for such a development in PPPs have been leveled out, with *$*-d-$ replacing *$*-δ-$ already in Proto-Italic, as occurred in PIE *$*Hi̯ud^h-to-$

> PItal. *i̯uδ-to- → *i̯ud-to- > *iussus* (Meiser 1998:124). In fact, all PPPs to roots ending in *-d^h- show up with *-ss-* or *-V̄s-*, e.g. *fīsus* 'trusted' < *b^hei̯d^h- and *gressus* 'stepped' < *g^hred^h-.[53] In fact, Sihler offers only *comēstus* 'eaten up' from *-*ed-to-* as the only other example besides *$h_2$ei̯d^h-to- of *-*dt-* giving Lat. *-st-*.[54] Hence, the circular argument that *aestās* continues *$h_2$ei̯d^h-to-tāt- can be cast aside, on the grounds that it is the only potentially reliable example of the very phonological rule it supposedly reflects.

Hill (2003:243-47) believed that PIE *-d^ht- should normally give *-ss-* in Latin, based on *iussus* 'order' (Andr.+) and *fīsus* 'split' (Pl.+), and thus rejected Leumann's solution. Instead, he concluded that the root *$h_2$ei̯d^h- built a reduplicated present *h_2é-h_2i̯d^h-e/o- 'kindle, ignite' that gave a Proto-Italic verb *áïd^{h}e/o- from which a new disyllabic root *aïd^h- was backformed. And to the new root a *tu*-stem abstract *aïd^h-tu- > *aestus* was built. According to Hill (2003:243-7), the regular outcome of *aïd^h-tu- and any *-d^ht-/*-dt- in the third syllable of a word in Italic is *-st-*, as also seen with *fūstis* 'stick, club' < *b^he$h_3$ud- and *caestus* 'boxing glove' < *keh_2i̯d-, but Kim (2007:13) rejects such a phonological development, on the grounds that it is "phonologically unnatural"

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[53] But cf. *EDLIL* (s.v. *gradior*) on projecting *gradior* back to *g^hred^h-. Go. *grid* 'step' and OCS *grędti* 'come' now are taken to reflect *g^hrid^h-. Italic and Celtic seem to suggest *g^hr̥d^h- (with or without aspirates), from which Italic built an *i*-present and Celtic a nasal present. The root difficulties do not have any implications for the PPP in Latin, however.

[54] *comēstus* is likely analogical itself, however. As Weiss (2009:431) observes, 3sg. *comēst* is analogical, since *$h_1$ed-ti should give *ēs*. Likewise, *ed-to- should give *-essus, but *-estus* must have been restored by analogy.

and that it requires unsupported trisyllabic verbal stems in PIE. For the root *$h_2e\underset{\frown}{i}d^h$- (*LIV* s.v.), there is no evidence whatsoever for a reduplicated present *$h_2é$-$h_2\underset{\frown}{i}d^h$-*e/o*- anywhere in Italic or, for that matter, PIE.

Differently, de Vaan (*EDLIL* s.v.) also rejects Leumann's view, arguing that such a development *-d^h*t*- > -*st*- is "unlikely" (but without providing his reasoning for the rejection). Instead, he favors a preform **aissāt*- < *$h_2e\underset{\frown}{i}d^h$-*tāt*-, in which the suffix *-*tāt*- was restored, with a similar process giving the noun *aestus* from a *tu*-stem *$h_2e\underset{\frown}{i}d^h$-*tu*-. It is extremely difficult to see how **aissāt*- could be remodeled to *aestāt*- without any driving factor. It seems to be the case that *aestās* is an old *tās*-abstract, built before the suffix was well grounded in Latin nominal derivation.

Nussbaum (1998b:527) has put forth the best solution, thinking that the *u*-stem abstract *aestus* is based on a *to*-adjective *$a\underset{\frown}{i}d^h$*sto*-, also found in OE *āst* 'kiln' and PDE *oast* 'id.'. An *s*-stem verbal abstract *$h_2e\underset{\frown}{i}d^h$-*es*-/*$h_2e\underset{\frown}{i}d^h$-*os*- must underlie Ved. *édhas*- 'firewood, kindle' and Gk. *αἰθος* 'glow, fire' (cf. Stüber 2002:94-5, 229), and the *s*-stem noun must have served as the basis for *$a\underset{\frown}{i}d^h$*sto*- (not mentioned in the above handbooks, but noticed by *DELL* and *WH* s.v. *aedēs*).[55] Hence, it is not too difficult to suppose that, much like *$\underset{\frown}{u}$*etus-to*- and **hones-to*- above, the *s*-stem noun *$h_2e\underset{\frown}{i}d^h$-*es*-/*$h_2e\underset{\frown}{i}d^h$-*os*- made a

[55] Gk. *αἰθος* is not found until Apollonius, but given his predilection for archaisms, the lateness is not too much of a problem.

to-adjective *$*h_2ei̯d^h$-*s-to*-, as also suggested by *aestīvus* 'of summer' (Pl.+), following the pattern whereby *s*-stems make adjectives in *-*sto*-.[56] Of great interest, *$*h_2ei̯d^h$-*s-to*- shows zero grade of the *s*-stem suffix, which may reflect the more archaic situation, compared to the more recent full-grade *-*es-to*- and *-*os-to*- in the examples discussed above. The *s*-stem *$*h_2ei̯d^h$-*es*-/*$*h_2ei̯d^h$-*os*- could have built a very old *to*-adjective *$*h_2ei̯d^h$-*s-to*-, beside a *tās*-abstract *$*h_2ei̯d^h$-*s-tās* and a *u*-substantivization of the *to*-adjective *$*h_2ei̯d^h$-*s-tu*-.[57] The *tu*-abstract *aestus* 'glow, heat; boiling' has the semantics of a general abstract noun, whereas the *tās*-abstract *aestās* 'summer' reflects a specialization in referring to the "hot" time of the year, namely "summer" (*OLD* s.v.; cf. the abstract *autumnitās* 'autumn'). *aestās*, likely backformed from the adjective **aestus*, continues an abstract in -*tās* with a technical use that sets it apart from a more traditional abstract formation (cf. also on *honestās* beside *honor* above in **II.D.3.**).

II.F.3. *aevitās/aetās* 'lifetime, age'

aetās 'lifetime, age' (Pl.+, beside *aevitās* in *XII Tab.*), alongside Osc. **aítateís** 'lifetime' (Si 7), Pg. *aetate* 'lifetime' (Pg 10), etc. (cf. discussion below), sits beside the noun *aevum* (sometimes masc. *aevus*) 'period of time' (Pl.+). *aetās* is thought to

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[56] Thus, *aestās*, beside the adjective **aestus*, really belonges above in **II.F.1.** with the other *tās*-abstracts and *to*-adjectives built to *s*-stems.

[57] For a similar development, cf. the *u*-stem substantivization in OIr. *lin* 'multitude' (< *plēnu*-) from the adjective **plēno*-.

continue **aiu̯o-tāt-*, built to the PIE *u*-stem **h₂oi̯-u-/*h₂ei̯-u-*, continued in Ved. neut. *áyu-* 'life, lifetime' (beside adj. *āyú-* 'full of life'), OAv./YAv. *āiiu-* 'life, lifetime' (n.), Gk. αἰϝεί, αἰές, αἰῶ, αἰέν 'always' (adv.) and αἰών 'lifetime, time' (m.), Go. *aiwam* 'times' (dat. pl.), and OHG *ēwīg* 'eternal' (*EDLIL* s.v. *aevus/aevum*). Based on Lat. *aevum*, Leumann (1977:374, but first suggested by Stolz 1895:554-5) assumed a thematic adjective **aiu̯o-* < **h₂eiu̯-ó-*, whence *aevitās* from **aiu̯o-tās*. Leumann, however, did not comment on the phonological development.

Rather than a thematic base, Untermann (2000:70-1) offered an original *i*-stem **aiu̯i-* (cf. Go. acc. pl. *aiwins* 'time, lifetime') or even an adverb **aeu̯i* to the root **h₂eiu̯-* (*DELG* s.v. αἰών), on which **aiu̯i-tās* was built.[58] He then assumed that **aiu̯i-tās* gave *aetās*, with **-iu̯i-* > *-ī-*, as also alleged in Osc. **biítam** 'life' and Lat. *vīta* 'life' from **gʷih₃-u̯o-teh₂* and similarly observed with *-īu̯ī-* > *-ī-* in *oblīvīscar* > *oblīscar* 'I will forget' (Acc. *Trag.* 448). But **aiu̯i-* is not the same phonological environment as **-ĭu̯ĭ-* and cannot at all be assumed to contract to *ae-* in the same way as in the examples above for intervocalic loss of **-u̯-* (i.e. between vowels of similar quality; cf. Weiss

[58] In support of Untermann's adverb **aeu̯i* may be Cypr. αιϝει/Lesb. αἴ 'always', which could be suggestive of an old loc. sg. **aiu̯-i* from an unattested root-noun **h₂eiu̯-/*h₂oiu̯-* (for this interpretation cf. *NIL* s.v. **h₂ei̯-u-* with n. 9).

2009:154).[59] Additionally, outside of the Gothic form, clear support for an *i*-stem **aiu̯i-* is lacking, and thus **aiu̯itās* can be dismissed. Instead, *aevitās/aetās* must certainly be built to the *o*-stem in *aevum*, with *-tās* used to distinguish the new abstract from its base, as is found with other *tās*-abstracts with abstract bases, e.g. *honestās* '(general) honor' beside *honor* '(specific, i.e. military) honor' (cf. **II.D.3.** above). *aevitās/aetās* 'period of a human life' can reflect a specific kind of timespan, beside the general abstract *aevum* '(general) period of time'.

Most promisingly, **aiu̯o-tās*, according to Schrijver (1991:39), can develop to **aio-tās*, with loss of **-u̯-* before **-o-*, which finally ends up as *aetās* with subsequent syncope of **-o-*. Weiss (2009:154) offers additional support for the loss of **-u̯-* before *-o-* in *parum* 'little' (Pl.+), *secundus* 'following' (Pl.+) < **sekʷondos*, etc., hence such a development for **aiu̯o-tās* seems preferable. In fact, the development might even be shared with Celtic, if OIr. *áes* 'age' and W. *oœs* 'id.' reflect a similar loss. *aevitās* in the *XII Tab.*, then, likely continues a secondary creation in *-itās* to *aevum* (EDLIL s.v. *aevus/aevum*), since *aevitās* cannot develop to *aetās* in the way described above. *aevitās* is only found a handful of times in Latin literature (*OLD* s.v.) and could well reflect a sort of a later "formal" version of *aetās*, with the full form of the abstract suffix in *-itās*.

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[59] It is thinkable that **aiu̯otās* was syncopated to **aiu̯tās*, in which the glide cluster is simplified by dropping **-u̯-* (cf. Nishimura 2008:127).

II.F.4. Osc. aítateís/Pg. *aetatu* 'age'

Beside Pg. *herentas*/Osc. **herentateís**, the only other *tās*-abstract found in Sabellian is Osc. gen. sg. **aítateís** (*ST* Si 7), acc. sg. **aitatúm** (*ST* Cm 13), Pg. acc. sg. *aetatu* (*ST* Pg 9) and abl. sg. *aetate* (*ST* Pg 10). But unlike Pg. *herentas*/Osc. **herentateís**, which seems to be a real Sabellian abstract in **-tās* (if the suffix was not borrowed from Latin; cf. **II.F.1.c.i.**), Pg. *aetatu* is best interpreted as a borrowing from Latin, given its locus in a text showing heavy Latin influence (cf. above on *herentas* also found in Pg 9) and its spelling in *ae-* (cf. also *praicime* in Pg 9).

On the other hand, Osc. **aitatúm** is also attested in an early curse tablet (*ST* Cm 13), used in parallel to **biítam** '*vitam*':

> **stenim.kalauiiúm.trí[2]aginss.urinss.úlleis[3]fakinss.fangvam[4]biass.biítam .aftiim.{a}[5]anamúm.aitatúm[6]amirikum.tíf[eí.** -?-]

The text dates to around 200 BCE, along with a few others from Cumae that belong to the 2nd or 1st BCE (cf. Vetter 1953:29). Dupraz (2010:313), assuming an underlying **aiu̯itās* without justification, believes **aitatúm** is most likely inherited, as seems likely, given that the curse tablet shows no traces of Latin influence in vocabulary and style. **aitatúm** may reflect the same development underlying Lat. *aetās* with **-u̯-* lost before **-o-* in **aiu̯o-tās*; or it could also simply reflect **aiu̯o-tās* > **aiu̯tās* with syncope and subsequent reduction of the glide cluster. Of greater interest, an inherited **aitatúm**

beside Lat. *aetās* shows one reasonably secure example of an abstract in **-tāt-* that belongs to Proto-Italic.

II.F.5. *iuventās* 'youth'

iuventās 'youth' (Lucr.+) is presumably built to the noun *iuvenis* 'young man' (Pl.+), but it, like the abstracts built to *s*-stem bases, reflects a denominal formation. *iuventās* 'the quality of a *iuvenis*' < **iuven-(i)tās* could reflect an abstract built to *iuvenis* with either plain *-tās* or *-itās* with syncope of the connecting vowel (but cf. n. 30 above on the likelihood of syncopation of the connecting vowel). If so, *iuventās* must reflect a fairly old creation in Latin, given that several abstracts that are built to stems in *-n-* in Latin reflect *-nitās*, e.g. *inānitās* 'emptiness' (Pl.+) to *inānis* 'empty' (Pl.+), etc. *iuventās*, however, does not have the attestation pattern of a very old abstract, in that it is not found until Lucretius. Moreover, the abstract is used to personify "Iuventas", the goddess of youth. Beside *iuventās* is found the *tūs*-abstract *iuventūs* 'youth' (Pl.+) and *iuventa* 'id.' (Laber.+) that could both point back to a possessive adjective **iuventus*, so it is very possible that the *tās*-abstract is just a backformation to **iuventus*, after the pattern of *honestās* : *honestus*, *venustās* : *venustus*, etc.

II.F.6. *maiestās* 'dignity'

maiestās 'majesty' (Andr.+), beside the *to*-adjective *maiestus* 'dignified' (Petr.), reflects an abstract traditionally assumed to be based on the comparative adjective *magis* 'more' (*XII Tab.*+), which continues a very old zero-grade neut. nom./acc. sg. *-*is* of the comparative suffix *-*i̯os-/-is-* (*EDLIL* s.v. *magnus*).[60] *maiestās* is thus potentially very interesting, based on the assumption that it seems to continue a *tās*-abstract built directly to a stem **maĝ-i̯es-*. If so, **maĝ-i̯es-tās* would continue the only known instance of the full-grade *-*i̯es-* of the comparative suffix (Meissner 2006:152n77), and as such, it would have to reflect the oblique stem of a very old comparative adjective **maĝ-is-/*m(a)ĝ-i̯es-* with PK ablaut.[61] With a total lack of evidence for *-*i̯es-* in the otherwise AK-ablauting comparative, however, it seems unlikely that *-*i̯es-* survived here alone (if it existed in this word at all), without any other support in PIE for an *e*-grade suffix.[62] The evidence

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[60] The *to*-adjective *maiestus* is only found in Petronius, and is most likely an analogical and secondary creation after *maiestās*.

[61] The *a*-vocalism in Latin *mag-* beside *e*-grade in Gk. μέζων is problematic. Schrijver (1991:480-1) has proposed that **m̥* may have developed to *ma-* immediately before a voiced stop and another consonant. If so, an original zero-grade **m̥ĝ-* could well have given Latin *mag-* (and Celtic **mag-* as in OIr. *mál* 'prince, chief', etc.). It is not clear whether *magis* continues **m̥ĝ-is* or if it copied vocalism from *magnus*, which could then reflect **m̥ĝ-no-*; but Schrijver's rule can account for the discrepant forms in **mag-* beside **meg-* found elsewhere.

[62] Sihler (1995:359) notes that some comparative adjectives in Indo-Iranian, e.g. Ved. dat. sg. *návyase* 'new' and gen. sg. *návyasas*, fail to show lengthening by Brugmann's Law in the suffix and have been alleged to continue a full-grade *-*i̯es-*, instead of the the usual *o*-grade *-*i̯os-*. Rather than take the forms at face value, though, Sihler dismisses them, attributing the short vowels to analogy with bahuvrīhis like *sumánasas* 'well-disposed' (gen. sg.). Sk. *mahīyas-* 'bigger' (Up.+) and YAv. *maziiah-* 'id.', along with

for the comparative adjective in PIE, moreover, shows that it inflected with AK ablaut, in which a full-grade *-i̯es- in the suffix is not expected in the first place.[63]

Abstract nouns in *-tāt- in Avestan could apparently be built to adverbial bases, e.g. OAv. *auuaētāt-* 'the word *auuōi*' to the interjection *auuōi* 'woe' (cf. **III.A.2.**), and a similar pattern may be found in Latin. *satietās* 'abundance' (cf. below) seems to be built to the comparative adv. *satius* 'better, preferable', and so it may well be the case that *magis* is at the base of *maiestās*; but it is immediately clear that **magis-tās* cannot be the direct source. Instead, a neuter *s*-stem **m(a)ĝ-es-* of the same sort discussed above may ultimately underlie *maiestās*. Such an *s*-stem is alleged for Ved. *máhas-* 'size', YAv. *mazah-* 'id.', and OIr. *mag* 'field' (*NIL* s.v. **még-*). Though there is no evidence for the *s*-stem in Italic, *maiestās* could be analyzed as a *tās*-abstract **maĝes-tās* built directly to the noun **m(a)ĝ-es-*, meaning something like "the quality of being great; superiority". It is thinkable that **magestās*, with its underlying comparative semantics, would have been remodeled to **maii̯estās* > *maiestās*, replacing **mag-* with **maii̯-* of the comparative

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maiestās, could also be used in support of a full-grade suffix *-i̯es- in PIE; but the Indo-Iranian material is late (*EWAia* s.v. *mahánt-*). Gk. μέζων (Myc. *me-zo*) reflects *o*-grade *-i̯os-, so it seems unlikely that Indo-Iranian would reflect *-i̯es-, and the lack of lengthening in the suffix could well be analogical. *maiestās* then remains entirely isolated.

[63] A full-grade *-i̯es- might be found in the loc. sg., however (Weiss 2009:356). It seems unlikely that *maiestās* continues **maĝ-i̯es-i-tās*, with *-tās* added directly to the loc. sg. in *-i*, but it is thinkable that it may continue **maĝ-i̯es-tās*, with *-tās* added to an endingless loc. sg. Decasuative derivatives are well known with certain suffixes, e.g. the *t*-suffix that will be explored later (cf. **V.A.**), but there is no evidence that abstracts in *-tās* could be built in this way to the locative case.

adjective *maior* 'greater' < **maĝ-i̯os-*. Driving the remodeling of the abstract was the close derivational and semantic connection with the comparative adjective, rather than an adverb, in that **magestās*, after the *s*-stem base **mag-es-* was lost, could have been perceived to belong more closely with the adjective **magi̯or-* than the adverb *magis*. By this line of reasoning, *maiestās* must reflect a very old abstract.

Due to the lack of evidence for an *s*-stem noun **m(a)ĝ-es-* in Italic, it may be more lkely that *maiestās* does indeed continue **mag-is-tās*. Given the fact that no *tās*-abstracts in *-istās* are found in Latin, it is likely that such a suffix shape would have been reformed to *-estās*, on the model of all the other abstracts built to *s*-stems. Thus, **magistās* could have been reshaped to **magestās* relatively early, since the *s*-stem forms seem to continue the older *tās*-material in Latin. And just as argued above, **magestās* may have again been reshaped to **maii̯estās* for the same reasons alleged above.

II.F.7. *satiās* 'abundance'

satiās 'abundance, plentifulness' (Pl.+) beside *satietās* 'id.' allegedly reflects another case of haplology (Leumann 1977:374), since the abstract is presumably based on the adv. *satis* 'sufficiently, enough'. *satiās* is interesting, in that the oblique forms are largely built to the stem *satietāt-*; only a few forms are built to the stem *satiāt-* (*OLD* s.v.). The nom. sg. *satiās* could reflect **sati-(i)tās*, if derived from the adverb, with a

dissimilatory loss of the second *-t-* in the nom. sg.; but the oblique stem *satietāt-* suggests an underlying **satio-* as the base, though no such positive adjective exists in Latin. The comparative *satius* 'better, preferable', however, could well have served as the base. A *tās*-abstract built to a comparative adjective stem has a parallel in Latin in *maiestās*, built to *magis* (cf. above). Though *satius* is historically an *s*-stem, *satietās*, with an original meaning 'betterness, preferableness' that could well have merged with **sati-tās* 'abundance' when **satio-tās* lost its comparative sense (as also did *maiestās*), must be analogical to other *tās*-abstracts built to adjectives in *-ius* that inevitably end up in *-ietās*, once the adv. *satius* became inflectionless. It is not too difficult to think that *satiās* was used predicatively in the nom. sg. in fixed phrases in a similar way to its derivational base *satis*, whereas *satietās* was used in the oblique. Eventually, the two words with nearly identical semantics collapsed into the same paradigm, with *satiās* providing the nom. sg. and *satietāt-* the stem for other case forms.

II.F.8. *pesestās* 'plague'

pesestās 'plague' is only attested in Paulus *ex Festo* (p. 231 L) as a synonym for *pestilentia*. The noun *pestis* 'plague, disease' (Pl.+) should build a normal abstract *ˣpestitās*, but no such form exists, nor does *s*-stem *ˣpesor/ˣpesus*. *pesestās* is highly problematic, in that it does not show rhotacism, despite the lack of any features that

normally prevent rhotacism, e.g. dissimilatory effects of a nearby *-r-* as in *miser* (cf. Weiss 2009:151 for further examples). Additionally, *pestis* is without a clear etymology (*EDLIL* s.v.). Given the etymological problems underlying *pesestās*, it can be of no use.

II.F.9. *potestās* 'power'

potestās 'ability; power' (*XII Tab.*+) has been considered to be a remodeling of **potis-tās*, built from the adverb *potis* 'able' (Naev.+) by analogy to *maiestās* 'majesty' (Leumann 1977:335-6, Brachet 2002:82n11). *potis* is restricted in the use as a predicate adjective in Latin in the phrase *potis esse* 'to be able, possible' (beside *pote esse* with loss of final *-s* and lowering of **poti* to *pote*; Weiss 2009:60n52). Given that *potis* did not originally inflect, it seems very likely that it built an abstract **potistās*; and it also seems very likely that such an abstract was remodeled to *potestās* on the model of *maiestās*, as Leumann suspected, as well as other abstracts with *s*-stem bases.

II.F.10. *ūbertās* 'abundance'

ūbertās 'abundance' (Enn.+) is related to the adjective *ūber, -eris* 'abundant' (Pl.+) and must be relatively old, given that a recent creation would likely turn up as **ūberitās*, e.g. *asperitās* 'roughness' (Lucr.+), *dexteritās*, 'adroitness' (Liv.+). *ūbertās* could be the same sort of plain *tās*-formation as suggested for *lībertās* (cf. **II.D.1.**) and *voluptās* (cf. **II.F.12.**), with *-tās* appended directly to *ūber-*, which continues the strong

stem of an old *r/n*-heteroclite *$*h_1o\underset{\smile}{u}Hd^h$-*r*-/*$*h_1o\underset{\smile}{u}Hd^h$-*n*- 'udder', as in Ved. *ūdhar/ūdhan-* 'id.' (*EDLIL* s.v. *ūber*). It is most likey the case that *ūbertās* is relatively old, continuing a formation in plain *-tās* to the synchronic adj. *ūber* (perhaps even aided by phonetic similarity to *līber* and *lībertās*), rather than the older noun *ūber* (cf. also on *vetustās* and *vetus* above in **II.F.1.e.**).

II.F.11. *voluntās* 'will'

The theonym Pg. *herentas*/Osc. **herentateís** finds a parallel in Latin *voluntās* 'will, choice' (Andr.+), which reflects a *tās*-abstract built to an *nt*-participle to the verb *volō* 'wish, want'. *volō* has comparanda in Go. *wili* 'wants', Lith. *velmi* 'wishes, wants', plus nasal presents in Ved. *vr̥ṇīté* 'chooses', OAv. *vərəṇtā* 'chooses', etc., and also has a few reflexes in Ven. **leno** 'voluntary act', Osc. **velliam** 'legal demand', Umb. **veltu** 'order', etc., pointing back to a PIE root *$*\underset{\smile}{u}elh_1$- (cf. *EDLIL* s.v. *volō*, *LIV* s.v. *$*\underset{\smile}{u}elh_1$-). The long-held consensus (i.e. Stolz 1895:555, von Planta 1892-97:70n1, Meiser 1998:226, Leumann 1977:374, Sihler 1995:616) is that *voluntās* reflects a deparicipial abstract in much the same way as does Pg. *herentas*/Osc. **herentateís**, but with *o*-grade of the pres. pple. suffix.[64] Hence, the stem **velont-*, beside the present participle proper

[64] **velont-* is one of only three examples in Latin with *o*-grade of the suffix, with *iēns*, *euntis* to *eō* 'go' and *sōns* 'guilty' being the others (cf. Watkins 1967), as already in Italic the zero-grade *-*n̥t-* was generalized, with *-*n̥t-* giving the familiar *-*ent-* (cf. the nearly ubiquitous *-ent-* in Italic, including Sabellian, e.g. Osc. *praesentid* '*praesente*' and Umb. **zeřef** '*sedēns*').

volēns to *volō* 'wish, want' with normal *-ent-*, must consequently preserve a very old piece of morphology, before the generalization of the weak stem of the suffix **-ont-/*-n̥t-* (Seldeslachts and Swiggers 2002:321).

Leumann (1977:374) assumed *voluntās* must either reflect **u̯elont-tās*, with simplification of **-nt-t- > -nt-*, or **u̯elonti-tās*, with syncope. Rix (1994:90n5), though, rightly rejected **u̯elonti-tās* on the grounds that there is no reason to assume an *i*-stem basis in the participle.[65] Conversely, *voluntās* can only reflect a suffixation in **-tās* directly to the stem **u̯elont-*, with post-PIE simplification of **-ntt- > -nt-*. Though double-dental clusters resulting after syncope in Latin remain, e.g. **ke-date > cette* 'give here!' (Weiss 2009:174), it is easy to imagine an instantaneous analogical simplification of an abstract in **-nttās* to **-ntās*, beside other *-tās* forms.

II.F.11.a. *voluntās* and Pg. *herentas*/Osc. herentateís

Lat. *voluntās*, built to the archaic *o*-grade of the present participle of *volō*, and Pg. *herentas*/Osc. **herentateís**, built to the old athematic verb stem of the "renewed" Osc. **heriiad**/Umb. **heri**, point back to archaic formations. The archaic formations, moreover, were built in parallel, reflecting *tās*-abstracts built to present participles of the verb

[65] Jasanoff (2006) has recently analyzed the gerundive in *-ndus* < **-nti-no-* (beside Osc. **sakrannas** 'to be consecrated' and Umb. **úpsannam** 'to be made') as continuing a *no*-adjective to a verbal abstract in **-i-* built to the present participle. Though Anatolian has abstracts in **-nti-*, e.g. Hitt. *tukkanzi-* 'cultivation' (cf. Melchert 1999b), no such forms are found in Italic. If, however, **u̯elontitās* underlies *velontās*, perhaps **u̯elonti-* reflects one such verbal abstract.

meaning 'wish, want' in both languages. Pg. *herentas*/Osc. **herentateís** reflects a substantivization with the *tās*-suffix with the semantics "that which is characterized by desire; Desire". Lat: *voluntās* was also specialized as a theonym "Voluntas" in the same personified sense of "Desire", though the noun was also used in the sense 'will, desire' (*OLD* s.v.). Untermann (2000:320) was of the opinion that Pg. *herentas*/Osc. **herentateís** cannot be a calque on Lat. *Venus*, since it is difficult to think that Sabellian knew of Venus before the historical period. On the other hand, it is possible but equally difficult to see the calque operating in the opposite direction, with Pg. *herentas*/Osc. **herentateís** inspiring the creation of Lat. *voluntās*, since the word is not used only as a divinity. Consequently, the abstracts must have been independently created, suggesting Sabellian also inherited the suffix **-tās* from PIE. It is possible that the early Italic tradition may have possessed a divinity "Desire", and when **ĝʰer-* replaced **u̯elh₁-* in Sabellian, **ĝʰer-ent-tās* was used for her theonym. Likewise, the reverse could have happened in Latin with the corresponding **u̯elh₁-ont-tās*.

II.F.12. *voluptās* 'pleasure'

voluptās 'pleasure' (Pl.+) is entirely isolated, with no underlying adjective and no corresponding adjective in *-tus* sitting beside it. It is generally thought to reflect a syncopated *tās*-abstract noun **u̯olupi-tāt-* built to an old and unattested *i*-stem adjective

*u̯olupi- 'pleasing', based on the root *u̯elh$_1$- (for the root, cf. above on *voluntās*; Leumann 1977:373; *DELL* and *EDLIL* s.v. *volup*,). Such a *u̯olupi-tās, however, is doubtful; by the normal rules of syncope, *u̯olupitās should be expected to reduce to ˣvolpitās, with deletion of the second syllable (especially after a resonant), not the third, e.g. Gk. Πολυδεύκης, which ends up as Lat. **Podlouqueique** (*CIL* I^2 2833, ca. 550-25 BCE with likely spelling error; cf. Leumann 1977:71), beside Praen. POLOVCES (*CIL* I^2 548), *Pollūcēs* (Pl., once at *Bacc.* 894)/CLat. *Pollux*, and Gk. βαλανεῖον gives Lat. *balineum* (Pl.+) and later *balneum*. Though direct evidence within Latin is lacking for an *i*-stem adjective *u̯olupi-, one is thought to underlie the adverb *volup*, meaning 'agreeably, to one's satisfaction' (Naev.+), most commonly found in the Plautine phrase *volup est* 'it's OK, fine by me'. *volup* is thought to continue an apocopated neut. nom./acc. sg. *volupe < *u̯elup-i built to the same original *i*-stem adjective underlying *voluptās* (*DELL* and *EDLIL* s.v.), but it could also reflect the same development seen in *mage est* and *pote est*, where final *-s* is lost and *-i* is subsequently lowered to *-e*: *volupis est > *volupi est > volup(e) est* (cf. Weiss 2009:60n52). Given its fixed usage in Plautus, the phrase *volup est* likely represents something old. *volup* must have originated in the elided phrase *volúpest(i) > volup est*, in which *volupe was reanalyzed as *volup* (*GEW* s.v. ἔλπομαι, Hamp 1980:158), after *volup(e) ceased to inflect on its own.[66]

[66] Cf. similar treatments with *pot(e) esse* 'to be able, possible', *sat(e) esse* 'to be sufficient, equal', etc.

Furthermore, it has gone unnoticed that additional support for an old phrase **volúpesti* can be gleaned from the fact that syncope would normally have affected the medial syllable in **volúpest(i)*, giving **volp*. By the accentuation pattern in Old Latin, however, enclisis with **est(i)* protected the "*u*" from syncopation, given that the syllable would have carried the accent (cf. Joseph and Wallace 1987:686-7 for more on the enclitic behavior of **esti* in Italic).[67] Because *volup* must have been created relatively early, the abstract *voluptās* can be best analyzed as a formation in plain *-tās* built directly to *volup* before generalization of *-itās* (cf. the other direct formations discussd above).

Taking a different approach, Olsen (2007:103-5) has recently proposed an alternate etymology for Lat. *voluptās*: she rejects the traditional connection of Lat. *lupa* 'she-wolf; prostitute' (Pl.+) with *lupus* 'wolf' based on the fact that wolves are not promiscuous animals, instead taking *lupa* back to the zero grade of a root **u̯elp-*, based on her hypothesis that initial **u̯l̥-* in Latin gives *lu-*. The stem *volup-*, then, is a "contamination" (Olsen 2007:104-5) of a full-grade stem with the zero-grade **lup-* < **u̯l̥p-*. *volup* continues either an *i*-stem adjective "**u̯l̥pi* → **u̯elupi* > *volup*...or perhaps better, the neuter of a basic *u*-stem undergoing the usual transfer to the *i*-stems in Latin, i.e. **u̯l̥pú* → **u̯l̥pu̯i* → **u̯elupu̯i* > *volup*" (Olsen *loc. cit.*). She offers *volūmen* 'scroll,

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(Hackstein 2008).

[67] Behaving much the same way is the enclitic *-que*, which Probert (2002) has recently argued fixed the accent on the syllable immediately preceding it, even if the syllable was light, e.g. *utráque*.

book' as another case of contamination of a full-grade *u̯elHu- > *volu- with a zero-grade *u̯lHu- > *u̯lū- > *lū- (by metathesis); but the source of the long vowel in *volūmen* is problematic: it is likely analogical in origin (cf. also *locūtus* 'said' and *secūtus* 'followed'), since there is no good reason to think the root contained a laryngeal (*LIV* s.v. 2. *u̯el-). The *i*-stem *u̯l̥p-i- or *u̯l̥pu̯i- (*sic*!; Olsen must imply a pre-contamination full-grade *u̯elpi- or *u̯elpu̯i-) existed beside the abstract *u̯l̥p-eh$_2$ 'desire, lust', which developed the concrete sense 'object of desire, desired one', according to her line of reasoning. While this sort of semantic transformation is certainly possible, it is hard to see how *lupa* could have contaminated *u̯elpi-, especially since *-lp-* is stable. *lupa*, moreover, meant 'she-wolf' as much as 'prostitute', and a semantic connection between 'she-wolf' and 'pleasure' is tenuous at best; the contamination theory has nothing to recommend it, and it is easiest to analyze *voluptās* as a formation in plain *-tās* to *volup*.

II.F.12.a. *volup* 'agreeably, to one's satisfaction'

As for *volup*, it seems best to connect it with the root *u̯elh$_1$-. *u̯elh$_1$-p-i-, operating with a "*p*"-extension to the root (on which cf. below), which would give Lat. *u̯elapi-, with vocalization of the laryngeal, then *u̯elapi- > *u̯olapi- (**e* > *o* before a "dark" *l*), and finally *volupi-, with customary inlaut weakening and rounding before

labials, e.g. *aucupō* 'go bird-catching' < **aucap-*.[68] But while the solution may work for Latin, the Greek forms cannot directly continue **u̯elh₁-p-i-*; ἐλπίς seems to continue **u̯el-p-*, without any trace of a laryngeal. In addition, the "*p*" element, which cannot be part of the root, needs further explanation.

Greek ἐλπίς and ἔλπομαι are fixtures of Homeric vocabulary, and, as mentioned above, have long been thought cognate with Latin *volup* and *voluptās*, in that they share the labial enlargement. Frisk (*GEW* s.v. ἔλπομαι) certainly believes in the association of the forms, but notes that the function of the extension is unknown, and *DELG* (s.v.) offers no more.[69] ἔλπομαι is the only verb to **u̯elh₁-* that shows the "*p*" enlargement and has all the appearances of a primary verb. As such, much like Lat. *volō*, it could continue a thematized present middle to **u̯elh₁-*, but if it indeed continues **u̯elh₁-p-e/o-*, laryngeal vocalization is expected. It is possible that ἔλπομαι continues a subjunctive built to an athematic present (on which cf. Vine 1998a:76 with references), which along with middle inflection would fit in nicely with the semantic development of "choose" → "hope". But there is no evidence for a root present **u̯élh₁-p-ti/*u̯l̥h₁-p-énti* with the labial enlargement. Complicating matters is the existence of ἔλδομαι 'desire, long for' (Hom.+), beside the

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[68] Velarization of **l* occurred before vowel weakening (Nussbaum 1999a:386-7). It is not beyond consideration, however, that *volup* continues an actual *o*-grade (cf. below).

[69] The semantics of ἐλπίς 'hope' and *voluptās* 'pleasure' can easily be reconciled with **u̯elh₁-* 'select, choose': the *i*-stem in ἐλπίς is simply an abstraction of the base meaning "choose" → "a choosing, desire, hope". The middle voice of ἔλπομαι < *"choose for oneself" reflects the same semantic development from "choose" to "wish, desire" found in Lat. *optō* 'choose, wish'. Latin substantivized the meaning by means of the *tās*-suffix via the development "desire" → "pleasure".

variant ἐέλδομαι, which looks like a formation to the same root but with a "δ"-extension, i.e. from *$\underset{\smile}{\mu}elh_1$-*d*-. Frisk (*GEW* s.v. ἔλπομαι) noted the connection, speculating that the extension must be a present stem formant of some kind. Nussbaum (1998a:142-3) has shown that the Homeric examples of initial *ee*CC- that alternate with *e*CC- most often reflect *$e\underset{\smile}{u}e$*RC*- > *ēRC*- > *eRC*-, with glide-loss and Osthoff-shortening, where the older form has often been preserved as a metrical variant, e.g. ἐέλδομαι beside ἔλδομαι from *$e\underset{\smile}{u}eld$-*e/o*-. The pattern was then extended to roots without etymological *e*($\underset{\smile}{u}$)*e*CC-, including ἐέλπομαι. *LIV* (s.v.) suggests, albeit with skepticism, that a root *$h_1\underset{\smile}{u}eld$-, with meaning similar to *$\underset{\smile}{\mu}elh_1$-, underlies ἐέλδομαι, which continues a reduplicated present *$h_1\underset{\smile}{u}e$-$\underset{\smile}{u}eld$-. Potential support for the root may be found in OIr. *fled* 'feast' and W. *gwledd* 'id.' < *$h_1\underset{\smile}{u}\underset{\circ}{l}d$-*eh*$_2$-, and perhaps even Gaul. PN *Vlidorīx*. If so, ἐέλδομαι cannot belong etymologically with ἔλπομαι, and *LIV* admits a connection is very tentative at best.[70] Without better evidence, there seems little justification for a root *$h_1\underset{\smile}{u}eld$-; but the fact that Greek had a reduplicated present to *$\underset{\smile}{\mu}elh_1$-, though one with a *d*-enlargement, in ἐέλδομαι beside ἔλπομαι requires justification. Given the behavior of *$\underset{\smile}{\mu}elh_1$- (cf. below), it is preferable to take ἐέλδομαι along with ἔλπομαι as an enlargement to *$\underset{\smile}{\mu}elh_1$-.

[70] But there is also no reason ἐέλδομαι cannot continue a similar reduplicated present *$\underset{\smile}{\mu}e\underset{\smile}{u}elh_1$-*d*- to *$\underset{\smile}{\mu}elh_1$-, and so it remains unclear whether taking ἐέλδομαι with *volup* and ἔλπομαι is justifiable. *LIV*'s lemma *$\underset{\smile}{\mu}elh_1$-, with h_1, is based on Dor. λέω/λῶ 'want' < *$\underset{\smile}{\mu}\underset{\circ}{l}h_1$-$\underset{\smile}{i}e$/*o*- (Harðarson 1993a:83-4 with nn. 103 and 104).

The *d*-stem of ἐλπίς is based on an *i*-stem that strengthens the connection between ἐλπίς and *volup* (< *$\underset{\sim}{\mu}elh_1$-*p*-*i*-). But problematically, all the Greek forms are *aniṭ*-looking, lacking any trace of the laryngeal in the root *$\underset{\sim}{\mu}elh_1$-; no *ἐλεπίς (*vel sim.*) exists anywhere in Greek. The *p*-extension to *$\underset{\sim}{\mu}elh_1$- in *volup* and ἐλπίς seems to be a root enlargement of some kind, given the fact that it follows the behavior of other root enlargements, in which *seṭ* roots seem to have enlarged *aniṭ* alternates.

PIE *$\underset{\sim}{\mu}elh_1$- beside Lat. *$\underset{\sim}{\mu}elh_1$-*p*- and Gk. *$\underset{\sim}{\mu}el$-*p*- (and perhaps even *$\underset{\sim}{\mu}el$-*d*-) seem to show a general pattern whereby *TeRH*- roots alternate with *TeR-T*- extensions, suggesting a possible segmentation *TeR-H*- for the *seṭ* forms (cf. Vine 1982:228-9 for a discussion of root structure alternations of this type). The alternation between *$\underset{\sim}{\mu}elh_1$-, *$\underset{\sim}{\mu}el$-*p*- (and *$\underset{\sim}{\mu}el$-*d*-), and *$\underset{\sim}{\mu}elh_1$-*p*- has parallels: Lat. PPP *perculsus* 'beaten down', for example, suggests an *aniṭ* *-*k̥ld-to*-, *clādēs* 'ruin' a *seṭ* *k̥lH-d*-, and *calamitās* 'misfortune' a *d*-less *k̥lH-mo*-. Lat. *tendō* 'stretch' < **ten-d*- beside Gk. τείνω 'id.' < **ten-i̯e/o*- has been alleged to show evidence of a *d*-suffix that might also be seen in *$\underset{\sim}{\mu}elh_1$-*d*-. If the **tenh*$_2$- suggested by Lat. *tenuis* 'thin', Gk. ταναός 'stretched', and Ved. *uttānā*- 'id.' belongs here with **ten-d*-, it would give further support for a similar alternation of a *seṭ* root beside an *aniṭ* root with

consonantal extension.[71] Greek might also attest a similar-looking *d*-enlargement alongside the *p*-enlargement in ἔλπομαι and ἐλπίς, if the connection of ἔλδομαι to **h₁u̯eld-* is rejected. If the root **u̯el-h₁-* had three alternating enlargements **-h₁-*, **-d-*, and **-p-*, *volup* must then show both h_1- and *p*-enlargements together (cf. *clādēs* < **kl̥-H-d-*). Greek would only show "*p*" and perhaps "*d*", forcing the assumption of a PIE **u̯elh₁-p-* next to **u̯el-p-* and **u̯el-d-*.

Michael Weiss (p.c.) has recently proposed that laryngeals may be subject to loss after *-ēC-*, whereby he views the laryngeal loss as a result of syllabic weight, e.g. Gk. κρημνός 'steep river bank' (*Il.*+) < **k̂rēm(h₂)-no-*, a deverbal noun in **-no-* built to a Narten present to **k̂remh₂-* 'be limp' (cf. *LIV* s.v.). Many of the most reliable examples of laryngeal loss in the roots occur in extra-heavy roots, e.g. Lat. *collis* 'hill' < **kolHni-* or **kolHu̯i-* (Nussbaum 1997:195). For this reason, the extra-heavy syllables of the shape **CoRH.C-* and **HRoC.C-* are liable to a similar reduction by the "Saussure Effect". Based on the effect of syllabic weight in these environments, Weiss has suggested that there is no reason why a long vowel, namely **ē*, should not have the same effect as an *o* vowel in triggering laryngeal loss, and so **CēRH.C-* and **HRēC.C-* could also be theoretical environments for laryngeal loss. Recalling that the root **u̯elh₁-* possessed Narten

[71] Sihler (1995:510), for his part, dismisses such a **-d-* in the root, based on his belief that Lat. **-nu̯i-* > *-nd-*. Meiser (2003:186), on the other hand, believes **-d-* arose prevocalically as a hiatus-breaker in roots of the shape *CN̥H-*. Hence, the interpretation of *-d-* in roots like **ten-d-* is still controversial.

properties, it is thinkable that the *i*-stem paradigm was also of the Narten type *$\underset{\sim}{\mu}\acute{e}lh_1$-p-i-/*$\underset{\sim}{\mu}\acute{e}lh_1$-p-i-, with lengthened grade in the strong forms. If Schindler's (1994) idea that the Narten properties belong to the root is valid, an *i*-stem built from a Narten root could conceivably inflect this way. Given the Narten *i*-stem *$\underset{\sim}{\mu}\acute{e}lh_1$-p-i-/*$\underset{\sim}{\mu}\acute{e}lh_1$-p-i-, the strong stem could be an environment for laryngeal loss in the extra-heavy syllable *$\underset{\sim}{\mu}\acute{e}lh_1$-p-i- > *$\underset{\sim}{\mu}\bar{e}l$-p-i-. Greek could have generalized the strong stem without laryngeal, whence ἐλπίς with shortening by Osthoff's Law. Latin, on the other hand, must have generalized the weak stem *$\underset{\sim}{\mu}\acute{e}lh_1$-p-i-, with laryngeal preserved in *volup*.[72]

An *i*-stem *$\underset{\sim}{\mu}\acute{e}lh_1$-p-i- is not without morphological parallel: Greek has another trace of a lengthened-grade *i*-stem in δῆρις 'strife, conflict; battle', based on *$d\acute{e}r$-i- 'splitting' to the root **der*- 'split'. In addition to δῆρις, Vine (2006a) has suggested that SP **ímih** 'devotion, love' (*ST* AP 2) reflects an acrostatic *i*-stem noun *$h_2\acute{e}mh_3$-i- (or *$h_1\acute{e}mh_3$-i-) of the deverbal action/abstract type, built to the root of Lat. *amō* 'love' and *amīcus* 'friend' < *h_2emh_3- (or *h_1emh_3-) 'touch, seize; swear'. Vine offered other potential examples of the noun type, first collected by Brugmann (1906:168): Gmc. *$\underset{\sim}{\mu}\acute{e}n$-i- 'hope' (Go. *wens*, ON *ván*, OE *wēn*) to *$\underset{\sim}{\mu}en$- 'win, strive' and/or *$\underset{\sim}{\mu}enH$- 'desire'; Gmc. *$\underset{\sim}{\mu}\acute{e}d$-i- 'binding' (OHG

[72] ἔλπομαι too might fit nicely into the picture, if it continues a Narten subjunctive *$\underset{\sim}{\mu}\acute{e}lh_1$-p-e/o- built to an athematic present *$\underset{\sim}{\mu}\acute{e}lh_1$-p-ti/*$\underset{\sim}{\mu}\acute{e}lh_1$-p-$\underset{\circ}{n}$ti, based on the strong stem *$\underset{\sim}{\mu}\acute{e}lh_1$-p- (though again no trace of such a present is found). The root vocalism of Narten subjunctives is unclear, as the data are inconclusive and subject to early analogical remodeling that has obscured their history.

wāt, OE *wǣd* 'garment, dress') to **u̯edʰ-* 'lead'; OCS *rěčĭ* 'speech' < **rḗk-i-* to **rek-* 'speak'. To the list might be added another example offered by Janda (1999) in Gk. ἧνις, an epithet for oxen, from **h₁ḗnh₃-i-* 'das Bewegen der Last' built to **h₁enh₃-* 'eine Last bewegen' (cf. Lat. *onus* 'load'). In light of the δῆρις-type of acrostatic noun, perhaps Lat. *volup* may be better conceived as continuing an old *i*-stem noun (rather than the assumed adjective), which may actually have meant something like 'pleasure' in the phrase **volúp(e) est* "it's a pleasure".

II.G. Anomalous Bases

There exist a very small number of abstracts in Latin that either reflect consonant-stem abstracts with no transparent base, e.g. *facultās* 'ease' beside *facilis* 'easy' or have no base adjective at all, e.g. *calamitās* 'loss, misfortune'. *facultās*, etc. may be interesting, in that the derivative in plain *-tās* would seem to have been made before the generalization of *-itās*.

II.G.1. Remaining Abstracts Built to C-Stem Bases

facultās 'ability' (Ter.+) next to *facilitās* 'ease' (Cic.+), *famultās* 'slavery' (Laev.) next to *famulitās* 'id.' (Pac.+), and *simultās* 'quarrel, contest' (Pl.+) descriptively reflect abstracts in plain *-tās*. *facultās* may reflect an older formation in *-tās* or a more recent one in *-itās* built to the original adjective stem **fakli-*, whence **fakli-(i)tās* > **fak̥ltās* (cf. Weiss 2009:193), while *facilitās* reflects a later byform built to *facilis*. Likewise,

simultās could continue **semli-tās* and *famultās *famli-tās*. But while the semantics of *facultās* reflect the quality of the adjective *facilis* 'easy', *simultās* does not seem to be directly derived to its base adjective *similis* 'similar, like'. If indeed continuing **semli-tās*, one could expect *simultās* 'similarity' (*vel sim.*), as in *similitās* 'likeness' (Caecil.+).

Beside *facultās* and *simultās* are the adverbs *facul* 'easily' (Pac.+) and *simul* 'together' (Pl.+), continuing an old neut. nom.acc. sg. **fakli* and **semli* to the *i*-stem adjectives (Weiss 2009:147). Perhaps, then, *simultās* is analogical, based on the model *facul* : *faultās* :: *simul* : *x*, where *x* = *simultās*. With the meaning 'together', it is reasonable to expect the new abstract *simultās* 'togetherness' might acquire a sense of hostility and develop to 'quarrel, strife'.

famultās is based on the the noun *famul* (Enn./Lucr.)/*famulus* (Pl.+) 'slave'. Beside *famulus* are found Osc./Pg. *famel*, Osc. *famelo*, Umb. **fameřias** < **fameliā-* (cf. Untermann 2000:263-4), which suggest a Proto-Italic thematic noun **famelo-* 'slave' (*EDLIL* s.v. *famulus*). *famultās* must continue **famo̯ltās* 'the quality of being a slave; slavery' to the stem **famo̯lo-* underlying *famulus*. Significantly, **famo̯ltās* reflects another denominal use of *-tās* in Latin, which built *tās*-abstracts to nouns to a much higher degree than did Indo-Iranian (cf. **III.A.1.** on Vedic bases and **III.A.2.** on Avestan bases) and Greek (cf. **IV.A.**).

Other than the abstracts discussed above, no other forms in *-tās* in Latin are found to consonant-stem bases. All other *tās*-abstracts are built with the generalized suffix form *-itās*.

II.G.2. Abstracts without Synchronic Bases

A small number of abstracts have no synchronic base to which they were built. Neither an *i*-stem adjective *ˣcalamis* nor an *o*-stem *ˣcalamos* underlying *calamitās* 'loss, misfortune' (Pl.+) is attested; only *incolumis* 'uninjured, safe' is found, which is ambiguous, given that both an *o*-stem and *i*-stem can be the source of a compound *i*-stem adjective. Based on *incolumis* Leumann (1977:374) opts for the *i*-stem *ˣcalamis*, though it need not be the case. **kalami-/*kalamo-* reflects either **kl̥h₂-mi-* or **kl̥h₂-mo-*, with the outcome **-ala-* from **-l̥H-* (but cf. *EDLIL* s.v. *calamitās* for the claim that **kalami-/*kalamo-* reflects **kl̥h₂-emi-/*kl̥h₂-emo-* and n. 137 on **sl̥h₂-u̯-o-* in **V.B.1.**). *incolumis* reflects different vocalism from *calamitās*, suggesting that the abstract, no longer with a synchronic base, could well be relatively old, reflecting either **kalami-tās* or **kalamo-tās*.

II.H. Conclusions

Upon examination of the abstracts in Latin, two broad categories can be identified: abstracts in *-itās* and abstracts in plain *-tās*. The first cagetory is a result of abstracts built to *o*-stem or *i*-stem adjectives giving *-itās*, where the vowel then became reanalyzed as a part of the suffix itself. *-itās* subsequently spread as the productive suffix

variant, and the generalization seems to have happened already by the *XII Tables* (cf. *aevitās* and *auctōritās*). Synchronically, *-itās* builds simple, qualitative abstracts from adjectives without deviations from the pattern (cf. **II.A.**).

Forms built with *-tās*, on the other hand, appear to constitute an older layer of abstracts in Latin, when plain *-tās* could still be added directly to a fairly diverse set of consonant-stem bases (cf. Fig. 4). Though it has been argued that haplology and/or syncope accounts for the majority of the forms, such explanations are *ad hoc*, in that no ironclad evidence for such developments can be found, and moreover, the use of haplology to explain the forms then becomes circular. Abstracts like *egestās*, *honestās*, *lībertās*, etc. are thus older than the class of forms in *-itās* and as such reveal more about the history of the *tāt*-suffix in Latin (cf. **II.F.**). Some of the other abstracts in plain *-tās*, however, are of indeterminate age, in that they could have been formed at any time within the history of Latin before the generalization of *-itās*.

A very large percentage of the consonant-stem abstracts are built to nouns, rather than adjectives. In fact, Latin is in possession of more denominal *tās*-abstracts than are found in any other language also attesting the suffix. *-tās* came to be closely associated with *s*-stem nouns, as also did *to*-adjectives built to them, and an ensuing relationship emerged between abstracts in *-stās* and adjectives in *-stus* (cf. **II.D.2.**). Significantly, plain *-tās* was used with an "individualizing" function to distinguish one abstract type

from another, serving either to build a qualitative abstract beside the substantivized abstract base, e.g. *honestās* beside *honor* (cf. **II.D.3.**), or to individualize the base abstract, which could then be used to name an abstract divinity, e.g. *tempestās* (cf. **II.F.1.b.**).

Fig. 4. Summary of the development of Latin *tās*-abstracts in plain *-tās*

• *aetās* < **aiu̯o-tās*	• *aestās* ← *aestus*
• *calamitās* < **kalami-tās* or **kalamo-tās*	• *angustās* ← *angustus*
• *egestās* < **eg-es-tās*	• *honestās* ← *honestus*
• *famultās* < **famlt̥ās*	• *iuventās* ← *iuventus*
• *facultās* < **fakli-(i)tās*	• *paupertās* by analogy
• Pg. *herentas*/Osc. **herentateís** < **ĝʰer-n̥t-tās*	• *pesestās* by analogy?
• *lībertās* < **leu̯dʰerotās* or **līber-tās*	• *potestās* ← **potistās*
• *maiestās* ← **maĝis-tās*	• *pūbertās* by analogy
• *tempestās* < **temp-es-tās*	• *satiās* < **sati-(i)tās/satietās* ← *satius*
• *voluntās* < **u̯elont-tās*	• *simultās* < *simul* + *-tās*
• *voluptās* < *volup* + *-tās*	• *ūbertās* < *ūber* + *-tās*
	• *viduertās* by analogy
	• *venustās* < *venus* + *-tās*
	• *vetustās* ← *vetustus*

N.B.: Sabellian forms not listed are assumed to be borrowings from Latin.

III. Indo-Iranian *-tāt-* and *-tāti-*

III.A. Descriptive Facts

Indo-Iranian *-tāt-* makes feminine abstracts from adjectives and a few nouns, with the oldest material found in Vedic and Old Avestan. In Vedic there exists an *i*-stem variant *-tāti-* that is more common than plain *-tāt-* in the Rig Veda and even replaces it altogether in later Vedic texts (Renou and Benveniste 1960:10). *tāt-* and *tāti*-abstracts in Vedic are rather rare, in comparison to other abstract types, with only 11 abstracts found in the Rig Veda (cf. Fig. 5), with both a few repetitions and additional abstracts found in later Vedic texts (cf. Fig. 9 in **III.C.**), in which they are already beginning to be replaced with abstracts in *-tā-* (Renou and Benveniste 1960:17). Unlike Latin *-tās* and Greek *-τηs*, *-tāt-* at no time became productive in Indic.

Fig. 5. Rig Vedic *tāt-/tāti*-abstracts (after Grassmann)

ariṣṭátāti- 'integrity, security' • dat. sg. *ariṣṭátātaye* (2x) • dat. pl. *ariṣṭátātibhis* (1x)	• 10.60.8-10d (repeated), 10.97.7d • 10.137.4b
ástatāti- 'home' • acc. sg. *ástatātim* (1x)	• 5.7.6d
uparátāt- 'superiority' • loc. sg. *uparátāti* (2x)	• 1.151.5c, 7.48.3b
g̥rbhītátāti- 'captivity' • dat. sg. *g̥rbhītátātaye* (1x)	• 5.74.4c
jyeṣṭhátāti- 'sovereignty' • acc. sg. *jyeṣṭhátātim* (1x)	• 5.44.1b

Fig. 5. Rig Vedic *tāt-/tāti*-abstracts (after Grassmann), Cont.

devátāt-/devátāti- 'collection of the gods' • acc. sg. *devátātim* (7x) • dat. sg. *devátātaye* (11x) • dat. sg. *devátāte* (3x) • loc./instr. sg. *devátātā* (13x) • loc. sg. *devátāti* (2x)	• (*-tātim*) 1.141.10b, 3.19.2c, 3.19.4c, 4.6.3b, 4.6.9d, 7.1.18b, 7.39.1b • (*-tātaye*) 1.127.9b, 1.127.9c, 3.26.2c, 8.3.5a, 8.60.10c, 8.62.8b, 8.101.1b, 9.15.2b, 9.17.7c, 9.65.27b, 10.141.6c • (*-tāte*) 9.96.3a, 9.97.19a, 9.97.27a • (*-tātā*) 1.34.5b, 1.58.1d, 1.95.8d, 1.128.2c, 3.19.1c, 4.6.1b, 5.29.1a, 6.4.1a, 6.68.2a, 7.2.5b, 7.38.7b, 7.43.3d, 10.53.1c • (*-tāti*) 8.74.3b, 10.8.2c
vasútāti- 'collection of Vasus' • nom. sg. *vasútātis* (1x) • acc.sg. *vasútātim* (1x)	• 1.122.12c • 1.122.5d
vr̥kátāt- 'pack of wolves' • loc. sg. *vr̥kátāti* (1x)	• 2.34.9a
śáṃtāti- 'health; luck' • acc. sg. *śáṃtāti* (1x) • nom. du. *śáṃtātī* (1x) • dat. pl. *śáṃtātibhis* (1x) • loc./instr. sg. *śáṃtātā*	• 8.18.7c • 1.112.20a • 10.137.4a • SV 1.2.1.1.6b *śáṃtātā* = RV 8.18.7c
satyátāt-/satyátāti- 'truth' • instr. sg. *satyátātā* (1x) • voc. sg. *satyatāte* (1x)	• 10.111.4d • 4.4.14c
sarvátāt-/sarvátāti- 'wholeness, health' • acc. sg. *sarvátātim* (3x) • dat. sg. *sarvátātaye* (4x) • loc./instr. sg. *sarvátātā* (9x)	• 3.54.11d, 10.36.14c, 10.100.1-11d (refrain) • 1.106.2a, 9.96.4b, 6.56.6c, 10.35.11a • 1.94.15b, 3.54.19b, 4.26.3c, 5.69.3c, 6.12.2b, 6.15.18b, 7.18.19b, 7.57.7b, 10.74.3b

In post-Rig Vedic texts, only *-tāti-* is found, including the AV forms *ayakṣmátātim* (AV-Ś 4.25.5c) 'health' and *dákṣatātim* 'capability' (AV-Ś 8.1.6b); otherwise, the AV repeats *ariṣṭátātaye* (9x; cf. Fig. 9 in **III.C.**), *jyeṣṭhátātaye*

'sovereignty' (AV-Ś 6.39.1d), *devátātaye* (AV-Ś 20.118.3a), and *sarvátātaye* (AV-Ś 6.3.3d). After Vedic, *-tāti-* ceases to make qualitative abstracts (cf. Debrunner 1954:621). A few forms are found in Classical drama, e.g. *śivatāti-* 'beneficial; welfare' (*Mālatīm.*) and *śubhatāti-* 'welfare' (*Śatruñj.*). None of the later abstracts differ in any way from the earlier forms, aside from a late adjectival use of *śivatāti-* (cf. **IV.D.4.**).

27 *tāt*-abstracts are found in both Old and Young Avestan, and *-tāt-* even becomes relatively productive in later Avestan texts. No *i*-stem variant is found in Old Iranian. Other than *amərətatāt-/amərətāt-* 'immortality' and *hauruuatāt-/hauruuāt-* 'wholeness', which are frequently found together, most of the 10 total abstracts found in Old Avestan are hapaxes, though half of them are also found later in Young Avestan (cf. Fig. 6).

Fig. 6. Old Avestan *tāt*-abstracts (after Bartholomae)

	[illegible]
amərətatāt-/amərətāt- 'immortality'†	14x
auuaētāt- 'the word *auuōi*'	hapax; Y. 31.20
astəntāt- 'liveliness'	hapax; Y. 41.3
karapō.tāt- 'class, group of *karapan*s'	hapax; Y. 32.15
kəuuītāt- 'class, group of *kavis*'	hapax; Y. 32.15
pauruuatāt- 'priority'†	2x; Y. 33.14, Y. 37.2
yauuaētāt- 'eternity'†	hapax; Y. 28.11
hauruuatāt-/hauruuāt- 'wholeness'†	11x
hunarətāt- 'talent'	hapax; Y. 50.8
x^{v}aētāt- 'membership'†	hapax; Y. 31.20

† Also appears in Young Avestan.

Young Avestan has a larger number of abstracts, with 22 total forms found. 10 of the forms are only found once, but some of the others find relatively productive use in the texts (cf. Fig. 7).

Fig. 7. Young Avestan *tāt*-abstracts (after Bartholomae)

amərətatāt-/amərətāt- 'immortality'	19x
arštāt- name of a goddess "Uprightness"	10x
uxšiiąstāt- 'condition of growth; waxing'	hapax; Yt. 7.2 (repeated)
uiθe.tāt- ?	hapax; N. 61
uθō.tāt- 'fatty substance'	hapax; V. 6.10
uparatāt- 'superiority'	13x
uštatāt- 'the word *ušta-*'	5x
uštatāt- 'what is desirable; luck, success'	10x
kahrkatāt- name of a rooster	hapax; V. 18.15
daēuuō.tāt- '*Daeva*-hood'	hapax; Yt. 13.90
pauruuatāt- 'priority'	2x; Y. 57.20, Vr. 9.4
pourutāt- 'multiplicity, quantity'	3x; Y. 62.10, Ny. 3.10, [+]Vyt. 6
fratəmatāt- 'authority'	2x; Yt. 10.18, Yt. 13.95
nərəfsąstāt- 'condition of decline; waning'	hapax; Yt. 7.2 (repeated)
vaŋhutāt- 'blood'	2x; V. 6.7 and 29
rasąstāt- name of deity?	hapax; Y. 1.14
[i]riθiiąstāt- 'mortality'	hapax; Y. 19.10
stitāt- 'essence'	hapax; Vr. 10.2
šiiaoθnōtāt- 'the word *šiiaoθna-*'	2x; Y. 19.9, N. 81
yauuaētāt- 'eternity'	13x ([+]VD. 9)
hauruuatāt-/hauruuāt- 'wholeness'	24x
x[v]aētāt- 'membership'	hapax; Y. 20.1

[+] Bartholomae questioned the form as found in the passage (*AiW* s.v.).

Middle Iranian evidence may point to an *i*-stem *-tāti-* that seems to have become productive to some degree. Benveniste (1960:18-9) argued that Middle Iranian evidence

reflects *-tātä* < **-tāti-*, with support for such a variant in Khot. *ttuśśāttātä* 'emptiness' and *ttāḍātä* 'obscurity'.[73] Modern Iranian may have preserved **-tāti-* as well: Pashto *təštiyā* 'emptiness' might reflect **-tāti-* > *-t(i)yā*, providing additional support that Iranian had a suffix *-tāti-* that was not continued into Avestan but found in other dialects of Iranian. Benveniste took the Middle and Modern Iranian data, together with Vedic *-tāti-*, as evidence for an Indo-Iranian suffix **-tāti-*, which reflects a "resuffixation" of abstracts in *-tā-* with *-ti-* within Indo-Iranian, though he rejected the connection with Lat. *-tātium*, rightly thinking of it as a secondary formation particular to Classical Latin, in agreement with Leumann (1977:375). Benveniste, nevertheless, did not pursue his hypothesis in any detail. As will be shown below in **III.C.**, *-tāti-* is an inner-Vedic creation, resulting from a reanalysis of forms in *-tātā*, so Indic provides no foundation for an Indo-Iranian *i*-stem suffix **-tāti-*.

Khotanese attests several abstracts in OKh. *-tātä*/LKh. *-ā* (Bailey 1942:27-8, Dresden 1955:409), which phonologically does seem to reflect **-tāti-*. Though Khotanese is rich in nominal vocalic stems, it does not possess many consonant stems, and no plain *t*-stems seem to exist in the language. It is thinkable that Iranian **-tāt-* was also inherited in Khotanese, but plain *t*-stems were lost and replaced with *i*-stem

[73] Khotanese Buddhist texts were heavily based on Sanskrit models, and so it is thinkable that the abstracts in *-tātä* are borrowings from Sanskrit texts. But Buddhist Hybrid Sanskrit does not seem to have used a suffix-*tāti-*, so a borrowing theory for *-tātä* seems to be a very remote possibility.

inflection, giving *-*tāti*-. Otherwise, it would be very difficult to justify *-*tāti*- in Proto-Iranian, when such a suffix is wholly absent from Avestan. And if Eastern Iranian indeed remodeled *-*tāt*- to *-*tāti*-, it is quite possible that Pashto -*t(i)yā* also continues a secondary suffix *-*tāt*-.

A book-by-book distribution within the Rig Veda, with the frequency of occurrence of each abstract found in each book, is given below in Fig. 8.

Fig. 8. Book-by-book rundown of Rig Vedic *tāt-/tāti*-abstracts

ariṣṭátāti- 'integrity, security' (3x)									10^{3}
ástatāti- 'home' (1x)				5^{1}					
uparátāt- 'superiority' (2x)	1^{1}					7^{1}			
gr̥bhītátāti- 'captivity' (1x)				5^{1}					
jyeṣṭhátāti- 'sovereignty' (1x)				5^{1}					
devátāt-/devátāti- 'collection of the gods' (35x)	1^{6}		3^{4}	4^{3}	5^{1}	6^{2}	7^{5}	8^{5}	9^{6}	10^{3}
vasútāti- 'collection of Vasus' (2x)	1^{2}								
vr̥kátāt- 'pack of wolves' (1x)		2^{1}							
satyátāt-/satyátāti- 'truth' (2x)				4^{1}					10^{1}
sarvátāt-/sarvátāti- 'entirety, health' (16x)	1^{2}		3^{2}	4^{1}	5^{1}	6^{3}	7^{2}		9^{1}	10^{4}
śáṃtāti- 'health; luck' (3x)	1^{1}							8^{1}		10^{1}

III.A.1. Vedic Base Types

As in Italic, Vedic built *tāt-* and *tāti*-abstracts to thematic adjectives, with 3 examples in the Rig Veda: *uparátāt-* 'superiority' to *úpara-* 'neighboring', *satyátāt-/satyátāti-* 'truth' to *satyá-* 'real; true', *sarvátāt-/sarvátāti-* 'wholeness, health' to *sárva-* 'all, whole'.[74] One abstract built to a *u*-stem adjective occurs, *vasútāti-* 'collection of Vasus' to *vásu-* 'good', likely built to the neut. substantive *vásu-* 'wealth, goods' (cf. **III.D.1.**). In addition to building abstracts to ordinary adjectives, abstracts in *-tāt-/-tāti-* are based on *tá*-participles in *ariṣṭátāti-* 'integrity, security', built to *áriṣṭa-* 'unharmed', and *gr̥bhītátāti-* 'captivity', built to *gr̥bhītá-* 'seized', *jyeṣṭhátāti-* 'sovereignty', based on the superlative to *jyeṣṭhá-* 'highest, chief'. The remaining few are built to nouns: *vr̥kátāt-* 'collection of wolves', *devátāt-/devátāti-* 'collection of the gods', and *śáṃtāti-* 'luck' to *vŕ̥ka-* 'wolf' (m), *devá-* 'god' (m), and the indeclinable noun *śám* 'luck, health' (n), respectively.

III.A.2. Avestan Base Types

Avestan also built *tāt*-abstracts to thematic stems, without any evidence for an *i*-stem variant *-tāti-* in Old Iranian (Debrunner 1954:620 and Renou and Benveniste

[74] Thieme (1938:53) believed *uparátāt-* securely meant 'superiority', based on YAv. *uparatāt-* 'id.'. YAv. *upara-* means 'higher', which can easily make an abstract noun 'higherness; superiority', but Ved. *úpara-* has a different meaning, reflecting 'behind; neighboring', while the adv. *úpari* means 'above, on top', as also does YAv. *upairi*. If **uparátāt-* reflects a Proto-Indo-Iranian form, as seems very likely, Ved. *úpara-* must have drifted from an original meaning of 'higher'. More importantly, the abstract *uparátāt-* preserves older semantics than the adj. *úpara-*.

1960:16), e.g. YAv. *fratəmatāt-* 'authority' to *fratəma-* 'first', *hauruuatāt-* 'wholeness' to *hauura-* 'whole', *pauruuatāt-* 'priority' to *pauura-* 'prior, former', YAv. *uparatāt-* 'superiority' (cf. Ved. *uparátāt-* 'superiority') to *upara-* 'higher', etc. YAv. *aršt̰āt-*, the name of a goddess "Uprightness", is thought to continue **aršt̰atāt-* with haplology (*AiW* s.v.) of the same sort in *amərətāt-* beside *amərətatāt-* (cf. **III.A.5.** below). As in Vedic, *-tāt-* was extended to past participles, e.g. YAv. *uštatāt-* 'what is desirable; luck, success', built to the past participle of *vas-* 'wish'. Other base types are found too, including *i*-stems, e.g. OAv. *kəuuītāt-* 'class, group of *kavis*', and *u*-stems, e.g. YAv. *pourutāt-* 'multiplicity, quantity' to *pouru-* 'much, many', *vaŋhutāt-* 'blood' beside YAv. *vohunī-* 'blood' (f), suggesting an underlying *u*-stem (cf. **III.D.2.**), and even OAv. *astəntāt-* 'liveliness' built to an *n*-stem **astan-* no longer found in Avestan (cf. Ved. *asthán-* 'bone'). OAv. *hunarətāt-* 'talent' might even continue an abstract built directly to a consonant-stem **hunar-* (*AiW* s.v. *hunarətāt-*; for a discussion of the possible background of *hunarətāt-*, cf. **IV.C.3.a.**). As in Vedic, nominal bases exist too, as in YAv. *daēuuō.tāt-* '*Daeva*-hood' to *daēuua-* 'evil god, demon'.[75]

[75] YAv. *daēuuō.tāt-* '*Daeva*-hood', beside Ved. *devátāt-/devátāti-*, may not be as clear as it seems. *daēuuō.tāt-* is only found once at Yt. 13.90, and the MSS all have *daēuuō.dātəm*, from the adjective *daēuuō.dāta-* 'made by *Daevas*' (11x in YAv.). Similarly, YAv. *fratəmatāt-* 'authority' is found twice at Yt. 10.18 and Yt. 13.95, but *fratəmaδātō* is transmitted at 10.18. The emendation of *dahiiuṇam fratəmaδātō* to *fratəmatātō* seems justifiable, given the similar passage at 13.95, in which *fratəmatātō* is

Avestan possesses a *tāt*-abstract built to a pronominal stem in OAv./YAv. *xᵛaētāt-* 'membership', related to the possessive adjective *xᵛa-* 'one's own'. Given that *xᵛaētāt-* unusual with respect to its formation, according to Narten (1986:265-6, but without explaining why), she suggested that *xᵛaētāt-* < **hu̯ai-tu-tāt-* might be haplologized from the *tu*-stem OAv./YAv. *xᵛaētu-* 'affiliated'. In Y. 20.1, the two words are used in a sort of *figura etymologica*, in which *xᵛaētāt-* 'family membership' reflected a partially collective sense beside *xᵛaētu-* 'family member', so *xᵛaētāt-* could well be based on *xᵛaētu-*. But perhaps one can also consider *xᵛaētāt-* nothing more than a replacement of the suffix *-tu-* in *xᵛaētu-* with plain *-tāt-* added directly to the stem *xᵛaē-*, based on the level of productivity of *-tāt-* in Avestan in its "technical" uses (cf. below). Supporting the idea of *xᵛaē-tāt-* is the fact that the older-looking Latin abstracts also were built with plain *-tās*, before generalization of *-itās* (cf. **II.H.** for a summary of the findings). As also witnessed in the Latin *tās*-abstracts (cf. **II.D.2.**), and as will also be seen with some Greek forms (cf. on ποτής 'drink' in **IV.C.2.**), not all abstracts seem to have such a mechanical derivational pattern to a base stem.

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also going with *dahiiunąm*. The phrase *yō vispąm daēuuō.dātəm vauuača aiiesniiąm auuahmiiąm* is ambiguous, in that *daēuuō.dātəm* can be understood in the sense "everything made by the *Daevas*", or if emended to *daēuuō.tātəm* in a sense similar to Ved. *devátāt-/devátāti-* "the entire assemblage of the gods", as Bartholomae preferred (*AiW* s.v. *daēuuō.tāt-*).

Benveniste (1960:19) noticed that Iranian and Italic both make use of *-tāt-* with present participles: YAv. *ⁱriθiiąstāt-* 'death, dying' to *raēθ-* 'die', YAv. *nərəfsąstāt-* 'condition of decline' to *narəp-* 'wane', YAv. *rasąstāt-* 'name of deity' (on which cf. below in **III.E.**), and YAv. *uxšiiąstāt* 'condition of growth' to *vaxš-* 'grow' are all built to present participles, much like Lat. *voluntās* 'will', Pg. *herentas* 'Venus', and Osc. **herentateís** 'Venus'. Unlike the Italic abstracts, on the other hand, which have been thought to reflect a syncopated **-nti-tās* or **-nto-tās* (with simplification of **-ntt-* to *-nt-*; cf. **II.F.1.d.** for discussion of the Italic material, including the suggestion of **-nt-tās*), the Avestan abstracts must continue **-nt-tāt-*, with the Iranian treatment of the PIE dental cluster **-tt-* > *-st-*.

Aside from *vasútāti-* and *śáṃtāti-*, which both seem to be inner-Vedic creations (on which cf. below), Vedic did not build abstracts in *-tāt-/-tāti-* to non-thematic bases; but Avestan attests a much wider number of base types. Both Italic and Greek built *tāt-* abstracts to multiple base types too, suggesting that PIE **-tāt-* may have been used with more than just thematic adjectives. Avestan preserves the original use of **-tāt-* more faithfully than Vedic, which has already restricted it to thematic adjectives, for the most part.

III.A.3. Semantics of Vedic *-tāt-/-tāti-*

The Vedic abstracts built to thematic adjectives, including *to*-participles, e.g. *uparátāt-* 'superiority', *satyátāt-/satyátāti-* 'truth', *sarvátāt-/sarvátāti-* 'entirety, wholeness', *gr̥bhītátāti-* 'captivity', *jyeṣṭhátāti-* 'sovereignty', and *ariṣṭátāti-* 'integrity, security', reflect qualitative abstract semantics, as do most of the abstracts in *-tās* in Latin. It has been claimed by Renou (1960:11-13) that the abstracts in *-tāt-* in the Rig Veda show a discrepant meaning from those in *-tāti-*, e.g. *sarvátāt-* 'en nombre total, au grand complet' beside *sarvátāti-* 'complétion, Vollständigkeit' (but cf. below for a refutation). The remaining forms that are built to nouns, *vr̥kátāt-* 'collection of wolves', *devátāt-/devátāti-* 'collection of the gods', *śáṃtāti-* 'luck', and possibly even *vasútāti-* 'collection of Vasus', have collective semantics (cf. discussion below in **III.C.1.b.**).

III.A.4. Semantics of Avestan *-tāt-*

In addition to having more base types, the Avestan *tāt*-abstracts have a much wider semantic range than the *-tāt-* abstracts found in other branches. Renou and Benveniste (1960:16) remark that "[s]ensiblement plus riche, morphologiquement mieux assise, la formation iranienne fournit des noms d'abstrait indiquant de préférence la catégorisation, le fait d'être tel, de porter tel nom. On y trouve comparativement beaucoup de noms impliquant une sorte de personnification, d'hypostase, des sortes d'abstraits à majuscule". Some forms in *-tāt-*, e.g. OAv. *karapō.tāt-* 'class, group of

karapans' and OAv. *kəuuītāt-* 'class, group of *kavis*', never have the semantics of a qualitative abstract, but rather seem to serve as a category heading or collective. Other rather odd abstracts are also found possessing a "technical" sense, e.g. OAv. *auuaētāt-* 'the word *auuōi*' to the interjection *auuōi* 'woe!', YAv. *ušťatāt-* 'the word *ušta-*' to *ušta-* 'welfare, health, luck', and YAv. *kahrkatāt-* 'the word *kahrka-*' to *kahrka-* 'chicken'.

III.A.5. Accentuation

In Vedic accentuation behaves identically to the Greek abstracts in *-τηϛ* built to thematic adjectives (cf. p. **IV.C.1.** for discussion of the Greek accentuation), with accent falling immediately before the *tāt*-suffix in the abstract, regardless of the position of accent in the base, just as with the *tā*-stems (Debrunner 1954:621). No accentual anomalies are found in the Rig Veda, with the exception of *ástatāti-* 'home' (and AV *dákṣatāt-* 'capability') and *śáṃtāti-* 'luck, health'.

śáṃtāti- is built to a consonant stem, unlike all other *tāt-* and *tāti*-abstracts in Vedic (cf. discussion on *śáṃtāti-* below), with accent on the base. Unlike the Homeric abstracts not directly derived from thematic stems that carry accent on the suffix, e.g. *ἀνδροτήϛ* 'manliness' (cf. **IV.C.3.a**) and *δηϊοτήϛ* 'battle strife', (cf. **IV.C.3.c.**), abstracts in Vedic are never accented on the suffix. *śáṃtāti-* likely has just copied its accent from the base noun *śám*.

Though it continues a *tāti*-abstract, *ástatāti-*, beside the neut. noun *ásta-* with the same meaning 'home', is also accented on the base, rather than the thematic vowel.[76] It seems probable that *ásta-* reflects a substantivization of an unattested adjective **astá-* 'returning' (*vel sim.*), with accent retraction. If so, *ástatāti-* may have been built to the adjective, copying the accent of the noun after the adjective fell out of use. Likewise, AV *dákṣatāt-* must have copied accent from the base *dákṣa-* 'powerful, strong', as well as from the compounds *dákṣapitr̥-* 'having mental powers', etc. and the *s*-stem *dákṣas-* 'power, capability'.

Because Avestan does not provide any direct evidence for accent, it is not possible to generalize where accent fell in the *tāt*-abstracts. But indirect evidence for accentuation is found in OAv./YAv. *amərətāt-/amərətatāt-*, an abstract built to the adjective OAv./YAv. *amaša-* 'undying' (< PIE **n̥-mŕ̥-to-*; cf. Ved. *amŕ̥ta-* 'immortal').[77] *amərətāt-* is thought to continue a haplologized *amərətatāt-* from PIE **n̥-mr̥-tó-tāt-* (cf. Barnes 2007, with refs.), with **-ŕ̥t-* giving *-š-* in Avestan (cf. Hoffmann-Forssman 2004:92, 112). PIE **n̥-mŕ̥-to-tāt-*, with root accent, would have given **amašatāt-*, and so Av. *amərətāt-/amərətatāt-* should point directly to PIE **n̥-mr̥-tó-tāt-*, with accentuation

[76] *ástam* is frequently used as an indeclinable adverb in the Rig Veda, especially with verbs of motion (Macdonell 1910:431).

[77] *amərətāt-/amərətatāt-* appears in frequent association with *hauruuatāt-/hauruuāt-* 'wholeness' in Avestan, particularly in the *Amәša Spənta*. For a detailed treatment of the pair, cf. Narten (1982:134-46).

on the thematic vowel immediately before *-tāt-* expected for an abstract built to a thematic adjective.[78]

III.B. Previous Scholarship

Previous work has paid relatively little attention to *-tāt-* and *-tāti-* in Indo-Iranian in comparison with *-tās* in Italic and *-τηs* in Greek. Debrunner (1954:620) noted that plain *-tāt-* was unknown to the Indian grammarians, though Pāṇini discussed *-tāti-* as a sort of "factitive" suffix in *śáṃtāti-* and *ariṣṭátāti-* in the sense of "he does *x*", while meaning "the condition of *x*" in other examples (*Aṣṭ.* 4.4.142-4, 5.4.41). Later, Mahīdhara (*Ved.* 18.4 = Panshikar 1992:444-5), in his 16th c. CE commentary on the Yajurveda, read *jyeṣṭhátātim* as a possessive compound (despite its accent), with *-tāti* signifying 'extension', thereby "strengthening" the original meaning of *jyéṣṭha-* 'best' to mean 'the best of all, notoriously best'. It was not until Aufrecht (1852) that the two suffixes *-tāt-* and *-tāti-* were given explicit attention in Vedic. Still, no systematic study of derivational patterns and semantics of the *tāt-* and *tāti*-suffixes in the level of detail approaching that of Paucker's for Latin (1877; cf. **II.B.** for discussion) and Petersen's for Greek (1922; cf. **IV.B.** for discussion) has been done for Indo-Iranian.

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[78] It is worth considering whether *amərətāt-* could continue **ń̥-mr̥-to-tāt-*, with accent on the privative. Possessive compounds with the negative *a-* tend to accent the base rather than the prefix (cf. Macdonell 1910:93), but plenty of examples of accented *á-* exist in the RV, e.g. *águ-* 'without kin', *ádyu-* 'not burning', and even *ámr̥tyu-* 'deathless', so **ń̥-mr̥-to-tāt-* cannot strictly be excluded. Given that Ved. *amŕ̥ta-* and OAv./YAv. *amašə-* reflect **n̥-mŕ̥-to-*, though, *amərətāt-* likely also reflects **n̥-mr̥-tó-tāt-*.

Aufrecht (1852:161-63) covered the Indo-Iranian evidence for *-tāt-* and *-tāti-* (for his treatment of the Italic material, cf. **II.B.**), considering *-tāti-* to be the primary suffix form that reflects a sort of extension of *-tā-* with *-ti-*. He gave some examples of *tāt-* and *tāti*-abstracts in Vedic and compared them to abstracts found in Avestan but made no systematic attempt to classify their base types. Aufrecht did note (1852:161) that in both Vedic and Greek, the accent falls immediately before the abstract suffix.

Bopp (1856:1131-40) included a discussion of *-tāt-* with his treatment of abstract nouns in *-tā*, thinking of *-tā-* as a version of *-tāt-* without *-t-* (1856:1132). He also added comparison of Vedic and Avestan material, but did not classify the data in any systematic way (1856:1137). He discussed the *i*-stem variant *-tāti-* (cf. on *-tāti-* below) found in Vedic (1856:1136), seeing *-tāti-* as a composite suffix continuing **-tā-* plus **-ti-* with Aufrecht (1852:162-3). Bopp then suggested that plain *-tāt-* might actually have been created first, with *-t-* added to *-tā-* "as a support" (1856:1136-7), as seen in compounds in Greek, e.g. *ἀγνώ-τ-* 'obscure' (*Od.*+) and *ὠμοβρώ-τ-* 'eating raw flesh' (Soph.+). If so, the *i*-stem variant *-tāti-* found in Vedic must be secondary. For reasons to be discussed below, Bopp will be seen to be essentially correct in thinking that the *i*-stem abstracts in Vedic *-tāti-* are secondary.

Brugmann (1906:451-2) was the first to discuss the distribution of *-tāt-* and *-tāti-* in a systematic way. He noticed that Vedic *tāt*-abstracts are limited to the dat. sg., instr.

sg., and loc. sg. cases (though *-tāti*-abstracts are found in more cases), whereas Avestan abstracts are found in all cases. Avestan abstracts also have a much wider range of meaning than their Vedic counterparts, as in e.g. *kəuuītāt-* 'class, group of *kavis*', *pourutāt-* 'multiplicity, quantity', and even *tāt*-abstracts built to adverbial forms like *auuaētāt-* 'the word *auuōi*' to *auuōi* 'woe' and *x^{v}aētāt-* 'belonging' to *x^{v}a-* 'one's own'. Brugmann, however, did not treat any of the forms in greater detail.

Repeating much of Brugmann's discussion (1906:451-2), Debrunner (1954:620-2) expanded on the distribution of *-tāt-* and *-tāti-* and observed that plain *-tāt-* is only found in Indic in the Rig Veda; the *i*-stem variant *-tāti-* replaced *-tāt-* everywhere else. Like all previous studies of *-tāt-* and *-tāti-*, Debrunner's does not offer much analysis of the diachronic formation of the suffix forms, other than reiterating the long-held assumption that *-tāt-* and *-tāti-* have a close association with abstracts in *-tā-*, with several abstracts in *-tāt-* and *-tāti-* found beside abstracts in *-tā-* in Sanskrit or in other languages, e.g. Ved. *devátāt-/devátāti-* 'Götterschar; Gottesdienst' beside *devátā-* 'Gottesdienst', OAv. *hunarətāt-* 'Inbegriff des Könnens' beside Ved. *sūnṛ́tā-* 'Freigebigkeit', and YAv. *aršt̰āt-*, the name of a goddess "Uprightness", beside OPer. *aršt̰ā-* 'Aufrichtigkeit', among others (cf. discussion of *tā*-abstracts at **I.D.**). Debrunner admitted (1954:620), though, that it is unclear how the connection between *-tāt-/-tāti-* and *-tā-* may have developed. Though Brugmann's discussion of *-tāt-* and *-tāti-* still

remains the most complete, many questions remain about *tāt-* and *tāti*-abstracts in Indo-Iranian, including the question of the *i*-stem *-tāti-* and their semantic development.

Based on their belief that a difference in semantic value between *-tāt-* and *-tāti-* is identifiable in the Rig Veda, Renou and Benveniste (1960) surveyed the data, claiming that abstracts in *-tāt-* showed collective semantics, while *-tāti-* behaved more like a true qualitative abstract (cf. below for an evaluation of the claims). Benveniste followed up Renou's treatment of the Vedic material with a short discussion of Middle Iranian evidence for an *i*-stem suffix *-tāti-* (on which cf. **III.A.** above).

It is clear enough from the data that there are no significant departures in semantics and base type derivation in the Indo-Iranian *tāt-* and *tāti*-abstracts, compared to *tāt*-abstracts in the other languages, and the previous scholarship has identified no other serious problems, aside from the issue of the *i*-stem suffix *-tāti-* beside plain *-tāt-*. The Vedic material nearly unanimously reflects abstract derivation to thematic stems, with accent on the thematic vowel of the base. Only *ástatāti-*, *vasútāti-*, and *śáṃtāti-* stray form the standard pattern, and as such will be treated below. Some of the abstracts in the Rig Veda seem relatively late, e.g. *gr̥bhītátāti-* and *jyeṣṭhátāti-*, but they still follow the usual patterns established by the older forms. Avestan attests a much wider range of base types and semantic uses of **-tāt-* than found in Vedic, but the oldest-looking Avestan material also offers little in the way of interesting forms in *-tāt-*, and the

remainder of the data, furthermore, has little to offer on the inherited accentuation of the *tāt*-abstracts. Additionally, the semantic uses of the abstract material are late and idiosyncratic, reflecting inner-Iranian developments that are not of direct relevance to the diachronic problems under investigation. Most of the Iranian material can thus be placed aside, except where helpful for elucidating the Vedic data. The most significant unresolved problem is that of the *i*-stem issue in Vedic.

III.C. *i*-Stem Variants

Though a few consonant-stem case forms are found in dat. sg. *devátāte* (3x) and loc. sg. *devátāti* (2x), the majority of case forms built to *-tāt-/-tāti-* in the Rig Veda reflect the *i*-stem endings in acc. sg. *-tātim*, dat. sg. *-tātaye* (with two pl. forms in *-ibhis*), and instr./loc. sg. *-tātā*, with the bulk of the forms attested to *devátāti-* 'collection of the gods' and *sarvátāti-* 'wholeness, health' (cf. Fig. 9).

Fig. 9. Vedic abstracts in *-tāti-* (RV forms unless marked; after Grassmann)

• *ástatātim* (1x)	• *ariṣṭátātaye* (2x RV; 9x AV[††])	• *devátātā* (13x)*
• *ayakṣmátātim* (1x AV)	• *ariṣṭátātibhis* (1x)	• *satyátātā* (1x)
• *jyeṣthátātim* (1x)[†]	• *gr̥bhītátātaye* (1x)	• *sarvátātā* (9x)
• *dákṣatātim* (1x AV)	• *jyeṣthátātaye* (1x AV)	• *śámtātā* (1x SV)
• *devátātim* (7x)	• *devátātaye* (11x RV; 1x AV)	
• *vasútātim* (1x)	• *sarvátātaye* (4x RV; 1x AV)	
• *sarvátātim* (3x)	• *śaṃtātibhis* (1x)	

[†] Also found at VS 7.12b, TS 1.4.9.1b, KS 4.3b, and ŚB 4.2.1.9b.
* Also found at VS 7.38.7b, VS 9.16b and 21.10b, TS 1.7.8.2b, MS 1.11.2b and 162.10, KS 13.14b, ŚB 5.1.5.22b, and N 12.44b.
[††] Found at AV-Ś 3.5.5b, 5.30.12d, 6.19.2c, 6.80.2d, 8.2.6d, 8.5.20b, 8.7.6d, 8.7.27d, and 19.38.3b.

The accusative and dative forms clearly belong to *i*-stem declensions in Vedic, but the ending *-tātā* is not as clear. Formally, it is ambiguous, in that it could continue a loc. sg. to an *i*-stem or an instr. sg. to a consonant stem. Secondary *i*-stems are rare but at least one made an instr. sg. in *-ī* (< PIE *$*-ih_1$*) in the Rig Veda, e.g. the hapax *supaptaní* 'by rapid flight' (cf. Wackernagel 1930:145-6 on secondary *i*-stems), but no instr. sg. in **-tātī* is ever found in a *tāt-* or *tāti*-abstract. Moreover, instr. sg. *^{x}-tātyā* continuing PIE *$*-i̯eh_1$* does not occur anywhere in Vedic; *-tātā* is the only instr. sg. found, so the instr. sg. *-tātā* can only belong to plain *-tāt-*. Consonant-stem loc. sg. in *-tāti* is found in the Rig Veda (cf. OAv. loc. sg. *amərətāitī*), but the usual loc. sg. in the abstracts is made with *-tātā*.[79] The loc. sg. ending in *-ā* can only belong to an *i*-stem declension, and thus loc. sg. *-tātā* can only belong to an *i*-stem suffix *-tāti-*. After Grassmann (1873 s.v.), *-tātā* has consequently been identified by several authorities as both instr. sg. and loc. sg. in the Rig Veda, depending on textual context (cf. below); but the question of what it reflects historically — and its implications for the *i*-stem variant *-tāti-* — has hitherto been ignored.

Based on the *i*-stem abstracts in Vedic like *ariṣṭátāti-* and *gr̥bhītátāti-*, beside the Latin genitives in *-tātium* (cf. **II.C.** for discussion of the *i*-stem forms in Latin),

[79] OAv. *amərətāitī* 'in immortality' at Y. 45.7 is loc. sg. in a parallel phrasing with loc. sg. *utaiiūtā* 'in perpetuity', as is commonly the case with the two words.

Debrunner (1954:621) posited a PIE *i*-stem variant **-tāti-* beside **-tāt-*, both ultimately based on **-tā-*; but he stopped short of comparing the two variants within Vedic. Despite the fact that no *i*-stem forms are found in Avestan and Greek and that the *i*-stem forms found in Latin are clearly secondary, it has been claimed that the suffix **-tāti-* somehow belongs to PIE, beside plain **-tāt-* (cf. **III.B.**), though two abstract suffixes **-tāt-* and an *i*-stem **-tāti-* in PIE, both with identical functions, seems rather surprising. In Latin, *i*-stem inflection spread to consonant stems including *-tās* at an early date (cf. **II.C.**), but while the spread of *i*-stem inflection is normal in Latin, it is not normal in Vedic. But perhaps it is worth considering whether Vedic *-tāti-* might also reflect a non-standard secondary development within Vedic, where *i*-stem inflection somehow spread to *-tāt-*.

III.C.1. Semantics of *-tātā* vs. *-tātim* and *-tātaye*

According to Grassmann (1873 s.v.), the consonant-stem and *i*-stem suffixes could sometimes be interpreted differently, as he argued for *devátāt-/devátāti-*, which sometimes means 'Gottesdienst' and other times 'Götterschar'. Grassmann made no semantic distinctions in the suffixes themselves, and his assignment to *-tāt-* or *-tāti-* was purely formal, in that abstracts in *-tātā* with instrumental semantics can only reflect a consonant-stem instr. sg. in *-ā*, while abstracts in *-tātā* with locative semantics can only reflect an *i*-stem loc. sg. in *-ā*. Geldner (1951) took Grassmann's interpretation one step

farther by setting up *devátāt-* 'Göttesdienst' vs. *devátāti-* 'Götterschaar' to reconcile the different semantics in the abstracts; but as will be shown below, a clean semantic division between suffixes cannot be maintained.

Given his proclivity for attaching specific semantics to suffixes, Renou argued that the two suffixes *-tāt-* and *-tāti-* are not semantically interchangeable (1960:11), and that the distinction between the two must point to two separate suffixes already in PIE. Focusing on the best-attested abstracts *sarvátāt-/sarvátāti-* and *devátāt-/devátāti-*, Renou (1960:11-3) thought that *-tātā* must only continue an instr. sg. of *-tāt-*, in that abstracts in *-tāt-* are characterized by a collective nuance, as in instr. sg. *sarvátātā* 'en nombre total, au grand complet', as opposed to abstracts in *-tāti-* that reflect a true qualitative abstract, e.g. acc. sg. *sarvátātim* 'complétion, Vollständigkeit'. *tā*-abstracts, which are also most often found in the instr. sg. in *-tā* with an adverbial sense, were interpreted as corroborating Renou's interpretation of *-tātā* as an instr. sg. (1960:11-13).

Given that *sarvátāt-/sarvátāti-* and *devátāt-/devátāti-* are the two best-attested abstracts in the Rig Veda, the hypothesis that the two suffixes *-tāt-* and *-tāti-* are semantically different will be investigated further by looking at how the two forms are used.

III.C.1.a. *sarvátātā* vs. *sarvátātim/sarvátātaye*

sarvátāt-/sarvátāti-, an abstract built to the thematic adjective *sárva-* 'all', which

reflects a *u̯o*-adjective to the PIE root **selh₂-* 'whole, all', is found 9 times in the instr./loc. sg. *sarvátātā* (cf. Fig. 5).[80] Renou believed that abstracts in *-tātā* reflect one of the trademark adverbial uses of the instrumental in marking a "modalité caractéristique" (1960:11-12), as does *sarvátātā* 'au grand complet' in *maruto víśva ūtí áchā sūrín sarvátātā* (7.57.7ab), where Renou read *sarvátātā* as reinforcing *víśve* 'vous tous'. But there seems to be no reason why a loc. sg. interpretation with qualitative semantics must be ruled out, as is clear from Brereton and Jamison's translation:[81]

> 7.57.7ab *á stutáso maruto víśva ūtí áchā sūrín* ***sarvátātā*** *jigāta*
> 7.57.7cd *yé nas tmánā śatíno vardháyanti yūyám pāta suastíbhiḥ sádā naḥ*
>
> When praised, o Maruts, come hither, all of you **in your totality**, with help to the patrons,
> who, possessing hundreds, strengthen us by themselves. — Protect us always with your blessings.

Renou's strict reading of *sarvátātā* as an instrumental adverb is not convincing, and the text admits locative semantics just as easily. In fact, the majority of the abstracts in *-tātā* are ambiguous in interpretation within the Rig Veda, resisting the rigid and formal classification Renou and others (e.g. Geldner) have attached to the forms.

Lubotsky (1997 s.v.), in his apparent rejection of a consonant-stem abstract *sarvátāt-*, cites all 4 examples of *sarvátātā* at pāda-end as loc. sg. to *sarvátāti-* (cf. below on the significance of *-tātā* at pāda-end):

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[80] For a detailed treatment of PIE **selh₂-* and some of the problems associated with its derivatives, cf. the discussion of Lat. *salūs* 'health' in **V.B.**

[81] All RV translations are taken from Brereton and Jamison's forthcoming RV translation. Translations here may be different from the revised published version.

1.94.15ab *yásmai tváṃ sudraviṇo dádāśo anāgāstvám adite* ***sarvátātā***
1.94.15cd *yám bhadréṇa śávasā codáyāsi prajāvatā rādhasā té siyāma*

For whom you will perform ritual service, o you possessing good property, for whom there is blamelessness **in his entirety**, o Aditi [= Innocence],
and whom you will rouse with your beneficial power and with generosity that provides offspring — might we be those

Brereton and Jamison take *sarvátātā* at 1.94.15b as a loc. sg. "in his entirety" with Renou (1960) and against Geldner's (1951 *ad loc.*) instr. sg. "with wholeness". The other examples also support the loc. sg. interpretation:

3.54.19ab *devānāṃ dūtáḥ purudhá prásūto ánāgān no vocatu* ***sarvátātā***
3.54.19cd *sr̥ṇótu naḥ pr̥thiví dyaúr utāpaḥ sūryo nákṣatrair urú antárikṣam*

The messenger of the gods, impelled forth in many forms — let him pronounce us blameless **in our entirety**.
Let Earth, Heaven, and the waters hear us, and the Sun with the heavenly bodies, and the broad midspace.

4.26.3ab *ahám púro mandasānó ví airaṃ náva sākáṃ navatíḥ śámbarasya*
4.26.3cd *śatatamáṃ veśíyaṃ* ***sarvátātā*** *dívodāsam atithigváṃ yád āvam*

I, acting with exhilaration, destroyed at the same time the ninety and nine fortifications of Śambara,
and the hundredth, his dwelling place, **in their entirety** when I helped Divodāsa Atithigva.

5.69.3ab *prātár devím áditiṃ johavīmi madhyáṃdina úditā súriyasya*
5.69.3cd *rāyé mitrāvaruṇā* ***sarvátātā*** *íḷe tokáya tánayāya śáṃ yóḥ*

In the early morning, again and again I call upon the goddess Aditi, also at midday and at the rising the sun.
For wealth **in its entirety**, Mitra and Varuna, I invoke (you), and for kith and kin, luck and life.

Beside the adverbial instr. sg. *sarvátātā* are the *i*-stem acc. sg. *sarvátātim* and dat. sg. *sarvátātaye* that do reflect actual abstract nouns with a sense of "completeness",

according to Renou: one example at 3.54.11d (cf. below) occurs within a few lines from *sarvátātā* (3.54.19b; cf. below for the significance of proximity), where "[i]l est clair qu'ici le sens du mot est bien différent de celui que est à la base de l'Instr. en *-tātā*; la nuance collective-numérique a disparu" (Renou 1960:12), and in this case, *sarvátātim* indeed reflects qualitative abstract semantics:

> 3.54.11ab *híraṇyapāṇiḥ savitā́ sujihvás trír á divó vidáthe pátyamānaḥ*
> 3.54.11cd *devéṣu ca savitaḥ ślókam áśrer ā́d asmábhyam ā́ suva* ***sarvátātim***
>
> Savitar of the golden hands and the lovely tongue, being master at the rite three times a day —
> and (when), Savitar, you have set your signal-call among the gods, then impel **wholeness** to us.

A qualitative abstract in *-tāti-* with *i*-stem inflection is hardly a surprise, beside the abstracts in *-ti-* with qualitative semantics, and its existence within Vedic need not imply that such a suffix also existed in PIE.

III.C.1.b. *devátātā* vs. *devátātim/devátātaye*

In the case of *devátāt-* and *devátāti-*, Renou (1960:14) observed that such a neat semantic division between a qualitative abstract *-tāti-* and a collective abstract *-tāt-* cannot be maintained. He rejected Geldner's *devátāti-* as 'Götterschar', admitting, though, that a loc. sg. interpretation *devátātā* with collective sense might be more semantically appropriate in some examples. According to him, the *i*-stem forms signify a kind of "divine service" implying "sacrifice", as at 3.26.2c *agním...havāmahe...mánuṣo*

devátātaye "nous appelons Agni pour le service divin de l'Homme". *devátāti-* could also mean simply "divinity", according to Renou (1960:13): *ṛ́dhag itthā́ sá mártyaḥ śaśamé devátātaye / yáḥ...* (8.101.1b) "il a pris peine ainsi (qu'on le voit), à bon escient, pour la divinité, le mortel qui (a appelé à lui Varuṇa et Mitra, etc.)". Despite Renou's readings in the two examples, Brereton and Jamison (*ad loc.*) take both instances of *devátātaye* as collectives meaning "divine assemblage, conclave of the gods".

Renou's readings again face difficulties in several examples, where his instr. sg. collective interpretation of *devátātā* does not work in the text:

> 1.34.5ab *trír no rayíṃ vahatam aśvinā yuvám trír **devátātā** trír utāvataṃ dhíyaḥ*
> 1.34.5cd *tríḥ saubhagatvám trír utá śrávāṃsi nas triṣṭhám vāṃ sūre duhitā ruhad rátham*
>
> Three times bring wealth to us, Aśvins, three times **to the divine conclave**; and three times aid our insights.
> Three times (bring) good fortune and three times acclamations for us. The Daughter of the Sun mounted your chariot with its three standing places.

> 3.19.1ab *agníṃ hótāram prá vṛṇe miyédhe gṛ́tsaṃ kavíṃ viśvavídam ámūram*
> 3.19.1cd *sá no yakṣad **devátātā** yájīyān rāyé vājāya vanate maghāni*
>
> I choose Agni as the Hotar at the ritual meal — him, the clever poet, knowing all, never fooled.
> As the better sacrificer, he will sacrifice for us **at the divine assembly**. For wealth and victory's prize, he will win rewards.

In the examples above, *devátātā* works best as a loc. sg., again with collective semantics. Likewise, *devátātā* at 5.29.1a is given under *devátāt-* by Lubotsky (1997 s.v.), suggesting

he interprets it as an instr. sg.; but Brereton and Jamison take it as a loc. sg.:[82]

> 5.29.1ab *trí aryamā́ mánuṣo **devátātā** trí rocanā́ divyā́ dhārayanta*
> 5.29.1cd *árcanti tvā marútaḥ pūtádakṣas tvám eṣām ṛ́ṣir indrāsi dhíraḥ*
>
> They uphold the three customs [= daily rituals] **at the divine assemblage** of Manu and the three heavenly realms of light.
> The Maruts of refined skill chant to you. You are their wise seer, Indra.

At 6.4.1a and 7.2.5b, pāda-end *devátātā* is traditionally taken as loc. sg. of *devátāti-*, as do Brereton and Jamison:

> 6.4.1ab *yáthā hotar mánuṣo **devátātā** yajñébhiḥ sūno sahaso yájāsi*
> 6.4.1cd *evā́ no adyá samanā́ samānā́n uśánn agna uśató yakṣi devā́n*
>
> Just as you, o Hotar priest, will (always) sacrifice with sacrifices **at** Manu's **attendance on the gods**, o son of strength,
> even so for us today sacrifice in the same way to the same gods — you willing, to them willing, o Agni.

> 7.2.5ab *svādhíyo ví dúro devayánto áśiśrayū rathayúr **devátātā***
> 7.2.5cd *pūrví śíśuṃ ná mātárā rihāṇé sám agrúvo ná sámaneṣu añjan*
>
> Serving the gods, they of good insight have laid the doors that seek the (divine) chariot open **to the gods' realm**.
> They join the many (doors), (each pair) like two mothers licking their calf, by unction, like marriage-ready girls in (marriage) assemblies.

Locative semantics make perfectly clear sense for *devátātā*, and, moreover, the collective force in *devátātā* is clearly evident.

Likewise, Brereton and Jamison take dat. sg. *devátātaye* at 3.26.2c and 8.101.1.b as "for the divine assemblage" with collective semantics that should only belong to the consonant-stem instr. sg. *devátātā*, according to Renou. He (1960:14) read similar

[82] Lanman (1880:386) did not identify *devátātā* at 5.29.1a in his list of *-ā* forms at pāda-end.

collective semantics with *devátātā* (again at pāda-end) in *tā́ hí śréṣṭhā devátātā* (6.68.2a) "vous êtes en effet les meilleurs d'entre les dieux", which Brereton and Jamison support with "For those two are the most excellent in the realm of the gods"; but an *i*-stem loc. sg. should show qualitative semantics, by Renou's theory. Again for 7.43.3d, *ágne mā́ no devátātā mŕ̥dhas kaḥ*, Renou appealed to a collective *devátātā* with locative semantics, reading it as "o Agni, ne nous mets pas en état de carence vis-à-vis des dieux!", though Brereton and Jamison's "o Agni, do not make us negligent in our attendance on the gods" appeals to the abstract sense of "divine service", which makes better sense in the context of the hymn. Hence, the actual uses of *devátātā* seem difficult to categorize neatly.

To account for the observed difficulties in his theory, Renou concluded that the two abstracts *devátāt-* and *devátāti-* must have undergone some sort of semantic "leveling" that failed to happen in *sarvátāt-* and *sarvátāti-* (1960:14). But he does not attempt to expand on how or why the leveling happened. There indeed exist some semantic differences in the use of *devátātā* in the Rig Veda, but the differences cannot be assigned to the use of *-tātā* vs. *-tātim/-tātaye*, as Renou argued. Abstracts in *-tātā* show both collective and qualitative semantics, and the unambiguous *i*-stem forms in *-tātim* and *-tātaye* also show both collective and qualitative semantics. Consequently, the overlap in semantics between the two suffixes *-tāt-* and *-tāti-* in Vedic cannot be suggestive that such a semantic difference between the suffixes was observed in PIE.

Moreover, it is still far from certain that PIE even had both suffixes; rather, the semantic differences Renou gave to the two suffixes likely reflect the different semantics possessed by a single suffix in PIE. In the other languages attesting *-*tāt*- both collective and qualitative abstract semantics are found, so the fact that Vedic (and Avestan; cf. below) also attest both collective and qualitative abstract semantics should not come as a surprise.

Generally, Brereton and Jamison (p.c.) are of the opinion that both *devátāt*- and *devátāti*- have collective semantics in the Rig Veda; there is no real functional difference between the two suffixes. In ambiguous passages, especially passages involving Manu, the secondary sense of "attendance on, service to the gods" can develop (p.c. Stephanie Jamison). The difference in meaning between *devátāt-/devátāti-* and *sarvátāt-/sarvátāti-* lies in the fact that their respective base types differ: *devátāt-/devátāti-* is a count noun, built to a noun, while *sarvátāt-/sarvátāti-* is an abstract with collective sense based on an adjective, and as such does not function as a collective. In fact, the *i*-stem forms in the acc. sg. *sarvátātim* and dat. sg. *sarvátātaye* only have qualitative abstract semantics in the Rig Veda.

III.C.2. *devátā* 'among the gods; godhead'

The adverb *devátā* 'among the gods; godhead', with identical meaning to *devátātā*, is found 12 times in the Rig Veda and has been thought by Wackernagel and

Debrunner (1930:116) to continue a haplologized *devátātā*.[83] Cardona (1968:47) followed Wackernagel and Debrunner in asking why only the instrumental case is subject to haplology, given the failure of other case forms to undergo a similar change.[84] He reasoned that the haplologized *devátā* "interacted" with other stems in *-tā-*, which helped reinforce its interpretation as a *tā*-stem rather than a *tāt*-abstract. In other words, *tā*-stems frequently in Vedic made instr. sg. in *-tā* (cf. Lanman 1880:358 for a list of instr. sg. in *-tā* to *tā*-stems), and so *devátā* haplologized from *devátātā* could have been reinterpreted as a *tā*-stem instr. sg. He adds that "[i]t is therefore inaccurate to say that....*devátā* resulted *uniquely* from haplology", seemingly thinking that *tā*-stem instrumentals and hapologized instrumentals to *-tāt-* conspired to form *devátā*. Cardona's explanation may be correct, but it is interesting that no other *tāt-/tāti*-abstract in the Rig Veda has a haplologized counterpart, e.g. there is no **sarvátā*; haplology in the Vedic *tāt-/tāti-* abstracts is only found with *devátāt-/devátāti-*. At any rate, Cardona's instr. sg. interpretation of *devátā* has important implications for the interpretation of *devátātā*.

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[83] 4 of the 12 examples of *devátā* are found at pāda-end, which may be suggestive of its metrical usefulness in the triṣṭubh. Cf. below for discussion.

[84] Cardona (1968:74n54), while disagreeing with Renou's (1960) semantic interpretation of *devátātā* as 'sacrifice, service to the gods', raises the possibility that *devátātā* is indeed loc. sg. to an *i*-stem suffix *-tāti-*, which then contributes to the locative function of haplologized *devátā*. He does not explore the possibility, but cf. below for a different historical interpretation of *devátātā*.

III.C.3. Interpretation of instr./loc. sg. *-tātā*

Abstracts in *-tātā* have been critical for the synchronic interpretation of the *tāt-/tāti*-suffix in Vedic, but no serious diachronic study of the ending itself has been made. Renou (1960:11) did not investigate the historical background of *-tātā*, since he felt it had little bearing on the synchronic semantic differences between *-tāt-* and *-tāti-*. Thus, the question of what *-tātā* is from a diachronic perspective remains open, namely, whether or not it must continue both an old consonant-stem instr. sg. and an *i*-stem loc. sg.

devátātā (13x) and *sarvátātā* (9x) comprise most of the *tātā*-forms (cf. Fig. 10).

Fig. 10. Rig Vedic loc. sg. vs. instr. sg. abstracts in *-tātā* (after Grassmann)

devátātā (13x) 'collection of the gods'	**10x:** 1.34.5b, 1.58.1d, 3.19.1c* = 10.53.1c*, 4.6.1b*, 6.4.1a†, 6.68.2a*, 7.2.5b†, 7.38.7b, 7.43.3d*	**3x:** 1.95.8d, 1.128.2c, 5.29.1a†
sarvátātā (9x) 'entirety, health'	**6x:** 1.94.15b†, 3.54.19b†, 5.69.3c†, 6.12.2b*, 6.15.18b, 10.74.3b	**3x:** 4.26.3c†, 7.18.19b*, 7.57.7b*
satyátātā (1x) 'truth'	[none]	**1x:** 10.111.4d†

† *-tātā* occurs at pāda-end

* *-tātā* occurs in the break

Lubotsky (1997 s.v. *sarvátāti-*) does not cite plain *sarvátāt-*, filing all 9 instances of *sarvátātā* with the *i*-stem *sarvátāti-* and thus evidently taking *sarvátātā* as loc. sg.; but as discussed above, abstracts in *-tātā* can be interpreted in the Rig Veda both as instr. sg.

and as loc. sg.[85] Given that there is no semantic difference between abstracts in *-tāt-* beside abstracts in *-tāti-*, there seems to be no good reason why *-tātā* must be interpreted as an instr. sg. to *-tātā* on the one hand, but a loc. sg. to *-tāti-* on the other. Instead, it may be possible to take *-tātā* back to a single PIE case form. The PIE instr. sg. **-tāteh$_1$* to **-tāt-* would directly give Ved. *-tātā* and fit the observable semantics of the abstracts in the text.[86] *-tātā* can reflect an *i*-stem loc. sg. in *-ā*, continuing the PIE *i*-stem "endingless" loc. sg. **-ēi̯* (cf. the recharacterized loc. sg. πόληϊ 'in the city' < **-ēi̯-i* at *Il.* 3.50), and though the *i*-stem loc. sg. in *-ā* was mostly replaced in Indic by *-au*, a good number of *i*-stem locatives in *-ā* are still found in the Rig Veda.[87] There are no good reasons, however, to project old *i*-stem inflection back to PIE. If *-tātā* does not in fact continue PIE loc. sg. **-tātēi̯*, it remains to be answered how *i*-stem inflection developed within the *tāt*-abstracts.

Of particular interest is the fact that the *i*-stem loc. sg. ending in *-au* is never

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[85] Grassmann was convinced that *sarvátātā* at 1.94.15b, 3.54.19b, 5.69.3c, and 6.15.18b was more appropriate as dat. sg. *sarvátāte* (cf. dat. sg. *devátāte*) and proposed the text be changed. Emendation in this case is neither justifiable nor necessary, since the texts make perfectly good sense as transmitted (cf. above for Brereton and Jamison's translation of some of the passages).

[86] For the sake of convenience, the hybrid citation form **-tāteh$_1$* is used here, until the source of **-tāt-* can be explored in greater detail in **V.D.** (on the general use of **-tāt-*, cf. n. 1 in **I.A.**).

[87] *-au* is thought to have replaced *-ā* by analogy to the *u*-stem loc. sg. in *-au* (cf. Dawson 2005:39 with refs).

found with *-tāti-*, even in positions where *-au* is much more common than *-ā*.[88] In his statistical account of the *i*-stem loc. sg. in *-ā* vs. *-au*, Lanman (1880:386) noted *-au* is nearly always used at pāda-end in the Rig Veda.[89] There are only 8 occurrences where *-ā* is found, 6 of which occur with *devátātā* (3x) and *sarvátātā* (3x), and the *i*-stem loc. sg. in *-ā* in general makes up only 29% of all loc. sg. forms in any position in the Rig Veda (Dawson 2005:65-6).[90] The unusual and repeated placement of *-tātā* at pāda-end in the Rig Veda suggests it may have had a special metrical status within the triṣṭubh, possibly continuing something old. The morphological characteristics of the forms in *-tātā* (vis-à-vis the loc. sg. in *-au*), with adverbial value in the Rig Veda passages in which it is found, may support the hypothesis that *-tātā* continues an old PIE instrumental **-eh$_1$* built to plain *tāt-*, as is also likely to be the case with the adverb *devátā*, as discussed above.[91] If so, it provides a contributing argument that the *i*-stem suffix *-tāti-* is indeed a secondary creation in Vedic and does not belong to PIE.

It may well be the case, in fact, that the abstracts in *-tātā* occurring in formulaic phrases in the Rig Veda became morphologically unidentifiable at some relatively early

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[88] Loc. sg. *-au* is twice as common as *-ā* in animate *i*-stems (Lanman 1880:385, Macdonell 1910:283). Consonant-stem loc. sg. *uparátāti* (1.151.5c, 7.48.3b), *vr̥kátāti* (2.34.9a), and *śáṃtāti* (8.18.7c) are found in the Rig Veda.

[89] Lanman (1880:386) offered 5 examples of *-ā* at pāda-end, but Lubotsky (1997 s.v. *devátāt-*, *sarvátāti-*) lists 2 additional ones at 5.29.1a (*devátātā*) and 4.26.3c (*sarvátātā*).

[90] The other two listed by Dawson (2005:66) are *yónā* (1.65.4c) and *śū́rasātā* (6.26.1c). Elsewhere, Dawson cites 9 cases of pāda-final loc. sg. *-ā*, but a ninth form is never given.

[91] Cf. García-Ramón 1997 for a discussion of PIE instrumentals used adverbially.

point, with the oldest inflection preserved in the "adverbial" instrumental ending *-tātā*. Abstracts in *-tātā* that became morphologically isolated were then subject to new interpretation by poets, and because *-tātā* was ambiguous in meaning, it could be interpreted sometimes as instr. sg. and other times as loc. sg. The collective semantics often found in *devátātā* aided in the loc. sg. interpretation, considering that locative semantics naturally work very well for collective nouns, e.g. "in a divine assemblage" vs. "with a divine assemblage". Once *-tātā* became synchronically analyzable as loc. sg. of an *i*-stem, new *i*-stem case forms could have been backformed, e.g. acc. sg. *-tātim* "divine assemblage (direct object)" and dat. sg. *-tātaye* "for the divine assemblage", when *-tāt-/-tāti-* reached its height in productivity, with the loc. sg. in *-ā* to *ti*-stems as an analogical model (cf. Macdonell 1910:284 on the *ti*-stem loc. sg. in *-ā*), as Renou suspected (1960:15-6).

Lanman (1880:359, 466), for his part, argued for an opposite direction of derivation, namely, that *-tāti-* adopted consonant-stem endings to produce abstracts in plain *-tāt-*. He likewise took the loc. sg. as the starting point, with "the forms *devátātā*, *satyátātā*, *sarvátātā* being easily referable to *t*-stems (as adverbial instrumentals), or to *i*-stems (as locatives)", though he did not expand on how such a transfer of inflection from the *i*-stem *-tāti-* to *-tāt-* may have happened via the loc. sg. He did offer an example of consonant-stem inflection being assumed by another stem type in the *tā*-stem *avíratā-*

'without sons', with the consonant-stem dat. sg. *avírate* (found only once at 7.1.19a; cf. also the normal dat. sg. hapax *avíratāyai* at 3.16.5b). It is difficult to see how the reanalysis of an *i*-stem loc. sg. *-tātā* as an instr. sg. to plain *-tāt-* could have happened in a similar way as *avírate*, as Lanman thought. Hence, his example cannot be used to support the spread of consonant-stem inflection to *i*-stems, especially to the number of forms in *-tāt-* found in the Rig Veda.

Clearly, *i*-stems never adopt consonant-stem inflection in Vedic. It is thus very difficult to see how *i*-stem inflection could be replaced by consonant-stem inflection in the way Lanman argued. But on the other hand, deverbal *ti*-stems became productive in Vedic, and despite the formal difference in base derivation between *tāt*-abstracts and *ti*-stem abstracts, *-ti-* also assumed an abstracting function in Vedic (beside their primary function as action nouns), e.g. *dídhiti-* 'devotion' and *sumatí-* 'benevolence'. *ti*-stems are thus a likely model, at least in part, on which a secondary *i*-stem suffix *-tāti-* could be extracted from an ambiguous form in *-tātā*, which could have been reanalyzed as a *ti*-stem abstract. That is not to say *all* forms in *-tātā* in the Rig Veda continue an old instr. sg., however. Rather, a certain number of old instrumentals must have existed, on which new *i*-stem forms like the acc. sg. in *-tātim*, dat. sg. in *-tātaye*, and even loc. sg. in *-tātā* were used by poets.

III.C.4. Grouping of Abstracts

A handful of abstracts are found clustered together, which may reveal something about their formation, especially the abstracts in *-tātā* that are found beside other case forms. Loc. sg. *devátātā* 'at the divine assembly' at 3.19.1c is beside acc. sg. *devátātim* also in the sense 'divine assembly' at 2c and 4c in the same hymn, and likewise loc. sg. *devátātā* 'at the divine assemblage' at 4.6.1b is next to the synonymous acc. sg. *devátātim* 'divine assemblage' in pādas 3b and 9d of the same hymn (in which the pāda *pradakṣiṇíd devátātim urāṇáḥ* at 3.19.2c is repeated again at 4.6.3b). And loc. sg. *devátātā* 'in the divine assembly' at 7.38.7b occurs immediately before acc. sg. *devátātim* 'towards the divine assemblage' at 7.39.1b, with another example of *devátātā* also at 7.43.3d, though here meaning 'in our attendance on the gods' (Brereton and Jamison *ad loc.*). The loc. sg. *sarvátātā* 'in our entirety' at 3.54.19b is found beside acc. sg. *sarvátātim* 'wholeness' at 11d in the same hymn.

3.19 to Agni, attributed to Gāthin, and 4.6 are very similar in thematic content and vocabulary (Brereton and Jamison *ad loc.*). *devátātā* in 3.19.1c *sá no yakṣad devátātā yájīyān* "He who sacrifices for us at the divine assembly" (with notable "iambic" break; cf. discussion below) is unambiguously loc. sg. (the pāda is repeated again at 10.53.1c). The acc. sg. forms *devátātim* later at 2c and 4c (also with "iambic" breaks) are used semantically identically to *devátātā*. Groupings of this sort around a clearly locative

abstract in *-tātā* may give support to the theory above, by which an original instr. sg. "adverbial" abstract in *-tātā* to plain *-tāt-* could be reinterpreted as an *i*-stem loc. sg., giving rise to new *i*-stem forms in *-tātim* and *-tātaye*. That is, *devátātā* in 1c with clear locative semantics was analyzed as a loc. sg. to a stem *devátāti-* within the poetic tradition, which then created acc. sg. *devátātim* with the same meaning. Hymn 4.6, also to Agni, may have simply copied the use of both case forms. A similar grouping of loc. sg. *devátātā* at 7.38.7b and acc. sg. *devátātim* 7.39.1b is found in two hymns composed to Savitar and to "All Gods", respectively, by the poet Vasiṣṭha who may have been using the *i*-stem suffix *-tāti-* in the same way.

In an analogous way, loc. sg. *sarvátātā* at 3.54.19b is found beside acc. sg. *sarvátātim* at 11d in the same hymn, a hymn to "All Gods". In the hymn, *sarvátātā* 'in our entirety' reflects collective semantics, though *sarvátātim* shows the qualitative abstract meaning of 'wholeness', as it has in all 3 of its occurrences in the acc. sg. in the Rig Veda (cf. Fig. 5 in **III.A.**).

III.C.5. Metrical behavior of *-tātā*

Metrically, the majority of the 13 occurrences of abstracts in *devátātā* tend to be placed in two positions (cf. Fig. 10): at the "cretic" break ⏑–⏑⏑ ‖ *devátātā* (5 times) and

at pāda-end (3 times).[92] Likewise, the majority of the 9 instances of *sarvátātā* occur at the break in ⏑⏑–– ‖ *sarvátātā* (3 times), and at pāda-end (4 times).[93] The examples found at pāda-end are understandable metrical positions for words of the shape –⏑–– for the cadence of the triṣṭubh, but the positions in the break are more interesting. The unusual cretic break ‖–⏑– (with secondary caesura after the 8th syllable characteristic of Vasiṣṭha hymns; cf. Arnold 1905:180), and iambic break ⏑‖–⏑ (with late caesura characteristic also of Vasiṣṭha hymns) "present the most striking features of the metre of the Rigveda" (Arnold 1905:184). Arnold commented that the "occasional" iambic break ⏑‖–⏑ is characteristic of the earlier Vedic period (though also used to a high degree later), whereas the "subnormal" cretic break ‖–⏑– belongs to the later period.

III.C.6. Metrical Behavior of Consonant-Stem Abstracts

The plain *t*-stem forms in the Rig Veda beside the ambiguous forms in *-tātā* (cf. Fig. 5) are dat. sg. *devátāte* in two adjacent hymns at 9.96.3a, 9.97.19a, and 9.97.27a, all meaning 'for the conclave of the gods'; loc. sg. *devátāti* in 8.74.3b and 10.8.2c in the phrases *devátāty údyatā* and *devátāty údyatāni*, respectively, both taken as 'im Götteropfer' and 'bei dem Götteropfer', respectively, by Geldner (1951 *ad loc.*), but 'at the conclave of the gods' by Brereton and Jamison; the lone instance of the loc. sg.

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[92] The other 5 examples of *devátātā* are found pāda-initially (2 times), with the comparatively rare "syncopated" opening characteristic of archaic lyric hymns (Arnold 1905:182), or beginning in the second syllable, i.e. – *devátātā* ‖ (3 times), with late caesura.

[93] The other 2 examples of *sarvátātā* also occur pāda-initially.

vṛ̥kátāti 'among wolves' at 2.34.9a (cf. below); and loc. sg. *uparátāti*, found only at 1.151.5c and 7.48.3b. Interestingly, the dat. sg. consonant-stem forms in the examples here are also positioned differently in comparison to the corresponding *i*-stem forms. All 11 examples of *devátātaye* are found pāda-finally in jagatī hymns, but *devátāte* is never found at line-end in the triṣṭubh, i.e. *sá no deva devátāte pavasva* (9.96.3a), *júṣṭo mádāya devátāta indo* (9.97.19a), and *evā́ deva devátāte pavasva* (9.97.27a). Moreover, the cretic break in 9.96.3a and 9.97.27a and "iambic" break in 9.97.19a and 10.8.2c, e.g. ––⏑–⏑ ‖ *devátātā*, are unusual in triṣṭubh lines.

In 8.74.3b the usual loc. sg. *-tātā* would be somewhat irregular in the gāyatrī line *yó devátāty údyatā*, with a heavy fifth syllable (Arnold 1905:159, 168); and the loc. sg. *vṛ̥kátāti* in the pāda *yó no maruto vṛ̥kátāti mártiyo* (2.34.9a) suits the jagatī cadence normally, whereas *vṛ̥kátātā* would make a very rare cadence with a heavy ninth syllable (cf. Arnold 1905:188 for a count of various metrical shapes in the Rig Veda). Similarly, *uparátāti* fills the break with ‖ ⏑⏑⏑, which is already "subnormal", and a loc. sg. *-tātā* in a heavy ninth syllable would be exceedingly rare (Arnold 1905:188). Hence, the *tāt-* forms seem to be metrically preferable to the *i*-stem forms in the few examples in which they are found.

In summary, the data for the suffix *-tāti-* in Vedic suggests that the *i*-stem variant may have been created secondarily from the difficult to interpret forms in *-tātā*. Once

created, *-tāti-* became semi-productive in Vedic texts, while plain *-tāt-* died out after the Rig Veda. In Avestan, *i*-stem inflection understandably never developed in *-tāt-*, given the less-restricted use of the abstract case forms in the texts. Avestan abstracts inflect in a greater number of cases than their Vedic counterparts, including the nom. sg., gen. sg. and abl. sg., as well as in all three numbers. Moreover, the instr. sg. and loc. sg. were declined with different endings, i.e. instr. sg. *amərətātā* and *xᵛaētātā* beside loc. sg. *amərətāitī* and *uštatāitii-a*, so the consonant-stem inflection of *-tāt-* was never confused with *i*-stem inflection elsewhere. Hence, the fact that Avestan, Latin, and Greek reflect no old evidence for an *i*-stem suffix in **-tāti-* verifies that PIE also had no *i*-stem variant **-tāti-* beside **-tāt-*.

III.C.7. Categories of attestation

The Vedic *tāt-* and *-tāti*-abstracts fall into 4 general categories (cf. Fig. 5 in **III.A.**): abstracts in *-tāt-* only (*uparátāt-*, *vr̥kátāt-*), abstracts in *-tāti-* only (*ástatāti-*, *ariṣṭátāti-*, *gr̥bītátāti-*, *jyeṣṭhátāti-*, *vasútāti-*, and *sáṃtāti-*), abstracts with both *-tāt-* and *-tāti-* forms (*devátāt-/devátāti*), and finally, abstracts only with ambiguous *-tāt-/-tāti-* forms in *-tātā* (*satyátāt-/satyátāti-* and *sarvátāt-/sarvátāti-*).

Though there is a significant amount of overlap in chronology in each of the 4 categories, the data is suggestive of a relative chronology of formation within the Rig Veda. Based on the arguments above that forms in *-tātā* continue an old adverbial instr.

sg., it is perhaps reasonable to consider that they represent one of the older layers of *tāt*-abstracts in the Rig Veda. Likewise, it may well be the case that the loc. sg. forms to plain *-tāt-* in *uparátāti* and *vr̥kátāti* also continue older material, given their somewhat atypical metrical behavior. If the *i*-stem abstracts in *-tāti-* were indeed backformed from ambiguous forms in *-tātā*, they must postdate forms in plain *-tāt-*. Consequently, the last two categories above, abstracts with both *-tāt-* and *-tāti-* forms and abstracts only with ambiguous *-tāt-/-tāti-* forms in *-tātā* must continue a mix of older material and recently-created material. Finally, the category of abstracts in *-tāti-* only likely represents the most recent layer of abstracts, as is also suggested by their secondary-looking formations, e.g. an abstract with anomalous accentuation in *ástatāti-* (cf. discussion above), an abstract built to a past participle in *gr̥bītátāti-*, an abstract built to a superlative in *jyeṣṭhátāti-*, the abstract *vasútāti-* created on the model of *devátāt-/devátāti-* (cf. discussion below), and the nonce abstract *sáṃtāti-* (cf. discussion below).

III.D. Individual Word Problems

III.D.1. Ved. *vasútāti-* 'collection of Vasus'

Brugmann (1906:451) took *vasútāti-*, meaning something like 'wealth', as a *tāti-* abstract built to the *u*-stem adj. *vásu-* 'good', with the parallel *u*-stem abstracts found in Hom. *βραδυτής* 'slowness' and *ταχυτής* 'quickness' (cf. **IV.A.**). But given the collective semantics of *vasútāti-* 'collection of the Vasus' (but cf. below on YAv.

vaŋhutāt- 'blood'), it is better thought of as a formation to the neut. noun *vásu-* 'Vasu (a god)':

> 1.122.5ab *ā́ vo ruvaṇyúm auśijó huvádhyai ghóṣeva śaṃsam árjunasya náṃśe*
> 1.122.5cd *prá vaḥ pūṣṇé dāvána ā́m̐ áchā voceya* ***vasútātim*** *agnéḥ*
>
> It is for (Kakṣīvant), son of Uśij [the fire-tending priest], to call the "screecher," the laud, for you as if with a shout, at the attainment of the silvery one [= soma?]. (Put him [= Pūṣan]?) forward for yourselves, for Pūṣan to give. I would call here Agni's **assemblage of the Vasus**.
>
> 1.122.12ab *etáṃ śárdhaṃ dhāma yásya sūrér íty avocan dáśatayasya náṃśe*
> 1.122.12cd *dyumnā́ni yéṣu* ***vasútātī*** *rārán víśve sanvantu prabhr̥théṣu vā́jam*
>
> "Whichever patron (we belong to), we shall establish him as a force," so did they say on the attainment of ten-part (gushing [= soma?]).
> The brilliant things **in which the assemblage of the Vasus** finds enjoyment — let them all [= gods] gain (them) as their victory prize.

The noun *vásu-* continues a PIE AS neut. *u*-stem noun *$*h_1u̯és$-u-/*$h_1u̯ós$-u-* (OAv./YAv. *vohu-* 'good', OIr. *fíu* 'worthy', etc.; cf. *NIL* s.v.), from which the Vedic adjective *vásu-* 'good' was derived (Nussbaum 1998a:130, 147 with refs.). *vasútāti-* is found twice in the same hymn at 1.122.5d (acc. sg. *vasútātim*) and 1.122.12c (nom. sg. *vasútātī*), which was composed by Kakṣīvant, a clever poet who likely created *vasútāti-* on the model of *devátāt-/devátāti-* 'collection of Vasus' (p.c. Stephanie Jamison), of which dat. sg. *devátātaye* occurs a few hymns later at 1.127.9b and 1.127.9c, as does a loc./instr. sg. *devátātā* at 1.128.2c (cf. above on the "clustering" of abstracts). If *vasútāti-* is indeed a relatively recent poetic neologism, it would show that Vedic, unlike the other languages

attesting forms in *-tāt-*, built no old-looking abstracts to non-thematic stems (except for the strange form *śáṃtāti-*, cf. below), including *u*-stems.

Arnold (1905:272) dated hymn 122 to the archaic period, based on its metrical variations that are characteristic of the archaic period, and by his count it contains 99 features that belong to the Rig Veda proper (as opposed to only 5 characteristic of the Atharva Veda). Arnold's statistical method of dating the hymn to the older layer of the Rig Veda is not certain proof that it indeed belongs in the archaic period, but it provides good evidence for an overall relative chronology of the hymns. If hymn 122, with the secondary abstract *vasútāti-* created on the model of *devátāti-*, is indeed archaic, it would show that the productive spread of the *i*-stem suffix *-tāti-* happened at a rather early date in the Rig Veda.

III.D.2. YAv. *vaŋhutāt-* 'blood'

YAv. *vaŋhutāt-* 'blood', found twice at Vendidad 6.7 and 29, seems at first sight to be related to Ved. *vasútāti-*, beside OAv./YAv. *vohu-* 'good', yet it has very different semantics and may belong to a different root altogether. Mayrhofer (*EWAia* s.v. *vásā-*) connects *vaŋhutāt-* with Ved. *vásā-* 'fat' and YAv. *vohunī-* 'blood', which appears as *vohuna-* in compounds, e.g. YAv. *vohunaŋhag-* 'bloodhound'. Beside *vásā-* and *vohunī-* are Khot. *hūnä*, MP *xōn*, and NP *xūn*, all meaning 'blood', and OE *wōs* 'moisture, juice' may provide a comparandum outside of Indo-Iranian. Perhaps a PIE root noun **u̯es-*

'slime' (*vel sim.*) underlies Ved. *vásā-*, which shows a feminine derivative in **-eh$_2$-*, and YAv. *vohunī-*, with a more complicated history: descriptively, Iranian appears to have made a *u*-stem to **u̯es-*, which was then extended with an *n*-stem suffix. An *n*-stem base is found in OAv. *astəntāt-* 'liveliness', but *vaŋhutāt-* looks as if it was built to the *u*-stem directly. Still, it seems unlikely that *vaŋhutāt-* continues an old abstract in **-u-tāt-*; more likely, it reflects a remodeling of **vasun-tāt-*, given that there are no synchronic *un*-stems in Avestan.

III.D.3. Ved. *vr̥kátāt-* 'collection of wolves'

vr̥kátāt- 'pack of wolves' is built to the noun *vŕ̥ka-* 'wolf' (PIE **u̯l̥kʷo-*), and is only found once in the Rig Veda as loc. sg. *vr̥kátāti* (with consonant-stem loc. sg. *-i*). Grassmann (1873 s.v.) understood it to mean 'Verderben, Raubanschlag' and Macdonell (1910:138, 188) something more like 'wolfishness'. Geldner (1951 *ad loc.*) and others, however, have taken *vr̥kátāti* as a collective meaning 'pack of wolves':

> 2.34.9ab *yó no maruto **vr̥kátāti** mártiyo ripúr dadhé vasavo rákṣatā riṣáḥ*
> 2.34.9cd *vartáyata tápuṣā cakríyābhí tám áva rudrā aśáso hantanā vádhaḥ*
>
> The cheating mortal who set us **among wolves** — o Maruts, good ones, protect (us) from his harm;
> roll over him with your searing wheel. O Rudras, strike down the murderous weapon of the one who lacks proper pronouncements.

Clearly *vr̥kátāti* is not reflecting qualitative abstract semantics and Geldner's collective reading is assured. *vr̥kátāti* also reflects the consonant-stem loc. sg. in *-i*, rather than the

more common *-ā*, perhaps to avoid a rare heavy ninth syllable in the line (cf. Arnold 1905:188).

III.D.4. Ved. *śáṃtāti-* 'luck, health'

śáṃtāti- 'luck, health', built to the indeclinable neut. noun *śám* 'id.', is found 3 times: nom. dual *śáṃtātī* (1.112.20a), acc. sg. *śáṃtāti* (8.18.7c), and instr. pl. *śáṃtātibhis* (10.137.4a; coordinated with *ariṣṭátātibhis*). *śám* can be dissimilated from **śvám* to PIE **$\hat{k}eu̯h_1$-* 'swell, rise', or it can also continue a root noun to the verbal root *śam-* 'work, act with intent' (*EWAia* s.v.). The phrase *sā́ śáṃtāti máyaḥ karad ápa srídhaḥ* (8.18.7c) is difficult, in that *śáṃtāti* seems to be used adjectivally modifying neut. acc. sg. *máyaḥ*, as done by both Grassmann (1873 s.v), who took it as 'heilbringend, wohltend', and Geldner (1951 *ad loc.*), translating the phrase "Sie soll uns beglückende Freunde schaffen", followed by Brereton and Jamison's (*ad loc.*) "She will create a luck-bringing joy".

An adjectival *tāti*-abstract is very hard to justify, given that this function does not have any precedent in any language attesting **-tāt-*. Alternatively, *śáṃtāti* could be interpreted as a regular abstract noun in asyndeton with *máyaḥ*. It seems best understood as a neut. acc. nonce form (p.c. Stephanie Jamison), which then helps to explain its unusual formation without thematic vowel vis-à-vis the other *tāt-* and *tāti*-abstracts (cf. above). Additionally, RV 8.18.7c is repeated in the Sāma Veda, though *śáṃtātā*

(1.2.1.1.6b) is found, which clearly reflects a nominal use of the abstract. Given that *-tāt-* disappeared after the Rig Veda, it makes sense to interpret SV *śáṃtātā* as an "updated" loc. sg. with the *i*-stem suffix *-tāti-*, beside RV *śámtāti*.

śivatāti- 'happiness, welfare' is found in the Chanda in a nearly-synonymous sense as *śáṃtāti-* and is later used adjectivally as 'causing good fortune, conferring happiness, propitious' in Bhavabhūti's *Mālatīmādhava* (Monier-Williams s.v. *śivá-*). But the early-8th c. CE play is very late, thus the adjectival use of *śivatāti-* cannot reflect anything about the use of the abstracts in Vedic.

III.E. Theonyms in *-tāt-*

Macdonell (1910:138) noted that *śáṃtāti-* and *satyátāt-/satyátāti-* also function as adjectives in the Rig Veda, but he does not identify where the forms are found or how many times the adjectival use can be observed. Presumably, he meant acc. sg. *śáṃtāti* at 8.18.7c discussed above and voc. sg. *satyatāte* at 4.4.14c (the only other instance of *satyátāt-/satyátāti-* is *satyátātā* at 10.111.4d, and its abstract meaning is secured; cf. above). Macdonell must be assuming that *satyatāte* is modifying some ellipsed noun, e.g "o truthful (one)!", but such an assumption need not be made. While it is may seem strange to find a *tāt*-abstract in the vocative case, *satyatāte* could reflect the personified use of the *tāt*-abstracts, in which the poet is addressing "Reality" directly, as done by Brereton and Jamison (*ad loc.*):

4.4.14ab *tváyā vayáṃ sadhaníyas tvótās táva práṇīti aśyāma vā́jān*
4.4.14cd *ubhā́ śáṃsā sūdaya* ***satyatāte*** *anuṣṭhuyā́ kṛṇuhi ahrayāṇa*

Through you may we — companions aided by you, in your guidance — attain prizes.
Sweeten both of the lauds [= for the gods and the patrons], **you who are reality itself**. Set them in order, you unabashed one.

Vedic does not personify the *tāt*-abstracts to the extent that Avestan or Latin did, but *satyatāte* could be understood in parallel to a number of forms in Avestan with a similar function. Instead, Ved. *devátāt-/devátāti-* 'collection of the gods' and *vasútāti-* 'collection of Vasus' are used in a collective sense to refer to a group of divinities.

In addition to the personified divinities *amərətāt-/amərətatāt-* and *hauruuatāt-/hauruuāt-* (cf. n. 77 above), Avestan reflects other examples of a naming function in YAv. *aršt̰āt-*, the name of a goddess "Uprightness", YAv. *kahrkatāt-*, the name of a rooster, and YAv. *rasąstāt-*, the name of a deity (cf. Hoffmann 1975-76:266n1), that are similar to Ved. *satyatāte*. Similarly, Lat. *-tās* is used in *tempestātēs* to name weather divinities (cf. **II.F.1.c.**), and Lat. *Voluptās*, beside Pg. *herentas*/Osc. **herentateís** 'Herentas', among other examples, were also used to signify deities (cf. **II.F.1.c.i.**). It may be the case that pl. *tempestātēs* reflects similar collectivizing semantics as in *devátāt-/devátāti-*, but with the morphological plural inflection modeled on Greek Ἔρωτες, given that at least the literary uses of "Tempestates" (cf. **II.F.1.b.**) may have been based on Greek ἀνέμοι (cf. Ἔρωτες whence Lat. *Veneres Cupidinesque* at Catul. 3

and 13, etc.). Hence, abstracts in *-*tāt*- seem to have a proclivity toward personification in both Indo-Iranian and Italic, which reflects the individualizing semantics of the suffix that may help understand how the suffix was formed (cf. **V.D.**).

IV. Greek -τηϛ

IV.A. Descriptive Facts

Unlike Latin and Indo-Iranian, which made old-looking collectives in *-tāt-* (e.g. Lat. *cīvitās* 'state; city'; cf. **II.A.**), as well as case forms used adverbially with a specialized and frequently technical sense, e.g. Ved. instr. sg. *devátātā* 'among the collection of the gods' (cf. the discussion at **III.C.1.b.**), Greek -τηϛ, -τητοϛ (Doric -τᾱϛ, -τᾱτοϛ) originally made simple qualitative abstracts, without any overt technical sense.[94] The suffix -τηϛ, with no *i*-stem variants, is certainly cognate with Latin *-tās* and Indo-Iranian **-tāt-/*-tāti-*, and was similarly generalized to thematic adjectives from an early date. Like Latin *-tās*, Greek -τηϛ becomes productive later, especially in Attic prose, and the majority of the Greek abstracts reflect formations to thematic adjectives, hence the generalized suffix form -ότηϛ (Buck and Petersen 1945:464-8). As also happened in Latin, -τηϛ even begins to appear in the Caland system beside *s*-stem abstracts, e.g. βαρύτηϛ 'weight; heaviness' (Th.+.) to βαρύϛ 'heavy', πλατύτηϛ 'breadth, bulk' (X.+) to πλατύϛ 'broad', etc. (Rau 2009b:122).

Homeric Greek possesses only 9 abstracts in -τηϛ, compared to the nearly 700 total abstracts found later in Greek literature (cf. Fig. 11 below): ἀνδροτήϛ 'manliness' (beside the manuscript variant cited as ἀδροτήϛ or δροτήϛ; cf. **IV.C.3.a.** for a full discussion), βραδυτήϛ 'slowness', δηϊοτήϛ 'battle strife', ἰότηϛ 'will, desire', κακότηϛ

[94] The development from simple abstract to one with more of a technical nuance occurred with -τηϛ in later philosophy and prose texts in general.

'badness', *νεότης* 'youth', *ποτής* 'drink', *ταχυτής* 'quickness', and *φιλότης* 'friendship; sexual love'; additionally, *βιότης* 'life' is found in the Homeric Hymn to Ares (cf. Risch 1974:149-50). Unlike Italic and Vedic, however, no abstracts are built to *i*- or consonant stems earlier in Greek, apart from the special case of *ἀνδροτής*; and outside of *-οτης* and *-υτης*, abstracts in plain *-της* without any connecting vowel are never found, given the early generalization of *-ότης* (cf. the same process with Lat. *-itās* and Ved. *-átāt-*). But in contrast to other suffixes that generalized the thematic vowel before the suffix, e.g. *-οεντ-* that built adjectives to *υ*-stems in *-υ-οεντ-*, some abstracts in *-της* are found even in Homer without the thematic vowel, e.g. *βραδυτής* (not **βραδυότης*). The absence of forms like *-υότης* is potentially due to the metrical restrictions in hexameter poetry, in that words with metrically light root syllables would make for an unallowable sequence of three successive shorts. No Mycenaean evidence reflects any sort of abstract with the suffix **-tāt-*, which is not surprising, given the inventorial nature of the texts (Mignot 1972:55n1).

Fig. 11. Homeric abstracts in *-της*

ἀνδροτής 'manliness' (3x)	• acc. sg. 3x	• 3x *Iliad*
βραδυτής 'slowness' (1x)	• dat. sg. 1x	• 1x *Iliad*
δηϊοτής 'battle strife' (30x)	• gen. sg. 9x • acc. sg. 8x • dat. sg. 13x	• 9x *Iliad* • 7x *Iliad*, 1x *Odyssey* • 10x *Iliad*, 3x *Odyssey*

Fig. 11. Homeric abstracts in *-τηϛ*, Cont.

ἰότηϛ 'will, desire' (12x)	• acc. sg. 1x • dat. sg. 11x	• 1x *Iliad* • 3x *Iliad*, 8x *Odyssey*
κακότηϛ 'badness' (25x)	• gen. sg. 8x • acc. sg. 9x • dat. sg. 8x	• 2x *Iliad*, 6x *Odyssey* • 2x *Iliad*, 7x *Odyssey* • 3x *Iliad*, 5x *Odyssey*
νεότηϛ 'youth; body of youth' (2x)	• gen. sg. 2x	• 2x *Iliad*
ποτήϛ 'drink' (10x)	• gen. sg. 9x • acc. sg. 1x	• 2x *Iliad*, 7x *Odyssey* • 1x *Odyssey*
ταχυτήϛ 'quickness' (2x)	• gen. sg. 1x • acc. sg. 1x	• 1x *Iliad* • 1x *Odyssey*
φιλότηϛ 'friendship; sexual love' (56x)	• nom. sg. 2x • gen. sg. 9x • acc. sg. 16x • dat. sg. 29x	• 2x *Iliad* • 4x *Iliad*, 5x *Odyssey* • 10x *Iliad*, 6x *Odyssey* • 17x *Iliad*, 12x *Odyssey*

The oldest of the *-τηϛ*-abstracts show a discrepancy in accentuation: *βιότηϛ*, *ἰότηϛ*, *κακότηϛ*, *νεότηϛ*, and *φιλότηϛ* show fixed presuffixal accent in *-ότηϛ*, but *ἀνδροτήϛ*, *δηϊοτήϛ*, and *ποτήϛ* have oxytone *-οτήϛ*, with accent fixed on the suffix vowel. The remaining 2 abstracts found in Homer are built with oxytone *-υτήϛ* (cf. *vasútāti-* 'collection of Vasus' to the noun *vásu-* 'Vasu (a god)', discussed at **III.D.1.**): *βραδυτήϛ* 'slowness' and *ταχυτήϛ* 'quickness', likely reflecting oppositional analogy, with one form built on the model of the other.

Hesiod essentially repeats Homer's *-τηϛ*-abstracts, though with a smaller variety of forms (cf. Fig. 12 below).

Fig. 12. Hesiodic abstracts in *-τηϛ*

δηϊοτήϛ 'battle strife' (3x)	• gen. sg. 1x • dat. sg. 2x
κακότηϛ 'badness' (5x)	• acc. sg. 3x • dat. sg. 2x
νεότηϛ 'youth' (1x)	• 1x (Frag 1.13; case?)
φιλότηϛ 'friendship; sexual love' (43x)	• gen. sg. 7x • acc. sg. 3x • dat. sg. 33x

Pindar and Aeschylus do nothing new with the *-τηϛ*-abstracts, each repeating the Homeric forms *κακότηϛ*, *νεότηϛ*, and *φιλότηϛ* (with Pindar using the Doric form *-τᾱϛ*). Pindar adds *ἁβρότᾱϛ* 'luxury', while Aeschylus uses *λειότηϛ* 'smoothness' and *τρᾱχυτήϛ* 'roughness' (on which cf. n. 102 below). It is clear that by the end of the archaic period the *-τηϛ*-abstracts have already assumed the productive form they would take in Attic prose, with generalized *-ότηϛ*.

IV.A.1. Metrical Behavior of *-τηϛ*-abstracts in Homer

Some of the more commonly found repeated phrases and metrical positions of the Homeric abstracts in *-τηϛ* are summarized below in Fig. 13. Given the restrictions forms in *-τηϛ* have within the hexameter, it should not be surprising that they tend to be found in only 3 locations within a line: spanning the biceps of the second foot, e.g. $\overset{1}{\text{–}}\breve{\breve{}}|\overset{2}{\text{–}}$ *ἰότητι*, spanning the biceps of the fourth foot through the first short of the fifth foot, e.g. *κακότητα* $\breve{}\overset{6}{\text{–}}$×#, and at absolute line end. Hesiod's metrical use of the abstracts exactly

matches Homer's practice, with most of his abstracts in *-τηs* falling into one of the three positions that Homeric abstracts occupy within the hexameter.

Fig. 13. Metrical positions of frequently used Homeric abstracts in *-τηs*

ἀνδροτήs 'manliness' (3x total)	• *ἀνδροτῆα καὶ ἥβην*# (2x)
δηϊοτήs 'battle strife' (30x total)	• *καὶ αἰνῆs δηϊοτῆτοs*# (3x) • *πόλεμον καὶ δηϊοτῆτα*# (4x) • *ἐν αἰνῇ δηϊοτῆτι*# (10x)
ἰότηs 'will, desire' (12x total)	• ⏑–⏑|– *ἰότητα* (1x) • ⏑–⏑|– *ἰότητι* (3x) • *θεῶν ἰότητι* ⏑–×#(7x) • *κακῆs ἰότητι* ⏑–×# (1x)
κακότηs 'badness' (25x total)	• *κακότητοs* ⏑–×# (3x) • *κακότητα* ⏑–×# (6x) • *κακότητι* ⏑–×# (1x) • *κακότητοs*# (5x) • *κακότητα*# (2x) • ⏑–⏑|– *κακότητα* (1x) • ⏑–⏑|– *κακότητι* (7x)
ποτήs 'drink' (10x total)	• *ἐδητύοs ἠδὲ ποτῆτοs*# (5x) • *αὐτὰρ ἐπει σίτοιό τ' ἐπασσάμεθ' ἠδὲ ποτῆτοs* (2x) • *μηδὲ ποτῆτοs*# (1x) • *οὐδὲ ποτῆτοs*# (1x)
φιλότηs 'friendship; sexual love' (56x total)	• ⏑–⏑|– *φιλότηs* (1x) • ⏑–⏑|– *φιλότητοs* (6x) • ⏑–⏑|– *φιλότητα* (9x) • ⏑–⏑|– *φιλότητι* (15x) • *φιλότηs τε καὶ εὐνῆ*# (1x) • *φιλότητι καὶ εὐνῆ*# (4x) • *φιλότητι*# (8x)

IV.A.2. Semantics

Petersen (1922:64-72) surveyed the semantic aspects of the abstracts throughout the history of Greek, though he does not give much attention to the Homeric forms. Generally, *-τηϛ* builds qualitative abstracts in Greek, just as *-tās* and *-tāt-/-tāti-* do in Latin and Indo-Iranian, respectively, and the Homeric corpus does exactly so too, e.g. *βραδυτής* 'slowness', *κακότηϛ* 'badness', and *φιλότηϛ* with basic meaning 'friendship'. But some forms in *-τηϛ* do not reflect simple qualitative abstract semantics, e.g. *νεότηϛ* 'youth' and later 'body of youths', *ποτής* with substantive meaning 'drink'. Such secondary developments to collective and/or substantive meanings from qualitative semantics are natural features of abstract nouns, and more discussion on semantic aspects of the abstracts will be given to individual forms below, where appropriate.

IV.A.3. *νεότηϛ* 'youth; collection of youths'

Both instances of *νεότηϛ* in Homer clearly mean 'youth' in the qualitative sense, but already in Pindar (*Isth.* 8.68) *νεότāϛ* has acquired the collective meaning of 'collection of youths; the "youth"', and subsequent Attic prose authors also used it in the collective sense. Such a collective use of the abstract is typologically regular (cf. **I.C.**) and is also seen in Ved. *devátāt-/devátāt-* 'collection of the gods' (cf. **III.C.1.b.**). Cret. acc. sg. *νεοτα* (*IC* 4.162) and gen. sg. *νεοταϛ* in (*IC* 4.162 and 163) are found in two inscriptions from Gortyn that date to the 3rd c. BCE, also with the meaning 'body of young men' (cf. Bile 1988:340). Chantraine (1933:297) thought that the connection with

the τηϛ-suffix was obscure, but Hom. νεότηϛ 'youth' and Lat. *novitās* 'newness' give support to the interpretation of νεοτα and νεοταϛ as abstracts in -τηϛ.

νεοτα and νεοταϛ have been thought to continue a haplologized *νεοτᾱτα and *νεοτᾱταϛ, based on the stem *νεοτατ- presumably underlying pres. pple. νεοτατευοντα (*IC* 4.164; Gortyn, 3rd c. BCE) formed from the abstract (cf. Bile 1988:150n311), but the absence of the expected gen. sg. in -οϛ is striking. Bile, following Schwyzer (1939:528n5), saw νεοταϛ as a remodeling of an original *νεοτοϛ with an ᾱ-stem gen. sg. in -ᾱϛ, after *νεοτᾱτα was haplologized to νεοτα. It is strange that the acc. sg. νεοτα, with an acc. sg. in -α proper to consonant stems in Greek, could have been interpreted as an ᾱ-stem, but no better explanation for νεοταϛ is apparent.

IV.B. Previous Scholarship

Close study of Gk. -τηϛ, -τητοϛ began with Aufrecht (1852), who compared the suffix to Lat. *-tās*, Ved. *-tāt-/-tāti-*, and Av. *-tāt-*. Unlike Latin, which attached *-tās* to a larger number of base types, Greek built abstracts in only -οτηϛ or -υτηϛ. The pattern continued later in Greek, though Aufrecht did not sort the data by age of attestation. Based on an idea of Pāṇini, Aufrecht (1852:163), followed by Justi (1864 s.v. *tāṭ*), thought that the suffix *-*tāti-* reflects a composite of the suffixes *-*tā-*, which is derived from the root *tan-* 'stretch', plus the suffix *-*ti-* already in PIE, in which case Lat. *-tās* and Gk. -τηϛ reflect some sort of unspecified syncope of the final *-*i-* in *-*tāti-*. His

view, however, can be dismissed, under the current thinking about the strength of the testimony for original *i*-stem inflection supplied by Ved. *-tāti-* (cf. **II.B.** for criticism of Aufrecht's analysis of **-tāti-*).

Aufrecht (1852:160) made special mention of the problematic forms in *βιότης* 'life', *ἰότης* 'desire', *πινυτής* 'understanding, wisdom' (found once in gen. sg. *πινυτᾶς* in PalAnth.; cf. discussion below), and *ποτής* 'drink' that do not have synchronic adjectival bases within Greek from which they were derived. Because all *της*-abstracts were thought by Aufrecht to be de-adjectival, *βιότης* must assume an underlying and unattested adjective **βιό-* 'living', for which he compared to Go. *qius* 'alive' (but cf. Ringe 2006:68-9 on the problems with such a comparison to *qius* and Eng. *quick* 'id.'), though only the masc. noun *βίος* 'life' is found. *ἰότης* must also presume an unattested adjective **ἰός* 'wanting, wishing'. *πινυτής* and *ποτής*, in his view, are deverbal abstracts rather than de-adjectival, though he does not expand on the idea. While it is conceivable that *βιότης* and *ἰότης* are built to old, unattested adjectives (to be evaluated below), there is no support for deverbal **-tāt-* anywhere in PIE, and similar formations in Greek are not found until very late, e.g. *δακνότης* 'property of stinging' found in medical texts seemingly built to to *δάκνω* 'bite' (cf. Petersen 1922:62). *πινυτής* and *ποτής* must have other explanations and will be treated below (cf. **IV.C.2.a.**).

Angermann (1870:122-6) confessed that he was not able to see Aufrecht's paper before publishing his own, but he rejected the earlier view that **-tāt-* continues some kind

of *ti*-stem built to the root *tan-* 'stretch', in that such an early specialization of a root as an abstract suffix did not seem probable. And, after considering the significant number of abstracts in *-*tā*- beside abstracts in *-*tāt*- found in multiple languages (cf. Fig. 2 in **I.C.**), e.g. Gk. πλατύτης 'width, breadth' (Xen.+) beside Skt. *pṛthutā-* 'id.' (Kālidāsa), Gk. βαρύτης 'weight' (Att.+) and Lat. *gravitās* 'id.' (Lucil.+) beside Skt. *gurutā-* 'id.' (Suśruta), Angermann also concluded that *-*tāt*- itself must consist of *-*tā*- plus *-*ti*-, as did Aufrecht (1852:161).[95] The dates when the above forms are first attested, however, show that old *-*tā*- : *-*tāt*- word-equations posited for them are of no value. Angermann had nothing to say about the *i*-stem variant *-*tāti*- in Indic.

Angermann did not discuss the problem with βιότης, but he did give special attention to the problems raised by Aufrecht (1852) in ἰότης, πινυτής, and ποτής, in addition to ἑσθής 'clothing' (*Od.*+), which to his mind reflects a της-form with aspiration of -τ- by the preceding -σ-. PIE *-*st*- never develops to -σθ- in Greek, and so ἑσθής need not be thought of as a της-abstract.[96] Regarding ἰότης, πινυτής, and ποτής, Angermann (1870:123) rejected the assumption of unattested adjectives, e.g. *πό-, given that an abstract built to such an adjective should surface as ˣποότης, following the assumed rules of abstract formation with -ότης. Angermann concluded that πινυτής and ποτής must

[95] Though the forms Angermann offered in support of the relationship between *-*tāt*- and *-*tā*- are late and thus of little use, other data indeed bear out such a relationship. The data are examined in greater detail in **I.D.**

[96] ἑσθής, beside the less frequent ἕσθος, belongs to a small class of nouns in -θος, including ἄχθος 'burden; grief' and πλῆθος 'crowd, multitude' (*EDG* s.v. ἔννυμι with refs.).

have been built directly from their roots, rather than derived from adjectives, but he did not elaborate. Old abstracts derived directly to roots with *-*tāt*- are unparalleled, leaving questions surrounding πινυτής and ποτής still open; both forms will be discussed below.

Brugmann (1906:452) believed *-*tāt*-/*-*tāti*- continued a combination of *-*tā*- and *-*ti*- but said little about -τηϛ in Greek, apart from its generalization in -οτηϛ and -υτηϛ on the model provided by abstracts built to to *o*- and *υ*-stem adjectives, in which -*o*-τηϛ and -*υ*-τηϛ were reanalyzed as -οτηϛ and -υτηϛ, respectively, in a similar way to the suffixes -οσύνη and -οεντ- in Greek (though without generalized ^{x}-υσύνη and ^{x}-υεντ-). Brugmann analyzed πινυτής and ποτής as the products of haplology from earlier *πινυτοτής and *ποτοτής, but because *-*tāt*- replaced denominative abstracts in *-*tā*-, Brugmann suggested it also might have made primary abstracts to verbal roots, if πινυτής and ποτής could be interpreted as such. But as Petersen (1922:54) observed, no deverbal pattern for building abstracts in *-*tāt*- is established anywhere, so a deverbal origin for πινυτής and ποτής can be dismissed.

Petersen's study (1922) stands as the most extensive of its kind for -τηϛ in Greek. His survey includes a list of all abstracts in -τηϛ, though his study was done strictly from a synchronic perspective. He did, however, include a useful summary of previous work on the suffix, though without evaluating the theories in detail (1922:44). He discussed Aufrecht's idea (1852:163) of a composite *-*tā*- plus *-*ti*- as the basis for *-*tāt*-, which he himself endorsed, as well as mentioning Hirt's (1912:292-3) view that *-*tāt*- is related to

the root "**tewā-* 'power'" (PIE **teu̯h₂-*), and that both **-tāt-* and **-tūt-* may ultimately go back to **-twāt-*. Traditionally in Greek, **-tuh₂-t-* would be expected to give *-τῡτ-*, but such a suffix is not found at all in Greek. On the other hand, the controversial "Lex Francis-Normier", according to which the PIE sequence **-tuh₂-t-* might alternately develop to **-tu̯āt-* in Greek, suggests that a suffix **-tuh₂-t-* derived from the root **teu̯h₂-* could give **-tu̯āt-* in the right environment.[97] Metrically heavy syllables are never found before *-τϝυτ-* in Homer, however; instead, a Siever's Law variant *-τυϝᾱτ-*, also never found, might actually be expected. Hence, it is very difficult to see how *-τητ-* could come from PIE **-tuh₂-t-*.

Petersen's extensive survey detailed the synchronic formation of abstracts in *-της* in Greek, including a thorough classification by stem types, and while the inner-Greek history of the suffix is extremely interesting, especially its later development in philosophical prose, it has little bearing on the prehistory of *-της*. Petersen did, however, treat a number of problematic forms discussed above. He (1922:51) followed Aufrecht (1852:160) in thinking *βιότης* must continue an abstract built to an unattested adjective **βιό-* 'living', since it must follow the normal pattern of de-adjectival derivation. Derivation of an abstract in *-της* from a substantive, though, is only paralleled much later

[97] Namely, PIE **-tuh₂-t-* could develop to **-tu̯āt-* in Greek in unaccented syllables, according to Olsen 2009. The Lex Francis-Normier affects PIE *-CiHC-* and *-CuHC-* sequences, with the best evidence found with h_2 or h_3. Its results are far from established, however, and the data remain controversial. Cf. Olsen 2009 (with refs.) for the most recent treatment of the data and Vine 2002 (with refs.) for discussion of the problems associated with it.

in Greek, and no denominal pattern had yet been established at the time βιότης was formed. The background of βιότης is much less interesting than Petersen thought, however, and will be explored below (cf. **IV.C.3.b.**).

In addition to the unusual accentuation of ἀνδροτής (cf. the discussion on accent below), it is one of a small number of της-abstracts built to nouns (cf. below on ποτής 'drink' besode ποτόν 'id.'), and it may be quite old, based on the similar-looking OAv. *hunərətāt-*. It is the only denominal της-abstract found in Homeric Greek (depending on the interpretation of βιότης; cf. discussion below), until the end of the 5th century BCE, when -της became particularly productive in Attic. After dismissing the manuscript variants ἀδροτής and δροτής as scribal substitutes for the unmetrical ἀνδροτῆτα, Petersen (1922:59-60) explained ἀνδροτής as a derivative in -της to the noun ἀνήρ, continuing "**nr̥-tāt-*" (in prelaryngeal notation).[98] He did not provide details for deriving ἀνδροτής from **nr̥-tāt-*, nor did he comment on the historical significance for such a denominal derivative in plain **-tāt-*, and so ἀνδροτής will be explored in detail below.

Chantraine's survey (1933:293-8) was primarily synchronic too, and, together with Petersen (1922), served as the basis for Buck and Petersen's collection of forms

[98] Petersen (1922:59) noted that the metrical need for an initial short syllable in ἀνδροτῆτα led to the emendation ἀδροτῆτα (beside ἀδρός 'thick, ripe, full-grown'), though the form lacks MS authority beside the *lectio difficilior* ἀνδροτῆτα. To obviate the metrical problem, Clemm (1877), followed by Brugmann (1906:418, 451) among others, emended ἀνδροτῆτα to δροτῆτα, which reflects **nr̥-tāt-* with replacement of expected **δρα-ταᾱτ-* by δρ-οτᾱτ-, as is common with ᾱ-stems later in Greek (cf. Petersen 1922:58) with the productive -οτης. δροτῆτα, however, does not work at *Il.* 24.6, which forced Clemm to assume that the poet was imitating an earlier line that already had ἀνδροτῆτα. Clemm's ad hoc solution is unnecessary, given that at the time of composition, -οτης likely had not yet become this productive and **-r̥-* was likely still intact. So, ἀδροτής and δροτής do not seem to be real words.

(1945:464-8). Chantraine suggested that *-*tāt*- is an enlargement of *-*tā*-, noting that this denominal *-*tā*- no longer exists in Greek, though not pursuing this problem further.[99] He never doubted the antiquity of *-*tāt*-, but he challenged Brugmann's word-equation (1906:451) Gk. *ὁλότης* 'wholenes' :: Ved. *sarvátāt*- 'wholeness, health' :: OAv. *hauruuatāt*- 'health', on the grounds that *ὁλότης* is not found until Aristotle and must be independently formed (cf. **I.A.**). He also stressed the fact that *της*-abstracts were built primarily to thematic stems in Greek, with 492 of the 658 examples cited by Buck and Petersen (1945:464-8) appearing in *-ότης*, which even spread to consonant-stems later in Greek, e.g. *χαριεντότης* 'grace, favor' (Plu.).

Of the Homeric corpus, Chantraine (1933:294) presumed that *ἰότης* comes from **istotāt*- as the "produit d'une dissimilation", based on comparison with Skt. *iṣṭa*- 'desired', and that *ποτής* is also "dissimilé" from an older **ποτοτής*, based on *ποτός* 'for drinking'. Chantraine's dissimilatiory explanation for *ἰότης* will be examined in **IV.C.3.d.**, and the haplology argued for *ποτής* (and even some other *της*-abstracts) will be investigated below in **IV.C.2**. He also added that *ἀνδροτής* 'manliness' must be based on the noun *ἀνήρ* 'man' and built with *-οτητ*-, but did not comment on the potential significance of such a denominal formation. As noted above, *ἀνδροτής* has the formulaic

[99] *βιοτή* 'life' (beside *βιότης* 'life') is the only secondary denominal abstract of this type in Greek and has a rather murky prehistory (cf. **I.D.** with n. 4 for a brief discussion of *βιοτή*).

appearance suggestive of an old formation likely before *-οτητ-* was generalized, thus requiring more attention than given by Chantraine.

Schwyzer (1939:528-9) did not add much to the earlier treatments of *-της*. He cited Aufrecht's connection (1852:160) with Ved. *iṣá-* 'desiring' to *ἰότης*. He (1939:528n8) also suggested that *ἰότης* may be related to Ved. *iṣanyáti* 'stimulate, urge' and Gk. *ἰαίνω* 'warm, melt', with the Aeolic development *-n̥- > -ο-, but did not pursue the idea. It seems very unlikely, however, that *-n̥- could develop to *-ο-* in this way; instead, *ἰότης* could simply reflect a productive abstract formation *-οτης* (cf. **IV.C.3.d.**). Hom. *ποτῆτος* "ist ‚poetisch' verlängert aus **ποτής*" , Schwyzer (1939:529) added, after Wackernagel (1914:35n1) first proposed the rather convincing idea (cf. further discussion below in **IV.C.2.a.**).

After a short but thorough survey of *-*tāt-* throughout PIE, Mignot's (1972) monograph handled the distribution and use of *-της* synchronically, though he did not explore the prehistory of *-*tāt-*. In fact, Mignot (1972:13) clearly stated at the outset that he was not interested in the prehistory of *-*tāt-*.

Risch (1974:149-50) had nothing to say about the history of *-της*, instead referring to previous discussion of Homeric *της*-abstracts and offering no new insights into the problems.

Meissner (2006:113), in his study of *s*-stem nouns in Greek, notes that abstracts in *-τηϛ*, along with those in *-οσύνη* and *-μα*, were in competition with the *s*-stems in Greek, and developed a sort of relationship with each other later in Greek. Regarding the relationship between *s*-stems and *τηϛ*-abstracts, Meissner (2006:111) comments that in secondary derivatives the *s*-stems usually took the role of a substantivized abstract noun, e.g. *ἧδος* 'vinegar; pleasure delight' (Ath.), while *-τηϛ* was used for qualitative abstracts, e.g. *ἡδύτηϛ* 'sweetness' (Sch. Ar. *Av.*). Given Meissner's purposes, though, he does not comment further on *τηϛ*-abstracts.

Though Homeric Greek attests just 9 abstracts in *-τηϛ*, the literature on the forms is rather dense, and many of the problems with the forms still remain unsettled, despite repeated attempts at solving them. The oxytone accent in *ἀνδροτής*, *βραδυτής*, *δηϊοτής*, *ποτής*, and *ταχυτής*, beside the generalized abstract form *-ότηϛ*, has also resisted explanation, and *ἀνδροτής* beside the noun *ἀνήρ* 'man', moreover, poses additional derivational problems. Haplology has been alleged for *πινυτής*, *ποτής*, and even perhaps Cret. *νεοτα*, though it is not certain that haplology need be the only explanation for the forms. *βιότηϛ* and the root **$g^{w}ih_3$-* 'live' present several difficult phonological problems; in addition to anomalous accentuation, *δηϊοτής*, presumably derived from *δήϊος* 'blazing, hostile', has a problematic root etymology; and *ἰότηϛ* lacks a synchronic adjectival base for derivation. Investigating the problems with the Homeric forms in detail may help

clarify how PIE *-*tāt*- was formed, so the problematic forms listed above will be examined in detail below.

IV.C. Special Problems

IV.C.1. Accentuation

Generally, *τηϛ*-abstracts in Greek have accent on the thematic vowel, as do those in Vedic (cf. **III.A.5.**); but of the Homeric abstracts, 5 show anomalous accent on the suffix: *ἀνδροτής*, *βραδυτής*, *δηϊοτής*, *ποτής*, and *ταχυτής*.[100] Angermann (1870:124) speculated that primary abstracts in *-*tāt*- preserved oxytone accent in Greek, beside secondary abstracts that were built with paroxytone accent. *βραδυτής* and *ταχυτής*, however, are clearly secondary abstracts built to *υ*-stem adjectives, but Angermann gave no justification, calling them Attic exceptions with oxytone accents, though both forms are attested already in Homer. Secondly, as mentioned above, *-*tāt*- did not make primary abstracts, so Angermann's account of accentuation can be discarded.

Petersen (1922:62-4) was the first to treat the divergent accentuation in a systematic way. He appealed to Wackernagel's (1914:59) claim that the PIE accent could not originally fall on the zero grades *-*i*-, *-*u*-, or *-*r̥*- when immediately before a suffix, e.g. Ved. *purutáma*- 'most' beside *purú*- 'much'.[101] *ἀνδροτής* (reflecting an original "**nr̥-tās*") and the abstracts built to *u*-stems *βραδυτής* and *ταχυτής*, according to

[100] N.B.: None of the forms are actually found in the nominative, and so the suffix *-τής* with oxytone accent has been reconstructed by grammarians based on accentuation on the suffix in the oblique forms of the abstracts, e.g. dat. sg. *δηϊοτῆτι* implies nom. sg. *δηϊοτής* (cf. Probert 2006:39).

[101] The superlative suffix *-tama*- normally leaves the accent on the derivational base in Vedic (cf. Probert 2006:43 with n. 70).

Petersen's view, can indeed reflect inherited accentuation, in that accent on the vowel preceding the suffix was shifted onto the suffix in abstracts built to non-thematic adjectives.[102] Petersen's explanation may carry some weight, but the oxytone accent in several other forms, e.g. *ἀνδροτής* and *δηϊοτής*, cannot be convincingly explained by such a rule. Moreover, Ved. *vasútāti-* 'collection of the Vasus' would be expected to show accent on the suffix too, suggesting that Petersen's idea may not be the correct interpretation.

Petersen himself admitted that "we need not expect a difference of this kind was ever maintained consistently in IE or Greek times" (1922:64) and that analogy was a powerful factor in the development of the accentuation of the abstracts. Under his view that *-της* is ultimately based on *-τᾱ-*, he suggested that association of an abstract in *-της* with an abstract in *-τή* may have helped remodel *-της* to oxytone *-τής*. Petersen thus suggested the existence of **δηϊοτή* and **ταχυτή* on which the accentuation of *δηϊοτής* and *ταχυτής* could be based, but there is no evidence supporting such forms. The accentuation of *ποτής*, moreover, cannot be copied from *ποτή*, given that it is late (cf. discussion below), but semantically associated oxytone words may have also influenced accentuation, e.g. *βρωτύς* 'eating; meat' (*Il.*+) and *ἐδωδή* 'food' (*Il.*+), according to Petersen.[103] It is difficult, however, to justify how an analogical reshaping could happen

[102] A few other oxytone abstracts in *-της* are found in Attic. Aeschylus attests oxytone acc. sg. *τρᾱχυτῆτα* (*PB* 80) that is certainly a repetition of Homeric *ταχυτής*. Only oxytone *τρᾱχυτής* is found later in Attic philosophy. The grammarians report that the Athenians said *κουφοτής* 'lightness' (cf. Probert 2006:39 with refs.), though *κουφότης* is found in Koine.

[103] Petersen also offered *ἐδωδύς as a potential accentual model for *ποτής*, but no such word exists.

in words that share nothing in common other than their general semantic field, especially given that presuffixal accent in the *τηϛ*-abstracts seems to have been a generalized fixture from a relatively early date.

It thus seems to be the case that there is no single explanation for the oxytone accentuation of the Homeric abstracts *ἀνδροτήϛ*, *βραδυτήϛ*, *δηϊοτήϛ*, *ποτήϛ*, and *ταχυτήϛ*, since each of the forms reflect different types of formations. Rather, explanations of the oxytonesis in the forms may be a result of their individual histories, and so each form will be explored in greater detail below.

IV.C.2. Haplology

In words where "*-ότηϛ* was preceded by a *τ*, the existence of two syllables with the same consonant might lead to simplification by haplology", according to Petersen (1922:53), and so oblique cases of *τηϛ*-abstracts built to stems in *-το-*, e.g. *ποτήτοϛ* < **ποτοτήτοϛ*, were assumed to be subject to such a simplification, much like the *tās*-abstracts built to *ti-* or *to*-stems in Latin (cf. **II.D.3.**). The few forms in which haplology is thought to have occurred (aside from the later case of *νεότηϛ*; on which cf. **IV.A.3.**), however, might have alternative explanations not requiring such a simplification.

IV.C.2.a. *πινυτήϛ* 'understanding, wisdom' and *ποτήϛ* 'drink'

πινυτήϛ, beside Homeric *πινυτή* 'understanding, wisdom', found once in the gen. sg. *πινυτᾶτοϛ* in the Palatine Anthology, must, according to Petersen (1922:54), continue a haplologized **πινυτότᾱτοϛ*. Such a *πινυτότηϛ* is indeed found, though late, as a secondary creation by Eustathius (*EDG* s.v. *πινυτόϛ*). On the other hand, Schwyzer

(1939:529), following Wackernagel (1914:35n1), saw πινυτής (along with ποτής) as a poetic replacement of an abstract in -τη by one in -της, rather than a haplologized πινυτοτής, which seems likely, given the extent to which Homer remade previously existing but unmetrical words with new suffixes, e.g. acc. sg. ἡνίοχον 'charioteer' remodeled to the ες-stem ἡνιοχῆα for use at line-end in the hexameter (cf. Mercado 2001 for discussion and further examples). Thus, πινυτής is of no importance for the history of *-tāt-.

Petersen (1922:53-4) also believed ποτής 'drink' was the product of halplology, not a deverbal formation in -της, given that a deverbal pattern for της-abstracts was not established in Greek until very late (cf. Brugmann 1906:452). Even so, deverbal της-abstracts were very marginal. According to Petersen's thinking, gen. sg. ποτῆτος, statistically the most common form found in Homer (found 9x; cf. Fig. 11 in **IV.A.**), continues *ποτοτῆτος, built to the *to*-adjective ποτός 'for drinking', but ποτός does not appear until Aeschylus. Homer does have a neuter substantive ποτόν 'drink' that could likely continue a neuter substantivization of the adjective ποτός, but the metrical use of ποτής suggests it might have another explanation altogether. Gen. sg. ποτῆτος make up 9 of the 10 instances of ποτής in Homer, and they are all found at line end, including 5 times in the phrase ἐδητύος ἠδὲ ποτῆτος# (cf. Fig. 13 in **IV.A.1.**). *ποτοτῆτος on its own is perfectly acceptable in the hexameter, with its phonetically identical shape to

κακοτῆτος, etc.; but if ποτῆτος indeed reflects a formulaic usage of the abstract, it is difficult to see how *ποτοτῆτος could have worked in this particular formula.

Alternatively, ποτής could have the same origin as πινυτής: Risch (1974:150) conceived of ποτῆτα a sort of poeticism for *ποτήν, and Chantraine (*DELG* s.v. πίνω), after Schwyzer (1939:529) and Wackernagel (1914:35), saw ποτῆτα as an "arrangement métriquement commode" for ποτή. While ποτή is late and only found on papyrus, Chantraine was essentially right, in that ποτής most likely reflects an epic remodeling of ποτόν, with copied oxytone accent. It was not the acc. sg. ποτῆτα that was the driving force behind its creation, however, but rather the gen. sg. found at line end: ποτοῦ cannot work at the end of a hexameter line, and so the more metrically useful ποτῆτος was created, aided by the fact that several της-abstracts are found in the oblique at line end in Homer (cf. Fig. 11). Hackstein (2010:412) also sees ποτῆτος as exactly this sort of poetic replacement, though replacing ποτῆς rather than ποτοῦ, which is equally as likely. Such a backformation of *o*-stems *metri causa* is known with other suffix types too, as mentioned above for the remodeled ἡνιοχῆα at line-end. In fact, 7 of the 10 total attestations of ποτῆτος occur at line end in the *Odyssey* (cf. Fig. 11 and 13), supporting the hypothesis that such a placement may have been formulaically advantageous.

IV.C.3. Individual Word Problems

IV.C.3.a. *ἀνδροτῆτα* 'manliness'

Much has been made of the unmetrical Homeric *ἀνδροτῆτα* 'manliness', an abstract built with the *tāt*-suffix to the noun *ἀνήρ* 'man', from PIE **h₂ner-*, traditionally glossed 'man' (cf. *NIL* s.v. **h₂ner-/*h₂nr̥-*). *ἀνδροτῆτα* is found three times in the *Iliad* (cf. Fig. 11 in **I.A.**): twice in the formula *ἀνδροτῆτα καὶ ἥβην#* 'manliness and youth' (16.857, 22.363) in reference to the death of Patroclus, and Hector, respectively; and once in the phrase *ἀνδροτῆτα τε καὶ μένος ἠύ#* 'manliness and good spirit' (24.6), expressing Achilles' longing for the departed Patroclus. As written, *ἀνδροτῆτα* scans —⏑—⏑, and so the cretic sequence —⏑— cannot fit into the dactylic hexameter. Consequently, it has remained a long standing textual problem and served as a key piece of evidence for the evolution of the hexameter as a combination of two smaller metrical units that allow cretic sequences at an earlier point in its history, according to the "Tichy-Berg" theory of the protohexameter.[104] While the attention surrounding *ἀνδροτῆτα* has concerned its age within the poetic tradition, the implications of this form for the morphological background of the *tāt*-suffix itself, with which *ἀνδροτῆτα* was built, have gone unremarked. Not only does *ἀνδροτῆτα* present a serious metrical problem, but it also

[104] The metrically-based "Tichy-Berg" theory of the history of the hexameter follows the hypothesis that the dactylic hexameter as found in Homer is relatively recent, having developed from an original combination of a glyconic (⏑⏑—⏑⏑—⏑—) and pherecretian (⏑⏑—⏑⏑—×) unit. The details of the theory, however, are controversial and complicated, and so they cannot be pursued at this time. No opinions about the prehistory of the hexameter will be offered here, but cf. Tichy (2010) for the most recent treatment of the data supporting the theory.

fails to follow the common accentual pattern observed with the *tāt*-suffix, with accentuation on the suffix as if built to a consonant stem. If ἀνδροτῆτα continues something old, it may help shed light on the otherwise obscure prehistory of this suffix.

IV.C.3.a.i. Interpretations of ἀνδροτῆτα

It is now widely assumed that ἀνδροτῆτα reflects */a.nr̥.tā.ta/* (cf. Hajnal 2003:66), scanning ⏑⏑—⏑ without the excrescent dental and preserving *-*r̥*-, which must date to no later than 1400 BCE (cf. most recently Hackstein 2010:413-4, with further references). Already in Mycenaean, *-*r̥*- has developed to *-or-/-ro-*, e.g. *to-pe-za /torpeza/* (KN V 280.5+) from PIE *(k^w)*tr̥-ped-ih*$_2$, so ἀνδροτῆτα, reflecting */a.nr̥.tā.ta/*, must have been versified in epic poetry at a rather early point before *-*r̥*- > *-or-/-ro-* (cf. Wathelet 1970:169-71). Moreover, the excrescent dental in ἀνδροτῆτα is also already found in Mycenaean, suggesting that ἀνδροτῆτα */a.nr̥.tā.ta/* was versified at some pre-Mycenaean date into the hexameter, which already had its classic shape.[105]

Alternatively, Tichy (1981), following Berg's (1978) views on the origin of the dactylic hexameter, takes ἀνδροτῆτα */androtāta/* at face value, with a cretic sequence —⏑— that reflects an earlier but still relatively recent stage of the hexameter. In Tichy's view, the versification of the formula ἀνδροτῆτα καὶ ἥβην# postdates the Aeolic change of *-*r̥*- to *-*or-/-ro-* and so must preserve a relatively recent Aeolicism. As a parallel,

[105] Some examples of excrescent dentals include *a-di-ri-ja-pi /andriantphi/* 'image of a man' (instr. sg.; PY Ta 708) and PN *a-di-ri-jo /andrios/* or */andriōn/* (KN Do 7613).

Tichy offers ἀμβροτάξομεν 'we miss' (Hom. *Il.* 10.65) and Hesiod's ἀεξομένοιο βροτήσια 'mortal (deeds) of the waxing (month)' (*WD* 773) as parallels for the preservation of a cretic sequence —⏑— in epic that provides support for Berg's proto-hexameter, though she remains skeptical of ἀσπίδος ἀμφιβρότης 'man-protecting shield' (Hom. *Il.* 2.389, 11.32, 12.402, 20.281), ἀβροτή 'divine' (*Il.* 14.78), and Ἐνυαλίωι ἀνδρειφόντηι 'Enualios the man-slayer' (Hom. *Il.* 2.651+), assuming an original *ἀνδροφόντηι, in that they can simply reflect *muta cum liquida* treatments (on which cf. Hackstein 2010:416-7), rather than preserved *-r̥-.

West (1988:157), on the other hand, offers ἀμβροτάξομεν and βροτήσια, as well as βροτοῖσι, τραπέζηι, and θρόνους (though for the latter, cf. de Lamberterie 2004:243-50 for a different view) as additional examples of *-r̥- preserved in the hexameter without reflecting an actual cretic sequence, though "wearing the disguise of Attic correption", in that the mute-liquid combination in the phrases does not make position.[106]

Recently, Barnes (2007) proposed an etymology for the phrase with ἀνδροτῆτα καὶ ἥβην, connecting it with Av. *hauruuatāt- amərətatāt-*, traditionally taken as 'health and immortality'. In support of the Tichy-Berg theory of the hexameter, he views ἀνδροτῆτα as recent and secondary, rejecting the traditional explanation of ἀνδροτῆτα, in that it cannot continue *anr̥tāta*. To his mind, it is difficult to see how *anr̥tāta*, with (pre-)Mycenaean *-r̥-, as well as the hexameter itself in its Homeric form, could be

[106] That is, assuming δειλοῖσι βροτοῖσι < *dṷeiḷóisi mr̥tóisi*, ἐπένειμε τραπέζηι < *tr̥pézāi̯*, κατὰ κλισμούς τε θρόνους τε < *t^{h}ŕ̥nons- k^{w}e* (cf. West 1988:157 with n. 45).

preserved unchanged for several hundred years up to the time of Homer.[107] He instead proposes that the adjective *βροτήσιος* 'mortal, human' (Hes.+) must be based on **ἀμβροτήσιος*, itself dependent upon an unattested *tāt*-abstract **ἀμβρότης*, haplologized from **n̥-mr̥tó-tāt-* and thus cognate with OAv./YAv. *amərətatāt-/amərətāt-*.[108] Such a relationship between an adjective in *-τήσιος* and an abstract in *-της* echoes the patterning found in the pair *φιλοτήσιος* 'of friendship, love' and *φιλότης* 'friendship, sexual love'. **ἀμβρότης*, which continues a haplologized **n̥-mr̥tó-tāt-* with Aeolic **-r̥-* > *-ρο-*, finds a direct cognate in OAv./YAv. *amərətāt-* (haplologized from *amərətatāt-*; cf. **III.A.2.**), suggesting that **n̥-mr̥tó-tāt-* is quite old.[109] It is thinkable, according to Barnes, that **ἀμβροτῆτα* provided a direct model on which *ἀνδροτῆτα* was built, even copying its accent.

It is not the case that a metrically preserved **anr̥tāta* necessarily implies the static hexameter remained unchanged for hundreds of years. There is no reason to think that the hexameter did not evolve with time, despite preservation of an archaism in *ἀνδροτῆτα*. And though the accentuation of the *της*-abstracts was a point of debate already in the Homeric scholia, the accentuation of *ἀνδροτῆτα* may not be as problematic

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[107] Barnes' argument makes no claims about the prehistory of the hexameter and the place of *ἀνδροτῆτα* in it.

[108] **ἀμβρότης*, haplologized from **ἀμβροτότης*, is cited by Barnes, with presuffixal accent based on a preform **n̥mr̥tótāt-*. But given the accentual behavior of abstracts for which haplology is alleged, e.g. *ποτής* and even *ἀνδροτής* (on which cf. below), oxytone **ἀμβροτής* should be assumed. As argued above, however, *ποτής* is a poetic remodeling, rather than a continuation of a haplologized **ποτοτής*.

[109] **n̥-mr̥-tó-tāt-* > (with haplology) **amr̥tāt-* with **-r̥-* should be expected to give Av. **amaš̰āt-* (beside **amaš̰ā-* 'undying' < **n̥-mr̥-teh₂-*; cf. **III.A.5.**), as Barnes observed (cf. Hoffmann 1992:837-57).

as it seems. No explicit statements about the accentuation of *ἀνδροτής* (never found in the nom.) are made, but it is reasonable to assume it must reflect the same accentuation pattern followed by the other consonant-stem *τησ*-abstracts, e.g. *ταχυτῆτα* 'quickness' (Probert 2006:43-4).[110] It is conceivable, then, that the accentuation of *ἀνδροτῆτα* is actually expected, if based on **h₂nr̥-tā́t-*.

Furthermore, the lexical value of *βροτήσιος* for positing **ἀμβροτῆτα* may not be as strong as it seems. Frisk (*GEW* s.v. *βρότος*) takes *φιλοτήσιος* as the starting point for the creation of other adjectives in *-ήσιος*, including *βροτήσιος* itself. But the hapax *φιλοτήσιος* (*Od.* 11.246) is the only example of a denominal adjective in *-ιος* based on a *tāt*-abstract, suggesting that *φιλοτήσιος*, as well as the other adjectives in *-τήσιος*, are poetic creations. Similarly, *θελκτήριον* 'charm, spell' (*Il.* 14.215; *Od.* 1.337, 8.509) : *θελκτήρ* 'charmer' (*h.H.*), *θρεπτήριος* 'feeding, nourishing' (Hes., *h.H.*) : *θρεπτήρ* 'feeder' (Anth. P.), and *ψυκτήριος* 'cooling' (Hes.) : *ψυκτήρ* 'a cool place' (Eurip.) would seem to lend some weight to a pattern by which agent nouns in *-τήρ* make adjectives in *-τήριος*, and there is no reliable evidence that *τησ*-abstracts were original in this relationship.[111]

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[110] Though ostensibly built to thematic bases, *δηϊοτῆτα* 'hostility', built to *δήϊος* 'blazing, hostile', and *ποτῆτα* 'drink' to *ποτόν* 'drink' show the oxytone accentuation characteristic of abstracts built to consonant-stem bases. Both forms, however, were not derived to thematic adjectives in the usual way (cf. **IV.C.2.a.** on *ποτής* and **IV.3.c.** on *δηϊοτής*).

[111] It is certainly thinkable that adjectives were derived directly to *tāt*-stems: cf. the similar-looking patter in Latin where *tāt*-abstracts served as the basis for tertiary adjectives in *-ārius*, e.g. *hērēditārius* 'of inheritance' (Cic.+) < **hērēditātārius* beside *hērēditās* 'inheritance', *voluntārius* 'of one's accord' (Cic.+) < **voluntātārius* beside *voluntās* 'will', etc. For more discussion, cf. n. 19 in **II.D.1.a.**

The agent nouns in -τήρ, moreover, are in the process of being replaced by agentive -τής already in Homer, with an emerging relationship between -τήσιος and -τής, e.g. ἱκετήσιος 'of a suppliant' beside ἱκέτης 'suppliant'. The increasing number of agent nouns in -τής beside adjectives in -τήσιος, alongside -τήριος and -τήρ, may have created a pattern whereby abstracts in -τής could make adjectives in -τήσιος. Given forms like Ἰθακήσιος 'Ithacan' (*Il.*+) to Ἰθάκη 'Ithaca' (*Od.*+) and πρυμνήσιος 'of a ship's stern' (*Il.*+) to πρύμνη 'stern' (*Il.*+) that suggest adjectives in -ήσιος spread as useful poetic creations, βροτήσιος could simply be based on βροτός 'a human' (*Il.*+), rather than Barnes' *ἀμβροτήσιος and *ἀμβρότης. Thus, the existence of the abstract *βρότης is at best questionable and therefore an unlikely model for the creation of ἀνδροτῆτα, which would then reclaim its antiquity.

Barnes is correct that the metrical shape of ἀνδροτῆτα need not make any implications about the prehistory of the hexameter, and furthermore there is no good reason to assume it reflects an Aeolic metrical archaism (i.e. as part of a pherecratean) here and here alone in epic.[112] If so, it remains to be answered, then, why as an unmetrical Aeolicism it should be preserved as such, when metrically useful creations of the sort ἀνδρει-φόντης (replacing a putative *ἀνδρο-φόντης) next to ἀνδρο-φόνος abound

[112] That is not to say that ἀνδροτῆτα is not Aeolic in origin, with diagnostic -ρο- < *-r̥-. In fact, the connection between this form and Achilles who himself was Thessalian does suggest this material is Aeolic in origin, but it still makes no claims about the origin of the hexameter.

in epic (cf. Nussbaum 2007b).[113] An ad hoc creation *ἀνδρειτῆτα would find support in the other poetic creations, but ἀνδρει- is only appropriate as the first member of a compound. Rather, no better alternative seems more persuasive than simply reading ἀνδροτῆτα as */a.nr̥.tā.ta/* and assuming a pre-Mycenaean *-r̥- is retained in the closely-related forms, as in the *communis opinio.*

Finally, if ἀνδροτῆτα, continuing an abstract *$h_2nr̥$-tāt-, is in fact old, it may have serious implications for the prehistory of the *tāt*-suffix, including the fact that *$h_2nr̥$-tāt- would fail to show the standard pattern mentioned above whereby *tāt*-abstracts are built to only thematic or *υ*-stem adjectives in Greek. Moreover, *tāt*-abstracts built to nouns seem to show collective semantics, e.g. Lat. *cīvitās* 'state; city' (cf. **II.A.**) and Ved. *devátāt-/devátāti-* 'collection of the gods' (cf. **III.C.1.b.**). ἀνδροτῆτα, however, never shows any such collective meaning, suggesting it may have a different history.

IV.C.3.a.ii. ἀνδροτῆτα < PIE *$h_2nr̥$-tāt-

That *$h_2nr̥$-tāt- fails to show the typical pattern of a de-adjectival abstract of this type may be illusory: Nussbaum (2007a) has suggested that Ved. *sū́nr̥ta-* 'magnanimous; in full vigor, unfaded' may reflect an original simplex substantive *$nr̥ta$- 'spirit, vigour', continuing a PIE *to*-adjective *$h_2nr̥tó$- 'having strength, strong' that also underlies OIr.

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[113] ἀνδρειφόντης is a later substitution for *ἀνδροφόντης (itself continuing *$anr̥p^hontās$) modeled on ἀργειφόντης, with ἀργει- replacing ἀργι- for metrical reasons (cf. ἐγχεσ-ί-μωρος with restored intervocalic -σ-), but the source for ἀργει- remains unclear (Watkins 1995:384).

nert 'strength'. The *to*-adjective *$*h_2nr̥to$- may itself assume as its basis a root adnominal with *e/z* ablaut *$*h_2ner$-/*$*h_2nr̥$- 'vigorous, spirited', corroborated by Ved. *nṛvánt*- 'full of men; strong', OAv. *hunarətāt*- 'talent', Gk. -*ήνωρ* 'strength, courage' (e.g. PN 'Αγήνωρ), Hitt. *innara*- 'strong, willful', and CLuv. *annari*- 'forceful'.[114]

The root noun *$*h_2ner$-/*$*h_2nr̥$-, moreover, may be based on an originally adjectival root. Nussbaum (2007a) has outlined several other good examples of adjectival substantivization of this sort, e.g. *$*b^herĝ^h$- 'high' → masc. *$*b^herĝ^h$-/*$*b^hr̥ĝ^h$- 'high one', *$*h_2r̥ĝó$- 'shining' (Gk. ἀργός) → masc. *$*h_2órĝi$-/ *$*h_2érĝi$- 'bright, shining (one)' (Hitt. *ḫarki*-) beside fem. *$*h_2órĝi$-/ *$*h_2érĝi$- 'brightness' (Gk. ἀργι-ποδας 'silver-footed', Ved. PN *ṛji-śvan*-), etc. The question is whether ἀνδροτῆτα can reflect *$*h_2nr̥$-*tāt*-, with *-*tāt*- appended to the adnominal *$*h_2ner$-/*$*h_2nr̥$-, or if it continues a haplologized *$*h_2nr̥$-*to*-*tāt*-, with the *tāt*-suffix added to the *to*-adjective.[115]

IV.C.3.a.iii. ἀνδροτῆτα < PIE *$*h_2nr̥to$-*tāt*-

Given that OAv. *amərətatāt*- sits beside the haplologized variant *amərətāt*-, it is reasonable to explore the possibility that ἀνδροτῆτα continues *ἀνδροτοτῆτα from

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[114] The Anatolian forms could continue an "ἔνθεος"-type compound *$*en$-$h_2nr̥r$-*o*-, with compositional *-*o*-; but based on the similar patterning in *$*ĝ^helh_3$- 'yellow-green' : *$*ĝ^hólh_3$-*o*- (Gk. χόλος 'green'), *$*ḱeu̯H$- 'swollen, big, hollow' : *$*ḱóu̯H$-*o*- (Gk. κόος/κῶος 'cavity'), etc. Hitt. *innara*-/CLuv. *annari*-, along with Gk. νωρεῖν· ἐνεργεῖ 'be at work', more likely continue a τόμος-noun in *$*R$(ó)-*o*- *$*en$-$h_2nór$-*o*- or *$*en$-$h_2nór$-*o*- (Nussbaum 2007a).

[115] Cf. **V.D.3.** for a more detailed analysis of how ἀνδροτῆτα and OAv. *hunarətāt*- may have been derived in PIE.

*$h_2nr̥to$-$t\bar{a}t$-, with the *tāt*-suffix added to the *to*-adjective *$h_2nr̥$-*tó*-. Greek does not have any traces of such an adjective *ἀνδροτός, but as discussed above Ved. *sū́nr̥ta* and OIr. *nert* give support for a PIE *$h_2nr̥$-*tó*-. Closely related to ἀνδροτῆτα is OAv. *hunarətāt*- 'talent' (cf. Brugmann 1906:418), a *tāt*-abstract built to an unattested adjective **hunar*- 'skill' (still found in MP and NP *hunar*- 'id.'), to which the noun OAv./YAv. *hunara*- 'art, prowess' (beside the Ved. adj. *sūnára*- 'beautiful') was made, according to Bartholomae (*AiW* s.v. *hunara*-).[116]

Though *hunarətāt*- is a hapax, found at Y. 50.8, it may shed light on ἀνδροτῆτα. *hunarətāt*- cannot be built to *hunara*-, in that *hunarətāt*- reflects regular *ə*-epenthesis after *-r*- (cf. Hoffmann and Forssman 2004:51). Much as ἀνδροτῆτα could continue a haplologized *$h_2nr̥$-*tó*-*tāt*-, *hunarətāt*- might instead be built to a *to*-stem *h_1su-h_2ner-*tó*-, in which case it continues the full grade of *h_2ner- (cf. OIr. *nert*). **hūnarta-tāt*- would give *ˣ**hunarətatāt*-, which could be shortened by haplology to ˣ*hunarətāt*- in Old Avestan. But *hunarətāt*- reflecting *h_1su-h_2ner-*tó*-*tāt*- beside *$h_2nr̥$-*tó*-*tāt*- found in Greek with zero grade of the adjective *h_2ner-, seems hard to reconcile, because the solution assumes that *to*-adjectives were independently built to the full and zero grades of

[116] The noun *hunara*- 'art; having skill' reflects a possessive thematic derivation to **hunar*- 'skill', while Ved. *sūnára*- 'beautiful' can simply be a thematization of the full grade of the adjective *h_1su-h_2ner-/*h_1su-$h_2nr̥$-.

the adjective *h_2ner-/*$h_2n\underset{\circ}{r}$- in Avestan and Greek, respectively, and then *tāt*-abstracts were subsequently derived, also independently. Finally, haplology must be assumed for both Avestan and Greek, though there is no direct evidence in support of such a development.

On the other hand, it is much simpler to project the solution back to PIE. *hunarətāt-* can continue **hūnartāt-*, with **-tāt-* added to the full grade adjective *h_1su-h_2ner-, much as ἀνδροτῆτα could reflect zero-grade *$h_2n\underset{\circ}{r}$- plus **-tāt-* , as argued above. The base adjective Av. **hunar-*, likely reflecting *h_1su-$h_2n\underset{\circ}{r}$- might provide an Iranian reflex of the root adjective *h_2ner- thought to underlie ἀνδροτῆτα, though continuing the full grade of the adjective. *hunarətāt-* < *h_1su-h_2ner-tāt-* could then reflect a relatively old *tāt*-abstract built directly to the adjective stem without any intervening vowel, just like Hom. ἀνδροτῆτα < *$h_2n\underset{\circ}{r}$-tāt-*. There is no need to invoke independent haplological developments in the preforms *h_1su-h_2ner-to-tāt-* and *$h_2n\underset{\circ}{r}$-to-tāt-*. Instead, *hunarətāt-* and ἀνδροτῆτα could be very old examples of the *tāt*-suffix added directly to a consonant stem. In fact, ἀνδροτῆτα < *$h_2n\underset{\circ}{r}$-tāt-* and *hunarətāt-* < *h_1su-h_2ner-tāt-* might be the only *tāt*-abstract word-equation datable to PIE, though showing different root vocalism (cf. Fig. 11 in **I.A.**).

Hence, despite the pattern in which some *tāt*-abstracts reflect nominal bases, e.g. Lat. *tempestās* 'period of time, season; storm' (Pl.+) and Ved. *devátāt-/devátāti-*

'collection of the gods' (cf. **II.F.1.b** and **III.C.2.**, respectively) there is good support that *$h_2nr̥$-$t\bar{a}t$- was built to a root adjective *h_2ner-/*$h_2nr̥$-, rather than a noun, as is commonly thought for ἀνδροτῆτα. The denominal use of *-$t\bar{a}t$- will be explored later (cf. **V.D.4.**). Additionally, though *-$t\bar{a}t$- was specialized to thematic adjectives at a relatively early date, ἀνδροτῆτα shows that the suffix could be added to non-thematic adjectives too. The details of the specialization remain opaque and will also be handled later (cf. **V.E.**).[117] Some answers to the development of *-$t\bar{a}t$-, however, may be viewed in a new light if *$h_2nr̥t\bar{a}t$- has the sort of background sketched above.

IV.C.3.b. βιότης 'life'

βιότης 'life' has been assumed to be based on an unattested adjective *βιός 'living', as Aufrecht (1852:160) thought, followed by Petersen (1922:51). βιότης, however, may not be as old as previously thought: it is only found twice in Greek, once in an epigrammatic inscription (*IG*^{14} 1449 = *IGUR* 4.1702 = Kaibel 588), and once in the Homeric Hymn to Ares (*h.H.* 8.10). Epigraphic βιότης, found in Rome, dates to the 3rd century CE, and the Homeric Hymn to Ares, which does not match the rest of the corpus in style and content, has been thought to be a later addition. In fact, West (1970 and 2003:17) has dated the hymn to as late as the 5th century CE, authored or strongly influenced by the Neoplatonist Proclus. Consequently, it is highly unlikely that βιότης

[117] Other similar generalizations (but with different chronology) of suffixes to thematic stems in Greek are well known, e.g. Myc. *-$\underset{\sim}{u}ent$-, as in *ped-$\underset{\sim}{u}ent$- → Hom. *-ό-ϝεντ, and already by Homer -σύνη → -ο-σύνη (cf. Vine 1999:576-8).

was formed directly to *βιός 'living', given that such an adjective is never found in Greek. βιότης, then, most likely reflects an analogical creation. It could be a sort of poetic (or "technical") formation derived from the noun βίος, but it is more likely that βιότης was modeled on βιοτή 'life' (Hom.+) and βίοτος 'life' (Hom.+), owing to the fact that abstracts in -της are closely related to those in -τη.[118] βιότης can add little to the picture of early της-abstract formation in Greek.

IV.C.3.c. δηϊοτής 'battle strife'

Hom. δηϊοτής 'battle strife' is well attested in several formulaic phrases, occurring 30 times in Homer (8x acc.sg., 13x dat. sg., 9x gen. sg.; cf. Fig. 11 in **IV.A.**) almost entirely at line end (cf. Fig. 13 in **IV.A.1.**). It is assumed to be based on the thematic adjective δήϊος 'blazing, hostile', reflecting a "normal" de-adjectival abstract in -οτης to it; but the assumption might not be valid, given that δήϊος has a complicated background. Additionally, δηϊοτής has consistent accent on the suffix, rather than accent on the thematic vowel, unlike all other της-abstracts to thematic adjectives (cf. the discussion of accent in **IV.C.1.**). Schwyzer (1939:382, 528) viewed the oxytone accent as analogical to the consonant-stem βραδυτής, but it is difficult to see how βραδυτής 'slowness' could have served as an analogical model for δηϊοτής. Schwyzer's idea is not very satisfying, and the accentual anomaly deserves more attention than it has hitherto been given. Treatment of the accentuation of δηϊοτής will thus be addressed in greater detail below.

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[118] Petersen (1922:51) offered the possibility that βιότης in the sense of "plentiful livelihood" was invented by the poet of *h.H.* 8 by "congeneric assimilation" beside κακότης 'wretchedness, poverty' by forming it with -της to the already-existing nouns βίος, βιοτή, or βίοτος, all of which mean 'life'.

IV.C.3.c.i. δήϊος 'burning' and *δήϊς 'division; battle'

The epic adjective δήϊος 'burning; hostile' (beside Hsch. τὸ δήϊον· τὸ πολέμου) is found in the phrase δηΐου ἐκ πολέμου (*Il.* 7.119)/ δήϊον ἐς πόλεμον (*Il.* 4.281) and δήϊον ἄνδρα (*Il.* 6.481, 22.84), but it mainly appears in the *Iliad* in frequent association with πῦρ 'fire' in the formulaic phrases πυρὸς δηίοιο and δήιον πῦρ, in which the initial syllable of δήϊος should occupy a short syllable and thus fails to scan as spelled (Chantraine 1958:107). In post-Homeric literature the adjective is found in its Doric form δᾱ́ϊος/δᾳ̂ος in Pindar and tragedy (often substantivized as δᾱ́ϊοι/δᾳ̂οι 'enemies'), beside Alcman's δάβιον 'burning' (i.e. "δάϝιον"), along with the denominative verb δηϊόω 'slay, kill'. The Hesychian gloss δῄειν· πολεμεῖν, φονεύειν implies a noun *δήϊς 'hostility' that is ultimately at the base of δήϊος, as suggested in the etymological dictionaries (*DELG*, *GEW*, *EDG* s.v. δήϊος), which all offer the personal names Δηίφοβος (*Il.*+). According to some scholars (cf. *EDG* s.v. δήϊος), Myc. *da-i-qo-ta* (KN Da 1164) can be read as "Δηιφόντης" (though not actually found in alphabetic Greek) and adds further support for *δήϊς. The formulaic association of δήϊος with πῦρ in the *Iliad* suggests a connection with δαίω 'burn' from PIE *deh_2u- 'catch fire' (*EDG* s.v. δήιος), but Myc. *da-i-qo-ta* shows no evidence of -ϝ-, e.g. *x**da-wi-qo-ta* (*vel sim.*), as Ruijgh (1970:318) observed (but cf. Alcman's δᾱ́βιον below). In fact, none of the Myc. PNs beginning with *da-i-* show any evidence for -ϝ- (Chadwick and Baumbach

1963:181).[119] Beekes (*EDG* s.v.), as he is accustomed to do, thus opts for a non-IE origin of *δήϊς from *dāhi-, with non-Greek -h-. But δήϊος has all the appearance of a good IE word, with familiar morphology and plenty of comparanda, especially in compounds, and so Beekes' conclusion is unnecessary.

Potentially related to *δήϊς is epic δαΐ 'in battle' that is thought to continue the loc. sg. of the noun δαΐς 'battle' (*DELG*, *EDG* s.v. δαΐ). But δαΐς seems to be found only in δαΐ and a clearly secondary acc. sg. δάϊν in Callimachus (*EDG* s.v. δαΐ), which leaves its existence questionable. If δαΐς is indeed related to δήϊος/*δήϊς, the root vocalism with short -α- needs explanation (cf. below). Despite Beekes' and Chadwick and Baumbach's (1963:181) assumption that Myc. *da-i-qo-ta* reflects */dā$^{(h)}$ik^{wh}ontās/*, there is no direct evidence for *dā-*.[120] It may rather be the case that *da-i-qo-ta* actually does reflect */da$^{(h)}$ik^{wh}ontās/*, with the dat. sg. δαΐ as the first member of the compound.[121]

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[119] Other names include *da-i-ta-ra-ro* (KN De 1231), *da-i-wo-wo* (V 1043), and *da-i-ze-to* (Da 1317). With regard to the names in *da-i-*, Chadwick and Baumbach (1963:181) comment that none of the material points to PNs in *da-wi-*, for which there is no evidence in Mycenaean.

[120] In principle, the root shape *dă̄s- cannot be excluded in underlying */da$^{(h)}$ik^{wh}ontās/*. Beekes (*EDG* s.v.) takes δᾰΐφρων 'warlike, fiery; wise, prudent' (*Il.*+) with PIE *dens- 'high mind, power', attributing the meaning 'brave' to a secondary connection with δαΐ. Beekes connects δαι- < *dahi- to Ved. *dasrá-* 'effecting miracles', thinking that Gk. *das-í-* and Ved. *das-ró-* reflect Caland variants and assuming that *dn̥s-i- > PGk. *dahi-*. Sihler (1995:171), on the other hand, offers examples for *-CṇsV- > *-CasV- in Greek, e.g. Gk. δασύς 'thick, rough' < *dn̥s-, beside Lat. *dēnsus* 'thick; close'. Regardless of the Greek outcome of *-CṇsV-, it seems to be the case that δήϊος and related forms belong better with *deh$_2$i- and *deh$_2$u-, so there is no need to assume *-h- in the PN */da$^{(h)}$ik^{wh}ontās/*.

[121] Cf. Risch (1974:220) for discussion of case forms as the first member of compounds in Homer, including δᾰΐφρων 'warlike, fiery; wise, prudent' (*Il.*+).

Corroborating **dai-* in PNs, Beekes offers Δάϊππος (Miletus), Δαϊκράτης (Olbia), and Δαϊλέων/Δαϊμένης (Athens).

Homer attests several PNs in Δηϊ-, all of which have identical or similar-looking names without the first member: Δηϊκόων (1x) beside Κόων, Δηϊοπίτης (1x) beside Οπίτης, Δηΐοχος (1x) beside Οχήσιος, Δηΐπυλος (1x) beside Πύλος/Πύλων, Δηΐπυρος (4x) beside Πύρις, and Δηΐφοβος (21x) beside Φόβος. "Δαϊφόντης" is perfectly acceptable in the hexameter, though never found. Other PNs built with δαι-, however, cannot work, if the second member of the compound begins with a light first syllable, e.g. Δαΐοχος (Sparta) and Δαΐπυλος (Corinth). No PNs in Homer begin with Δᾰϊ- (Δαίδαλος has the diphthong Δαι-). It may well be the case, then, that the PNs in Δηϊ- reflect a metrical lengthening of the sort offered by Chantraine (1958:98), e.g. *ἀνεμόεις (not actually attested) lengthened to ἠνεμόεις 'windy, breezy', facilitating the use of the PNs within the hexameter by lengthening the initial of the shape ⏑⏑⏑ (as opposed to closing the final syllable; cf. also Horrocks 1997:211, West 1997:226, and Hackstein 2010:409 for further discussion). Once the PNs Δηΐφοβος, etc. were established within the tradition, it is thinkable that δηϊ- could have ousted δαϊ- in subsequent derivatives, e.g. Δηιάνειρα (Sapph.+) modeled on ἀντιάνειρα and κυδιάνειρα (*DELG* s.v. δήϊος). Given the confusion between Δηι- and Δαι-, it is thinkable that the adjective δήϊος might also be a substitute of sorts for an earlier *δᾰ́ϊος.

*δᾰ́ϊος could reflect a thematic adjective to the PIE "long-diphthong root" *deh_2i- 'split, divide' (cf. *LIV* s.v.), with the same semantic relationship underlying δέ͜ρω 'flay' (*Il.*+) and δῆρις 'strife' (*Il.*+) to *der-.[122] It is easy to see how the semantics of "split, divide" can develop to "(hostile) rift", as found in δαΐ; but δήϊος, with its formulaic connection to πῦρ in Homer, also reflects the semantics of δαίω 'burn' from PIE *deh_2u- 'catch fire'. Though no Myc. PNs in *da-i-* reflect evidence for -$\digamma$-, Alcman's δᾰ́βιον, found in the Homeric-looking phrase καὶ χεῖμα πῦρ τε δᾰ́βιον (cf. Hinge 2006:112-3, 243), in which -β- represents -$\underline{u}$-, may provide a connection to *deh_2u- for an adjective δήϊος from *$deh_2\underline{u}$-$i\underline{i}o$-. But the two phrases δηΐου ἐκ πολέμου and δήϊον ἄνδρα (as well as the τηϛ-abstract formulae πόλεμον καὶ δηϊοτῆτα# and ἐν αἰνῇ δηϊοτῆτι; cf. Fig. 13 in **IV.A.1.**) have nothing to do with fire, instead belonging beside δαΐ and the semantics of "hostile", and so may continue an original adjective *δᾰ́ϊος from a different root *deh_2i- beside δήϊος 'burning'. It may be the case that δή(͜)ϊος belongs with *deh_2u-, originally, while δαΐ, PNs in Δαι-, and even perhaps the adjective *δᾰ́ϊος 'hostile' (beside *δᾰ́ϊος) ultimately belong to *deh_2i-. In Homer, the adjectives δή(͜)ϊος 'burning' and

[122] Other forms built to the root *deh_2i- reflect a diphthong in the root, e.g. δαίϛ, δαιτός 'banquet', so it can be questioned why *δᾰ́ϊος, etc. do not behave the same way. If *δᾰ́ϊος and other related forms indeed were contaminated with δή(͜)ϊος 'burning' < *$deh_2\underline{u}$-$i\underline{i}o$- (as argued below), perhaps the analogical pressure exerted by δή(͜)ϊος, with hiatus from -$\digamma$-, may have initially prevented diphthongization in δᾰ́ϊος.

*δᾰ́ϊος/*δᾰ̄́ϊος 'hostile', given their general semantic overlap in the context of battle, were easily confused.

It would be quite bizarre, however, that Greek would have inherited two synonymous adjectives *δᾰ́ϊος and *δᾰ̄́ϊος 'hostile'. Instead, it may be possible to combine the data into an acrostatic noun *$d\bar{e}\acute{h}_2i$-/*$d\acute{e}h_2i$- 'division; battle' with "Narten" ablaut. Though reflecting a long-diphthong root, *$d\bar{e}\acute{h}_2i$-/*$d\acute{e}h_2i$- may well have behaved as an *i*-stem noun (cf. δῆμος 'country, township' that does not reflect *i*-stem material), which have Narten parallels elsewhere, including the semantically closely-related noun δῆρις 'strife' < *$d\acute{e}r$-*i*-/*$d\acute{e}r$-*i*- and perhaps also Lat. *volup* 'agreeably, to one's satisfaction' < *$\underset{\sim}{u}\bar{e}\acute{l}h_1$-*p*-*i*-/*$\underset{\sim}{u}\acute{e}lh_1$-*p*-*i*- (cf. **II.F.12.a.** and Vine 2006a for further examples). Based on such a semantic relationship between δῆρις and *δήις, it is certainly thinkable that they may have also shared a morphological relationship with Narten ablaut, with the strong stem of *$d\bar{e}\acute{h}_2i$- leveled in *δήις and continued in δῆειν. If so, a $\underset{\sim}{i}o$-adjective *deh_2i-$\underset{\sim}{i}\acute{o}$- of essentially the same sort as in δήϝϊος (< *$deh_2\underset{\sim}{u}$-$i\underset{\sim}{i}o$-) built to the weak stem *$d\acute{e}h_2i$- and meaning something like "having a rift; hostile" could underlie *δᾰ́ϊος. Finally, the weak stem *$d\acute{e}h_2i$- could survive in the adverb δαΐ, which may continue an endingless loc. sg. *deh_2i (but with full grade in the root) of the root noun, formed before the strong stem was leveled in the noun.

But while **déh₂i-* with a Narten lengthened grade can explain *δήις, *δᾱ́ϊος, and δαΐ́, the adjective δᾱ́ϊος 'hostile' must have a different origin. To date, no satisfying explanation for the required metrical shortening of δη- in the Homeric formulae πυρὸς δηΐοιο and δήϊον πῦρ has been offered.[123] *LSJ* (s.v.) opts for disyllabic scansion of δήιον, with synizesis of -ήι-, which is unlikely for a formula that has all the appearances of being old. Rather than reflecting any Narten properties, the long vowel variant *δᾱ́ϊος (possibly continued in Hsch. τὸ δήϊον· τὸ πολέμου) may actually reflect an earlier metrical lengthening of *δᾰ́ϊος (**deh₂i-i̯ó-*), e.g. in the phrase *δᾰ́ϊον ἄνδρα, after ἄνδρα itself became a fixture of Homeric vocabulary (similarly in ἀθάνατος → ἀ̆θάνατος and ἄνερι → ἄνερι; cf. Hackstein 2010:409). The fact that Homer has no PNs in Δᾰϊ- may also suggest an early lengthening of δᾰϊ- to δᾱϊ- in onomastic material. In addition, the alternations of δηϊ- and δᾰϊ- found in PNs and the Homeric formulae might be suggestive of a contamination of the adjective δήϊος 'burning' (< **deh₂u-*) with *δᾰ́ϊος 'hostile', facilitated by the metrically lengthened *δᾱ́ϊος. In fact, based on the usage in Homer and subsequent texts, the original adjectives δήϊος 'burning' (**deh₂u̯-i̯o-*) and *δᾱ́ϊος/*δᾰ́ϊος 'hostile' (< **deh₂i-i̯ó-*) seemed to have merged into a single adjective δᾰ́ϊος 'burning; hostile' already in the epic tradition, with fluctuating vowel quantity in the initial syllable that was taken advantage of in hexameter poetry, e.g. in the phrases πυρὸς δηΐοιο and

[123] Cf. Rau (2009a) for a different type of metrical shortening in ευ-stem patronymics in Homer, e.g. Πηλέι beside Πηλῆι.

δήιον πῦρ reflecting the morphology of *δᾰ́ϊος 'hostile' but the semantics of δήϊος 'burning'. After Homer, the short-vowel variant *δᾰ́ϊος, though preserved in πυρὸς δηίοιο and δήιον πῦρ, was lost in favor of δᾱ́ϊος that subsequently became entrenched in the epic tradition and inherited by Pindar and the tragedians.

IV.C.3.c.ii. δηϊοτής ← *δηϊτής

The noun *δήϊς 'division; battle' < *$dḗh_2i$-/*$déh_2i$- could have implications for δηϊοτής too: it has been thought to reflect a mechanical derivative in -της to the thematic adjective δήϊος, but its oxytone accent might suggest otherwise. Oxytone accent is only attested in Greek for της-abstracts not directly built to thematic stems, e.g. ἀνδροτής (cf. discussion in **IV.C.3.a.**), the pair βραδυτής/ταχυτής, and ποτής (cf. discussion in **IV.C.2.a.**). In addition, δηϊοτής never means "burning" like its supposed base δήϊος, so it may not have been formed directly to the adjective, as is usually thought. In only 2 of its 30 attestations does δηϊοτής not appear at absolute line end, and 1 of the 2 instances has δηϊοτής line-initially (cf. Fig. 11). No other abstract in -της has such a fixed pattern of use in Homer. Line end, moreover, is perfect for a word of the shape —⏑⏑—×#, suggesting that δηϊοτής, then, may have been a poetic creation at least partially conditioned by metrical utility. The noun *δήϊς 'division; battle' may have originally built a της-abstract *δηϊτής 'hostility', which was unusable in the hexameter. Its metrical restriction, combined with the fact that it no longer followed the familiar pattern in -οτης, led to its reshaping to the more metrically accommodating δηϊοτής, with accentuation

retained on the suffix, just as ποτής did after it was remodeled from ποτόν (on which cf. **IV.C.2.a.** above). A similar remodeling can be seen in *φοινῑκϝεντ-, which was subsequently reshaped to φοινῑκο(ϝ)εντ-, after the suffix -ϝεντ- was generalized to -οϝεντ- (cf. **V.E.**). *φοινῑκο(ϝ)εντ-, however, contains an unusable cretic sequence, hence it was shortened to φοινῑκοεντ- to accommodate the meter (cf. Hackstein 2010:410 for other reshapings, including ἱπποχαίτην → ἱππιοχαίτην, etc.). Similarly, *δηϊτής may have been remodeled to δηϊοτής, retaining accent on the suffix, after -της was generalized to -οτης, just as did ποτής 'drink', after gen. sg. ποτοῦ or ποτῆς was remodeled to ποτῆτος (cf. **IV.C.2.a.**).

If δηϊοτής has the background described above, it would provide evidence that the suffix -οτης was generalized from -της already by Homer. Additionally, analyzing δηϊοτής as a Homeric reshaping of an original *δηϊτής is attractive, in that it offers an explanation for its otherwise anomalous oxytone accent and furthermore shows that earlier in Greek -της could originally be used with *i*-stem bases (and also consonant-stem bases, if ἀνδροτῆτα continues *$h_2$nr̥-tāt-; cf. **IV.C.3.a.ii.**), in addition to the well-known thematic and *u*-stem bases.

IV.C.3.d. ἰότης 'will, desire'

ἰότης 'will, desire' occurs almost entirely in the dat. sg., mostly in the Homeric phrase θεῶν ἰότητι ⏑—×# 'by the will of the gods', which has all the appearances of being an old formula (cf. Fig. 13 in **IV.A.1.**). Aufrecht (1852) first identified ἰότης as

problematic, in that it lacks a synchronic adjectival base in Greek, and Schwyzer's (1939:528n8) idea that ἰότης may belong with Ved. *iṣanyáti* 'stimulate, urge' and Gk. ἰαίνω 'warm, melt', was dismissed earlier in **IV.B**. Leumann's (1950:127-8) analysis of ἰότης as a product of reanalysis of δηϊοτής as δὴ ἰότης can also be rejected on semantic grounds. Hence to date, two reasonable sources have been offered (*EDG* s.v. ἰότης): it may be based on an unattested adjective *ἰός 'wanting, wishing', belonging with Skt. *iṣa-* 'id.' (SiraUp.) to PIE *$h_2e\underset{\wedge}{i}s$- 'seek', according to Petersen (1922:51 with refs.); or it may be related to the verb ἵεμαι 'desire, want' (< PIE *$\underset{\wedge}{u}e\underset{\wedge}{i}H$- 'move towards, desire'), in which case the short vowel in ἰότης vis-à-vis ἵεμαι warrants explanation.[124] Either way, it seems to be the case that ἰότης must have been formed to an unattested adjective *ἰός early in Greek.

Chantraine (1933:294) assumed that ἰότης < **isotāt-* reflects **istotāt-* as the "produit d'une dissimilation", based on comparison with Skt. *iṣṭá-* 'desired'; but it is hard to see how such a dissimilation of **istotāt-* could give **isotāt-* in Greek, when similar sequences, e.g. πιστότης 'good faith, honesty' (Hdt.+), are otherwise unaffected. In addition, Homeric Greek — unlike Indo-Iranian — did not seem able to build της-abstracts to *to*-participles. If ἰότης is indeed related to Skt. *iṣa-* and PIE *$h_2e\underset{\wedge}{i}s$-, it would have to be a direct formation to an adjective *ἰός (< *h_2is-ó-); but there is no evidence for

[124] *LIV* (s.v. *$\underset{\wedge}{u}e\underset{\wedge}{i}h_1$-), followed by *EDLIL* (s.v. *via*), cites the root as *$\underset{\wedge}{u}e\underset{\wedge}{i}h_1$- with *-$h_1$-, but there seems to be no evidence supporting it.

an adjective in Greek to this particular root, and even the Indic evidence is late, throwing the likelihood of PIE *h_2is-ó- into doubt.

But there is evidence, in fact, suggesting *ἰός belongs to *$\underset{\sim}{\mu}e\underset{\sim}{i}H$-. Petersen (1922:51) rejected the idea of Fick (1890-1909:543) that ἰότης reflects *ϝιότης and is thus related to ἵεμαι, based on both the lack of evidence for ϝ- in any dialect and the discrepancy in vowel length in the verb.[125] Beekes (*EDG* s.v. ἰότης) offered support for initial ϝ- in Boeot. PN θειοϝιοτο (*IG*^{7} 3511; probably 6th/5th c. BCE), beside which can also be added θειο[.]ιοτ[--] (*SEG* 19.348; 6th/5th c. BCE) that may also reflect "θειοϝιοτος" (but the reading of -ϝ- is unclear; cf. Vénencie 1960:603), implying that ἰότης indeed belongs with *$\underset{\sim}{\mu}e\underset{\sim}{i}H$-. But Chantraine (*DELG* s.v. ἰότης) objected that *ἰότης should be expected, if related to ἵεμαι (cf. Lat. *invītus* 'unwilling'). His objection, though, can be dismissed if *ἰός reflects a zero-grade *$\underset{\sim}{\mu}iH$-ó- to the root *$\underset{\sim}{\mu}e\underset{\sim}{i}H$- and Harðarson's idea for explaining the vowel length is correct.

Other nominal forms to *$\underset{\sim}{\mu}e\underset{\sim}{i}H$- include Lat. *via* 'road' (*XII Tab.*), Osc. **víú** and **amvían(n)ud**, Umb. **vea**, **via**/*via,* SP *viam*, Ved. *deva-ví-* 'invigorating the gods', *pada-ví-* 'guiding, driving the pace', and *pratī-ví-* 'gladly accepting, receiving acceptance', with verbal forms Ved. *vī-* 'go, drive', YAv. *viia-* 'to pursue' (*EWAia* s.v.*vay-*[1]), and

[125] Harðarson's (1993b) idea that ἰ- in ἵεμαι results from vowel contraction of a 3pl. middle *$\underset{\sim}{\mu}i$-*uiH-entoi*, with aspiration copied from ἵημι 'send, hurl', is the best explanation for its length.

Lith. *výti* 'to drive' (*EDLIL* s.v. *via*).[126] Ved. *(-)vī-* suggests that **u̯eiH-* built a root noun **u̯eiH-/*u̯iH-*, to which Italic made a zero-grade derivative **u̯iH-éh₂-* of the "*fuga*"-type (Vine 1998b:259). It is then thinkable that **ἰός* continues a thematic possessive adjective **u̯iH-ó-* 'having desire; direction' built to the same zero-grade of the root. If so, *ἰότης* reflects a simple derivative in *-της* built to **u̯iH-ó-* > **ἰός* before it was lost.

Though *ἰότης* reflects a standard qualitative abstract built to a thematic adjective, rather than an unusual derivational pattern as argued for *ἀνδροτῆτα* and *δηϊοτής*, it still seems to continue old material. The abstract is well grounded in Homeric poetry and could reflect one of the older abstracts built to a thematic stem in Greek.

[126] Gk. *οἶμος* 'streak; path, road' (Hom.+) has a murky history (cf. *EDG* s.v.), but, given the fact that the Homeric phrase *δέκα (ϝ)οἶμοι κυάνοιο* (*Il.* 11.24) requires *ϝ-* in hiatus (*EDG* s.v. *οἶμος*), it may well belong here too, continuing **u̯óiH-mo-* with loss of **-H-* by the Saussure effect.

V. Secondary *t*-Stems and PIE "*-*tāt*-"

All the languages attesting *tāt*-abstracts reflect formations primarily to thematic adjectives, with accentuation on the thematic vowel of the base adjective (where accentual information is available), e.g. Ved. *devátāt-/devátāti-* 'collection of the gods' (cf. **III.A.5.**), perhaps OAv./YAv. *amərətatāt-/amərətāt-* 'immortality' (cf. **III.A.5.**), and Gk. φιλότης 'love; sexual love' (cf. **IV.A.**). Only a handful of old-looking abstracts are built to athematic stems, which show accentual anomalies of sorts, e.g. Ved. *śáṃtāti-* 'health; luck', with accent on the base (cf. **III.D.4.**), and Gk. ἀνδροτής 'manliness', with accent on the suffix (cf. **IV.C.3.a.**). It is not immediately clear whether *-*tāt*- was specialized as an abstract suffix to thematic adjectives already in PIE, or if the pattern was independently formed in each language, and so the nature of the relationship between *-*tāt*- and thematic stems must be investigated.

Additionally, *-*tāt*- may have had a collectivizing-individualizing function (cf. **V.D.4.**), based on some of the material found in Italic (e.g. *cīvitās* 'state; city' beside *cīvis* 'citizen'), in Indo-Iranian (e.g. *devátāt-/devátāti-* 'collection of the gods' beside *devá-* 'god' and OAv. *kəuuītāt-* 'class of, group of *kavis*' beside OAv. *kəuui-* '*kavi*') and even in Greek (e.g. νεότης 'youth; collection of youths' beside νεός 'new'). The individualizing function (e.g. Lat. *Voluntās* 'Desire' and Pg. *herentas*/Osc. **herentateís** 'Herentas (i.e. "Desire")' in **II.F.1.c.i.** and OAv./YAv. *amərətāt-/amərətatāt- hauruuatāt-*

/hauruuāt- 'Immortality and Wholeness' in **III.E.**) which lends itself well to the collective semantics underlying **-tāt-* may be suggestive of its history as a complex suffix, since collectivizing-individualizing semantics are found with other simple suffixes, namely *t*-stems.

As suspected by all who have studied the suffix, **-tāt-* can be analyzed as a composite of **-tā-* plus **-t-* or **-ti-*. But given that the *i*-stem variants found in Latin, e.g. *cīvitātium* 'of citizens' (cf. **II.C.**) and Vedic, e.g. *devátāti-* 'collection of the gods' (cf. **III.C.**) are secondary inner-Latin and inner-Vedic creations, there is no evidence for a PIE *i*-stem suffix **-tāti-*. PIE **-ti-*, moreover, built verbal abstracts, making it very difficult to see how it could have been involved in a suffix **-tāti-*, especially given the fact that there is very little secondary *t*-stems in PIE to begin with. Plain **-t-*, on the other hand, had the sort of collectivizing-individualizing semantics seen in some *tāt*-abstracts. Hence, **-tāt-* might best be analyzed as a kind of complex suffix built with a plain *t*-stem. PIE **-tāt-* could, in principle, be segmented in any number of ways, including **-tā-* + **-(o)t-*, **-(o)t-* + **-ā-* + **-(o)t-*, etc., which will be explored below.

The relationship assumed by scholars between *tā*-abstracts and *tāt*-abstracts suggests that **-tāt-* is best and most simply segmented as an underlying **-tā-* plus a *t*-stem suffix (cf. **I.D.**), but it is still unclear what the derivational base **-tā-* reflects and

why it was augmented with a *t*-suffix. The long vowel in **-tāt-* deserves attention too, in that it is not immediately clear what it reflects. It may simply be **-teh$_2$-*, or it might also continue the sort of "presuffixal" lengthening common in "decasuative" derivatives, in which suffixes are added directly to case forms (cf. Weiss 2009:266-324 *passim*). Analysis of athematic **-tāt-* in this way, however, is at first glance problematic, given the fact that a large number of the well-known decasuative derivatives are thematic, e.g. Lat. *-īko-* and *-ōno-*. A few athematic examples are found too, though, e.g. some *d*-stems like Lat. *hērēs, hērēd-* 'heir' and *-iō-/-tiō-*. If **-tā-* underlying **-tāt-* continues the PIE *teh$_2$-* suffix, as seems probable for the *tā*-abstracts, it may also be the case, then, that **-tāt-* can be thought of as a de-instrumental derivative built with a *t*-suffix. Before investigating such a *t*-stem suffixation to *teh$_2$*-abstracts, it will be helpful to examine secondary *t*-stems in greater detail, including examples built to athematic instrumental case forms.

V.A. PIE Secondary *t*-Stems

The study of PIE *t*-stems has been somewhat of a growth industry in recent years, with several recent detailed treatments of the suffix. Rieken (1999:83-169) surveyed the *t*-stem formations in Anatolian, focusing on primary *t*-stems, Irslinger (2002:37-68) reviewed all the dental suffixes in Old Irish, including *t*-stems, and Vijūnas (2009) examined primary *t*-stems in Indo-European. The two best-known *t*-stem varieties are primary ablauting stems, variously with **-ot-/*-et-/*-t-*, e.g. Lat. *nepōt-* 'grandson', *neptis*

'granddaughter' and cognates; and *t*-stems in second members of compounds, e.g. Ved. *devastút-* 'praised by the gods' and Hitt. *šāudišt-* 'suckling; yearling' (but cf. **V.C.3.** for a different view). Relatively little attention has been paid to secondary *t*-stems.[127]

Nussbaum (1996, 2004b) has recently sketched the background of secondary *t*-stems to both thematic and athematic stems, which fall into three main types: 1.) masculine *t*-stem substantives, 2.) feminine *t*-stem abstracts, and 3.) deverbative *t*-stems (which are not of interest here). The first class is exemplified in Fig. 14.

**steiṗo-* (Gmc. **steifa-* > OE/MHG *stīf-* 'stiff') → Lat. *stīpes* 'branch'
Gk. *ἀργός* 'shining' → Hom. acc. *ἀργῆτα/ἀργέτα* 'dazzling white, glistening' and dat. *ἀργῆτι/ἀργέτι*[128]
Gk. *γυμνός* 'naked' → *γυμνής, -ῆτος* 'light-armed soldier'
Lat. *dīus* 'shining' → *dīves, -itis* 'rich'
i-stem Ved. *hári-* 'yellow' → *harít-* 'tawny; bay horse'

The second class of feminine *t*-stem abstracts is illustrated in Fig. 15.

**neiu̯o-* 'low' (Gk. *νειός/νειόθεν* 'from deep') → **niu̯et-* (Ved. *nivát-* 'depth')
**pro-u̯o-* 'going forth' (ON *frár* 'fast', OHG *frō* 'happy') → **prou̯et-* (Ved. *pravát-* 'way forward')
Hitt. *ḫappina-* 'rich' → *ḫappinatt-* 'wealth' (c)
i-stem Hitt. *karuili-* 'former' → *karuiliyatt-* 'status quo ante' (c)

[127] Rau (1998:159) has also analyzed the perfect active participle in **-u̯ot-* in terms of an underlying amphikinetic *t*-stem derivative to a *u*-stem, but he did not discuss secondary *t*-stems in any detail.

[128] *ἀργός*, dissimilated from **ἀργρός*, is traditionally analyzed as a Caland *ro*-adjective, beside Ved. *ṛjrá-* 'shining'. Vine (2011), however, has recently shown that the dissimilation assumed for **ἀργρός* is rather unlikely, based on the phonological environments expected for dissimilation in Greek. *ἀργός* may be better thought of as actually continuing a full-grade or zero-grade thematic adjective **h_2(e)rĝ-ó-*, as conceived by Nussbaum.

The data show that secondary *t*-stems are primarily built to thematic adjectives and seem to show suffix ablaut, namely **-ēt-/*-et-* and **-ot-/*-et-*, after contraction with the thematic vowel; the *t*-stems built to athematic (*i*-stems) reflect only plain **-t-*.

Nussbaum (2004b) reasons that the thematic adjective **h_2(e)rĝe/o-* must have built an ablauting *t*-stem **h_2(e)rĝe-ot-/*h_2(e)rĝe-t-*, where the contraction product ***-eo-* gives **-ē-*, just as ***-oe-* gives **-ō-*. *t*-suffix substantivizations to adnominals were apparently built with amphikinetic **-ot-/*-t-* (cf. Yoshida 2000), based on the archaic-looking *t*-stem derivative to **di̯éu̯-/*diu̯-* 'day; heaven' **di̯éu̯-ot-* (> Hitt. *šīwatt-* 'day, Sun God', CLuv. *tiwat-* 'id.') that can only point back to **-ot-*, beside the weak stem **di̯u-t-* (> Ved. *dyút-* 'light, brilliance').

V.B. Latin *salūs* 'health'

Lat. *salūs, -ūtis* 'health' is attested from the beginning of Latin literature, notably in the phrase *salutem servare* (e.g. Pl. *Capt.* 530, etc.), and the deity "Salus" (beside Pg. **saluta**; cf. below) plays a significant role in Roman religion, with several dedicatory inscriptions to her, e.g. SALVTE (*CIL* I^2.373, 2nd c. BCE), SALVTES.POCOLOM (*CIL* I^2.450, 3rd c. BCE), ARA.SALVTVS (*CIL* I^2.62, 3rd/2nd c. BCE), among others (cf. Carandini 2002:402-3 for some additional religious connections). *salūs* sits beside Ved. *sarvátāt-/ sarvátāti-*, OAv./YAv. *hauruuatāt-*, and Gk. ὁλότης (Arist.+), all built to the root **selh$_2$-*

'whole' and meaning something like 'wholeness';[129] but *salūs* reflects a stem in *-ūt-* rather than the productive abstract suffix in *-tāt-*. *-ūt-* has long been identified as morphologically problematic, in that *-ū-* with a secondary *t*-stem is unparalleled elsewhere within Latin.[130]

Meillet (1928:42) first suspected that *salūs* continues something morphologically old, identifying the religious and poetic formula in *salutem servare* and further semantic and morphological parallels in the etymologically unrelated OPr. feminine abstract noun *kailūstiskun* 'health' (acc. sg.), continuing a suffix **-ūt-to-* also with a segment **-ūt-* (on which cf. below); but he could not offer a source for **-ūt-* itself. Watkins (1993:470) further connected the *ūt*-suffix in *salūs* with Luv. *ḫīrūn, ḫīrūt-* 'oath' (n). Following Leumann (1977:349, 543) that **salvere* must be denominative to a *u*-stem **salu-*, however, Schrijver (1991:296) and Klingenschmitt (1992:126n63) independently adduced the existence of an unattested verb **salvere* with a perfect passive participle (PPP) **salūtus*, based on the existence of the adjective *salūbris* 'healthful; wholesome' (Pl.+). *salūs*, then, can reflect what Schrijver (1991:196) called a deverbative *ti*-stem built to the verb **salvere*, implying that *salūs* is built to the PPP of **salvere*. Likewise,

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[129] Cf. **I.A.** on the very low likelihood that the word-equation Ved. *sarvátāt-* : OAv./YAv. *hauruuatāt-* : Gk. ὁλότης (Arist.+) actually belongs to PIE.

[130] Cf. also *d*-stems like Lat. *hērēs, hērēdis* 'heir' (m), etc., with presuffixal lengthening (Weiss 2009:304-5). The question of *d*-stems and their relationship to *t*-stems remains open but cannot be pursued here (cf. **VI.C.**). The *s*-stem *tellūs, ūris* 'earth' (f) also seems to show lengthening before the suffix but lacks a good explanation (*EDLIL* s.v.).

Schrijver (1991:296) compared the similar patterning found in *statuere* 'to fix' (Pl.+) with PPP *statūtus* (Pl.+) from *status*, *-ūs* (m) 'position; state' (Pl.+). Mignot (1981:338) offered *volvere* 'to roll' beside PPP *volūtus* and *volūbilis* 'rolling' (Verg.+) as a parallel to **salvere* ~ **salūtus* ~ *salūbris* that would seem to show the pattern by which *volvere*, with PPP *volūtus*, is the basis for the adjective *volūbilis*;[131] but *volūbilis* is late, as revealed by the lack of dissimilation of *-lis* to *-ris* (cf. Weiss 2009:155-6). Furthermore, there is no denominative **volūt-* in parallel to *salūs*, and so *volūbilis* is consequently of little use in establishing the derivational pattern under consideration.

There are, additionally, two larger problems with Schrijver's approach: First, there is no pattern by which PPPs serve as the basis for denominative *ti*-stems in Latin, e.g. **salū-ti-*; for example, there is no **statūti-* built to *statūtus*. Equally important, *salūs* does not show any trace of original *i*-stem inflection.[132]

Secondly, *salūbris* (with archaic dissimilation from **-blis*; cf. Leumann 1977:348) need not depend upon the existence of a verb **salvere*. Adjectives in *-lis/-bilis/-bris* tend to be deverbative and passive in sense, though not all of them are connected to verbs.

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[131] The discussions of Latin verbs in *-uō* have not been clear in distinguishing verbs built to denominative *u*-stems in **-i̯o-* from thematized *u*-presents. Verbs like *salvere* < **salu-i̯e/o-* built to *u*-stems should have been phonetically realized as */saluō/*, while a thematized *u*-present in *volvō* < **u̯olu̯-e/o-* (*EDLIL* s.v.) should come out */u̯olu̯ō/*. Despite the phonetic difference between verbs in *-uō* and *-u̯ō* in Latin, there were apparently no divergent effects for the analogical patterns discussed above.

[132] PRO SALVTI (*ISc*M^5 23; Capidava, mid-2nd c. CE) is of no value, given the proclivity of consonant stems to adopt *i*-stem inflection, e.g. COVENTIONID (SCdB), AIRID (CIL I^2.38), etc., as happened even in some *tāt*-abstracts in Latin, e.g. gen. pl. *cīvitātium* (cf. **II.C.**).

Some are apparently derived from PPP stems, e.g. *fossilis* 'obtained by digging' (Var.+) from *fossus* and *volātilis* 'able to fly' (Lucr.+) from *volātus*. But *lūgubris* 'of mourning' (Ter.+) ~ *lūgēre*, rather than *ˣlūgēbris*, and *terribilis* 'frightful' (Enn.+) ~ *terrēre*, rather than *ˣterrēbilis*, seem to show that the pattern involved root + pseudo-thematic stem vowel + suffix. *ˣsalvebris* or *ˣsalvibris* might be expected in place of *salūbris*, if built from a verb **salvere*. Even if *salūs* is somehow based on a stem *salūto-*, it need not be the PPP base to a verb *ˣsaluere* (on which cf. below).

A large number of adjectives in *-bilis* are based on 1st-conjugation stems (cf. Leumann 1977:349), e.g. *exōrābilis* 'placable' (Pl.+) from *exōrābulum* 'means of winning over' (Pl.+), including ones like *voluptābilis* (Pl.+), which, without a verb *ˣvoluptāre*, is clearly based on the noun *voluptās* (on which cf. **II.F.10.**). It is conceivable that an analogical relationship of the sort *-āre* (beside PPP *-ātus*) : *-ābilis* :: *-ūtus* : *x* was involved, where *x* = *-ūbilis* (*-ūbris*), thereby doing away with the need for a parallel verb in *-uere*. In this way, one could imagine that *salūbris* could be based on an old adjective **salūtus*, even in the absence of a thematic verb **salvere*. But despite the derivational productivity of the family of words directly related to *salūs*, no expected derivatives to the stem can be found in Latin, e.g. a productive *tiō*-abstract *ˣsalūtiō*. Hence, there is no good reason to assume the existence of **salūtus*. And consequently, *salūbris* has no

analogical model on which it can be based. In fact, *salūbris* is isolated within Latin and cannot thus shed light on *salūs*, which could then quite possibly reflect something old, as Meillet and Watkins suspected.

The *t*-stem suffix in *salūs*, along with the descriptive presuffixal lengthening before the suffix, may help in understanding how **-tāt-* is related to **-teh$_2$-*, and so a detailed investigation of the background of *salūs* and its potential ramifications for **-tāt-* will now be explored.

V.B.1. PIE **sol(h$_2$)-u̯-o-/*sḷh$_2$-eu̯-o-*

The root etymology of *salūs* is clear Nussbaum (1997:186-7): it belongs with the thematic adjective *salvus* 'safe', next to Osc. **salavs**, Mc. *salaus*, U. *saluuom*, etc. (cf. *EDLIL* s.v. *salvus*, Untermann 2000:652-4), along with Gk. ὅλος/Ion. οὖλος 'whole, all', Ved. *sárva-* 'all', OAv. *hauruua-* 'all', and perhaps Alb. *gjallë* 'powerful, fat, lively'.[133] TA *salu*/TB *solme* 'wholly' (adv.) likey also belong here (cf. below on TA *salu*).[134]

[133] Beekes (*EDG* s.v.) suggests ὁλοός·φρόνιμος καὶ ἀγαθός 'clever and able' (Suid., Hsch.) and ὁλοεῖται· ὑγιαίνει 'is healthy' (Hsch.) may belong with ὅλος. If so, the stem ὁλο- must reflect **solh$_2$-ou̯-o-* rather than **sḷh$_2$-eu̯-o-* found in Italic.

[134] MW *holl*/Bret. *oll*/Corn. *oll* 'all, entire' continue **sol(h$_2$)-i̯o-* vis-à-vis OIr. *huile/uile* 'all' (cf. Schrijver 1995:323-4). CLuv. *šalḫitti-* 'growth' and Hitt. *šalli-* 'great'; OIr. *slán* 'sound'/'soundness' (n) point to a *no*-stem **sḷh$_2$-no-*. The same *no*-stem with *o*-grade **sol(h$_2$)-no-* perhaps underlies Osc. **sullus**/Pg. *solois* 'all' (cf. Untermann 2000:714-5) and Lat. *sollistimus* 'entirely adequate', though they could also reflect **-u̯o-* (Nussbaum 1997:192). TB *solme* 'wholly', beside Khot. *harma-* 'all, any', continue a *mo*-stem.

Descriptively, the data point to a thematized *u*-stem adjective to the root *$*selh_2$*- 'whole' with PK$^{e/z}$ ablaut.[135] *$*sol(h_2)$-$\underset{\sim}{u}$-*o*-, built to the strong stem, with *o*-grade in the root, is found everywhere but in Latin, which continues the zero-grade *$*s\underset{\circ}{l}h_2$-$\underset{\sim}{u}$-*o*- or *$*s\underset{\circ}{l}h_2$-*e*$\underset{\sim}{u}$-*o*- .[136] *salvus*, and even perhaps Osc. **salavs**, Mc. *salaus*, and U. *saluuom*, reflect the prevocalic development of *$*\underset{\circ}{l}H$ (cf. Weiss 2009:104-5), which, in principle, could continue the weak stem of a thematic derivative *$*s\underset{\circ}{l}h_2$-*e*$\underset{\sim}{u}$-*o*- to the PK$^{e/z}$ noun, with full grade of the suffix subsequently lost by syncope (Nussbaum 1997:186-7).

Alternatively, *salvus* could also reflect *$*s\underset{\circ}{l}h_2$-$\underset{\sim}{u}$-*o*-, with zero grade in both the root and *u*-stem suffix. There is some evidence for an alternate development of the "long syllabic resonants", whereby interconsonantal *-$\underset{\circ}{R}H$- gives *-*aRa*-, e.g. *palma* 'palm of the hand' < *$*palama$* < *$*p\underset{\circ}{l}h_2$-*mo*-; cf. Gk. παλάμη (Weiss 2009:110), which may well be the case with *$*s\underset{\circ}{l}h_2$-$\underset{\sim}{u}$-*o*-, assuming accent retraction. If so, *$*s\underset{\circ}{l}h_2$-$\underset{\sim}{u}$-*o*- would give an Italic *$*salau$*-, whence Lat. *salvus*, etc. by syncope.[137]

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[135] The root *$*selh_2$*-, often cited with h_2, has no clearly diagnostic forms suggesting the laryngeal h_2. If CLuv. *šalḫitti*- 'growth' is indeed related to Lat. *salūs*, as Craig Melchert (p.c.) believes, it would secure h_2 in *$*selh_2$*-.

[136] In the environment -*oRHC*- laryngeals are expected to be lost by the "Saussure Effect". *$*sol(h_2)$-$\underset{\sim}{u}$*o- with *o*-grade is hence a candidate for laryngeal loss and as such *h_2 will be put in parenthesis in subsequent citations of the form (cf. Nussbaum 1997:186-92).

[137] For the "Leiden School" and others who do not believe in the secondary development *-$\underset{\circ}{R}H$- > *-*aRa*-, the otherwise regular outcome of *$*s\underset{\circ}{l}h_2$-$\underset{\sim}{u}$*o- should be *$*slā\underset{\sim}{u}$*o-, which could be expected to give Lat. *^{x}lāvus*, though good examples of inherited *$*sl$*- are lacking (i.e. those not mired in "*s*-mobile" problems; cf. Southern 1999:79-82). Instead of assuming, *-*aRa*- from long syllabic resonants, Leiden followers posit

It is very difficult to determine whether *salvus* reflects *$s\underset{\circ}{l}h_2$-$\underset{\frown}{u}$-o- or *$s\underset{\circ}{l}h_2$-$e\underset{\frown}{u}$-o-, and the evidence suggests that plain *-$\underset{\frown}{u}o$- existed beside *-$e\underset{\frown}{u}o$- (cf. Nussbaum 1997:187). In Latin, some forms continue a suffix *-$V\underset{\frown}{u}o$-, e.g. *calvus* 'bald' (Pl.+) < **kal-*$e\underset{\frown}{u}o$-, *fulvus* 'tawny' (En.+) < **fol-*$e\underset{\frown}{u}o$-, while others reflect plain *-$\underset{\frown}{u}o$-, e.g. *clīvus* 'slope' (Pl.+) < **kle*$\underset{\frown}{i}$-$\underset{\frown}{u}o$-, *rīvus* 'stream' (Pl.+) < **re*$\underset{\frown}{i}$-$\underset{\frown}{u}o$- (but cf. Eichner 1988-90:214 with nn. 91 and 94 for **ri*-$\underset{\frown}{u}o$- based on his metrical reading of RIVOIS in the "Duenos" inscription).[138] Like *$s\underset{\circ}{l}h_2$-$\underset{\frown}{u}$-o-, *set*-roots are ambiguous between *-$\underset{\frown}{u}o$- and *-$e\underset{\frown}{u}o$-, however, e.g. *flāvus* 'golden, pale yellow' (Enn.+) < *$b^h\underset{\circ}{l}h_1$-$\underset{\frown}{u}o$-, *rāvus* 'grey' (Var.+) < *$\hat{g}^h\underset{\circ}{r}h_1$-$\underset{\frown}{u}o$- (cf. Weiss 2009:297-8 for a list of forms). While it is unclear exactly what *salvus* continues, it is clear enough that an older PK$^{o/z}$ *u*-stem noun underlies it.

The evidence suggests an ablaut in the $\underset{\frown}{u}o$-adjective *$sol(h_2)$-$\underset{\frown}{u}$-o-/*$s\underset{\circ}{l}h_2$-$e\underset{\frown}{u}$-o- that may reflect a "neo"-proterokinetic *u*-stem noun *$sól(h_2)$-u-/*$s\underset{\circ}{l}h_2$-eú-, which was thematized in PIE (cf. Watkins 1993:197). In turn, *$sólh_2$-u-/*$s\underset{\circ}{l}h_2$-eú- can reflect an earlier *o/e* acrostatic noun *$sólh_2$-u-/*$sélh_2$-u-, with reshaped ablaut common to roots in *CVR(H)*, e.g. Ved. nom. *dā́ru*, gen. *dróḥ* 'tree' (n) < **dór-u/*dr-éu-s*. The general pattern whereby thematized *u*-stem adjectives in *-$\underset{\frown}{u}o$- sit beside *u*-stem nouns has parallels (cf.

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full-grade suffixes *-$e\underset{\frown}{u}$-, and because *$s\underset{\circ}{l}h_2$-$\underset{\frown}{u}o$-/*$s\underset{\circ}{l}h_2$-$e\underset{\frown}{u}$-o- may actually reflect such a full grade in the suffix, there are no problems with the word by their way of thinking.

[138] A form like **kal*-$\underset{\frown}{u}o$- with plain *-$\underset{\frown}{u}o$- would be expected to give **kallo*- in Latin, with assimilation in the consonant cluster. The cluster *-lv-* without assimilation can only come from *-*lVv*-, with syncopation of the vowel (cf. Nussbaum 1997:190).

Chantraine 1933:122-4), e.g. **k̂ér-u-/*k̂ór-u-* 'horn' beside **k̂er-u̯-ó-* 'horned' (cf. Neri 2003:79 n. 195 with references) and Gk. ἄστυ 'town' beside ἀστός 'townsman'.

salūs would seem to continue a secondary *t*-stem of the feminine abstracting type built to the weak stem of the PK$^{o/z}$ *u*-stem **sólh$_2$-u-/*s̥lh$_2$-eú-*. Such a *t*-stem built to the expected weak stem **s̥lh$_2$-eu̯-t-* would give Lat. **saleu̯t- > salūt-* directly. There are certain traces of weak stems with full grade of the suffix serving as the base for thematic derivatives, e.g. Gmc. **dr-eu̯-o-* 'tree' (Go. *triu*, OIc. *tré*, OE *tréow*) and Gk. δένδρεον (cf. Pinault 2003:173-4), based on the weak stem **dr-éu̯-*. But while such a pattern may give support for **s̥lh$_2$-eu̯-o-*, which may already exist as such in Lat. *salvus*, no secondary plain *t*-stems built to weak stems with full-grade suffix are known.

V.B.2. PIE **s̥lh$_2$-u-t-* → **s̥l-uh$_2$-t-* and Laryngeal Metathesis

If Lat. *salūs* preserves a secondary *t*-stem built to the zero grade of the PK$^{o/z}$ *u*-stem **s̥lh$_2$-u-*, the source of the long vowel in the suffix still remains opaque. Watkins (1993:197) speculated that laryngeal metathesis may be the source, e.g. **s̥lh$_2$-u-t-* → **s̥luh$_2$t- > salūt-*, though he admits that this phonological development for Latin is uncertain, since there is little direct evidence for the development. There is good support for metathesis in the environments *CHiC* and *CHuC* (cf. Weiss 2009:114), but a similar development for *R̥HiC* and *R̥HuC* is not as certain. Rasmussen (1989:84, 87) offered

some support for metathesis in *$g^{w}rh_2$-$\underset{\cdot}{u}$-*tó-s* > *$g^{w}r\underset{\cdot}{u}h_2$*tós* > *$g^{w}ruh_2$*tós*, as found in Lat. *brūtus*, but Neri (2003:95n244) is skeptical, citing counterexamples against phonological metathesis, e.g. Ved. *purú-* 'full' < *$p\underset{\circ}{l}h_1$-*ú-*, *gurú-* 'heavy' < *$g^{w}\underset{\circ}{r}h_2$-*ú-*, and *urú-* 'wide' < *$\underset{\cdot}{u}\underset{\circ}{r}h_1$-*ú-* that meet the *C*$\underset{\circ}{R}$*HuC* environment requirement for metathesis in cases with consonantal desinences. Schrijver (1991:249) is more optimistic, though still with reservations, for a Latin (and possibly Italo-Celtic) development of *-*Hú-* > *-ū-*, if under stress, e.g. Lat. *grūs* 'crane' (beside Lith. *gérvė*, OPr. *gerwe* < *$\hat{g}rh_2u$-s*; cf. Schrijver *loc. cit.* for other examples).

Rasmussen's example of Lat. *brūtus* 'heavy; dumb', which may ultimately be based on *$g^{w}\underset{\circ}{r}h_2$-u-to-*, may offer indirect support for metathesis (cf. Vine 1982:316 with n. 3). *$g^{w}\underset{\circ}{r}h_2$-u-to-* should give **baruto-* in Greek; but Hom. *βρίθω* 'be heavy', *βρῑθύς* 'weight', and *βριαρός* 'strong' point to the existence of a *$g^{w}rih_2$-, possibly from a zero-grade Caland form *$g^{w}\underset{\circ}{r}h_2$-i-*. Perhaps there is reason to adduce a parallel *u*-stem variant *$g^{w}ruh_2$-, but such a form is only speculative, despite the assumed development in *EDLIL* (s.v. *brūtus*). Class. Skt. *gurutā-* 'weight' (Suśruta) and Hom. *βαρύτης* 'id.' seem to weigh against laryngeal metathesis, but *gurutā-* is late, and *βαρύτης* may have been an analogical creation in Homer beside *ταχυτής* 'quickness' (cf. **IV.A.**).

*$s\underset{\circ}{l}h_2$-*u-t-*, if accented on the suffix vowel *$s\underset{\circ}{l}h_2$-*ú-t-*, could be expected by Schrijver's rule to metathesize to *$s\underset{\circ}{l}úh_2t$- in Latin. But if so, it is not very likely that it could give *salūt-*. After metathesis, *$s\underset{\circ}{l}úh_2t$- should give *$slúh_2t$-, with non-syllabic *-*l*-, which then gives *$slūt$- and probably $^{x}lūt$- in Latin. *brūtus* must have undergone a similar development from *$g^w\underset{\circ}{r}h_2$-*u-tó*-.[139] Hence, even if metathesis could be predicted for *$s\underset{\circ}{l}h_2$-*u-t*-, it is not at all the case that *salūs* can be the ultimate outcome of a metathesis product of *$s\underset{\circ}{l}h_2$-*u-t*-.

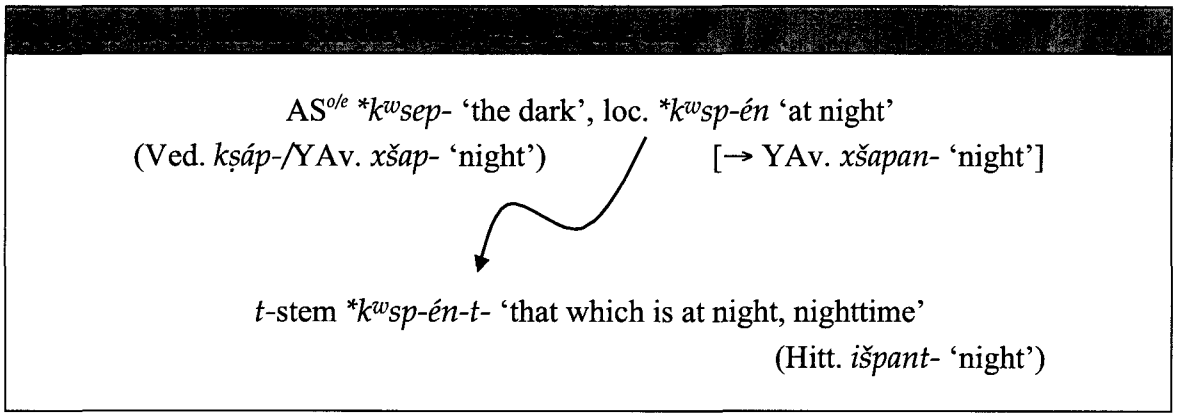
brūtus, moreover, assuming *$g^w\underset{\circ}{r}h_2$-*u-tó*- with the typical oxytone accent for denominal possessive adjectives in *-*to*-, cannot be explained by Schrijver's rule: it would have given *$\underset{\smile}{u}aruto$- (*vel sim.*) in Latin. While laryngeal metathesis is a potentially interesting idea, it creates more problems than it solves in the above examples. The vowel length in *salūs*, with descriptive "presuffixal" lengthening, may have an easier solution. Other similar lengthenings before derivational suffixes have been explained via "decasuative" derivation, in which suffixes are added directly to case forms, giving the observed vowel length immediately before the suffix. *salūs*, along with perhaps *mātūta* (cf. **V.C.5.** below), may be one such derivative built with a secondary *t*-stem whose derivational history may also have implications for PIE *-*tāt*-.

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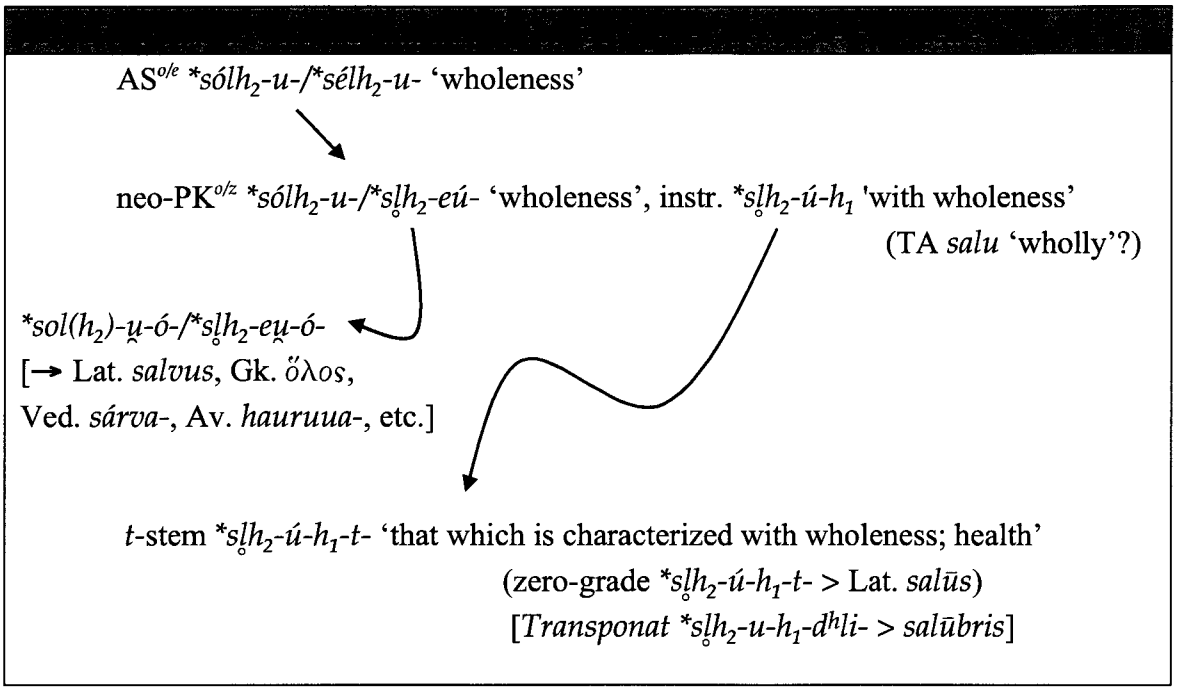
[139] Rasmussen's (1989:87) example above does not include "syllabic *r*" notation, but he presumably begins with *$g^w\underset{\circ}{r}h_2$-*u-tó-s*, which after metathesis gives *$g^w\underset{\circ}{r}uh_2tós$ > *$g^wruh_2tós$ with non-syllabic *-*r*-.

V.B.3. PIE instr. *$s\underset{\circ}{l}h_2$-u-h_1 → *$s\underset{\circ}{l}h_2$-u-h_1-t-

salūs may point to the type of "adnominal" derivative discussed by Nussbaum (1996, 2004b), whereby old case forms serve as the basis for secondary (external) derivation. There are several examples of a process by which various case forms are extended with derivational suffixes, including *t*-stems with a substantivizing and abstracting function, e.g. in Fig. 16.



In the present case, the instr. sg. of the PK$^{o/z}$ *u*-stem noun *$s\underset{\circ}{l}h_2$-*ú*-h_1 (cf. Ved. *ūtí* 'with help' < *h_2uh_1-*tí*-h_1, with zero grade of both root and suffix) might serve as the base for a similar *t*-stem substantivization, with generalized zero grade of the suffix. In addition, instr. sg. *$s\underset{\circ}{l}h_2$-*ú*-h_1 could also have been the base on which the isolated adjective *salūbris* < (*Transponat*) *$s\underset{\circ}{l}h_2$-u-h_1-d^hli- was built (Fig. 17).



Thematic derivatives to de-adnominal *t*-stems are well-attested, e.g. *k^wsp-én-t- 'that which is at night, nighttime' → *k^wsp-en-t-ó- 'at night', and such a derivative may underlie the Pg. cognomen **saluta** (cf. Untermann 2000:653). Alternatively, Lat. *aurōra* 'Dawn' (f), built from the PIE *s*-stem *h$_2$éu̯s-ōs/*h$_2$us-s- (cf. Ved. *úṣās-*, Hom. ἠώς), is a representative example within Italic of a femininization of an original athematic stem for onomastic purposes to remark the feminine gender (cf. Weiss 2009:300).[140] Pg. **saluta** may also be one such inner-Italic femininization, rather than a form of PIE date (p.c. Michael Weiss). Even if not of PIE date, **saluta** provides support for the *t*-stem *salūt-* outside of Latin.

[140] Cf. also the several names reflecting *-ōn-eh$_2$ based on original *n*-stems, e.g. Umb. **puemune** and **vesune**, both with uncertain etymology (cf. Weiss 2010:217-23 with nn. 264 and 236-44, respectively).

TA *salu* 'wholly' (adv.) may provide direct evidence for the PIE *u*-stem noun **sólh$_2$-u-/*s̥lh$_2$-ú-*: it is found three times and could represent an old adverbial instrumental **solh$_2$-ú-h$_1$*, giving PT **səlū*, built to the noun **sólh$_2$-u-*, with leveled *o*-grade.

Though controversial, PIE **-uh$_2$-* and **-uh$_3$-* are generally thought to develop to PT **-u̯a-*. There may be some evidence, however, suggesting that **-uh$_2$* and **-uh$_3$* develop to PT **-ū* word-finally, e.g. nom. pl. TA *akrunt*, TB *akruna* 'tears' remodeled from PIE **dáḱru-h$_2$* (cf. Ringe 1996:30-1); iron-clad evidence for word-final **-uh$_1$*, however, is lacking (cf. Hackstein 1995:17, with references). It is conceivable that the instr. sg. **solh$_2$-u-h$_1$* to PK$^{o/z}$ **sólh$_2$-u-/*s̥lh$_2$-ú-* indeed gives PT **-ū* and thus TA *salu*.[141]

In conclusion, there seems to be good support for an old PIE *u*-stem noun PK$^{o/z}$ **sólh$_2$-u-/*s̥lh$_2$-ú-* that underlies Lat. *salvus* and *salūs*, the latter of which was built via an abstracting *t*-stem. The *t*-stem substantivization seems best explained as a decasuative formation to the instr. sg. of the *u*-stem noun, in that such a derivation explains both the phonology of the noun and its semantics.

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[141] It may also be thinkable that **-h$_1$* in **s̥lh$_2$-u-h$_1$* was lost by the Kuiper effect, thereby obviating the problems with the outcome of PT **-uh$_1$*. Though the three attestations of TA *salu* (KT 3.49, 4.18, 10.3) do not reflect the right environment, i.e. clause-final placement, the texts also do not rule out the possibility that *salu* was used clause-finally.

V.C. Further Support for PIE Decasuative Secondary *t*-Stems

As mentioned above, Latin *salūs* is by no means an isolated case of a decasuative secondary *t*-stem; there appear to be several other parallels in Latin and elsewhere in PIE, including de-instrumental derivatives and even a possible de-genitival form in Hitt. *šāudišt-* 'suckling'.

V.C.1. Luv. *ḫīrūn/ḫīrūt-* 'oath'

Watkins (1993) has connected the pattern whereby a thematic stem **sol(h_2)-u̯o-/*sl̥h_2-u̯o-* sits beside a *t*-stem **salūt-* with the similar-looking **$h_2$er-u̯-ó-* that underlies Gk. ἀρ(ϝ)ᾱ́/Ion. ἀρή 'prayer' and Arc. καταρϝος 'accursed' beside the secondary *t*-stem **h_2ēr-ūt-* found in Luv. *ḫīrūn/ḫīrūt-* 'oath'.[142] *ḫīrūt-* < **h_2ēr-u-h_1-t-* could be explained as continuing an instrumental of an original "Narten" *ē/e* acrostatic (AS$^{ē/e}$) *u*-stem noun **h_2ḗr-u-/*$h_2$ér-u-* 'shout; din' (*vel sim.*) extended with a *t*-suffix: **h_2ḗr-u-h_1-t-* 'that with din; shouting' (with the strong stem generalized).[143]

Melchert (2004:473-4) explains the final *-n* in the nom./acc. sg. *ḫīrūn* analogically after neuter nouns in *-is-*, which are reshaped with a thematic *-an*, e.g. Luv. *tanis-* '(memorial) stele' with remodeled nom./acc. sg. *tanisan-za* (beside original *tanis-*

[142] Hitt. *ḫarzaz/šūn, ḫarzazūt-* 'bread soup' (n) reflects the same oblique *-ūt-* found in *ḫīrūt-*, though without etymological connection to any known root (Watkins 1993:476-7) and may also continue a suffix **-ūt-* that perhaps has the same history as *-ūt-* in Luv. *ḫīrūn/ḫīrūt-*. But without a clear root etymology, it is difficult to analyze the word.

[143] The neut. gender of the "abstract" in this case is somewhat problematic; but because **h_2ḗr-u-h_1-t-* is not so much an abstract as much as a *nomen actionis*, perhaps the gender here is more appropriate.

sa). Rieken (2006:288-9), approving of Watkins' etymology of *ḫīrūt-* (though viewing it as a *vr̥ddhi*-derivative of **h₂er-u-*), rejects Melchert's analogical explanation for *-ūn*, since it would remain unexplained why the oblique stems were not remodeled. But if Melchert's analysis is correct, Rieken's "*vr̥ddhi*" simply represents the generalization of the strong stem **h₂ér-u-* (next to regular full grade in ἀρᾱ́ < **h₂er-u̯-ó-*). Furthermore, no generalizations of *-ū-* need be made, and the remaining remodeling of the nominative ending is trivial.

Thus, *ḫīrūn/ḫīrūt-* is best explained as a de-instrumental secondary *t*-stem, built in the same way Lat. *salūs* was formed. Moreover, it shows that the *t*-stem formations under discussion are very old and that they seem to reflect plain **-t-* in the suffix, rather than the ablauting *t*-suffix **-ot-/*-t-* found in the non-decasuative examples given by Nussbaum (2004b) above.

V.C.2. OPr. *kailūstiskun* 'health'

As first noticed by Meillet (1928:42), OPr. acc. sg. *kailūstiskun* 'health' (f), with suffix *-ūst-* possibly reflecting **-ūt-* but built to a different root (cf. OCS *cělŭ/cělostŭ* 'safe', Go. *hails* 'all, safe', Welsh *coel* 'sign', and perhaps Hsch. κοῖλυ· τὸ καλόν), may also belong here. The attested forms are suggestive of an original *u*-stem noun, at least in Baltic and perhaps in Greek, meaning something like 'safety' (but with problematic etymology; cf. Beekes *EDG* s.v. κοῖλυ) that may have replaced an inherited **sólH-u-h₁-t-*

in Baltic. *kailūstiskun* assumes an underlying **kailūstas* 'healthy' to the *u*-stem *noun*, and the suffix *-*ūsto*- seems to continue *-*ūt*- augmented with another adjective-making dental suffix **koil-ūt-to*-.[144]

Smoczyński (2000:40-1) offers further Baltic parallels to OPr. *kailūstiskun* in the Lith. adjectives *vėlústas* 'late' beside *vėlùs*, *-aũs* 'late', Lith. *ligústas* 'sickly' (perhaps built in opposition to **kailūstas* with the spread of *-*ūsto*-) beside *ligà*, *-õs* 'illness' (f), and Lith. *miegústas* 'sleepy' beside *miẽgas*, *-o* 'sleep' (m). The forms continue a suffix in *-*ūt*- extended with *-*to*-, but not all of them point back to original *u*-stems. It may well be the case that *-*ūt-to*- (or perhaps already Baltic *-*ūsto*-) was extracted and spread as a semi-productive adjective-forming suffix in Baltic (cf. below for a possibly similar development with PIE *-*tāt*-).

V.C.3. Hitt. *šāudišt*- 'suckling; yearling'

Hitt. *šāudišt*- 'suckling' (on the semantics of the word, cf. Rieken 1999:147-50) has also been recently demonstrated to continue a decasuative formation in *-*t*- by Vine (2009:241). He argues convincingly that the apparent *s*-stem underlying PIE **u̯et-es*- and its derivatives (including Lat. *vetus*; cf. **II.F.e.**) actually continues a gen. sg. to the root noun **u̯et*-. *šāudišt*- must then reflect a degenitival *t*-stem substantivization **s(o)m-u̯et*-

[144] It remains to be determined whether the Lith. dim. suffix in *-ūtis*, e.g. *kaútis* 'little cat', *varliútis* 'small frog', etc., is of any relation (cf. Schmalstieg and Jēgers 1971:127).

es-t- 'that which belongs to the same year', in much the same way *salūs*, etc. continue de-instrumental formations in **-t-*.

V.C.4. Lat. *-ās, -ātis*

The suffix *-ās, -ātis*, with very similar structure to *-tās, -tātis*, was used to make ethnic adjectives in Latin, forming place adjectives to mostly thematic stems, e.g. *Arpīnās* 'of Arpinum' beside *Arpīnātis* (Cat.), some pronominal abstracts, e.g. *cuiās/quoiās* 'of what country, town' (Pl.+), *nostrās* 'of our country, native' (Cat.+), and even some adjectives, e.g. *optimās* 'of the best class' (Enn.+), etc. (cf. Neue and Wagener 1892:47, Leumann 1977:345-6). Unlike **-tāt-*, to which correspond nouns in **-teh$_2$-*, the majority of the base forms to which adjectives in **-āti-* are built are thematic *o*-stems, e.g. *Arpīnum, noster, optimus*, etc. Several adjectives appear in *-nās* in Latin, built to bases that lack the nasal, e.g. *Aletrīnātēs* 'inhabitants of Aletrium' (Cic.) beside *Aletrium* (Ernout 1965:35). Moreover, apparently related forms can be found in Sabellian, e.g. Osc. dat. sg. **kluvatiui** '*Cluvātī*' (*ST* Cp 37), etc. and Celtic, e.g. the "Atrebates" (Caes. *BG* 2.4), suggesting a rather widespread use of the suffix (cf. Ernout 1965). Based on the *i*-stem inflection in *Arpīnātis*, Osc. gen. sg. **Lúvkanateís** 'Lucanian' (*ST* Fr 2) and Gaul. ναυμασατις 'of Namausos', Weiss (2009:316-7) reconstructs the suffix as **-āti-* (but cf.

Dupraz 2010:260-1 for the opinion that *-*āti*- is Etruscan in origin).[145] There seems to be a relationship to *-*tāt*-, given their structures with presuffixal lengthening, though *-*āti*- may reflect an extension with *-*ti*- rather than plain *-*t*-.

Interestingly, forms built with *-*āti*- function adjectivally, rather than nominally as abstracts. Ved. *śáṃtāti*- and later Skt. *śivatāti*- have also been thought to function as adjectives but this is likely not the case for *śáṃtāti*- (cf. **III.D.4.**). Otherwise, there is no support at all for an adjectival *-*t*- in PIE, and *-*āti*- cannot somehow continue *-*tāt*- in any way. Furthermore, there is no evidence for decasuative *ti*-stems in PIE. *-*āti*- may rather reflect an inner Italic (or Italo-Celtic) development from an inherited *-*āt*-, if the *i*-stem inflection is secondary. Italic remade in mass all athematic adjectives with *i*-stem inflection at a very early date, and perhaps *-*āt*- was converted to an adjective and thus reshaped to *-*āti*-, not unlike the *t*-stem *dīves* 'rich' (Pl.+) < **deiu̯-et*- that subsequently became an *i*-stem adjective in Latin.

Potentially related to *-*āti*-, perhaps, are the adjectives *Picēns* 'Picene' (Cic.+) to *Picēnum* (Cic.+) and *Samnīs* 'Samnite' (Naev.+) beside *Samnītis* (Cat.) to *Samnium* (on

[145] Ernout (1965:31 with n. 1) was of the opinion that the consonant-stem case forms, e.g. abl. sg. -*āte* and gen. pl. -*ātum*, were older than the *i*-stem forms, and thus reconstructed the suffix as *-*āt*-. But the nom. sg. *-*ātis* reflected in both Italic and Celtic is indicative that the suffix had *i*-stem inflection at a very early date. Though the adverbial forms *tuātim* 'in your (sg.) manner' (Pl.+), *vestrātim* 'in your (pl.) manner' (Var.), *nostrātim* 'in our manner' (Var.) are suggestive of *i*-stem inflection, adverbs in -*ātim* became marginally productive, so the forms could have easily been secondary formations, as Leumann (1977:501) thought.

which cf. Rix 1957), reflecting *-*ti*- (Weiss 2009:317) and suggesting that the vowel preceding the *-*t*- did not belong to the original suffix and supporting the secondary *t*-stem analysis of *-*āti*-. It is thinkable that a secondary *t*-stem was built to an eh_2-stem used as a place name, giving *-*āti*- from *-eh_2-*ti*-. For instance, a place name like *Interamna* could have built an adjective **Interamnā-ti*- 'of Interamna' with plain *-*ti*-, from which could have been extracted a suffix *-*āti*- (and *-*nāti*-) that was used to build newer place adjectives.

Though it seems theoretically possible that a place name like *Interamna* could be involved in the creation of *-*āt*-, most of the underlying place names reflect *o*-stem nouns (cf. the collection of names in Ernout 1965). It may well be the case, then, that the supposed eh_2-stem is not a feminine place name, but rather an old collective in *-eh_2- built to the place name. A collective of this sort could have meant something like "that which belongs to *x*" to a place name *x*. A secondary *t*-stem in *-eh_2-*t*- built to the collective noun would serve to "individualize" the noun in the sense "the one who belongs to *x*", and the subsequent *-eh_2-*t*- > *-*āt*- could have easily been reanalyzed as a suffix meaning "belonging to", with the addition of adjectival *i*-stem inflection, well after the collective noun in *-eh_2- fell out of use.

If *-*āti*- indeed continues a secondary *t*-stem formation recharacterized with adjectival *-*i*-, it remains to be seen if it can also continue a decasuative formation. On the other hand, *-*eh*$_2$-*t*- need not point back to one in the first place, due to the fact that instrumental semantics are not particularly evident in the usage of *-*āti*-, in that it does not mean anything like "that which is characterized with/by *x*", where *x* is an abstract or adjective. In addition, it is difficult to justify phonologically any other case form underlying a suffix shape *-*āti*- in Latin (and Celtic, for that matter). Hence, it may be most practical to consider *-*āti*- not to reflect any decasuative formation of the "*salūs*" type above, but rather to continue a sort of "possessive" *t*-stem substantive built to an underlying collective noun of the type described above that had the semantics of "the one who belongs to *x*", where *x* is a place name. A derivative built in this way to an *eh*$_2$-stem would have produced *-*āt*-, which finally gave rise to an *i*-stem adjective in *-*āti*- that productively spread to other place names already in both Italic and Celtic, including *o*-stem nouns, such as *Penātēs* 'the Penates; those who live in the *penus*' (Pl.+) to *penus* 'pantry' (Pl.+).

V.C.4.a. *-*āti*- < *-*h*$_2$*e-(o)t*-

The sort of pattern whereby a thematic adjective is used as the base for further derivation has parallels in PIE, but such thematic derivation has generally been neglected,

in comparison to the much better understood athematic system of derivation.[146] Rau (*forthcoming*) has outlined a pattern of thematic derivation, by which **nome-h_2-* '(act of) pasturing', e.g. Gk. *νομή* 'pasture; grazing' (Soph.+), makes a poss. adj. **nomh$_2$-ó-* 'pasturing', e.g. Gk. *νομός* 'pasture; food' (*Il.*+), that can then be substantivized to **nomh$_2$e-(e)d-* 'one connected with pasturing', e.g. Gk. *νομάς, -άδος* 'roaming for pasture' (Pi.+). The pattern, furthermore, is found in other thematic derivatives without **-h_2-*, e.g. **dolo-* > Gk. *δολός* 'trick' (*Il.*+) → **dole-ro-* > Gk. *δολερός* 'deceitful' (Soph.+).

It is worth considering a similar pattern with a thematic derivative to underlie the "affiliative" suffix **-āti-*, which does not seem to reflect the semantics of the decasuative abstract noun, as argued above. Starting with a base in **-eh$_2$-*, which could well continue an old collective, a thematic derivative in **-h$_2$-e-* could have served as the foundation for an ablauting *t*-stem formation **-h$_2$e-ot-/*-h$_2$e-t-* > **-aot-/*-at-* that could also have been leveled to **-āt-*, before being outfitted with *i*-stem adjective endings. Semantically, the masculine *t*-stem substantive built with **-ot-/*-t-* well suits the actual use of the *āti-* suffix in building ethnic adjectives, e.g. "those who belong to *x*". Consequently, **-āti-* could be explained either as **-eh$_2$-ot-* > **-aot-* (with contraction of **-aot-* to **-āt-*), reflecting a secondary *t*-stem built to a collective base in **-eh$_2$-*; or as **-h$_2$e-ot-* > **-aot-* (again with

[146] For more on the developments in athematic nominal morphology and internal derivation, cf. especially Schindler (1967), Eichner (1973), Rix (1976:121-8), Nussbaum (1986), and Widmer (2004, with refs).

contraction to *-*āt*-), with a secondary *t*-stem substantivization of the "*νομάς*"-type built to the collective base.

Though *-*āti*- became fairly productive, it is difficult to explain its use with adjectives that do not denote a place name, e.g. *optimās* 'of or belonging to the best' (Pl.+) to *optimus* 'best', as well as its use with pronomial forms, e.g. *cuiās/quoiās* 'of or belonging to what country' to *cuius/quoius* 'whose?' (Pl.+) and *nostrās* 'of or belonging to our country' (Cat.+) to *noster* 'our'. The forms may in fact continue archaisms, clearly reflecting the original individualizing function of the *t*-stem suffix. While *-*āti*- became productive as an affiliative suffix with place names, its use with other adjectives fell out of use. And though *-*āti*- does not have a direct connection to decasuative *t*-stems as argued for *salūs*, the archaic examples reflecting the suffix do show that secondary *t*-stem derivation was well established already in Proto-Italic (and may even belong to Italo-Celtic).

V.C.5. Summary of Secondary *t*-Stem Data

If the analyses above are correct, *salūs*, alongside perhaps Luv. *ḫirūt*- and OPr. *kailūstiskun*, continue examples of instrumentals (possibly with adverbial functions) substantivized with a *t*-stem, while Hitt. *šāudišt*- 'suckling' continues a degenitival substantivization. Moreover, the Baltic forms in *-*ūt-to*- may continue an instrumental adnominal *-*uh*$_1$*-t-to*- that subsequently spread, after reanalysis as a new suffix itself,

which may find a parallel in PIE *-*tāt*-. Another potential example of a secondary *t*-stem built to a *u*-stem instrumental may be briefly noted: Lat. *mātūta*, as in *Mater Matuta*, a goddess of dawn and childbirth, beside *mātūrus* 'ripe, full-grown; early, quick', built to *meh_2- 'good, at a good/right/early time' (importantly without a verb built to the root), may also follow a similar derivational pattern as *salūs*. A *tu*-stem abstract *$mā$-*tu*- underlies *mātūta* and *mātūrus* that may have made an adverb *$mātū$ 'with/at the proper/early time' using the instr. sg. *meh_2-*tu*-h_1 (Fortson 2007:88-9 with nn. 26 and 27). Rather than continuing a *to*-adjective, *mātūta* < *Transponat* *meh_2-*tu*-h_1-*t*-eh_2 could well be based on a secondary *t*-stem built to *meh_2-*tu*-h_1, as implied by de Vaan (*EDLIL* s.v. *mātūrus*), with the meantime 'the one with ripeness' and reflecting the same femininization in *-eh_2- underlying Pg. **saluta** (cf. **V.B.3.** above). With a thematic adjective *mātūrus* already found in Latin, there may have been no need to derive another thematic adjective in *-*to*- to the abstract *meh_2-*tu*-, perhaps adding justification for the use of a secondary *t*-stem substantivization.

The examples illustrated above conclusively point back to a secondary *t*-stem with an abstracting/individualizing function in PIE. Furthermore, decasuative abstracts built with the *t*-stem seem to continue the zero grade of the *t*-stem suffix, rather than *o*-grade underlying Hitt. *šīwatt*- 'day, Sun God' and CLuv. *tiwat*- 'id.' < *$d\underset{\smile}{i}\acute{e}\underset{\smile}{u}$-*ot*-. Hence,

secondary *t*-stems with an abstracting function are well based in PIE derivational morphology, with many abstract nouns continuing adnominal formations to case forms. It seems likely, then, that the suffix *-*tāt*-, with its own abstracting/individualizing function and presuffixal lengthening, could also continue a decasuative secondary *t*-stem formation of the type described above that underlie Lat. *salūs*, etc.

V.D. PIE "*-*tāt*-"

The *tāt*-suffix may well owe its origin to a secondary *t*-stem of the sort demonstrated above, with a decasuative formation to a stem in *-teh_2-. Vedic possesses a small number of old-looking adverbial instrumentals to stems in *-teh_2, e.g. *devátā* 'among the collection of the gods', that may continue an instr. sg. in *-eh_1/*-h_1 (cf. **III.C.2.**), and abstracts in *-teh_2- have synonymous *tāt*-abstracts elsewhere, e.g. Ved. *sūnr̥̊tā-* 'vigor' beside OAv. *hunarətāt-* 'talent' and Ved. *devátā-* 'among the gods' beside Ved. *devátāt-/devátāti-* 'collection of the gods'. Supporting the relationship between the *-*tāt*- and *-teh_2- is the fact that secondary Greek abstracts in -τη/-τᾱ, in fact, already by Homer were entirely replaced by -της/-τᾱς (Buck and Petersen 1945:541). Given this relationship between abstracts in * -teh_2- and -*tāt*- (on which cf. **I.D.**), it is likely the case that a *t*-stem derivative of the sort underlying Lat. *salūs* is at the base of the *tāt*-suffix, with the *t*-stem built to an abstract in *-*tā*-. It remains to be

demonstrated whether or not *-*tāt*- continues a decasuative formation, analyzable as a secondary *t*-stem built to a case form of a *teh*$_2$-abstract that perhaps continues the instr. sg. of *-*teh*$_2$-.

V.D.1. Derivation from *-*teh*$_2$-: *-*tāt*- < *-*teh*$_2$-*eh*$_1$-*t*- or *-*teh*$_2$-*h*$_1$-*t*-

*-*teh*$_2$- productively made abstract nouns to both adjectives and nouns in PIE (Brugmann 1906:416), but some justification is necessary in arguing that a *t*-stem suffix was used to to derive one abstract noun from another. But as seen above with Lat. *salūs*, PIE secondary *t*-stems did more than just derive an abstract noun: the *t*-suffix served to derive "individualized" nouns, when added to a case form, e.g. with the instr. sg. of the underlying base, abstracts that reflect the semantics "that which is characterized with/by *x*" are built. *t*-stem substantivizations of the sort demonstrated above, e.g. de-instrumental *$s\underset{\circ}{l}h_2$-*u*-*h*$_1$-*t*- and *$h_2\acute{e}r$-*u*-*h*$_1$-*t*-, and degenitival *$s(o)m$-$\underset{\sim}{u}$*et*-*es*-*t*-, always reflect zero grade of the *t*-suffix, moreover. Thus, a secondary *t*-stem of the adnominal type built to the instr. sg. of an abstract in *-*teh*$_2$- would give *-*teh*$_2$-*eh*$_1$-*t*- or *-*teh*$_2$-*h*$_1$-*t*- > *-*tāt*-, with the meaning 'that which is characterized with/by *x*'.

The Gāθās preserve laryngeal hiatus to a remarkable degree, but the OAv. instr. sg. to *teh*$_2$-stems in -*ā* never scans disyllabically. Moreover, the vowel within the suffix in Ved. -*tāt*-/-*tāti*- and Av. -*tāt*- never scans disyllabically, e.g. -*taat*-, in any of its

attestations. For these reasons, it seems preferable that the instr. sg. reflects *-teh_2-h_1 rather than *-teh_2-eh_1, and *-$t\bar{a}t$- should likewise reflect *-teh_2-h_1-t-.

V.D.2. Alternative Suggestion: *-teh_2- → *-th_2o- → *-th_2e-(o)t-

The decasuative derivatives studied above reflect consistent zero grade of the *t*-stem suffix, but as Nussbaum (2004b) identified, the *t*-suffix ablauted in PIE, reflecting AK ablaut *-*ot*-/*-*t*-. It is worth considering a similar sort of pattern demonstrated by Rau discussed above for *-*āti*- that need not appeal to a de-instrumental *t*-stem. Beginning with a PIE abstract in *-teh_2-, the abstract may have built a thematic derivative *-th_2-*o*- (cf. Ved. -*tha*-, e.g. *ukthá*- 'song' and *yūthá*- 'herd'; cf. Rau *forthcoming* for further discussion) that was substantivized to *-th_2e-*ot*-/*-th_2e-*t*- by an ablauting *t*-stem. Supposing the vowel contraction **-*ao*- > *-*ā*-, based on assumption that the color of the first vowel is retained after contraction (cf. Sihler 1995:80; cf. also **-*eo*- > *-*ē*- and **-*oe*- > *-*ō*- above), *-th_2e-*ot*-/*-th_2e-*t*- gives *-*taot*-/*-*tat*- that contracts to *-*tāt*-/*-*tat*-, which could have been leveled to *-*tāt*-.[147] Such a derivation, though, must assume that the leveling of *-*tāt*- took place already in PIE, since no evidence whatsoever for *-*tat*- exists

[147] The assumption of leveling *-*tāt*- from *-th_2e-*ot*-/*-th_2e-*t*- cannot be taken for granted, given that secondary *t*-stems of the sort described above built to thematic stems show both variants. For instance, the long vowel variant is seen in Gk. *γυμνής* 'light-armed soldier', beside the short vowel stem in Lat. *dīves* 'rich' (cf. Fig. 14 in **V.A.**). One example, Hom. acc. sg. *ἀργῆτα* 'dazzling white, glistening' beside *ἀργέτα* 'id.', shows both suffix variants (cf. Nussbaum 2004b).

in any of the languages attesting *-*tāt*-. Also, Vedic might be expected to reflect a suffix *x*-*thāt*-/*x*-*thāti*- with laryngeal aspiration, if continuing PIE *-*th*$_2$*e-ot*-/*-*th*$_2$*e-t*-.

Though the morphological and phonological questions surrounding a PIE suffix of the shape *-*th*$_2$*e-ot*-/*-*th*$_2$*e-t*- make it difficult to see how it can underlie *-*tāt*-, the semantics of the suffix make it extremely unlikely that it is the source. The type of secondary *t*-stem reflected in the suffix *-*ot*-/*-*t*- tends to build masculine substantives to adjectives, rather than abstract nouns (cf. Nussbaum 2004b). The hypothetical *-*th*$_2$*e-ot*-/*-*th*$_2$*e-t*-, if following the customary pattern for derivatives of this type, should mean something like "the one who is *x*", where *x* is a possessive adjective built to the original underlying abstract noun, e.g. Lat. *stīpes* 'the one that is stiff; branch, stick' (Catul.+) to **steip*o- 'stiff' (cf. **V.A.** above for further examples).[148] Furthermore, there is good support in Lat. *salūs*, etc. for decasuative formations underlying *t*-stem abstracts, and as a result, the derivative *-*th*$_2$*e-ot*-/*-*th*$_2$*e-t*- should be disfavored next to the decasuative *t*-stem *-*teh*$_2$-*h*$_1$-*t*- on semantic grounds.

V.D.3. Reanalysis of *-*tā-t*- to *-*tāt*-

If *-*tāt*- has the sort of origin described above, reflecting *-*teh*$_2$-*h*$_1$-*t*- formed from a secondary *t*-stem built to the instr. sg. case of an abstract noun in *-*teh*$_2$-, it remains to

[148] N.B.: Hitt. *šāudišt*- 'suckling', argued by Vine to reflect a degenitival *t*-stem, reflects an animate substantivization rather than an abstract noun. Such an animatization could well be secondary, given that the word is applied to an animal, which naturally assumes animate gender.

be seen how the secondary *t*-stem formation to *-teh_2- was generalized to a suffix of the shape *-*tāt*-. It is not too difficult to imagine how the resulting *-*tāt*- could have been reanalyzed as a new suffix, however, especially as denominal abstracts in *-teh_2- began to be replaced by other abstract noun suffixes. An abstract *x*-teh_2 built an individualized abstract *x*-teh_2-h_1-*t*-, and after the new abstract found use, *-*tāt*- was extracted beside the base *x* preserving the same function of the original *t*-suffix 'that which is characterized with/by *x*'.

Reliable PIE word equations for *tāt*-abstracts are very difficult to identify, but one pseudo-word-equation may be found in Hom. ἀνδροτῆτα 'manliness' (< *$h_2nr̥$-*tāt*-) beside OAv. *hu-narətāt*- 'talent' (< *h_1su-h_2ner-*tāt*-; cf. **IV.C.3.a.iii.**). The PIE abstract noun *h_2ner-*tāt*-/*$h_2nr̥$-*tāt*- that may underlie the two forms is ultimately built to PIE *h_2ner-/*$h_2nr̥$- 'vigorous', which may serve to illustrate exactly how a *tāt*-abstract was built. Beside the adjective *h_2ner-/*$h_2nr̥$- was a *to*-adjective *$h_2n(e)rtó$-, which is continued by Ved. *sū́nr̥ta*- 'magnanimous; in full vigor, unfaded' and OIr. *nert* 'strength'. It is thinkable that *$h_2n(e)rtó$- made an abstract noun *$h_2n(e)rteh_2$- 'vigor, strength' that served as the basis for a secondary *t*-stem as described above. The instr. sg. *$h_2n(e)rtéh_2$-h_1 'with vigor, strength' built a de-instrumental abstract *$h_2n(e)rtéh_2$-h_1-*t*- meaning "that which is characterized with/by vigor, strength". The new substantive

*$h_2n(e)rt\acute{e}h_2$-h_1-t- gave *$h_2n(e)rt\bar{a}t$-, which resulted in pre-Gk. *$an̥r\acute{t}\bar{a}t$-, with -$r̥$- still preserved in Homer, and *hu-narətāt-* (< *h_1su-$h_2nert\bar{a}t$-) in Avestan. The abstract noun *$h_2n(e)rt\bar{a}$-t-*, next to the base *$h_2n(e)r$-*, was subsequently reanalyzed as *$h_2n(e)r$-$t\bar{a}t$-*, from which was extracted a new suffix *-tāt-* with the semantics "that which is characterized with/by *x*". The abstract *$h_2n(e)rteh_2$- was then replaced by *$h_2n(e)r$-$t\bar{a}t$- in Greek and Avestan (but not in Indic), as *-tāt- generally began to replace the older abstracts in *-teh_2-.

The *$h_2n(e)r$-$t\bar{a}t$- example illustrates a denominal formation in *-tāt-, and Latin likewise built a good deal of *tāt*-abstracts to nouns, especially athematic nouns, as did Old Avestan; but the great majority of abstracts in *-tāt- are built to thematic adjectives. In fact, in Latin and Greek, where *-tāt- became extremely productive, the suffix was associated with thematic adjectives at a very early date, when the suffix was reanalyzed as *-otāt-. How and why *-otāt-, for all intents and purposes, became associated primarily with thematic adjectives remains as one of the final questions surrounding the suffix. Before considering how and why *-tāt- was generalized to *-otāt-, the denominal use of *-tāt- will be examined, since it may reflect an older use of the suffix.

V.D.4. **-tāt-* as a Denominal Suffix

Given the semantic function of the secondary *t*-stem formation underlying **-tāt-*, it is rather easy to see how it could build abstract nouns to both adjectives and nouns. The denominal use is very apparent in Latin, which builds more abstracts to nouns than any of the other languages, especially the archaic abstracts built to *s*-stems that reflect the expected semantics of **-tāt-*, e.g. *egestās* 'that which is characterized by **egos/*eges-*; destitution' (cf. **II.F.1.a.**) and *venustās* 'that which is characterized by *venus*; charm, grace, beauty' (cf. **II.F.1.c.**). Such a semantic use of the suffix easily finds employment in building names for personified qualities, especially divine names, as in Lat. *voluntās* and Pg. *herentas*/Osc. **herentateís** (cf. **II.F.11.a.**). Avestan also uses **-tāt-* in similar ways to build abstracts to athematic nouns at the oldest layer of the language, e.g. *astəntāt-* 'liveliness' and *xᵛaētāt-* 'membership' (cf. **III.A.2.**).

Elsewhere, **-tāt-* made collectives when added to nominal bases everywhere it is attested, e.g. Lat. *cīvitas* 'state; city' (cf. **II.A.**), Ved. *devátāt-/devátāti-* 'collection of the gods' (cf. **III.C.1.b.**), OAv. *kəuuītāt-* 'class, group of *kavis*' (cf. **III.A.4.**), and Gk. *νεότης* 'youth; body of youth' (cf. **IV.A.3.**). Despite the fact that some of the oldest *tāt-* abstracts are denominal, the great majority of the abstracts are built to thematic adjectives everywhere the suffix is found. The generalization seemed to have occurred very early in each language too and may even have happened already in PIE.

V.E. Thematic Stems and *-*ótāt*-

The languages attesting the *tāt*-suffix all show a strong tie to thematic stems, even in the earliest available material. Vedic mostly reflects abstracts to thematic stems, with 9 of the 11 attested abstracts continuing *-átāt-/-átāti-* (cf. Fig. 5 in **III.A.**), though of the 10 Old Avestan *tāt*-abstracts, only two are built to thematic stems (cf. Fig. 6 in **III.A.**), but Young Avestan attests a much larger percentage of "thematic" abstracts (cf. Fig. 7 in **III.A.**). Likewise, 6 of the 10 Homeric abstracts in *-τηϛ* have thematic bases (cf. Fig. 11 in **IV.A.**), though in Greek some of the abstracts were the result of metrical remodelings of earlier abstracts, e.g. *ποτήϛ* 'drink' (on which cf. **IV.C.2.a.**). The picture in Latin is much more complicated, due to the lack of comparatively old texts like Homer's poems or Vedic poetry, though the data suggest that *-itās* was lexicalized as the productive form of the abstract suffix already by the *XII Tables* (cf. **II.A.**). Despite the early generalization to thematic stems, it is not entirely clear that such was the state of affairs already in PIE, allowing for the possibility that the generalization of *-*tāt*- to *-*ó-tāt*- was at least partly *einzelsprachlich.*

PIE *-*teh*$_2$- built abstract nouns to all stem types, including thematic stems (Brugmann 1906:414-22), but it does not seem at all likely that *-*otāt*- developed solely from abstracts in *-*oteh*$_2$- in the way described above, given the large number of consonant-stem bases attested in both PIE *teh*$_2$-stems and early *tāt*-material. While the

initial development of *-*otāt*- could have been in place in PIE, its generalization seems to have occurred later. Hence, it is reasonable to think that *-*tāt*- was only associated with thematic stems secondarily, without assuming that it necessarily depends on an earlier association of *-*teh*$_2$- with *o*-stems.

It is the case that many of the older Latin abstracts built with -*tās* reflect formations to consonant stems (cf. Fig. 4 in **II.H.**) and that consonant-stem formation continued to be productive for a lengthy period of time. -*tās* could be added directly to consonant stems, particularly *s*-stems, without the need of a linking vowel, e.g. *tempestās* (cf. **II.F.1.b.**), and the pattern spread by analogy to other abstracts like *aestās*, *iuventās*, etc. (cf. Fig. 4 in **II.H.**). Some old-looking abstracts built to thematic stems do exist, including *aetās* < **a̯i̯u̯o-tās* and *lībertās* < **leu̯dʰero-tās* or perhaps **līber-(o)tās*. Already by the *XII Tab.*, -*itās* was generalized, based on the form *auctōritās*.

In Indic, the oldest material already points to a connection to thematic stems (cf. **III.A.1.**), e.g. Ved. *devátāt-/devátāti-* 'collection of the gods' and Ved. *sarvátāt-/sarvátāti-* 'wholeness, health'; but Iranian reflects a state of affairs closer to that of Latin, with a number of abstracs built to consonant stems (cf. **III.A.2.**), e.g. OAv. *yauuaētāt-* 'eternity' and *xᵛaētāt-* 'membership'. Later in Avestan, though, -*tāt*- indeed becomes increasingly associated with thematic stems, e.g. YAv. *uštatāt-* 'what is desirable; luck, success' and *fratəmatāt-* 'authority'.

Greek abstracts also reflect close ties with thematic stems in Homer, e.g. *ἰότης* 'will, desire' and *φιλότης* 'friendship; sexual love', but athematic forms are also found, e.g. *ἀνδροτής* 'manliness' and *βραδυτής* 'slowness'. Already in Homer, the suffix *-ότης* seems to have been generalized, given the possible remodelings underlying *δηϊοτής* 'battle strife' (cf. **IV.C.3.c.**) and *ποτής* 'drink' (cf. **IV.C.2.a.**).

The data indicated that PIE used **-tāt-* with both thematic and athematic stems, including a wide variety of athematic stems, as is reflected in Latin, e.g. *s*-stems like *tempestās*, and Avestan, with an *i*-stem OAv. *kəuuītāt-* 'class, group of *kavis*', an *n*-stem OAv. *astəntāt-* 'liveliness', and a *u*-stem YAv. *vaŋhutāt-* 'blood'. Hence, it seems to be the case that the generalization to thematic stems happened individually. Generalizations of suffixes to thematic stems are typologically common in many languages. Within Latin, other abstract suffixes acquire the "linking vowel" *-i-* before the suffix, including the other complex suffixes *-itia*, *-itiēs*, and *-itūdō*, etc., and a similar generalization can be seen in Gk. *-οϝεντ-* and *-οσύνη*.

o-stems originally could build abstract nouns using the "*cvi*-construction", e.g. Lat. *ravis* 'hoarseness' to *ravus* 'hoarse' and Ved. *rúci-* 'light, spendor' < **leu̯ki-* or **lou̯ki-* beside Hom. *λευκός* 'white' < **leu̯kó-* (cf. Balles 2006), but the *i*-stem abstracts in these examples are mostly relic forms. Once the *cvi*-construction fell out of use, *o*-stem adjectives thus had no certain way to build abstract nouns, compared to consonant stems,

which could make abstract derivatives with *-eh_2-. *o*-stems thus needed a clearly identifiable abstract suffix, and the newly emerging *-*tāt*- served the role, along with other derived suffixes mentioned above that also show a generalized linking vowel before the suffix. Already in PIE, abstracts built to *o*-stems reflecting *-*o-tāt*-, e.g. Lat. *vēritās* 'truth', Ved. *sarvátāt-/sarvátāti-* 'wholeness, health', and Hom. φιλότης 'friendship; sexual love', could easily have been reanalyzed as showing adjective stem plus "*-*otāt*-", from which a new suffix form *-*otāt*- was extracted that then started to spread to athematic stems, with accentuation fixed on the thematic vowel immediately before the suffix. Thus, the spread of *-*otāt*- in Latin in the weakened form -*itās* and in Greek -ότης typologically fits in with other suffixes found in the languages in which the thematic vowel has become part of the suffix itself. While the reanalysis from *-*ó-tāt*- to *-*ótāt*- may have begun in PIE, it seemed to have progressed at different rates in each individual language, happening most quickly in Greek and most slowly in Latin.

In Indo-Iranian, *-*tāt*- had a slightly different history. For the most part, Vedic had already restricted *-*tāt*- to thematic stems, aside from the secondary abstracts *vasútāti-* 'collection of Vasus' (cf. **III.D.1.**) and *śáṃtāti-* 'health; luck' (cf. **III.D.4.**). After the Rig Veda, -*tāti*- replaced -*tāt*-, and non-Rig Vedic abstracts that were built with the suffix were still connected only with thematic adjectives, e.g. AV *ayakṣmátāti-* 'health' and Classical Skt. *śivatāti-* 'beneficial; welfare' and *śubhatāti-* 'welfare'. In

Iranian, *-tāt-* became increasingly attached to thematic stems, but it also was used in a much more specialized semantic sense than found in the other languages, building both qualitative abstracts, in addition to technical terms like YAv. *nərəfsąstāt-* 'condition of decline; waning' and YAv. *šiiaoθnōtāt-* 'the word *šiiaoθna-*', and names YAv. *arštāt-*, the name of goddess "Uprightness", and YAv. *kahrkatāt-*, the name of a rooster (cf. **III.E.**). **-tāt-* may even have survived in Khotanese, which attests a number of abstracts built to thematic adjectives with a suffix *-tātä* from a secondary *i*-stem suffix **-tāti-*, e.g. *ttuśśāttātä* 'emptiness' and *ttāḍātä* 'obscurity'. Moreover, Pashto *təštiyā* 'emptiness' may also continue *-tāti-* (cf. **III.A.**), suggesting that *-tāt-* may have been almost as productive in Iranian as it was in Latin and Greek.

VI. Concluding Remarks

VI.A. Summary of Results

Despite the lack of secure word-equations for *-*tāt*-, its PIE pedigree is secured by its appearance in Italic, Indo-Iranian, and Greek. And based on the various uses of the suffix in each of the branches, some conclusions can be drawn. First, PIE *-*tāt*- generally built qualitative abstract nouns. At the oldest layer of formation, the abstract nouns could be derived from both adjectives and nouns, regardless of whether the base was thematic or not. Given the close typological relationship between abstract nouns and collective nouns, *-*tāt*- may well have built some collectives already in PIE, especially when added to a nominal base. Ved. *devátāt-/devátāti-* 'collection of the gods' and OAv. *kəuuītāt-* 'class, group of *kavis*' may reflect such a collectivizing use (cf. **III.C.1.b.** and **III.A.4.**).

In the individual languages, *-*tāt*- continued to evolve. Latin *-tās* was generalized to *-itās* (abstracted from its use with thematic bases), which then became one of the more productive abstract-forming suffixes in the language, forming qualitative abstracts to both thematic and athematic adjectives. But the older abstracts that can clearly be identified in Latin show that plain *-tās* could be used until relatively late in the language, especially with athematic nouns, including *s*-stem nouns, with which *-tās* developed a special relationship for a short period of time (cf. **II.H.** with Fig. 4). *-tās* seemed to have an "individualizing" function in Italic that was inherited from PIE, which was used to build abstracts with different semantics from other abstracts built to the same base. For

instance, the *tās*-abstract *tempestās* was used in the sense '(specific) period of time' beside *tempus* '(general) period of time', and *tempestās* was even used (in the plural) to personify the weather gods, the "Tempestates" (cf. **II.F.1.b.**).

Indo-Iranian continues similar uses of **-tāt-* as found in Italic, reflecting both qualitative abstracts and collectives, though Indic generalized the suffix to thematic stems much more quickly than Latin. Vedic also attests an *i*-stem variant *-tāti-* that eventually replaced *-tāt-* altogether in post-RV texts. Upon examining the use of *-tāti-* in the Rig Veda, it seems clear that *i*-stem inflection in the suffix is secondary, having arisen by reanalysis of an instr. sg. *-tātā* (to *-tāt-*) as loc. sg. of an *i*-stem suffix *-tāti-*, which then spread productively to the acc. sg. *-tātim* and dat. sg. *-tātaye* (cf. **III.C.**). Iranian *-tāt-* was more persistent than in Indic. Avestan used the suffix with a wider variety of base types than found in Indic, and it also became semi-productive to an extent in Young Avestan texts, when *-tāt-* was used with more thematic adjectives than found in Old Avestan (cf. Fig. 6 and 7 in **III.A.**). As in Italic, Avestan also used the individualizing function of *-tāt-* to build theonyms with the sense "the one who possesses the quality of *x*", e.g. *aršt̰āt-*, which signifies the name of a goddess "Uprightness".

Although Greek *-τηϛ* became productive in the form *-ότηϛ* in Attic prose, it shows the most restricted use of **-tāt-* in earlier literature. Aside from the two marginally attested *u*-stems *βραδυτής* 'slowness' and *ταχυτής* 'quickness', Homeric abstracts

already superficially reflect *-οτης* (cf. Fig. 11 in **IV.A.**). But the situation is much more complicated, in that earlier abstracts built to athematic stems may have been remodeled to *-οτης* to take advantage of is metrical shape in the hexameter, if the original abstract was unusable; this may have occurred with *δηϊοτής* 'battle strife', with its unusual oxytone accentuation pointing to athematic derivation in Greek, perhaps ultimately from **δηϊτής* (cf. **IV.C.3.c.**). Additionally, *ἀνδροτής* 'manliness', though apparently displaying the ordinary suffix *-οτης*, may thus exhibit an abstract formation to a very old adjective **h₂ner-* (cf. **IV.C.3.a.**). The Greek abstracts, seemingly not very interesting on the surface, may actually continue some of the same features found in both Latin and Indo-Iranian. Nevertheless, aside from a relatively late use of *νεότᾱς* in Pindar in the collective sense 'body of youth' (cf. **IV.A.3.**), there is no old evidence that *-της* had the same sort of individualizing function found elsewhere.

The individualizing and abstracting functions found in **-tāt-* strongly suggest that the suffix is a sort of secondary *t*-stem formation to an abstract in **-teh₂-*, as has long been thought (cf. **I.D.**). Secondary *t*-stems are well documented in both individualizing and collectivizing functions in PIE (cf. **V.A.**), so such an individualizing formation to a base **-teh₂* is understandable. But rather than simply continuing **-teh₂-t-*, the semantics indicate that **-tāt-* is a "decasuative" formation, in which the *t*-stem is built to the instr. sg., in much the same way as argued for Lat. *salūs* in **V.B.** In this way, PIE **-tāt-*

continues *-*teh*$_2$-*h*$_1$-*t*-, an analysis that accounts for its qualitative semantics of "that which is characterized by *x*" (cf. **V.D.**).

Finally, *-*tāt*- seems to have been usable with a wide variety of base types early on, but everywhere it is found, it shows a relatively strong connection to thematic adjectives. Though the relationship may have begun in PIE, it seems to have been taken to completion in each language individually, rather than already in PIE. Once thematic stems could no longer build *i*-stem abstracts with the "*cvi*"-construction, they had to resort to other suffixes to build abstract nouns, and *-*tāt*- conveniently filled such a role. The resulting relationship of *-*tāt*- to thematic stems thus led to the reanalysis of *-*o-tāt*- to *-*otāt*-, which also fixed accentuation on the thematic vowel, based on the evidence in Indic and Greek, giving the productive suffix forms in Lat. -*itās*, Ved. -*átāt*-/-*átāti*- and Av. -*atāt*-, and Gk. -*ότης* familiar from the handbooks (cf. **V.E.**).

VI.B. Theoretical Import

The analysis of *-*tāt*- as a decasuative secondary *t*-stem *-*teh*$_2$-*h*$_1$-*t*- has interesting implications for the study of PIE word-formation. Beside primary *t*-stems (which are relatively well-studied), secondary *t*-stems are now becoming better understood (cf. especially Nussbaum 1996 and 2004b), and it is now known that formations of this sort played a larger role in PIE word-formation than previously thought. Moreover, it seems to be the case that secondary *t*-stems were one of several

suffix types that could build decasuative derivatives directly to case endings; these give the subsequent derivative the semantics particular to that case form, as found with de-instrumental substantivizations with the meaning "that which is characterized with/by *x*", delocatival substantivizations with the meaning "that which is at/in *x*", e.g. Hitt. *išpant-* 'night' from **k^{w}sp-én-t-* 'that which is at night, nighttime' built to loc. sg. **k^{w}sp-én* 'at night' (cf. **V.B.3.**), and even degenitival substantivizations meaning something like "that which belongs to/in/during *x*", as in Hitt. *šāudišt-* 'suckling' from **s(o)m-u̯et-es-t-* 'that which belongs to the same year' to gen. sg. **s(o)m-u̯et-es* '(born) during the same/current year' (Vine 2009:241).

Unlike the secondary *t*-stems in Hitt. *išpant-* and *šāudišt-* that reflect nominal substantivizations, however, **-tāt-* is a suffix, rather than an independent word. The fact that a complex suffix like **-tāt-* owes its origin ultimately to a reanalyzed secondary *t*-stem based on the existing abstract suffix **-teh$_2$-* may provide insight into how more complex suffix forms arose in late PIE. Despite the lack of word-equations for **-tāt-*, it is clearly the case that the suffix belongs to PIE, rather than reflecting independent creations in three separate branches. The evidence suggests that **-tāt-* was formed relatively late and eventually replaced abstracts in **-teh$_2$-*, before becoming a generalized abstract suffix on its own. Therefore, the secondary *t*-stem machinery that created such a suffix may

have given rise to similar suffix formations in late PIE that have yet to be identified as such, not all of which are found in multiple branches.

VI.C. Further Work

Latin has a number of complex abstract suffixes similar to *-tās*, including the very similar-looking feminine abstracting suffix *-tūs*, e.g. *iuventūs* 'youth' (Pl.+), *virtūs* 'manliness, courage' (Andr.+), and *senectūs* 'old age' (Pl.+). Beside *-tūs* are some Celtic masculine abstracts in **-tūt-*, e.g. OIr. *oítiu* 'youth', *sentu* 'age', and MW *gwendid* 'weakness'. Gothic has a feminine suffix *-dūþs* < **-tūti-* found in *ajukdūþs* 'eternity' and *mikildūþs* 'greatness'. YAv. *gaδōtūt-* 'band of robbers' may also reflect **-tūt-* (but cf. n. 5 in **I.E.**), and so the suffix **-tūt-* may belong in three or even four branches of PIE (Brugmann 1906:453) and may well have a similar origin to that of **-tāt-*.[149] Though the *tūt*-suffix never gained the same level of productivity of **-tāt-* in any of the languages in which it is found, it may ultimately be explained via a de-instrumental derivative to an underlying *tu*-stem. Interestingly, though, Celtic **-tūt-* is masculine, and so the hypothetical decasuative *t*-stem formation needs to be explored in greater detail.

Similar in structure to **-tūt-* is the Latin complex suffix *-itūdō/-itūdinis*, e.g. *altitūdō* 'height' (Cat.+) and *fortitūdō* 'courage' (Ter.+). Much like *-tās*, the suffix *-itūdō* was also very productive in Latin, including reflecting a generalized *-i-* before the suffix,

[149] Cf. **II.B.** for Olsen's (2009) idea that **-tūt-* is based on the root noun **tuh_2-*.

and seems to be relatively old. But unlike *-tās*, *-itūdō* seems to reflect an enlargement with two suffixes, rather than plain **-t-*, including both a dental suffix ending in **-d-* and an *n*-stem suffix **-on/-en-* added to **-tūd-*. A secondary *n*-stem suffix is well represented in Latin *-tiō*, which continues a *ti*-stem abstract augmented with the *n*-stem **-on/-en-* (cf. Nussbaum 2006); but not much is understood regarding *d*-stems and their relationship to *t*-stems. An intermediate stage **-tū-d-* looks very similar to **-tū-t-* and **-tā-t-*, and so the study of secondary *d*-stems beside secondary *t*-stems may yield interesting results for the study of PIE word-formation.[150]

Nussbaum (2004a) has recently examined secondary *d*-stems underlying the Latin abstract suffix *-ēdō* (beside *-īdō* and *-ūdō*), e.g. *absūmēdō* 'consumption' (Pl.) and *gravēdō* 'heaviness, head cold' (Pl.+), which finds a parallel in Greek *-ηδών/-εδών*, e.g. *ἀλγηδών* 'pain' (Soph.+) and *τηκεδών* 'wasting away' (Hom.+), and related adjectives in *-εδανός*, e.g. *ῥῑγεδανός* 'shudder-inducing' (Hom.+). According to Nussbaum, the data in Latin and Greek points to an ablauting suffix **-ed-/*-od-* that built endocentric masculine substantives and feminine abstract nouns in a very similar way to the secondary *t*-stem suffix **-et-/*-ot-* outlined in **V.A.** Since *-itūdō* has a very similar internal structure to **-tāt-* (but with the *n*-stem extension), it may well reflect another

[150] But cf. Olsen (1994) and Pinault (2000, 2002) for the idea that *-tūdō* ultimately goes back to **-tūt-* by the way of **-tūt-$h_3$on-*, with voicing of **-t-* by h_3, which crucially depends on the (controversial) reconstruction of the "Hoffmann" suffix as **-$h_3$on-* with h_3 (cf. Fortson's 2010:124 reconstruction **-Hon-* without specifying laryngeal color).

decasuative formation like the one behind *-*tāt*- and perhaps *-*tūt*-, though built with *-*d*- rather than *-*t*-. In the same way that a de-instrumental origin of *-*tāt*- is supported by Lat. *salūs*, a de-instrumental origin for the suffix *-*itūdō* may be supported by *d*-stems showing presuffixal lengthening (Weiss 2009:304-5), e.g. Lat. *mercēs*, *mercēdis* 'wages' (Pl.+) and *hērēs*, *hērēdis* 'heir' (Naev.+).

If PIE *-*tūt*- and Lat. -*itūdō* can be shown to have a similar origin to *-*tāt*-, the implications for PIE abstract formation would be significant. De-instrumental origins for the suffixes would show that *tu*-stems, in addition to *teh*$_2$-stems, were subject to further abstract derivation, as the suffixes themselves began to fall out of use beside the more characterized complex suffixes. The productive uses of *-*tūt*- and -*itūdō* also indicate that the two suffixes could be the product of reanalysis, much like *-*tāt*-, and that -*itūdō* was also generalized with a linking vowel before the original suffix.

Until now, decasuative formations have been best studied in secondary adjective derivation, e.g. of the de-instrumental type in *-*ih*$_1$*-to*- and *-*uh*$_1$*-to*- (cf. Nussbaum 1996). But the analysis of PIE "*-*tāt*-" in this work shows that decasuative and especially de-instrumental patterns (as with secondary *t*-stems), in addition to participating in adjectival derivation, can participate in secondary abstract derivation, building both masculine substantivizations and feminine qualitative abstracts, as suggested by Nussbaum (2004)

for Gk. πολίτης 'citizen' (Hom.+) < *$polh_1$-ih_1-t- 'the one who inhabits a πόλις', which is an area of study deserving further attention.

References

Conferences

APA	Annual Meeting of the American Philological Association
ECIEC	East Coast Indo-European Conference
WeCIEC	Annual UCLA Indo-European Conference (West Coast Indo-European Conference)

Journals

AJP	*American Journal of Philology*
BSL	*Bulletin de la Société de Linguistique de Paris*
CP	*Classical Philology*
CQ	*Classical Quarterly*
IBS	*Innsbrucker Beiträge zur Sprachwissenschaft*
IF	*Indogermanische Forschungen*
IJDL	*International Journal of Diachronic Linguistics and Linguistic Reconstruction*
JAOS	*Journal of the American Oriental Society*
JHS	*Journal of Hellenic Studies*
JIES	*Journal of Indo-European Studies*
KZ	*Kuhns Zeitschrift*
MSS	*Münchener Studien zur Sprachwissenschaft*
RhM	*Rheinisches Museum für Philologie*
ZPE	*Zeitschrift für Papyrologie und Epigraphik*

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